

Mathematical Biology

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Welcome

This is the webpage of **Mathematical Biology**! Here you will find all the material for the course. Stay tuned for regular updates!

Schedule

	Mon	Tue	Wed	Thu	Fri
13:30	Lecture A209				
14:30		Lecture A224			
15:30			Lab A202		
16:30					

Figure 1: Schedule of the course

Exam's schedule

- 1st call
 - written, Jan 8th, 2025, 10:30 - 13:30 (A208)
 - oral, Jan 10th, 2025, 08:30 - 13:30 (A210)
- 2nd call
 - written, Feb 12th, 2025, 14:00 - 17:00 (A204)
 - oral, Feb 14th, 2025, 08:30 - 13:30 (A208)

Office hours

Any time Mon-Wed, please make an appointment by email first Works best right after the lectures

Exams

The exam will consist of a written test (**50% of the final grade**), an oral exam (**20%**), and a project presentation (**30%**). Specifically:

- **Written Test:** Exercises similar to those covered in class and available on the course website.
- **Oral Exam:** One question at the blackboard on the theoretical part of the course.
- **Project:** Conducted in groups of up to 2 students. Typically involves reading an article, or implementing or studying (presentation of the model, qualitative analysis, simulations) a model not covered in class. The presentation will last about 25 minutes (total, not per student) and will be part of a mini-workshop open to other students in the course and the public. A selection of possible projects and articles will be available on the website.

Topics

The course “Mathematical Modeling” has a dual purpose: on one hand, to introduce students to some basic mathematical models in various areas of biology (demography, ecology, infectious diseases, enzyme reactions, physiology, molecular networks); on the other hand, to provide fundamental knowledge in the analysis and numerical simulation of ordinary and partial differential equations.

Specifically, the first part of the course is dedicated to modeling using ordinary differential equations and introduces various analytical techniques (linearization, equilibria and their stability, bifurcation, regular and singular perturbations).

- **Overview of ordinary differential equations (ODEs):** Solution of linear equations; equilibria and linearized stability; phase plane, limit cycles; numerical schemes for solving ODEs.
- **One- or two-dimensional models** in demography, ecology, epidemiology, and immunology. Non-dimensionalization of variables and parameters.
- **Slow-fast systems**, enzyme reaction models and their simplification using perturbative methods.
- **Bifurcation of equilibria** and application to predator-prey systems and molecular networks. Simplified models of important biological phenomena, such as the cell cycle and glucose-insulin oscillations.
- **Excitable systems:** Hodgkin-Huxley equations (overview) and FitzHugh-Nagumo equations.
- **Parameter estimation** for differential models.

In the second part, partial differential equation models and some techniques for constructing or approximating solutions will be studied. Additionally, some of the most interesting phenomena

of reaction-diffusion equations (traveling wave solutions, Turing mechanism) will be presented in a biological context (morphogenesis).

- **Dynamical systems on networks.** Examples in epidemiology.
- **Introduction to partial differential equations (PDEs):** Solutions by separation of variables. Fourier series. The heat equation and Brownian motion. Eigenfunctions of the Laplacian. Numerical approximation.
- **Skellam and Fisher equations:** Waveform solutions; stationary solutions of the boundary value problem.
- **Stability of stationary solutions** of reaction-diffusion systems and Turing's mechanism for morphogenesis. Conditions for its validity and examples. Chemotaxis: The Keller-Segel model.

Projects

A list of possible projects for the exam

i General remarks

1. First step is *understanding* the problem by reading the article, possibly skipping the most technical steps. Projects can then proceed to complete the analysis (sometimes only sketched in the paper), or discuss alternative assumptions and analyse the resulting models, or simplify the models discussed in the paper, or extend them. Project should include analytical computations, when possible, and numerical simulations.
2. A project on a different topic can certainly be suggested. As a rule of thumb, you can check references or books (Murray, Strogatz, Keener&Sneyd, Iannelli&Pugliese, ...), and find a topic that intrigues you.
3. Note that most articles should be accessible through the University Library. Go to the library web page, then choose Periodicals and look for the journal name. Using Google does not necessarily lead to finding the full-text of the article. If you have problems in getting some of them, please contact me.
4. We can use chatGPT or any other “co-pilot”, if you wish. Just acknowledge it. I use it myself for converting tables, drafting code or a text, and so on. For instance, the introductory lab on Matlab has been produced by chatGPT from some notes I wrote in italian several years ago (plus manual fine tuning).
5. [More possible projects here](#)
6. Enjoy the project!

Paradox of value and Allee effect

In ecology, the Allee effect implies that some species are at increased risk of extinction when their populations become very small. Low population sizes can make it harder for individuals

to find mates, gather food, or defend themselves from predators, thus risking a decline that is intrinsic to the species. While human actions can drive populations to dangerously low numbers, humans do not create the Allee effect itself.

Or could they? One possible mechanism is through the “paradox of value” in economics, where something’s value can drive it toward scarcity. In this project, analyze the model by Courchamp *et al*, which parallels prey-predator dynamics, by examining equilibria, stability, and bifurcation diagrams. Apply this model to scenarios described in their paper.

References

- Courchamp, Franck, *et al*. “[Rarity value and species extinction: the anthropogenic Allee effect.](#)” *PLoS biology* 4.12 (2006): e415.

Full spruce-budworm system

Spruce budworms (*Choristoneura occidentalis*) are a group of closely related pests of conifers, especially spruce. Since 1909, repeated budworm outbreaks have affected the eastern U.S. and Canada, with tree-ring studies indicating that such cycles have recurred approximately every 30 years since the 16th century.

In the 1970s, researchers proposed that these cycles arise from interactions between budworms, their avian predators, and spruce foliage. The system has two distinct timescales: budworm dynamics are fast, while bird populations and foliage vary more slowly.

Extend the lecture analysis by incorporating these slow dynamics and, if possible, diffusion effects. Show that, under certain assumptions, periodic oscillations can emerge.

References

- Ludwig, D., Dixon, D.J., and Holling, C.S. “[Qualitative analysis of insect outbreak systems: the spruce budworm and forest.](#)” *The Journal of Animal Ecology* (1978): 315-332.
- Ludwig, D., Aronson, D.G., and Weinberger, H.F. “[Spatial patterning of the spruce budworm.](#)” *Journal of Mathematical Biology* 8.3 (1979): 217-258.

Type-1 diabetes

Type I diabetes is a multifactorial disease of epidemic proportions. The disease consists in the loss of pancreatic β -cells, which are responsible for the secretion of insulin and in turn for the maintenance of glucose homeostasis. Diabetic patients need regular insulin injections and some care in their diet to avoid serious conditions.

A promising new treatment consists in the regeneration of β -cell mass from residual islet cells or replacement by new cells derived from stem cells. However, ensuring the correct mass transport and oxygenation during the preparation is a challenge.

This project aims at studying a simple model for diffusion and consumption of oxygen in the islets. The respiration of oxygen is modelled with a Michaelis-Menten type law. Insulin secretion is then dependent on the partial pressure of oxygen, and thus can be estimated from the model.

Possible implementation

1. Read the paper from Ernst *et al.* [1], specifically focusing on Section “Mass transport theory”
2. Derive a 1D model for the mass transport, using equation (7) and (8)
3. Simulate the model with appropriate boundary conditions
4. The authors eliminated the transient phase in the model, so the parabolic equation becomes an elliptic equation. Try to simulate the transient with some reasonable initial conditions for the oxygen.
5. Estimate insulin secretion from (9) and the average from the islet from (10).
6. The average insulin secretion will depend on the size L of the islet. Plot it versus L and see whether there is an optimal size for it (compare to [2]).
7. If you like scientific computing, go to the next level and try to make a 2D simulation like in the original paper.

References

- [1] Ernst *et al.* “[A predictive computational platform for optimizing the design of bioartificial pancreas devices](#)”, *Nat. Comm.* 13.6031 (2022).
- [2] Komatsu, Hirotake, *et al.* “[Oxygen environment and islet size are the primary limiting factors of isolated pancreatic islet survival.](#)” *PloS one* 12.8 (2017): e0183780.
- [3] Avgoustiniatos, Efstathios S., and Clark K. Colton. “[Effect of external oxygen mass transfer resistances on viability of immunoisolated tissue.](#)” *Annals of the New York Academy of Sciences* 831.1 (1997): 145-166.

Gate channels dynamics

Ion channels are large proteins embedded in the cellular membrane. They open and close depending on a number of factors, including the transmembrane voltage. An effective way to describe ion channels is through Markov models, where the channels (or each channel unit) can be in different states. The probability of transition from one state to another is then assigned, as a function of the voltage. The channel will be open (and so, a current will be present) only when it is in some specific states.

Since the gating is probabilistic, the single-channel current will fluctuate over time. With several channels, the total current will be close the mean, and the variance is small.

The idea of this project is to simulate the gating dynamics at single channel level, and then observe how the total current varies as the number of channel increases. It is an important topic, because drugs or genetic diseases can affect the gating dynamics.

Implementation

1. Study Sections 2.9 and 3.5 of Keener & Sneyd (2009).
2. Simulate the stochastic behavior of a single 2-state ion channel (the channel can only be open or closed), with fixed transition rates α and β . The output should be a function $S(t)$ that can only be 0 or 1 (closed or open).
3. Simulate N independent, identically distributed 2-state channels, and visualize the total fraction of open channels, $p(t) = \frac{1}{N} \sum S_i(t)$. This is proportional to the average current from a population of channels.
4. Show that for large N the function $p(t)$ converges to the solution of the Master equation derived in class.
5. Evaluate the variance of $p(t)$, and show that for N moderately small it can be quite large.

References

- Rudy, Yoram, and Jonathan R. Silva. “[Computational biology in the study of cardiac ion channels and cell electrophysiology](#).” *Quarterly reviews of biophysics* 39.1 (2006): 57-116.
- Tveito, Aslak, *et al.* “[Computing rates of Markov models of voltage-gated ion channels by inverting partial differential equations governing the probability density functions of the conducting and non-conducting states](#).” *Mathematical Biosciences* 277 (2016): 126-135.
- Tveito, Aslak, and Glenn T. Lines. *[Computing characterizations of drugs for ion channels and receptors using Markov models](#)*. Springer Nature, 2016.

Love dynamics

The overall structure of prey-predator and food-chain models is sometimes reused in completely different contexts: an example is love dynamics. Sergio Rinaldi proposed several models for describing the “emotional” dynamics in several famous novels (*Pride & Prejudice* by Jane Austen), movies (*Jules et Jim* by F. Truffaut or *the Beauty and the Beast*), and poetry (Laura and Petrarca)

The idea is to represent the “love” as a function of time $L(t)$, and then provide an ODE for describing its dynamic over time. When more actors are interacting, we have more variables. It is interesting that in the case of 3 or more individuals, the dynamic of love may be chaotic, as in the food-chain model. This is what happens in a “love triangle”, as in the movie from F. Truffaut.

The project consists in reading the literature listed below (or a selected subtopic) and present the results with some simulations. It is also possible to think about other novels, movies, comics, manga, games, not necessarily those listed above. There is also a [YouTube video](#) of the Christmas lecture from Prof. Rinaldi, where he presents the topic.

References

- Rinaldi, Sergio. “[Laura and Petrarch: An intriguing case of cyclical love dynamics.](#)” *SIAM Journal on Applied Mathematics* 58.4 (1998): 1205-1221.
- Rinaldi, Sergio, and Fabio Della Rossa. “[Modelli Matematici di Storie D’Amore.](#)” *Quaderni di Matematica PoliMI* (2017).
- Dercole, Fabio, and Sergio Rinaldi. “[Love stories can be unpredictable: Jules et Jim in the vortex of life.](#)” *Chaos: An Interdisciplinary Journal of Nonlinear Science* 24.2 (2014): 023134.
- Rinaldi, Sergio, Pietro Landi, and Fabio Della Rossa. “[Small discoveries can have great consequences in love affairs: the case of Beauty and the Beast.](#)” *International Journal of Bifurcation and Chaos* 23.11 (2013): 1330038.

SIR parameter identifiability

We have studied the SIR model of epidemics in great detail. In the real world application, when dealing with data, we need to fit or identify the parameters of the model, in order to make good prediction. Secondly, we also need to provide a confidence interval for our predictions, as data, used to fit the parameters, is always affected by uncertainty.

More generally, even before starting fitting the parameters, we should ask ourselves: can we really fit the parameters in a unique way? In other words, given a partial observation of the output (e.g., number of active cases over time), are we able to determine all the parameters?

Sometimes we can only deduce a combination of them. In this context we talk about *structural identifiability*.

This topic is studied in the paper by Piazzola *et al.* (2021). For this project, you should read the article and present some results and simulations.

References

- Piazzola, Chiara, Lorenzo Tamellini, and Raúl Tempone. “[A note on tools for prediction under uncertainty and identifiability of SIR-like dynamical systems for epidemiology.](#)” *Mathematical Biosciences* 332 (2021): 108514.

Regulation of gene control networks

The concept of [operon](#) was introduced by Jacob and coworkers in the 1960s to explain how gene regulation could work. Two important cases were identified in the regulation of tryptophan concentration and lactose metabolism in *E. coli*. The example of the *Lac operon* we have seen briefly during the lecture. A key aspect of the Lac operon is the switch-type regulation of the expression of the gene.

Gene regulation can be described with feedback networks, as for enzymatic reaction. There are several important example in the literature, such as the *trp*-operon. Some are listed below. There are also some general receipt for building oscillators, see the work by Novák and Tyson (2008). You can select one or more and present it, along with simulations and/or bifurcation diagrams.

References

- Bliss, Richard D., Page R. Painter, and Allen G. Marr. “[Role of feedback inhibition in stabilizing the classical operon.](#)” *Journal of theoretical biology* 97.2 (1982): 177-193.
- Santillán, Moisés, and Michael C. Mackey. “[Dynamic regulation of the tryptophan operon: a modeling study and comparison with experimental data.](#)” *Proceedings of the National Academy of Sciences* 98.4 (2001): 1364-1369.
- Yildirim, Necmettin, and Michael C. Mackey. “[Feedback regulation in the lactose operon: a mathematical modeling study and comparison with experimental data.](#)” *Biophysical journal* 84.5 (2003): 2841-2851.
- Yildirim, Necmettin, *et al.* “[Dynamics and bistability in a reduced model of the lac operon.](#)” *Chaos: An Interdisciplinary Journal of Nonlinear Science* 14.2 (2004): 279-292.
- Novák, Béla, and John J. Tyson. “[Design principles of biochemical oscillators.](#)” *Nature reviews Molecular cell biology* 9.12 (2008): 981-991.

Zombies

A common joke of SIR models is to substitute the infected individuals with zombies. The simplest model is however suboptimal, for instance because zombies never recover or are difficult to kill. Hence, the endpoint is always a large-scale invasion of zombie with no hope for humans. That's good of a B-movie but quite boring for a mathematician.

It would be interesting to present the topic in a more mathematical and “biological” manner, as done by Munz *et al.* [1] or by McGahan *et al.* [2]. Starting from the paper, you can include the effect of diffusion in the model [3].

References

- [1] Munz, Philip, *et al.* “[When zombies attack!: mathematical modelling of an outbreak of zombie infection.](#)” *Infectious disease modelling research progress* 4 (2009): 133-150.
- [2] McGahan, Ian, James Powell, and Elizabeth Spencer. “[28 models later: model competition and the zombie apocalypse.](#)” *Bulletin of Mathematical Biology* 83 (2021): 1-33.
- [3] Chinviriyasit, Settapat, and Wirawan Chinviriyasit. “[Numerical modelling of an SIR epidemic model with diffusion.](#)” *Applied Mathematics and Computation* 216.2 (2010): 395-409.

Cell cycle

The cell-cycle (growth and then mitosis) can be modeled quite effectively with a simple gene regulation that leads to a non-reversible switch (once divided, the cell cannot return to the original state.) There are various models, and some are presented also in Section 10.3 of Keener and Sneyd (2009).

From there, one can move either to understand the more biologically realistic models developed by the Tyson and Novák and collaborators, or to analyse in greater detail the basic model, also for different parameter values.

References

- Tyson, John J., and Bela Novak. “[Regulation of the eukaryotic cell cycle: molecular antagonism, hysteresis, and irreversible transitions.](#)” *Journal of theoretical biology* 210.2 (2001): 249-263.
- Csikász-Nagy, Attila, *et al.* “[Analysis of a generic model of eukaryotic cell-cycle regulation.](#)” *Biophysical journal* 90.12 (2006): 4361-4379.

- Novák, Béla, and John J. Tyson. “[Design principles of biochemical oscillators.](#)” *Nature reviews Molecular cell biology* 9.12 (2008): 981-991.

Glucose-insulin oscillations

An example of a negative feedback loop involves insulin and glucose in the bloodstream. Intake of glucose (say, as a result of a meal) causes the pancreas to secrete more insulin, which then lowers the level of glucose by helping the glucose to be metabolized in the body. This feedback loop causes oscillations like those seen in class, even when a person is hooked up to a constant intravenous glucose supply with no meals.

An interesting model has been proposed by Sturis *et al.*, presented also in the textbook by Keener and Sneyd (Section 1.5 and Section 16.3.2). It is also interesting the interaction between the circadian clock (seen during the lab) and the insuline production, as studied by Perelis *et al.* From there, one can modify/extend the model looking at the vast literature about glucose-insulin dynamics.

References

- Sturis, Jeppe, et al. “[Computer model for mechanisms underlying ultradian oscillations of insulin and glucose.](#)” *American Journal of Physiology-Endocrinology And Metabolism* 260.5 (1991): E801-E809.
- Tolić, Iva Marija, Erik Mosekilde, and Jeppe Sturis. “[Modeling the insulin–glucose feedback system: the significance of pulsatile insulin secretion.](#)” *Journal of theoretical biology* 207.3 (2000): 361-375.
- Perelis, Mark, et al. “[Pancreatic cell enhancers regulate rhythmic transcription of genes controlling insulin secretion.](#)” *Science* 350.6261 (2015): aac4250.

Self-organization and pattern formation

A remarkable case of self-organization, very much studied in mathematical biology, is shown by *Dictyostelium discoideum*. A quick introduction to the problem, and to the existing literature is provided by Painter (2019):

Cellular slime molds, the “social amoebae”, form a celebrated group of soil-dwelling microorganisms whose rise to prominence stems from a life cycle that straddles individuality and multicellularity. [...] In its vegetative phase individual cells swarm, consume and divide until food is depleted. Starvation heralds a remarkable shift, as the dispersed population accumulates into a multicellular entity that passes through various stages: differentiating and sorting into pre-spore and pre-stalk

types, transforming into a migrating “slug” and eventually a fruiting body. This final act invokes apparent altruism, with spore cells shaped into a ball, encased and suspended by a thin trunk of sacrificed stalk cells. Transport of spores to a new location restarts vegetative growth.

A project could start exploring some of the model proposed and the possible outcomes.

References

- Painter, Kevin J. “[Mathematical models for chemotaxis and their applications in self-organisation phenomena.](#)” *Journal of theoretical biology* 481 (2019): 162-182.

Lotka-Volterra with multiple prey species

Consider the system with two prey species (H_1 and H_2) and one predator with interactions of Lotka-Volterra type:

1. List the possible equilibria with 1 or 2 species present, and examine the conditions for their existence and stability. As for positive equilibria, find the equations it must satisfy, without discussing its existence (in the positive quadrant) or its stability.
2. Show by numerical examples that these two phenomena may occur
 - predator-mediated coexistence: without the predator the two preys do not coexist (not necessarily at equilibrium), but will coexist together with the predator;
 - apparent competition: without the predator the preys do not compete and could coexist, but the introduction of the predator causes extinction of one of them.
3. With the use of Matcont, explore numerically the system, finding parameter values when a positive equilibrium exists; try to see whether it can undergo Hopf bifurcation, giving rise to periodic solutions.

In Section 9.4 of Iannelli, Pugliese (2014) there are some results and literature references. It is however better to start the analysis from scratch, and possibly only afterwards check the literature.

Prey-predator models with diffusion

An article by Petrovskii and Malchow (2001) suggests that diffusion in prey-predator models leads to spatio-temporal chaos. A more general review by Sharrett and Smith also addresses the same model. One should be able to reproduce the results by Petrovskii and Malchow (care in numerical methods is required), understand their potential ecological relevance (looking also at recent references) and discuss what is meant by spatiotemporal chaos.

References

- Petrovskii, Sergei V., and Horst Malchow. “[Wave of chaos: new mechanism of pattern formation in spatio-temporal population dynamics.](#)” *Theoretical population biology* 59.2 (2001): 157-174.
- Medvinsky, A.B., *et al.* “[Spatiotemporal complexity of plankton and fish dynamics.](#)” *SIAM review* 44.3 (2002): 311-370.

Control of dengue infection

A technique that has been sometimes used and often proposed for the control of insect populations is that of sterile male release. There are several papers that present mathematical models of the problems; one reasonably clear is by Esteva and Yang (2005).

One may start from the model in Esteva and Yang (2005) and try to make it more realistic (a more recent paper is by Stugarek *et al.* (2019) but there should be others considering other biological details).

Alternatively, you could focus on controlling dengue. An interesting method for controlling dengue infection is via release of *Wolbachia*-infected mosquitoes that are less likely to get infected by the dengue virus, and thus to transmit the infection to humans. A nice review of the possible strategies and of their potential is provided by Bull and Turelli (2016), while a more mathematical analysis is provided by Hughes and Britton (2013). Schmidt *et al.* (2017) present the result of the first experimental introduction in Australia and dos Santos *et al.* (2022) the results in Brazil.

References

- Esteva, Lourdes, and Hyun Mo Yang. “[Mathematical model to assess the control of *Aedes aegypti* mosquitoes by the sterile insect technique.](#)” *Mathematical biosciences* 198.2 (2005): 132-147.
- Stugarek, Martin, Hervé Bossin, and Yves Dumont. “[On the use of the sterile insect release technique to reduce or eliminate mosquito populations.](#)” *Applied Mathematical Modelling* 68 (2019): 443-470.
- Bull, James J., and Michael Turelli. “[Wolbachia versus dengue: evolutionary forecasts.](#)” *Evolution, medicine, and public health* 2013.1 (2013): 197-207.
- Hughes, Harriet, and Nicholas F. Britton. “[Modelling the use of Wolbachia to control dengue fever transmission.](#)” *Bulletin of mathematical biology* 75.5 (2013): 796-818.
- Schmidt, Tom L., *et al.* “[Local introduction and heterogeneous spatial spread of dengue-suppressing *Wolbachia* through an urban population of *Aedes aegypti*.](#)” *PLoS biology* 15.5 (2017): e2001894.

- dos Santos, Gabriel Ribeiro, *et al.* “[Estimating the impact of the wMel release program on dengue and chikungunya incidence in Rio de Janeiro, Brazil.](#)” medRxiv (2022).

Spatial patterns in vegetation

In many semi-arid environments, vegetation cover is sparse, and is self-organized into large-scale spatial patterns. In particular, banded vegetation is typical on hillsides. Klausmeier proposed a very influential model to explain such patterns. Starting from a (partial) analysis and simulation of the model, you should explore some possible extensions, based also on the many papers referencing it.

A stunning example is provided by [Fairy circles](#), a common formation in Namibia. You can find online several discussions on how to explain the big “Mystery of Fairy Circles”. (Maybe also [aliens](#) are involved, who knows...) There are several (scientific) papers trying to address the mystery. The idea is to use self-organization in vegetation and pattern formation, a topic related to the Turing instability we have seen during the lectures. A recent work by Tarnita *et al.* (2020) may have solved the issue. (See also this [Quanta Magazine](#) interview of the first author, and you may also listen to the [Podcast](#) where Steven Strogatz interviews Corina Tarnita on the broader subject of mathematical biology.)

References

- Klausmeier, Christopher A. “[Regular and irregular patterns in semiarid vegetation.](#)” *Science* 284.5421 (1999): 1826-1828.
- Tarnita, Corina E., *et al.* “[A theoretical foundation for multi-scale regular vegetation patterns.](#)” *Nature* 541.7637 (2017): 398-401.
- Martinez-Garcia, Ricardo, *et al.* “[Integrating theory and experiments to link local mechanisms and ecosystem-level consequences of vegetation patterns in drylands.](#)” *Chaos, Solitons & Fractals* (2022): 112881.

Can we program our cells?

Synthetic biology is an interdisciplinary field that focuses on designing, constructing, and modifying biological systems or organisms to create new, artificial life forms or to improve existing ones for practical purposes, such as medicine, energy production, and more. It involves the application of engineering principles and genetic techniques to create customized biological functions or products.

The topic is nowadays incredibly vast [2]. You could spend some time listening to the podcast linked below [4], a nice interview with Steven Strogatz (expert in dynamical systems) as host

and Michael Elowitz (leading expert in synthetic biology) as guest. Then, you could try to study one application, for instance the genetic toggle [1] or cell differentiation [3] (Figure 2A from this paper and reported above should be familiar to you.)

References

- [1] Gardner, Timothy S., Charles R. Cantor, and James J. Collins. “[Construction of a genetic toggle switch in *Escherichia coli*](#).” *Nature* 403.6767 (2000): 339-342.
- [2] Khalil, Ahmad S., and James J. Collins. “[Synthetic biology: applications come of age](#).” *Nature Reviews Genetics* 11.5 (2010): 367-379.
- [3] Wu, Fuqing, *et al.* “[Engineering of a synthetic quadrastable gene network to approach Waddington landscape and cell fate determination](#).” *Elife* 6 (2017): e23702.
- [4] Podcast <https://www.quantamagazine.org/can-we-program-our-cells-20230308/>

Pattern formation in inflammation

Skin rash, such as hives, often comes in patterns. It has been hypothesised that rash could be related to a Turing-like instability of the inflammatory response. This can be done by incorporating inhibition (represented by an anti-inflammatory cytokine) into a classical Keller-Segel chemotaxis model. In this way, it is possible to produce a variety of spatial patterns.

This project should explore the 2D model proposed in the paper, and review the stability analysis. You could also focus on simulations in 1D or 2D.

References

- Penner, Kevin, Bard Ermentrout, and David Swigon. “[Pattern formation in a model of acute inflammation](#).” *SIAM Journal on Applied Dynamical Systems* 11.2 (2012): 629-660.

Cybergenetics

The [Kapitza pendulum](#), also known as the inverted pendulum, is a mechanical system in which an unstable equilibrium (the vertical position) becomes stable. The Kapitza pendulum is stabilized rapid vertical oscillations of the pivot point, causing the pendulum to remain balanced in an unstable equilibrium. The Kapitza pendulum is a classic example of a non-linear dynamical system and has applications in physics and engineering to study stability and control theory.

Recently, Lugagne and others suggested to use a similar principle to stabilize an unstable equilibrium of the genetic toggle switch. Rather than rapid oscillations, the control is performed via a feedback loop.

Try to repeat some numerical simulations from the paper.

References

- Lugagne, Jean-Baptiste, *et al.* “[Balancing a genetic toggle switch by real-time feedback control and periodic forcing.](#)” *Nature communications* 8.1 (2017): 1671.

Part I

Lectures

1 Bathtub model

1.1 The bathtub model

The models of Newtonian physics are made of differential equations built starting from the second law of the dynamics. The structure of the models discussed here is instead simpler; they are based on the “balance equation of the bathtub”: if $Q(t)$ is the quantity of a substance in the bathtub we have

$$\frac{dQ}{dt} = Q'(t) = I(t) - O(t),$$

where

- $I(t)$ is the *input rate* (quantity that enters per unit time)
- $O(t)$ is the *output rate* (quantity that leaves per unit time).

To be more precise, the assumption is that, if $I_{(t,t+\Delta t)}$ is the quantity that enters in the interval $(t, t + \Delta t)$, we have $I_{(t,t+\Delta t)} = I(t)\Delta t + o(\Delta t)$, where $o(\Delta t)$ is a higher order infinitesimal than Δt . Hence:

$$I(t) = \lim_{\Delta t \rightarrow 0} \frac{I_{(t,t+\Delta t)}}{\Delta t}.$$

The input rate $I(t)$ is like an instantaneous velocity: the quantity entered in a given time, when that time becomes very small. Hence $I(t)$ is measured in $[C][t^{-1}]$ units where $[C]$ represents the concentration of the quantity Q . Similarly for the exit rate $O(t)$.

Let us start from a very simple example. Assume $I(t) = \Lambda$ constant input flux; $O(t) = \gamma Q(t)$, i.e. exit flux is proportional to the quantity present at the moment; the proportionality constant γ is often called the *exit rate* and has the dimension $[t^{-1}]$, the inverse of time. From these assumptions we get:

$$Q'(t) = \Lambda - \gamma Q(t), \tag{1.1}$$

supplemented with some *initial condition*

$$Q(0) = Q_0.$$

The solution is:

$$Q(t) = e^{-\gamma t} Q_0 + \frac{\Lambda}{\gamma} (1 - e^{-\gamma t}).$$

 Exercise

Solve Equation 1.1 with the method you prefer.

 Exercise

Solve Equation 1.1 with the general formula for linear ODEs, by first defining the matrix exponential (here, just a scalar function).

Note that if $\Lambda = 0$ (no input), the solution is simply

$$Q(t) = Q_0 e^{-\gamma t}.$$

This means that the survival time of a molecule initially present follows the exponential distribution:

$$\mathbb{P}[\text{a molecule present at time 0 is present at time } t > 0] = \frac{Q(t)}{Q_0} = e^{-\gamma t}.$$

From the properties of the exponential distribution, we obtain that the mean survival time $\mathbb{E}[T] = 1/\gamma$; hence the exit rate γ can be interpreted as the inverse of the mean survival time.

To be more precise, let us define a continuous random variable T , which measures the lifetime of a particle present in the bathtub. Then, the cumulative distribution $F(t)$ of T is given by

$$\begin{aligned} F(t) &= \mathbb{P}[T \leq t] \\ &= 1 - \mathbb{P}[T > t] \\ &= 1 - \mathbb{P}[\text{a molecule present at time 0 is present at time } t > 0] \\ &= 1 - e^{-\gamma t}. \end{aligned}$$

So, we indeed have an exponential distribution. The probability density function is:

$$f(t) = F'(t) = \gamma e^{-\gamma t},$$

and the expectation is:

$$\mathbb{E}[T] = \int_0^\infty t f(t) dt = \frac{1}{\gamma}.$$

 Exercise

Compute the above integral explicitly.

1.2 Malthus equation

The metaphor of the bathtub can be used to model the dynamics of a population. Neglecting all differences among individuals (due to age, sex, genetic,...) we can represent a population through its size $N(t)$; this will increase through inputs due to births and outputs due to deaths (if immigration and emigration are not considered). Hence

$$N'(t) = B(t) - D(t),$$

where $B(t)$ = births and $D(t)$ = deaths.

Malthus model assumes

- within a (short) time period of length Δt , each individual gives, on average, birth to $\beta\Delta t$ new individuals; hence $B(t) = \beta N(t)$;
- within the same time period Δt , each individual has probability $\mu\Delta t$ of dying; hence $D(t) = \mu N(t)$.

We get the following equation

$$N'(t) = \beta N(t) - \mu N(t) = (\beta - \mu)N(t),$$

that represents the *Malthus model*. The parameter β is known as *fertility rate*, while μ is the *mortality rate*. Finally,

$$r = \beta - \mu$$

is the (instantaneous) *growth rate* and is also called *Malthus parameter* or *biological potential* of the population.

With the initial condition

$$N(0) = N_0,$$

the evolution of the population is completely determined. In fact, the solution is

$$N(t) = N_0 e^{rt},$$

and we see that the population will go to extinction or will grow without limits if $r < 0$ or $r > 0$, respectively. If instead $r = 0$, the population size is constant (births and deaths compensate.)

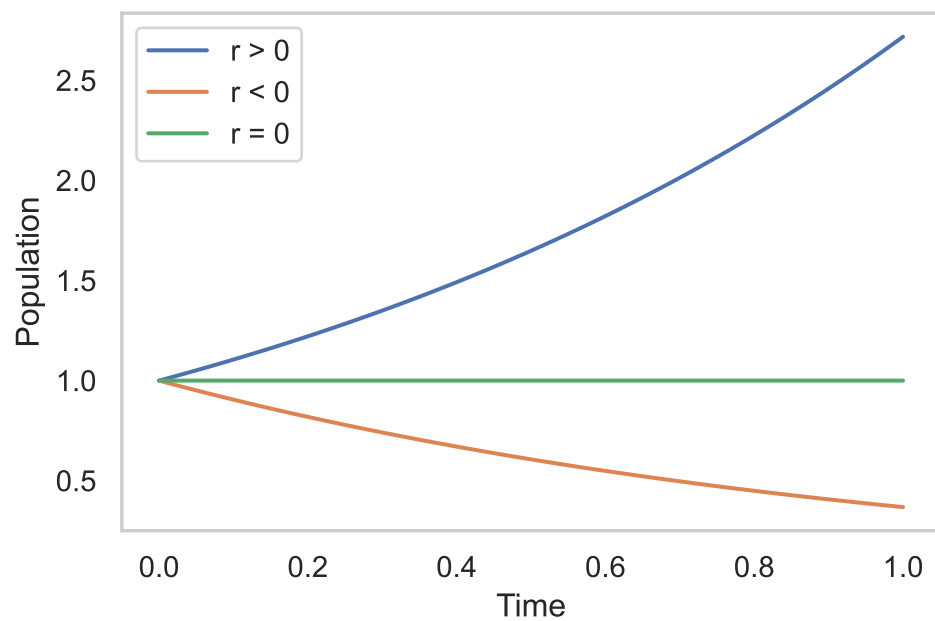


Figure 1.1: Example of solutions of Malthus equation

2 Population dynamics

This lecture is based on [ip14, Section 1.1-1.4]

2.1 Population growth model

We have seen, using the bathtub analogy, that the *fundamental balance equation of population dynamics* take the form

$$N'(t) = B(t) - D(t) + I(t) - E(t),$$

where at time t we have that

- $N(t)$ is the population,
- $B(t)$ is the birth rate,
- $D(t)$ is the death rate,
- $I(t)$ is the immigration rate, and
- $E(t)$ is the emigration rate.

There is also an integral interpretation, in fact:

$$N(t) - N(t_0) = \underbrace{\int_{t_0}^t B(s) \, ds}_{\text{births}} - \underbrace{\int_{t_0}^t D(s) \, ds}_{\text{deaths}} + \underbrace{\int_{t_0}^t I(s) \, ds}_{\text{immigration}} - \underbrace{\int_{t_0}^t E(s) \, ds}_{\text{emigration}}.$$

A population growth model is an ODE of the above form with some specific form for each term above. Note that we could provide functions above explicitly, as function of time alone. However, it should be clear that some of them like $B(t)$ or $D(t)$ should depend on N .

2.1.1 Malthus model

Quoting Thomas R. Malthus (1766-1834):

Population, when unchecked, increases in a geometrical ratio. Subsistence increases only in an arithmetical ratio. A slight acquaintance with numbers will show the immensity of the first power in comparison of the second.

[Essay on the Principle of Population, 1798](#)

Malthus' model is mathematical formulation of the above statement. We already derived Malthus model, but let us recall the hypotheses.

1. The population is *homogeneous*, that is all individuals are identical. We have a single class to represent them, that is $N(t)$.
2. The population is *isolated*, so $E(t) = I(t) = 0$.
3. The habitat is *invariant*, so resources and life conditions are not affected by the environment nor the population itself.
4. The population is *very large*, so we can consider continuous functions.
5. On a short time scale Δt , each individual gives birth to $\beta\Delta t$ new individuals, $B(t) = \beta N(t)$.
6. On a short time scale Δt , each individual has probability $\mu\Delta t$ of dying, $D(t) = \mu N(t)$.

The non-negative parameters β and μ are the fertility and mortality rate, respectively. From hypothesis 4, we have that μ and β are constant. We introduce the *growth rate*

$$r = \beta - \mu,$$

also called *Malthus parameter* or *biological potential*.

The Malthus' model reads:

$$N' = rN, \quad \Rightarrow \quad N(t) = e^{rt}N(0),$$

thus, when $r > 0$, the growth of population is geometrical and unbounded, as predicted by Malthus.

2.1.2 Expected life

We have seen that, in absence of births, the population goes like

$$N(t) = e^{-\mu t}N_0,$$

thus, we could say that the probability of surviving up to time t is $e^{-\mu t}$. Specifically, the *life expectancy* L is a random variable such that

$$\mathbb{P}[L > t] = e^{-\mu t}.$$

Thus, the cumulative distribution of L is

$$F_L(t) = \mathbb{P}[L < t] = 1 - \mathbb{P}[L > t] = 1 - e^{-\mu t},$$

and the probability density function is

$$f_L(t) = F'_L(t) = \mu e^{-\mu t}.$$

We conclude that $L \sim \text{Exp}[\mu]$, the exponential distribution. The *average life expectancy* is:

$$\mathbb{E}[L] = \int_0^\infty s f_L(s) \, ds = \int_0^\infty s \mu e^{-\mu s} \, ds = \frac{1}{\mu}.$$

We have another interpretation of μ : it is the reciprocal of the expected life time.

2.1.3 Basic reproduction number

Let us rescale the equation and put it in non-dimensionalized form. This is a fundamental step in general, because

1. it reduces the number of parameters,
2. it removed scale effects (units are removed),
3. it highlights the determining factors of the model (maybe what matters is not this or that parameter, by their ratio or sum).

Here, we rescale as follows:

$$\tau = \mu t, \quad u = N,$$

so that time is now in units of “expected life time”: $\tau = 1$ means $t = \mu^{-1} = \mathbb{E}[L]$. We have that

$$N' = \frac{dN}{dt} = \frac{du}{\mu^{-1}d\tau} = \mu \frac{du}{d\tau} = \mu \dot{u}.$$

We use the “dot” notation $\dot{u}$ for the derivative for the non-dimensional form, just to remember that now the time is τ and not t . We finally obtain:

$$\dot{u} = \mu^{-1} N' = \frac{\beta}{\mu} N - N = (R_0 - 1)u,$$

where we defined the *basic reproduction number*

$$R_0 = \frac{\beta}{\mu} = \beta \mathbb{E}[L].$$

We could interpret it as the average number of newborns produced by one individual during his whole life. Note that R_0 is non-dimensional.

Exercise

Why it does not make much sense to use the scaling $\tau = rt$?

2.1.4 Migration

In the presence of migration, say with a constant rate, we have the ODE:

$$N' = rN + m = f(N), \tag{2.1}$$

where $m = I - E$. If positive, there is a net immigration, otherwise emigration.

In order to study how the model will behave, we have 3 options:

1. Solve the problem analytically, that is finding $N(t) = \dots$ explicitly.
2. Solve the problem numerically, which is always possible.
3. Study the problem qualitatively.

The last option has the advantage that we can be generic, there is no need for a specific value of the parameters or the initial condition. The qualitative study consists in the following steps:

1) Fixed points of the system. A fixed point or equilibrium (see Definition A.12) is a constant solution of the ODE. We can find it by setting the right hand side to zero:

$$N' = 0 \quad \Leftrightarrow \quad rN + m = 0.$$

The model has a single equilibrium for

$$N = N^* = -\frac{m}{r} = \frac{m \mathbb{E}[L]}{1 - R_0}$$

2) Biological feasibility. Equilibria must be biologically feasible. For this model, we need to check that N^* is non-negative, otherwise it doesn't make sense biologically speaking. Therefore

$$N^* \geq 0, \quad \Leftrightarrow \quad m \text{ and } r \text{ have opposite sign and } r \neq 0.$$

3) Local stability. Informally, an equilibrium is *locally stable* when, starting from a neighborhood of it, the solution stays close to it for $t \rightarrow \infty$. It is *asymptotically stable* when the solution converges toward the equilibrium for $t \rightarrow \infty$. It is *unstable* otherwise. See Definition A.13 for a more precise statement.

The local stability is determined by the sign of the derivative of the right hand side, that is $f'(N^*)$ for $f(N) = rN + m$. In general (see Section A.5.5),

- when $f'(N^*) < 0$, the equilibrium is asymptotically stable, and
- when $f'(N^*) > 0$, the equilibrium is unstable.

To see this in this specific case, let us define $w(t) = N(t) - N^*$. Then,

$$w' = N' = rN + m = r(N - N^*) = rw.$$

Note that $f'(N) = r$. We have that:

$$w(t) = w_0 e^{rt},$$

hence,

- if $r < 0$, then $w \rightarrow 0$ for all w_0 , so $N(t) \rightarrow N^*$. The equilibrium is locally asymptotically stable.
- if $r > 0$, then $w \rightarrow \infty$ and the equilibrium is unstable.

4) Global stability. What if we start very far away from the equilibrium? In this particular case, with a linear ODE, the local stability argument applies also globally, thus the equilibrium is globally attractive. But for general, nonlinear ODEs this may not be the case, so we perform the analysis anyway. Note that if $N(0) = N_0 > N^*$, then

$$N'(0) = rN(0) + m = r(N_0 - N^*) < 0,$$

so the derivative of the solution is negative (assuming $r < 0$). Furthermore, for $N(t) > N^*$, the derivative is always negative. So, the solution must be monotonically decreasing. But the solution is bounded from below by the equilibrium $N(t) = N^*$, so we conclude that:

$$N(t) \rightarrow N^*$$

for all $N_0 \geq N^*$. Symmetrically, when $N_0 < N^*$, the derivative is positive and stays positive for all t , so the solution is monotonically increasing. Hence:

$$N(t) \rightarrow N^*$$

for all N^0 . The equilibrium is therefore *globally* stable when $r < 0$.

5) Phase portrait. The phase portrait of a dynamical system is the collection of all possible orbits. Here, the phase space is $\Omega = [0, \infty)$. The only equilibrium we have, N^* , is a *barrier* to other orbits, because orbits cannot intersect (See Proposition A.2). Therefore:

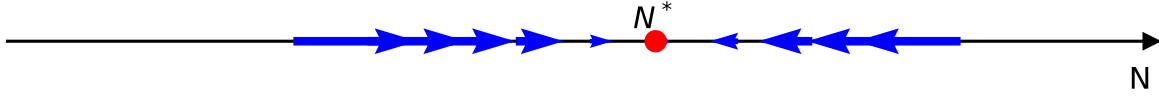


Figure 2.1: Phase space

For this problem we actually have the general solution

$$N(t) = (N_0 - N^*)e^{rt} + N^*.$$

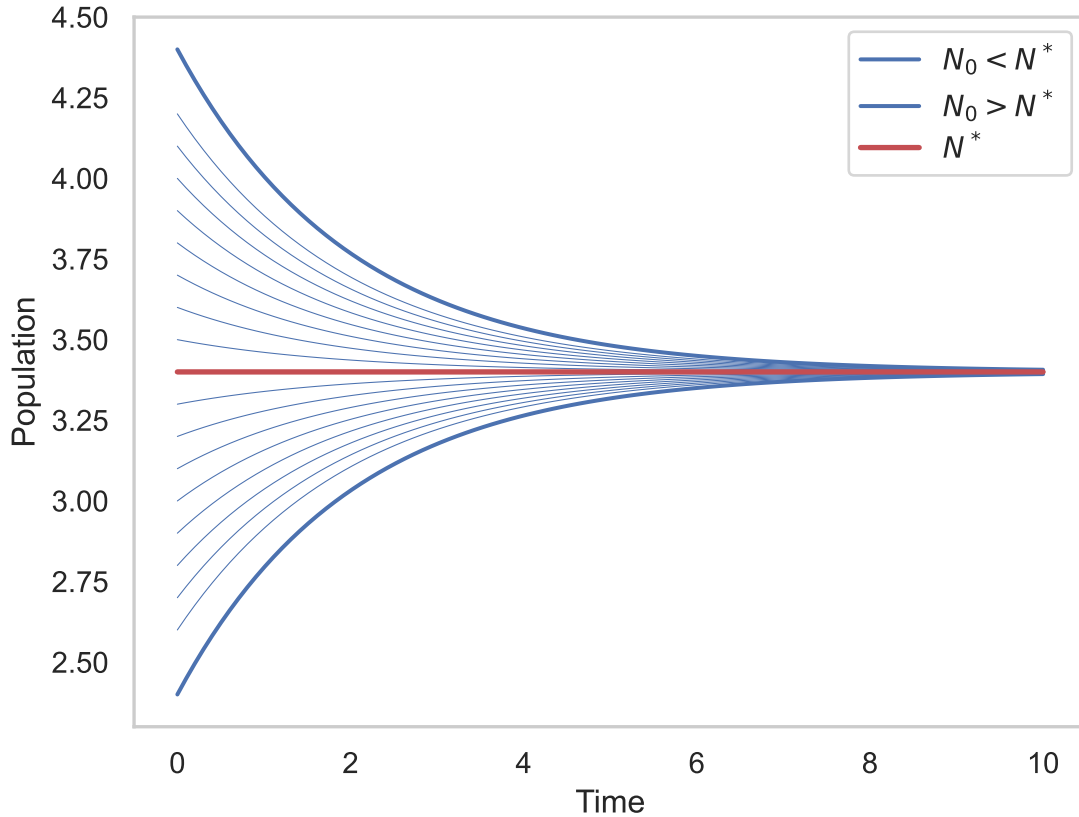


Figure 2.2: Example of trajectories

The case $m < 0$ and $r > 0$ is also interesting. Here, the equilibrium N^* is positive, but unstable. For $N_0 > N^*$ the population still grows exponentially, so emigration as no effect on the overall

population. But for $N_0 < N^*$ the solution will become negative in finite time: the model is not correct. The reason is simple: the emigration m cannot be constant (like the immigration), it must depend on N as well.

Exercise

Solve Equation 2.1 with separation of variables.

2.1.5 Exogenous variability

By exogenous variability we mean variability in the parameter r that does not depend on the population. (Endogenous variability is when the parameters depend on internal variables like N .) Thus, we consider the problem:

$$N' = r(t)N,$$

for $r(t)$ continuous function. An example could births and deaths that depend on climate or temperature. The solution is:

$$N(t) = N_0 e^{\int_{t_0}^t r(s) ds}$$

We can also rewrite the solution as:

$$N(t) = N_0 e^{(t-t_0) \frac{1}{t-t_0} \int_{t_0}^t r(s) ds},$$

showing that if the limit

$$r^* = \lim_{t \rightarrow \infty} \frac{1}{t-t_0} \int_{t_0}^t r(s) ds$$

exists then the asymptotic behavior of the solution is:

$$N(t) \approx e^{r^*(t-t_0)}, \quad t \gg 1.$$

An interesting case is when $r(t)$ is *periodic*, that is there exists $T > 0$ such that $r(t+T) = r(t)$. In this case the above formula still applies, but we can “forget” the limit (check Assignments for a proof) and take:

$$\bar{r} = \frac{1}{T} \int_0^T r(s) ds,$$

and write the solution for any $t \geq t_0$ as:

$$N(t) = e^{\bar{r}(t-t_0)} N_{\pi}(t),$$

for some T -periodic function N_π . Note that

$$N(t_0 + kT) = e^{\bar{r}kT} N_\pi(t_0) = e^{\bar{r}kT} N(t_0),$$

so after k periods the solution grows by a factor $e^{\bar{r}kT}$. The factor $e^{\bar{r}}$ is called *Floquet multiplier*.

3 Logistic model

This lecture is based on [aip14, Section 1.5-1.6]

3.1 Logistic model

While populations can follow a phase of exponential growth for a limited amount of time, it seems impossible that this can go forever and that populations can grow to infinity. Indeed, we expect that there exists a negative effect of crowding, which can be stated in words as follows:

An increase of the population size produces a fertility decrease and a mortality increase; since resources are limited, if the population size exceeds some threshold level, the habitat cannot support the growth.

This simple statement tries to summarize the complex phenomenology of *intraspecific competition* due to many factors such as resource availability, habitat pollution and waste, predation increase, energy consumption for social organization.

The simplest way to include this effect into a model, is to suppose that fertility decreases and mortality increases linearly with the number of individuals; namely

$$\begin{aligned}\beta(N) &= \beta_0 - \tilde{\beta}N, \\ \mu(N) &= \mu_0 + \tilde{\mu}N,\end{aligned}$$

where β_0 , μ_0 , $\tilde{\beta}$, and $\tilde{\mu}$ are non-negative constants. Hence:

$$\begin{aligned}B(N) &= \beta(N)N = \beta_0N - \tilde{\beta}N^2, \\ D(N) &= \mu(N)N = \mu_0N + \tilde{\mu}N^2.\end{aligned}$$

The resulting equation is generally written, after simple algebraic steps, as

$$\begin{cases} N' = r\left(1 - \frac{N}{K}\right)N, \\ N(0) = N_0, \end{cases}$$

where $r = \beta_0 - \mu_0$ and $K = r/(\tilde{\beta} + \tilde{\mu})$. These parameters are usually called *intrinsic growth rate* and *carrying capacity*.

A couple of comments are necessary: first of all, either $\tilde{\beta}$ or $\tilde{\mu}$ can be 0, but not both, otherwise there is no effect of crowding and K is not well-defined. Note also that, if $\tilde{\beta} > 0$, the birth rate $B(t)$ would become negative if $N(t)$ is too large, which does not make sense biologically. However, this does not cause mathematical problems and the biological nonsense would occur only at population levels not normally reached, so we neglect this problem.

We will generally assume that $r > 0$, so that also $K > 0$. In that case, the behaviour of solutions to that equation displays a first phase of exponential growth, followed by convergence to the limiting value K . The general solution is

$$N(t) = \frac{KN_0}{N_0 + (K - N_0)e^{-rt}}.$$

When $N_0 < K/2$, the resulting sigmoid curve have been called *logistic curve*, so that equation is also named *logistic equation*.

The logistic equation is extremely common in experimental biology. Below, data fitted to a logistic for micro-organisms.

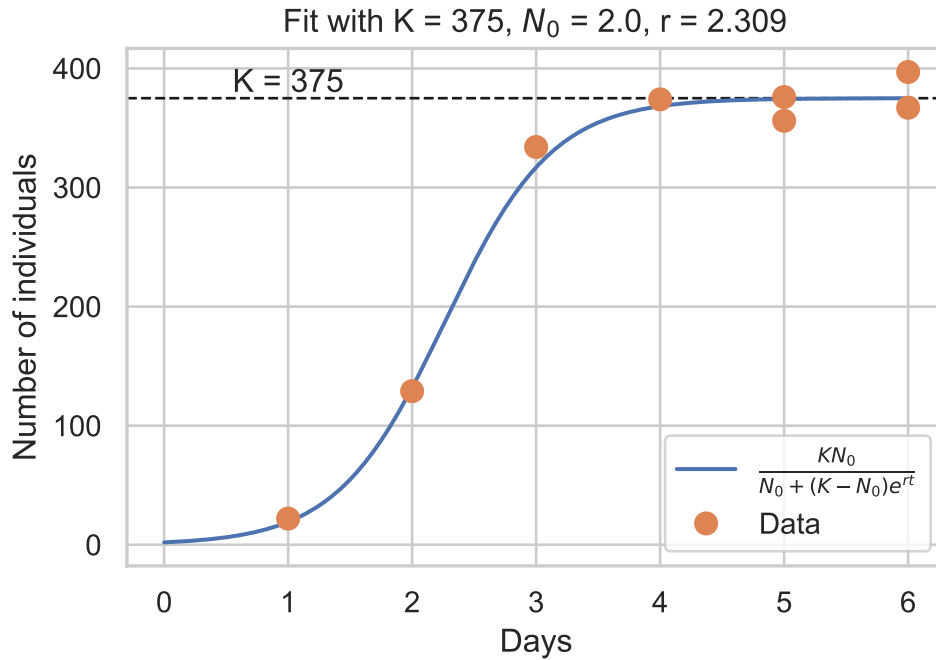


Figure 3.1: Fitting of *Paramecium caudatum* data using the logistic model. Data from (Gause 1934, fig. 4).

As before, we can non-dimensionalize the equation. Now we select:

$$\tau = rt, \quad u = \frac{N}{K},$$

so that $u = 1$ means that we are at carrying capacity. Substituting:

$$\dot{u} = u(1 - u).$$

The new equation has no parameters. Thus, the general solution of the logistic equation is just

$$N(t) = Ku(rt).$$

The absence of parameters in the non-dimensional equation means that its dynamic is always the same, up to a rescaling. We say that all parametric solutions (as we vary r and K) are topologically equivalent. Here, we cannot expect bifurcation, as we shall see.

3.2 Generalized logistic model

In general, we can take β and μ as generic functions of N , thus:

$$r(N) = \beta(N) - \mu(N)$$

is also a generic function of N . A generic growth model reads:

$$N' = r(N)N = f(N).$$

We can implicitly solve this equation, in fact:

$$N(t) = N(0)e^{\int_{t_0}^t r(N(s)) ds}.$$

Thus, if $N(0) > 0$, then $N(t) > 0$ for all time. This is important to verify, since a population cannot be negative. Here, we need no further restriction on $r(N)$ so to verify the condition.

How about equilibria? We have that $E_0 = 0$ is always an equilibrium. (Thus, since orbits cannot cross it, they positive stay positive forever. This is another possible proof.) It is called *extinction equilibrium*. Its stability follows from:

$$f'(N)|_{N=E_0} = r'(0) \cdot 0 + r(0) = r(0).$$

The stability of E_0 is given by $r(0)$, which is called *intrinsic growth rate*. It is the growth rate we observe for very small population size. When $r(0) > 0$ the extinction equilibrium is *unstable*. Equivalently, it is unstable when

$$R_0 = \frac{\beta(0)}{\mu(0)} > 1.$$

How to capture a general logistic effect? By general we mean what we quoted above: a population increase should correspond to a decrease of fertility and an increase of mortality. Thus:

$$r'(N) < 0, \quad \text{and} \quad \lim_{N \rightarrow \infty} r(N) < 0.$$

The second hypothesis avoids the existence of positive horizontal asymptotes. Biologically, a sufficiently large population has always a negative growth rate.

We can now study more equilibria, those corresponding to $r(N) = 0$.

- If $r'(0) < 0$, then by monotonicity we conclude that $r(N) < 0$ for all N , so $E_0 = 0$ is the only equilibrium and the population is doomed.
- If $r'(0) > 0$, we have one additional (unique) equilibrium N^* , that we denote by K : that is, $r(K) = 0$. Since

$$f'(K) = r'(K)K + \underbrace{r(K)}_{=0} = r'(K)K < 0,$$

this equilibrium is (globally) asymptotically stable. This equilibrium is also called *carrying capacity*.

The classic logistic equation has $r(N) = r(1 - N/K)$. The θ -logistic model (or Bernoulli model) has

$$r(N) = r(1 - (N/K)^\theta),$$

for $\theta > 0$. There are a multitude of models for $r(N)$, some we will explore in the assignments (see also figure below). Nonetheless, the above hypotheses always imply a sigmoid growth, when $N(0) \in (0, K)$.

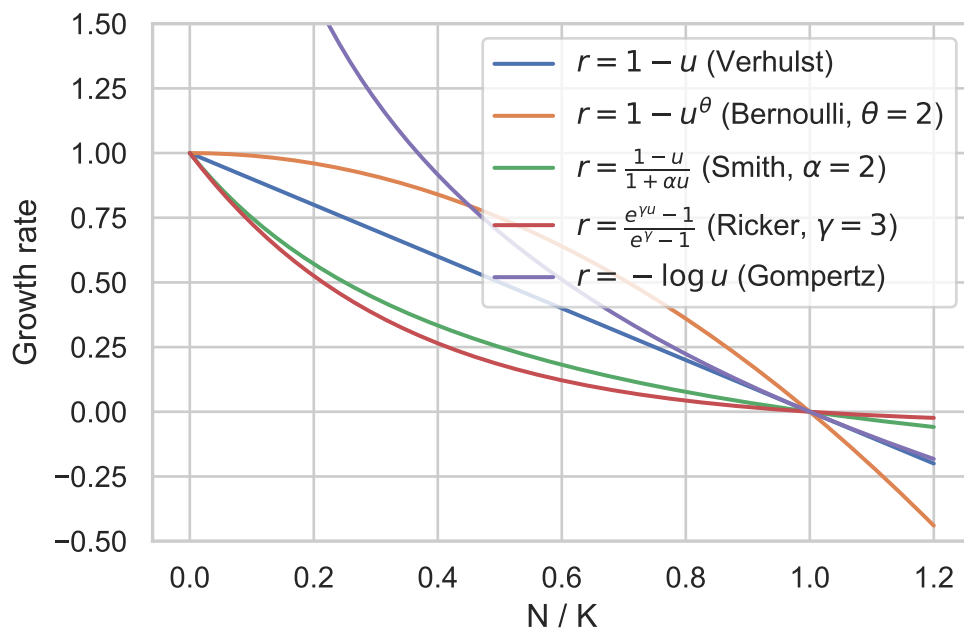


Figure 3.2: Different growth model giving a logistic effect. We set $u = N/K$, where K is the carrying capacity.

💡 Exercise

Verify that the above models are “logistic”, in the sense that they satisfy the hypotheses. Show that the Bernoulli model yields sigmoid solutions. (Hint: study N' and N'' .)

💡 Exercise

Integrate the Gompertz model, $\dot{u} = -u \log u$. This model is very common in the study of tumor cells proliferation. (Hint: set $u(t) = e^{w(t)}$.)

3.3 Allee effect

(Story time) The [great auk](#) was a bird that became extinct by the end of the 19th century. They were found at the northern Atlantic (Canada, Scotland, Iceland), usually on rocky islands. The overall population was composed by millions of individuals, before sailors and hunters started killing them for their meat, feathers, and oily fat.



Figure 3.3: The Great Auk (from Wikipedia)

As it could be expected, the auk population drastically reduced and the bird disappeared from many islands very quickly. Governments became aware of the situation in the 18th century, and some laws forbidden the hunt of the great auk, although with limited success.

Some species like the great auk are not able to fully recover, even in the absence of predation (or with a small one). In order for a species to go to extinction with no predation, say for

$$N' = r(N)N,$$

we would need $r(0) < 0$, otherwise the equilibrium $N = 0$ is not stable. But this would lead to no further equilibria under the hypotheses of general logistic growth. Hence, we can further generalize the hypotheses as follows: the function $r(N)$ is such that

1. there exists $N_m > 0$ such that $r'(N) > 0$ for $N < N_m$ and $r'(N) < 0$ otherwise;
2. $r(N_m) > 0$;
3. $\lim_{N \rightarrow \infty} r(N) < 0$.

Therefore, we certainly have the equilibrium $K > N_m$ (the carrying capacity). However, if $r(0) < 0$, we also have another equilibrium at $T \in (0, N_m)$. In this case, we say that we have a *strong Allee effect*. On the other hand, for $r(0) > 0$ there are no further equilibria, thus we have a *weak Allee effect*.

Let us study the stability for $r(0) < 0$. Remember that

$$f'(N) = r'(N)N + r(N).$$

We have 3 equilibria:

- $N = E_0 = 0$, which is asymptotically stable, since $f'(0) = r(0) < 0$.
- $N = T \in (0, N_m)$, which is unstable, since $f'(T) = r'(T)T > 0$.
- $N = K > N_m$, which is asymptotically stable, since $f'(K) = r(K)K < 0$.

Hence, we have the following result: if $N(0) < T$, then $N(t) \rightarrow 0$ (extinction), otherwise if $N(0) > T$, then $N(t) \rightarrow K$ (survival). We call T the threshold population for survival.

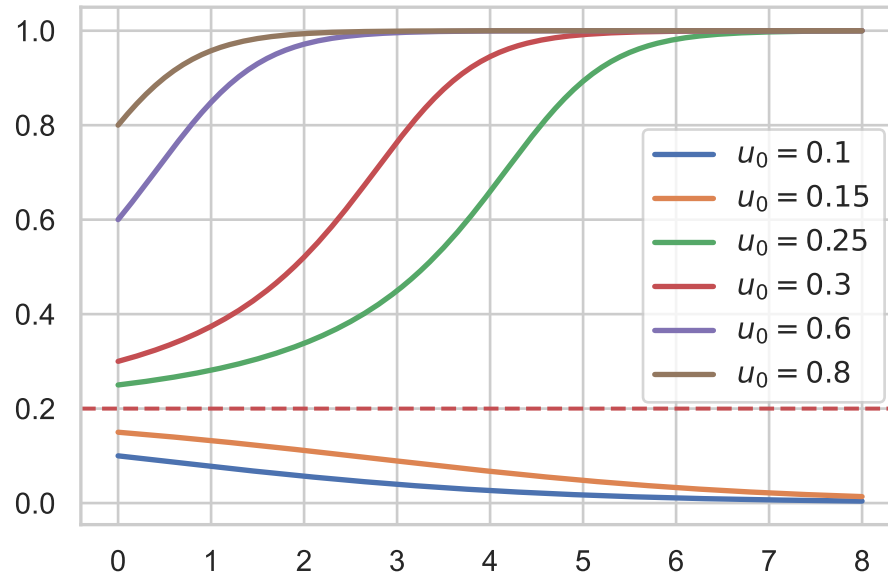


Figure 3.4: Solutions of the equation $N' = rN(N/T - 1)(1 - N/K)$. Note the threshold value for $N = T$.

The Allee effect may be found in other situations, for instance when the predation rate is a non-linear function of N .

4 Predation models

This lecture is based on [ip14, Section 1.8-1.9]

4.1 Generalist predation

Predation is a fundamental topic in ecology and the main interaction between species. It must be understood in a broader sense: fishing, harvesting, hunting are all form of predation.

The simplest model of predation consists in formally increasing the mortality rate of N by an extra rate aP^* , that is

$$N' = r(N)N - aP^*N = f(N),$$

where $P^* > 0$ is the number of predators and $a > 0$ is the *attack rate* or *effective killing rate*. We note that the death rate aP^*N is *proportional* to both the number of predators P^* and the number of preys N , according to the *law of mass action*, as suggested by Volterra with the method of encounters. The idea is similar to collision theory for ideal gases.

This type of predation is also called *generalist*, in the sense that predators' survival does not depend on the survival of the prey population N . If the preys go to extinction, the predator will hunt something else. A *specialized* predator, on the other hand, will suffer from a low level of N , as we shall see next week.

The equilibria of the equation are: $N = E_0 = 0$, as usual, and the zeros of:

$$r(N) = aP^*.$$

- If $aP^* > r(0)$, then we have no additional equilibria, because $r'(N) < 0$.
- If $aP^* < r(0)$, then there exists a second equilibrium $N^* > 0$.

The stability of the second equilibrium follows from

$$f'(N^*) = r'(N^*)N^* + r(N^*) - aP^* = r'(N^*)N^* - aP^* < 0,$$

because $r'(N) < 0$. So N^* is asymptotically stable.

The stability of E_0 is similar:

$$f'(E_0) = r'(0) \cdot 0 + r(0) - aP^* = r(0) - aP^* > 0,$$

meaning that it is unstable. Concluding:

- If $aP^* > r(0)$, we have one stable equilibrium $E_0 = 0$, thus the population will go to extinction. In fact, the predation level is high.
- If $aP^* < r(0)$, $E_0 = 0$ is unstable but we have another equilibrium $0 < N^* < K$ asymptotically stable. Thus, the predation is sustainable.

The case $aP^* = r(0)$ is delicate. We have a single equilibrium, $E_0 = 0$, but $f'(E_0) = 0$, so we cannot deduce the stability from the linearization (why?). By inspecting the sign of E_0 , we observe that $f(N) < 0$ for $N > 0$, so E_0 is attractive. However, for $N < 0$ we have that $f(N) < 0$, thus it is repulsive (biologically, we do not care because $N < 0$ is irrelevant.) This type of equilibrium is called *saddle-node*.

The point $aP^* = r(0)$ is a *bifurcation point*, specifically a *transcritical bifurcation* (See Section B.6). Roughly speaking, the dynamic before and after the bifurcation point is topologically different: in one case we have one equilibrium, in the other 2 equilibria. Actually, we still have 2 equilibria for $aP^* > r(0)$, one being negative. Thus, what really happens is that as we increase aP^* the two curves of equilibria ($N = 0$ and $N = E_0$) crosses and swap stability.

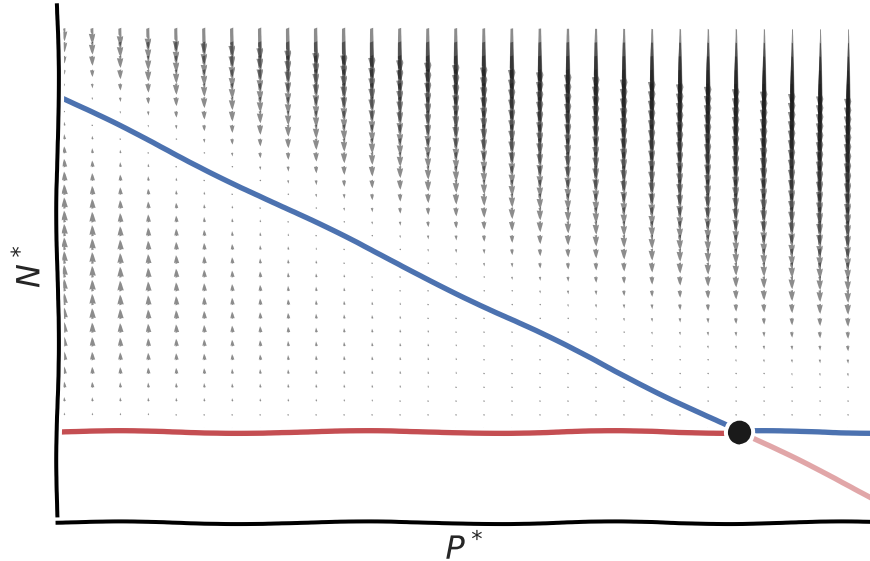


Figure 4.1: Bifurcation diagram for the model with generalist predation. In blue, the stable equilibrium, in red the unstable one. Beyond the bifurcation point (transcritical, in black) there is only one equilibrium.

4.2 Holling type predation models

More generally, we define the predation as:

$$N' = r(N)N - \pi(N)P^*,$$

where $\pi(N)$ is called *functional response*. The functional response can be interpreted as follows: $\pi(N)\Delta t$ is the number of preys killed in Δt units of time by a single predator.

For the simplest model, we have $\pi(N) = aN$. However, this is not realistic, because for very large N the predation rate cannot grow indefinitely, it must reach some limiting value. For instance, a predator needs some time to consume the prey, and this time cannot be reduced below a certain limit. Similarly, it is reasonable that at low density of preys, say when N is small, predation is harder.

In general, we can assume that:

1. $\pi(0) = 0$, no preys no predation, and
2. $\pi'(N) > 0$, the more preys, the higher the predation rate.

Holling (1965) proposed the following types of functional responses:

- **Holling type I**

$$\pi(N) = \begin{cases} aN, & 0 \leq N \leq N^*, \\ aN^*, & N > N^*. \end{cases}$$

This model is exactly like the linear one, but it assumes that for $N > N^*$ we have constant predation rate aN^* . The parameter a is called *attack rate*, and it measures, after an encounter between prey and predator, the success rate of predation. Note that the function $\pi(N)$ is not $\mathcal{C}^1$ but it is Lipschitz.

- **Holling type II**

$$\pi(N) = \frac{aN}{1 + a\tau N},$$

with $a, \tau > 0$. This is a smooth version of Holling type I. For small N , $\pi(N) \approx aN$, so the meaning of a is the same as above. For large N , we have that $\pi(N) \rightarrow 1/\tau =: \alpha$, which is called *maximum killing rate*. That is, α is the number of preys killed by one predator in a unit of time, when the number of preys is very large. Alternatively, we can interpret τ as the time required by the predator to consume the prey.

- **Holling type III**

$$\pi(N) = \frac{\alpha N^\theta}{\nu^\theta + N^\theta},$$

with $\alpha, \nu > 0$ and $\theta > 1$. The last type also accounts for a lower predation rate as low density of preys. In fact, for small N and $\theta > 1$ we have $\pi'(0) = 0$. For $N = \nu$, $\pi(\nu) = \frac{\alpha}{2}$, and for $N \rightarrow \infty$ we have $\pi(N) \rightarrow \alpha$, the maximum killing rate. Thus, ν is the number of preys at which the killing rate is exactly half of the maximum one.

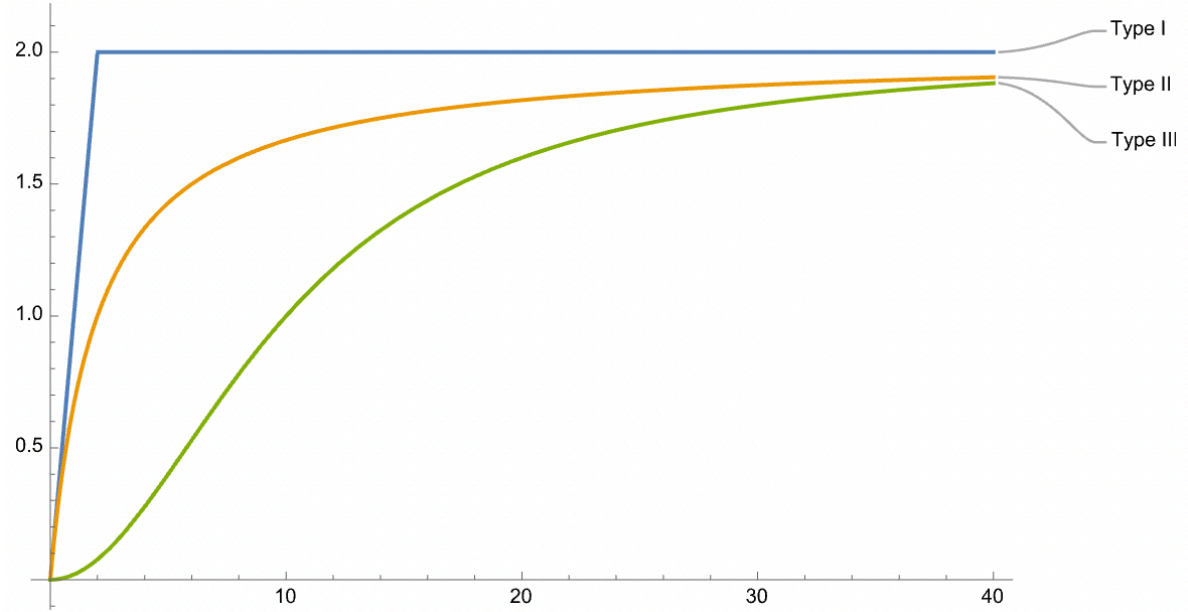


Figure 4.2: Different types of Holling predations

It is possible to justify Holling type II in a few ways. One, we will see in the completely different context of enzymatic reactions. A simpler one is as follows. In a time T , a single predator will spend $1/\pi(N)$ time in hunting. (Always keep in mind the bathtub example!) But the total time splits into T_s , the time spent seeking for a prey, and τ , the time needed to consume a prey. The time τ is fixed, no matter how large is N . The time T_s , however, is exactly $(aN)^{-1}$, because the more preys the easier is to catch them. Putting all together we have:

$$\frac{1}{\pi(N)} = T = \tau + T_s = \tau + \frac{1}{aN}, \quad \Rightarrow \quad \pi(N) = \frac{aN}{1 + a\tau N}.$$

4.3 Spruce budworm model



Figure 4.3: The effect of the spruce budworm on a forest.

The spruce budworm is an insect that feeds on needles of balsam fir trees (see [this website](#)). If needles are removed, the tree dies. Historical data in Canada evergreen forests, where the budworm is present, shows that in most years the budworm density is very low. However, in a outbreak year, the budworm population spikes and can kill up to 80% of mature trees in the forest. The period of the outbreak is roughly 30-70 years.

The spruce budworm model has been introduced in Ludwig et al. (1978). They proposed a system of 3 variables:

1. $N(t)$, the budworm density,
2. $S(t)$, the habitat space for larvae, and
3. $E(t)$, a measure of food energy reserves available to the budworm.

We focus here on the equation for $N(t)$. In fact, $S(t)$ and $E(t)$ will vary very slowly compared to $N(t)$, thus they can be assumed constant. (This argument can be made rigorous, as we shall see for enzymatic reactions.)

The equation for $N(t)$ reads as follows:

$$N' = rN\left(1 - \frac{N}{K}\right) - \frac{\alpha P^* N^2}{\nu^2 + N^2}.$$

The first term is the logistic growth. The second one is the predation of the budworms due to birds. This is Holling type III. Note that the parameters are fixed numbers (they do not depend on time), but they will depend on S and E , somehow. Thus, it will be interesting to see what happens to the system as we change them.

Since we have too many parameters, we proceed with non-dimensionalization. Here we select:

$$\tau = \frac{t}{T}, \quad u = \frac{N}{\nu},$$

for some $T > 0$ to be selected. We have:

$$\frac{\nu}{T} \dot{u} = r\nu u \left(1 - \frac{\nu u}{K}\right) - \frac{\alpha P^* u^2}{1 + u^2},$$

thus by selecting

$$T = \frac{\nu}{\alpha P^*}, \quad \rho = \frac{r\nu}{\alpha P^*}, \quad q = \frac{K}{\nu}$$

we arrive at

$$\dot{u} = \rho u \left(1 - \frac{u}{q}\right) - \frac{u^2}{1 + u^2},$$

where we are left with only 2 parameters:

- ρ is proportional to the intrinsic growth rate, while
- q is the carrying capacity normalized to the half-saturation population ν .

4.3.1 Equilibria and stability

As usual, we start by looking for equilibria of the system, that is solutions of

$$\rho u \left(1 - \frac{u}{q}\right) - \frac{u^2}{1 + u^2} = 0.$$

We have that $u = 0$ is an equilibrium. The others solve the equation

$$\rho \left(1 - \frac{u}{q}\right) - \frac{u}{1 + u^2} = 0,$$

which would lead to a 3rd-order polynomial equation, thus we can expect up to 3 real solutions. An analytical approach is not practical. However, equilibria are intersections of the two curves $f(u)$ and $g(u)$ where

$$f(u) = \rho \left(1 - \frac{u}{q}\right), \quad g(u) = \frac{u}{1 + u^2}.$$

The function $f(u)$ represents the per capita growth rate of u , whereas $g(u)$ is the per capita death rate due to predation. Thus, solutions of the equation are equilibria of the system. Since the function $g(u)$ does not depend on any parameters, we can fix it and simply change $f(u)$, which is a segment.

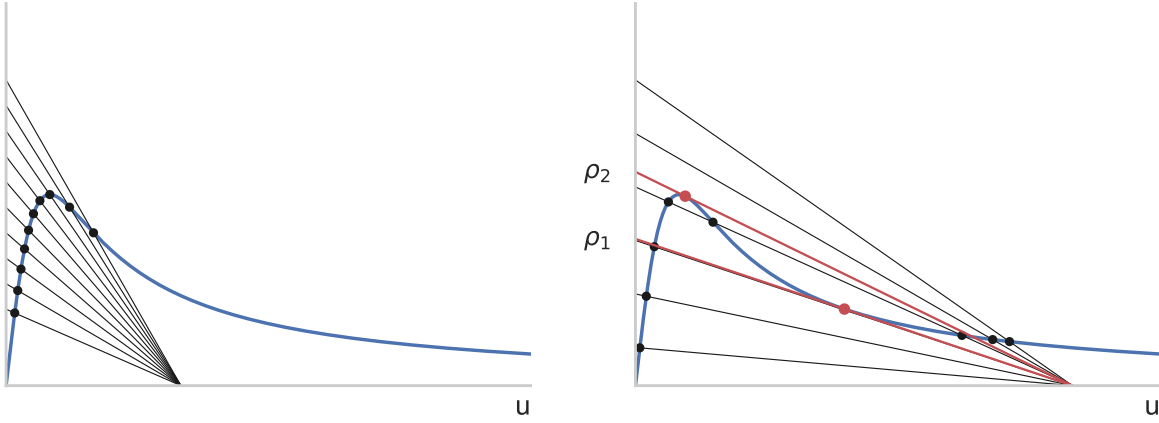


Figure 4.4: Equilibria for the spruce budworm model. On the left, the case of small q , on the right the case of large q .

We can see from the figure that when q is small, we only have a single equilibrium. When q is large, we can have 1, 2, or 3 additional equilibria, depending on ρ : given $\rho_1 < \rho_2$ we have:

1. If $\rho < \rho_1$, then there exists a single equilibrium $u_1^* \ll q$. This is the *refuge* equilibrium (low population). The equilibrium is globally stable, because $f(u) - g(u) > 0$ for $u < u_1^*$ and $f(u) - g(u) < 0$ for $u > u_1^*$.
2. If $\rho > \rho_2$, then there exists a single equilibrium u_3^* close to q . This is the *outbreak* equilibrium (large population). The equilibrium is globally stable, because $f(u) - g(u) > 0$ for $u < u_3^*$ and $f(u) - g(u) < 0$ for $u > u_3^*$.
3. If $\rho_1 < \rho < \rho_2$, then there exist 3 equilibria $u_1^* < u_2^* < u_3^*$. The stability of u_1^* and u_3^* is as above, but now locally. The equilibrium u_2^* is unstable.

The case 3. is the most interesting one. If $u(0) < u_2^*$, then $u \rightarrow u_1^*$, otherwise $u \rightarrow u_3^*$. This is again a threshold effect.

4.3.2 Bifurcation diagram

As we change ρ , we may have a different number of equilibria. Thus, there must be some bifurcation occurring. The equilibria are on the curve defined by

$$h(u, \rho) = \rho \left(1 - \frac{u}{q}\right) - \frac{u}{1 + u^2} = 0.$$

The function $h(u, \rho)$ is smooth in ρ , since $\partial_\rho h \neq 0$ for $u \in [0, q)$. Thus we can write ρ as a function of u :

$$\rho(u) = \frac{qu}{(1 + u^2)(q - u)}.$$

Since $h(u, \rho(u)) = 0$, the curve $(u, \rho(u))$ with $u \in [0, q)$ defines a curve of equilibria, shown below:

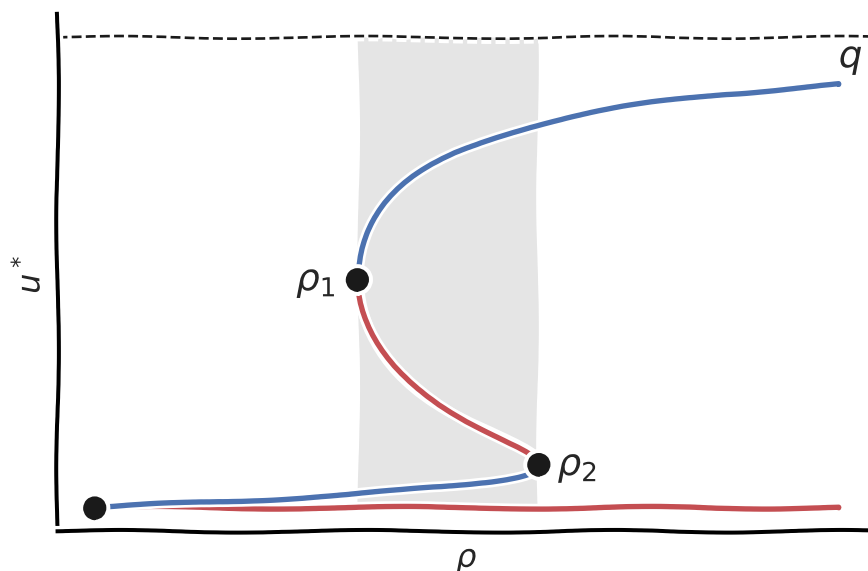


Figure 4.5: Bifurcation diagram with respect to ρ for the spruce budworm system, for $q = 12$. In blue, the stable equilibrium, in red the unstable one. The shaded region is bistable.

The plot above is a bifurcation diagram. We can interpret it as follows: given $\rho = \bar{\rho}$, the equilibria are those corresponding to the intersections between $\rho = \bar{\rho}$ and the curve $\rho = \rho(u)$. So, for $\rho < \rho_1$ we have one equilibrium, for $\rho_1 < \rho < \rho_2$ we have 3, and for $\rho > \rho_2$ we have one.

For $\rho = \rho_1$ or $\rho = \rho_2$, we have a *tangent bifurcation* (See Section B.5). As we can see, the curve of equilibria is always smooth, with no branching or crossing on another curve of equilibria,

like in the case of *transcritical bifurcation*. However, there is a change in stability: in fact, the branch between ρ_1 and ρ_2 corresponds to u_2^* , which is unstable.

When varying also q , the tangent bifurcations points ρ_1 and ρ_2 moves as well. In particular, as q is reduced, the two bifurcation points will get closer until they meet for $q = 3\sqrt{3}$. This point is another bifurcation, called *cusp bifurcation*. Beyond this point, the system is never bistable.

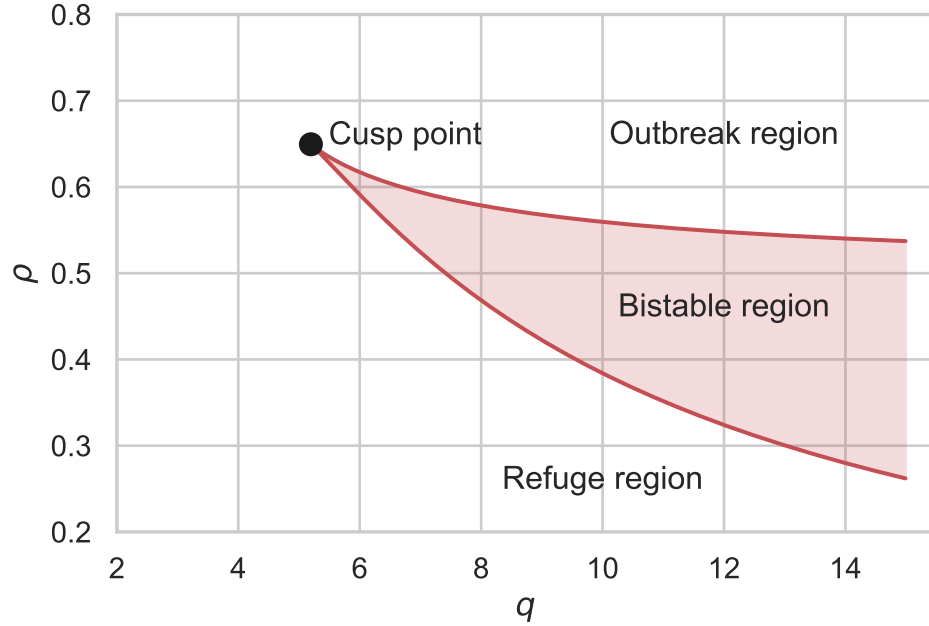


Figure 4.6: Bifurcation diagram with respect to (q, ρ) for the spruce budworm system. The shaded region is bistable. The cusp point is a bifurcation point of co-dimension 2. It occurs for $q = 3\sqrt{3}$.

If we plot the surface of equilibria in the space (q, ρ, u^*) , as the solution of the equation $h(u, \rho, q) = 0$, we obtain the plot below. This plot clearly shows the bistable region and the threshold value (in red).

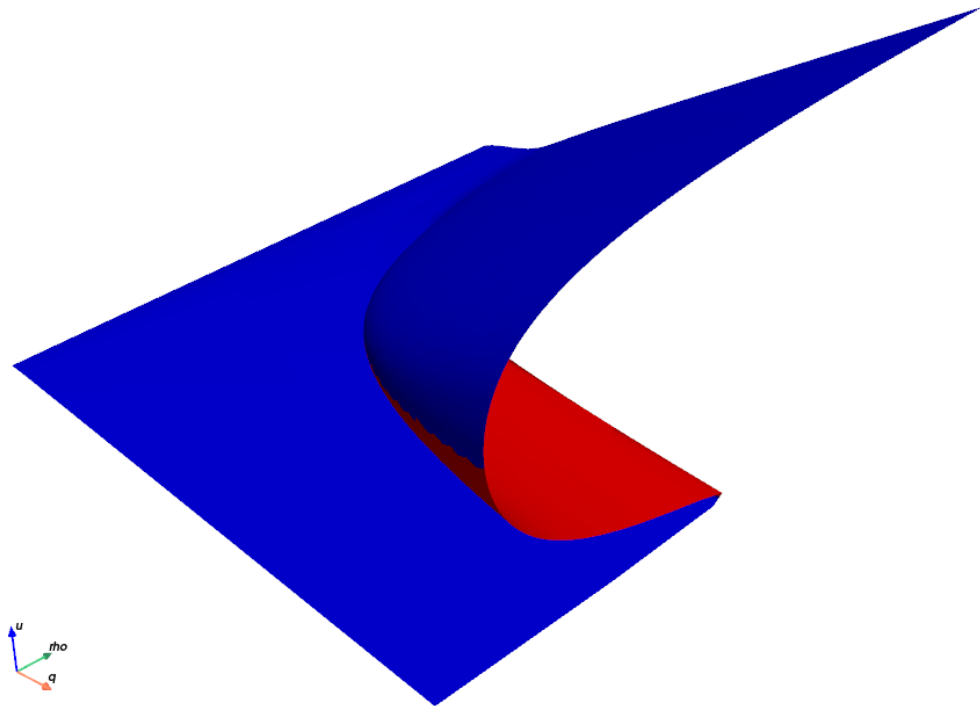


Figure 4.7: Bifurcation diagram with respect to (q, ρ) in 3D.

4.3.3 Hysteresis

Now that we have the bifurcation diagram, we can use it to find the periodic outbreaks. Remember that ρ and q are functions of S and E , but they vary very slowly compared to $N(t)$. On the other hand, when N is at equilibrium, it does not vary anymore, so even slow and small variations of E and S could matter.

Suppose to start with small ρ , say $\rho < \rho_1$. The spruce budworms is at refuge state u_1^* . Now we slowly increase ρ . The equilibrium $u_1^*(\rho)$ will only slightly increase. Once we reach the point $\rho = \rho_2$, the equilibrium u_1^* disappears, so $u \rightarrow u_3^*$ (outbreak), the only other stable equilibrium. Note that the outbreak is fast, even for a small change of ρ . For this reason, the tangent bifurcation is a *catastrophic bifurcation*. As we keep increasing ρ , the outbreak equilibrium keeps increasing, but again slowly.

Since ρ is the intrinsic growth rate, we can assume that ρ will now start to decrease, because

there are too many spruce budworm consuming the resources. As we go back, decreasing ρ , we follow a specular path, jumping at $\rho = \rho_1$ from the outbreak to the refuge equilibrium. We go back to the original situation.

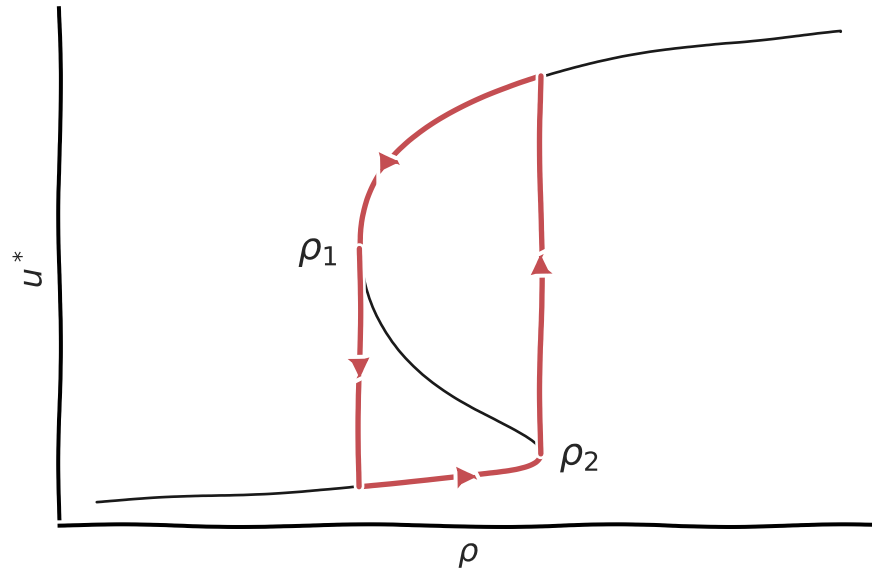


Figure 4.8: Bifurcation diagram with respect to ρ for the spruce budworm system. In blue, the stable equilibrium, in red the unstable one. The shaded region is bistable.

The result is a *hysteresis loop*, which could explain the periodic outbreaks.

We will analyze this model more in depth during the Lab session.

5 Lotka-Volterra model

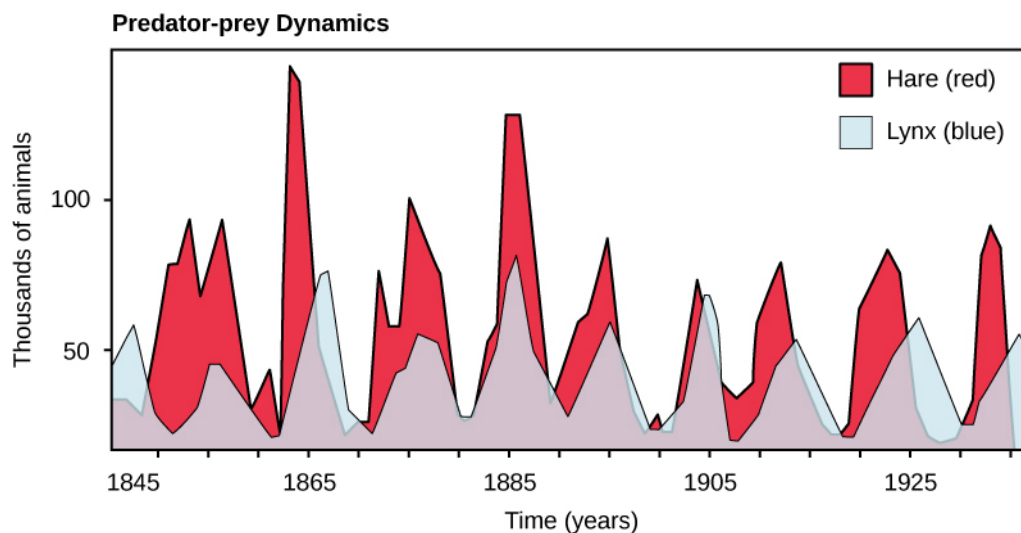
This lecture is based on [ip14, Section 6.1]

5.1 Prey-predator models

In all previous models, predation was only at cost of preys. Generalist predators, like humans, can survive even in absence of preys. Thus, P^* can be assumed constant.

What if the predator is a *specialist*, that is it can only hunt a specific prey? When the prey population gets too low, also survival of the predator could be at risk. One equation is not enough.

A classic example is the Hare-Lynx dynamics, shown below:



The data actually shows pelt-trading data taken from the Hudson's Bay Company Elton and Nicholson (1942). We observe that data is oscillatory with a 10-year period. How is that?

5.2 Lotka-Volterra model

(*Story time*) During the 1920s, the biologist Umberto D’Ancona (who married Volterra’s daughter) noticed that during World War I, in his fishery data in the Adriatic Sea, the frequency of predation increased significantly. He could not explain this phenomenon, but it should have been somehow related to a decrease of fishery during the war. Furthermore, the data was oscillatory.

In 1926, Vito Volterra proposed a model for the prey-predator interaction. (After the publication on *Nature*, Alfred Lotka wrote a short communication, still on *Nature*, where he points out that his book “Elements of Physical Biology”, published in 1924, covered the same model. See also the book Volterra (1932).) Suppose that

- $H(t)$ is the population of *preys* over time, and
- $P(t)$ is the population of *predators* over time.

Volterra made the following hypotheses:

1. The growth rate for prey and predator is Malthusian;
2. In absence of predators, $H(t)$ increases;
3. In absence of preys, $P(t)$ decreases;
4. In a unit of time, the number of preys consumed by predators is $-aH(t)P(t)$;
5. In a unit of time, successful predation increases the number of predators by $\gamma aH(t)P(t)$.

The *Lotka-Volterra model* readily follows:

$$\begin{cases} H' = rH - aHP, \\ P' = -\mu P + \gamma aHP. \end{cases}$$

We can also rewrite the system as:

$$\begin{cases} H' = HF(P), \\ P' = PG(H), \end{cases}$$

for $F(P) = r - aP$ and $G(H) = -\mu + \gamma aH$. The system is in Kolmogorov form: orbits cannot cross the axes $H = 0$ and $P = 0$, so trajectories starting in the positive quadrant remain in the positive quadrant.

5.2.1 Equilibria

We have the trivial equilibrium $(H, P) = (0, 0) = E_0$, and another one satisfying:

$$\begin{cases} r - aP = 0, \\ -\mu + \gamma aH = 0. \end{cases}$$

So, $E^* = (H^*, P^*) = (\frac{\mu}{\gamma a}, \frac{r}{a})$. It is positive, thus biologically feasible.

The stability follows from the Jacobian (see Section A.5.4.2):

$$J(H, P) = \begin{pmatrix} r - aP & -aH \\ \gamma aP & -\mu + \gamma aH \end{pmatrix}.$$

For E_0 , we have $J(0, 0)$ diagonal with eigenvalues r and $-\mu$. It is a saddle.

For E^* , we have

$$J(H^*, P^*) = \begin{pmatrix} 0 & -aH^* \\ \gamma aP^* & 0 \end{pmatrix}$$

which has eigenvalues with zero real part, because the determinant is positive but the trace is zero. Thus, we have a center. The solutions of the linear equation are periodic with a period

$$T_{\text{linear}} = \frac{2\pi}{\sqrt{\det J(H^*, P^*)}} = \frac{2\pi}{\sqrt{\mu r}}.$$

Unfortunately, the linearization does not help to conclude anything for the non-linear case.

5.2.2 Phase portrait

On the other hand, the system is in Hamiltonian form (see Section A.6.1). To find the Hamiltonian, here denoted by $\mathcal{H}(H, P)$ to avoid confusion with $H(t)$, we notice that:

$$\frac{dH}{dP} = \frac{dH/dt}{dP/dt} = \frac{HF(P)}{PG(H)},$$

and the ODE is in separable form, so we have:

$$\int_{H_0}^{H(t)} \frac{G(h)}{h} dh = \int_{P_0}^{P(t)} \frac{F(p)}{p} dp.$$

We can easily carry out the integration:

$$-\mu \ln h + \gamma a h \Big|_{H_0}^{H(t)} = r \ln p - ap \Big|_{P_0}^{P(t)},$$

and we find that:

$$-\mu \ln H(t) + \gamma a H(t) - r \ln P(t) + aP(t) = -\mu \ln H_0 + \gamma a H_0 - r \ln P_0 + aP_0.$$

Since the right hand side is a constant, we have found that along the trajectory parallel to flow dH/dP the quantity

$$\mathcal{H}(H(t), P(t)) = -\mu \ln H(t) + \gamma a H(t) - r \ln P(t) + aP(t)$$

remains constant. Thus, $\mathcal{H}$ is an Hamiltonian. We can check this again by noting that

$$\begin{aligned} \frac{d}{dt} \mathcal{H}(H(t), P(t)) &= \frac{\partial \mathcal{H}}{\partial H} H' + \frac{\partial \mathcal{H}}{\partial P} P' \\ &= \left(-\frac{\mu}{H} + \gamma a\right) HF(P) + \left(-\frac{r}{P} + a\right) PG(H) \\ &= G(H)F(P) - F(P)G(H) = 0. \end{aligned}$$

We conclude that level sets of $\mathcal{H}$ are orbits of the system. Are they closed? Yes, because the function $\mathcal{H}(H, P) \rightarrow +\infty$ as $(H, P) \rightarrow +\infty$ or $(H, P) \rightarrow 0^+$. Moreover, $\mathcal{H}$ is bounded from below, with minimum for $(H, P) = (H^*, P^*)$. In fact, since E^* is an equilibrium, $\nabla \mathcal{H}|_{E^*} = 0$. Furthermore, the Hessian $\nabla^2 \mathcal{H}|_{E^*}$ is positive definite.

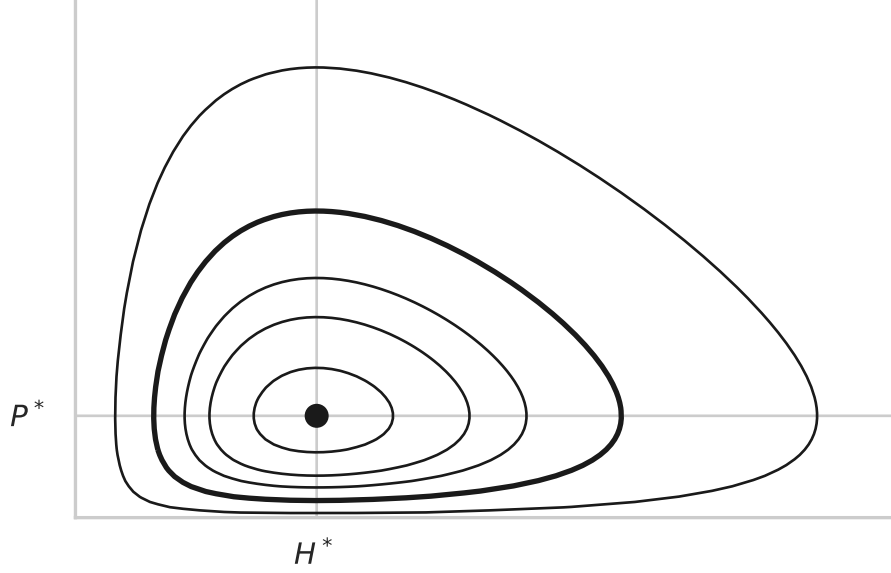


Figure 5.1: Level sets of the Hamiltonian function. All orbits are closed, and they correspond to periodic solutions of the Lotka-Volterra model.

Therefore, the curve

$$\mathcal{H}(H(t), P(t)) = \mathcal{H}(H(0), P(0)),$$

is a closed orbit of the system. Depending on the initial conditions, we end up with a different orbit, but all of them are closed. It still can be that as the period of revolution around the orbit is not finite. This can happen for homoclinic orbits, for instance (see Figure A.6.) We can exclude this option as follows: we define the angular coordinate as:

$$\theta(t) = \arctan\left(\frac{P(t) - P^*}{H(t) - H^*}\right),$$

which measures the angle of the vector $(H(t), P(t)) - E^*$ with respect to the abscissa. The angular velocity is:

$$\begin{aligned} \theta'(t) &= \frac{(H - H^*)P' - (P - P^*)H'}{(H - H^*)^2 - (P - P^*)^2} \\ &= \frac{\gamma a P (H - H^*)^2 + a H (P - P^*)^2}{(H - H^*)^2 - (P - P^*)^2} \\ &\geq \min\{\gamma a \min P, a \min H\} > 0, \end{aligned}$$

where $\min P$ and $\min H$ are taken on the curve and are certainly positive thanks to the logarithmic term in $\mathcal{H}$. Since the velocity is strictly positive, and the length of the curve is finite, the period must be finite as well.

In conclusion, all orbits are periodic. The temporal behavior of $H(t)$ and $P(t)$ is then periodic, as observed in data.

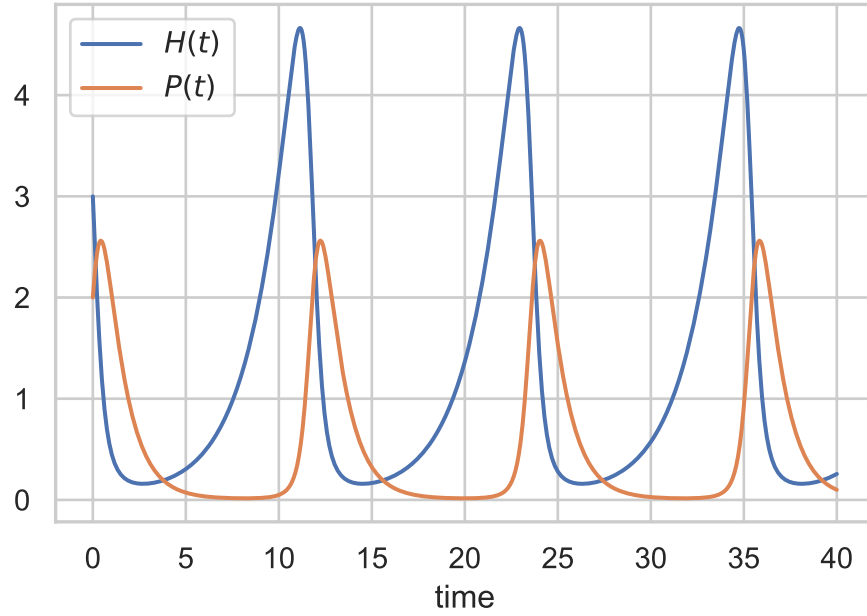


Figure 5.2: Example of solution of the Lotka-Volterra model

5.2.3 The fishery problem

Now we have a model that we can use to address D'Ancona question. If we regularly capture some predators and preys (both are fishes, we can catch both), their quantity is going to be proportional to the actual level of population. So, we can check the average value of H and P .

Let T be the period of an orbit. Then we'd like to compute

$$\bar{H} = \frac{1}{T} \int_0^T H(t) dt,$$

$$\bar{P} = \frac{1}{T} \int_0^T P(t) dt.$$

Moreover,

$$\begin{aligned}\int_0^T \frac{H'}{H} dt &= \int_0^T \frac{HF(P)}{H} dt = \int_0^T (r - aP) dt = rT - aT\bar{P}, \\ \int_0^T \frac{P'}{P} dt &= \int_0^T \frac{PG(H)}{P} dt = \int_0^T (-\mu + \gamma aH) dt = -\mu T + \gamma aT\bar{H}.\end{aligned}$$

Since the solution is periodic, $H(T) = H(0)$ and $P(T) = P(0)$. Hence:

$$\begin{aligned}\int_0^T \frac{H'}{H} dt &= \int_{H(0)}^{H(T)} \frac{1}{H} dH = \ln \frac{H(T)}{H(0)} = 0, \\ \int_0^T \frac{P'}{P} dt &= \int_{P(0)}^{P(T)} \frac{1}{P} dP = \ln \frac{P(T)}{P(0)} = 0.\end{aligned}$$

We end up with the system:

$$\begin{cases} r - a\bar{P} = 0, \\ -\mu + \gamma a\bar{H} = 0, \end{cases}$$

from which we deduce that

$$\bar{H} = \frac{\mu}{\gamma a} = H^*, \quad \bar{P} = \frac{r}{a} = P^*.$$

The average quantities match the value of the equilibrium, *irrespective* of the initial condition! This result is not obvious. Biologically, the system has an equilibrium, although non constant in time.

Now, we introduce harvesting (fishing). Consider a parameter $E \geq 0$, called the *effort*. This is an abstract concept mutuited from bioeconomics. A small or zero effort means that we do not harvest much, a large effort is the opposite. We modify the Lotka-Volterra system as follows:

$$\begin{cases} H' = rH - aHP - \alpha EH, \\ P' = -\mu P + \gamma aHP - \beta EP, \end{cases}$$

where $\alpha, \beta > 0$ are the *returns* per unit effort. That is, $\alpha H \Delta t$ is the number of preys obtained in Δt units of time per unit effort. If we rewrite the system as follows:

$$\begin{cases} H' = H(r - \alpha E - aP) = H(\tilde{r} - aP), \\ P' = P(-\mu P - \beta E + \gamma aH) = P(-\tilde{\mu} + \gamma aH), \end{cases}$$

with $\tilde{r} = r - \alpha E$ and $\tilde{\mu} = \mu + \beta E$, the system is identical to the original one, provided that $r > \alpha E$. The new equilibrium $E^{**} = (H^{**}, P^{**})$ reads:

$$H^{**} = \frac{\tilde{\mu}}{\gamma a} = \frac{\mu + \beta E}{\gamma a}, \quad P^{**} = \frac{\tilde{r}}{a} = \frac{r - \alpha E}{a}.$$

Thus, for $E > 0$ we are simply shifting the equilibrium. But the equilibrium is also the average value of $H(t)$ and $P(t)$ over the period, thus we should observe the same shift in the data. As a matter of fact, for $E > 0$, the number of prey diminishes, but the number of predator increases! During the World War I, fishery reduced, so E decreased: an increase of predators occurred, as observed by D'Ancona.

In conclusion, Volterra stated the following laws:

1. the fluctuations of the two species are periodic;
2. the average number of individuals over the period is conserved and does not depend on the initial datum;
3. (also called *Volterra's principle*) with $E > 0$, the average number of preys increases and the number of predators decreases.

Volterra's principle is important in ecology. It somewhat paradoxically explain why, when applying non-specific killing (e.g., pesticides) some species may actually increase instead of decreasing.

6 Gause type models

This lecture is based on [@ip14, Section 6.2-6.4]

6.1 Gause type models

The Lotka-Volterra model is powerful, in spite of its simplicity. However, the model fails to predict many situations observed in experiments, such as those performed on bacteria by G.F. Gause in 1930s. He observed that with preys (*Paramecium*) and predators (*Didinium*) in the same test tube, first predators killed all preys, and then predators starved to death. No oscillations. This is *never* predicted by Lotka-Volterra original model. On the other hand, when he artificially introduced an immigration of preys into the tube, oscillations were present. Thus, stable oscillations may result from spatial structure of a population (*metapopulation*).

Here we focus on an alternative scenario, where oscillations may arise from nonlinear trophic interaction, such as more complex growth rates for the preys, or more complex predation types.

The most general form for a prey-predator model, called of Gause type, is as follows:

$$\begin{cases} H' = r(H)H - \pi(H)P, \\ P' = -\mu P + \omega(H)P. \end{cases}$$

We already analyzed $r(H)$, the growth rate, and $\pi(H)$, the functional response. The function $\omega(H)$ is the *numerical response*, that is the number of new predators produced in a unit time by a single predator. Usually, it is an increasing function of $\pi(H)$, so $\omega'(H) \geq 0$ and $\lim_{H \rightarrow \infty} \omega(H) > \mu$ (with a lot of preys, predators survive.) An example is $\omega(H) = \gamma\pi(H)$.

6.2 Lotka-Volterra model with logistic growth

We start from the simplest modification:

$$\begin{cases} H' = rH(1 - H/K) - aHP, \\ P' = -\mu P + \gamma aHP. \end{cases}$$

Here, preys grow according to the logistic equation, but predation is only linear.

6.2.1 Well-posedness

Before moving into equilibria and phase portrait, we should check the well-posedness. The right hand side is polynomial, thus local well-posedness applies. It is also possible to show that trajectories are bounded, since the triangular region $\mathcal{T}$ formed by the axes and the line $P = m(q - H)$, with q, m sufficiently large, is trapping the orbits.

The axes are both orbits, so they act as barriers. The diagonal side of $\mathcal{T}$ comes from the observation that for large H and P , we have $H' < 0$ and $P' < 0$. The actual computation is

$$\begin{aligned} 0 &> (H', P') \cdot (m, 1) = mH' + P' \\ &= mrH - mr\frac{H^2}{K} - \mu P + a(\gamma - m)HP \\ &= mr\left(1 - \frac{q}{K}\right)H + m\frac{r}{K}(q - H)H - \mu P + a(\gamma - m)HP \\ &= mr\left(1 - \frac{q}{K}\right)H + \frac{r}{K}HP - \mu P + a(\gamma - m)HP \\ &= mr\left(1 - \frac{q}{K}\right)H - \mu P + \left(a(\gamma - m) + \frac{r}{K}\right)HP. \end{aligned}$$

Since $H, P > 0$, all coefficient must be negative so that the function is always negative. Thus we require:

$$\begin{aligned} a(\gamma - m) + \frac{r}{K} < 0 &\Rightarrow m > \frac{r}{aK} + \gamma, \\ 1 - \frac{q}{K} < 0 &\Rightarrow q > K. \end{aligned}$$

6.2.2 Equilibria

As usual, we have the *extinction equilibrium* $E_0 = (0, 0)$. Additionally, we have $E_K = (K, 0)$, called *exclusion state*, and $E^* = (H^*, P^*)$, the *coexistence equilibrium*. The last solves the equation:

$$\begin{cases} 0 = r\left(1 - \frac{H^*}{K}\right) - aP^*, & \Rightarrow P^* = \frac{r}{a}\left(1 - \frac{H^*}{K}\right), \\ 0 = -\mu + \gamma aH^*, & \Rightarrow H^* = \frac{\mu}{\gamma a}. \end{cases}$$

Note that P^* can be negative, if

$$1 - \frac{H^*}{K} < 0, \quad \Rightarrow \quad H^* = \frac{\mu}{\gamma a} > K.$$

Thus, we have 2 situations:

- $H^* > K$: we only have 2 equilibria, E_0 and E_K ;
- $H^* < K$: we additionally have E^* .

The stability follows from the Jacobian:

$$J(H, P) = \begin{pmatrix} r - 2rH/K - aP & -aH \\ \gamma aP & -\mu + \gamma aH \end{pmatrix}.$$

We have:

$$\begin{aligned} J(0, 0) &= \begin{pmatrix} r & 0 \\ 0 & -\mu \end{pmatrix}, \\ J(K, 0) &= \begin{pmatrix} -r & -aK \\ 0 & -\mu + \gamma aK \end{pmatrix}, \\ J(H^*, P^*) &= \begin{pmatrix} -rH^*/K & -aH^* \\ \gamma aP^* & 0 \end{pmatrix}. \end{aligned}$$

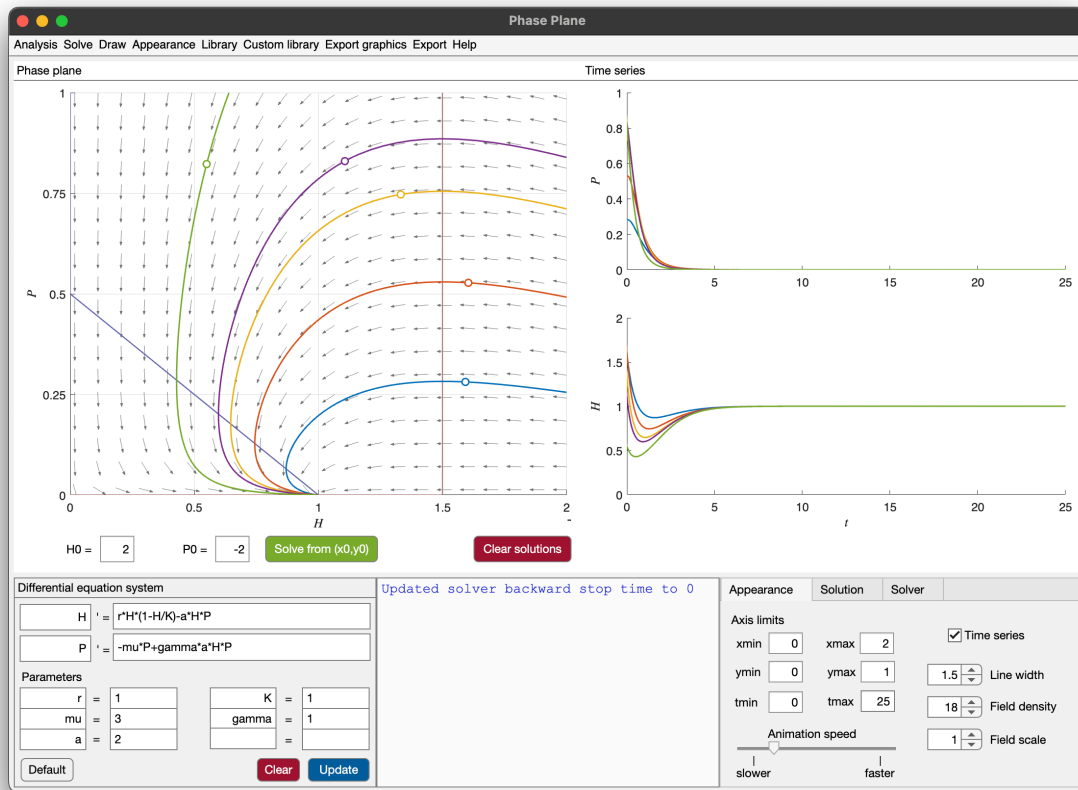
Thus:

- E_0 is always a saddle;
- E_K is asymptotically stable when $H^* > K$, otherwise is a saddle, as we can see from the eigenvalues on the diagonal;
- E^* , when there exists, is asymptotically stable. In fact, $\det J(H^*, P^*) = \gamma a^2 H^* P^* > 0$ and $\text{tr } J(H^*, P^*) = -rH^*/K < 0$.

In conclusion, we have 2 cases: $H^* > K$ and $H^* < K$.

6.2.3 Phase portrait: the case $H^* > K$

We have 3 regions of the positive quadrant, as depicted below:



On the right of $H = H^*$, orbits always flow to the left, since $H' < 0$. Thus, starting from there, we enter into the region between $H = H^*$ and $P = \frac{r}{a}(1 - \frac{H}{K})$. Again, orbits either flow into the triangle, or approach the only available equilibrium E_K . Finally, within the triangle, orbits flow towards E_K . We conclude that E_K is globally stable.

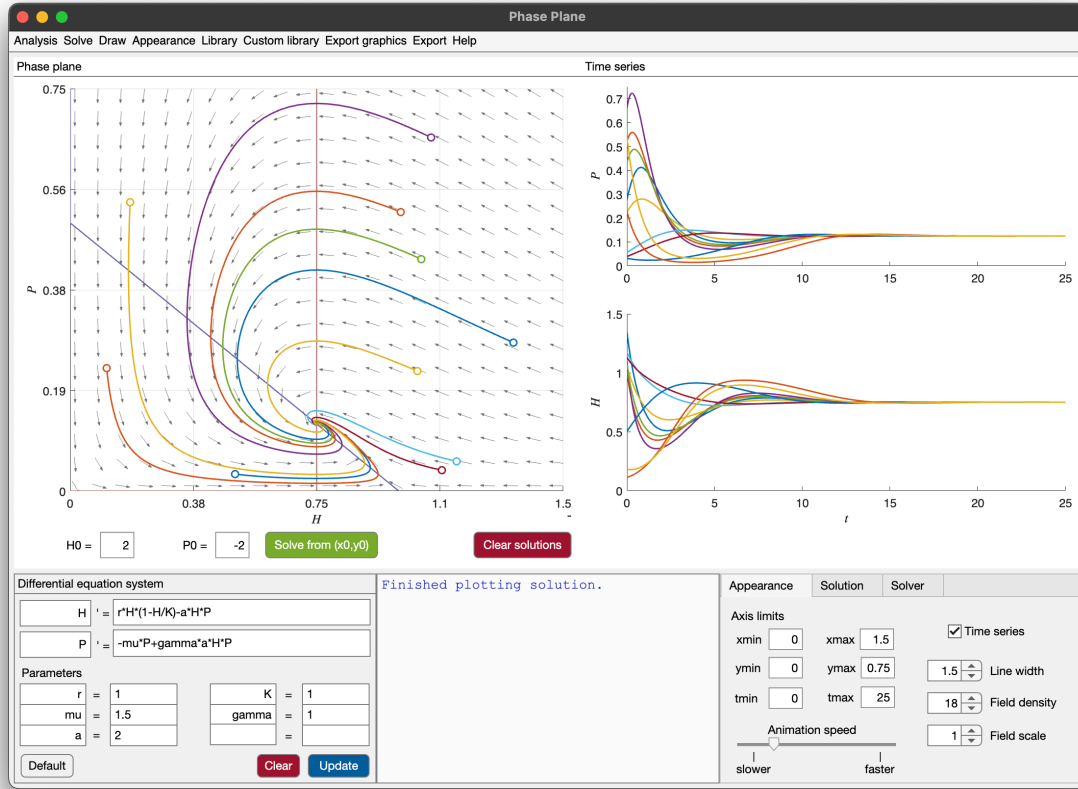
If $K \leq H^*$, then for any $(H(0), P(0)) > 0$ we have

$$\lim_{t \rightarrow \infty} (H, P) = (K, 0).$$

The predators go to extinction, while preys survive.

6.2.4 Phase portrait: the case $H^* < K$

We have 4 regions, as follows:



First, we show that we cannot have a limit cycle. Using Dulac's criterium with factor $1/(HP)$ we have:

$$\operatorname{div}\left(\frac{\mathbf{f}(H, P)}{HP}\right) = \frac{\partial}{\partial H}\left(\frac{r(1 - H/K) - aP}{P}\right) + \frac{\partial}{\partial P}\left(\frac{-\mu + \gamma aH}{H}\right) = -\frac{r}{KP} < 0.$$

Since $\mathcal{T}$ is a trapping region with no limit cycles, the ω -limit set is an equilibrium point. The only stable one is E^* , a spiral, thus all trajectories should converge there.

An alternative proof is based on the Lyapunov function

$$V(H, P) = \gamma(H - H^* \log H) + (P - P^* \log P).$$

We recall that a Lyapunov function has the property that is positive, zero at the equilibrium,

and (semi-)negative definite, then the equilibrium is stable. This is the case, since:

$$\begin{aligned}
\frac{d}{dt}V(H(t), P(t)) &= \gamma\left(1 - \frac{H^*}{H}\right)H\left(r\left(1 - \frac{H}{K}\right) - aP\right) + \left(1 - \frac{P^*}{P}\right)P(-\mu + \gamma aH) \\
&= \gamma a(H - H^*)\left(\frac{r}{a}\left(1 - \frac{H}{K}\right) - P\right) + \gamma a(P - P^*)\left(-\frac{\mu}{\gamma a} + H\right) \\
&= \gamma a(H - H^*)\left(P^* - \frac{r}{a}\frac{H - H^*}{K} - P\right) + \gamma a(P - P^*)(H - H^*) \\
&= -\frac{r\gamma}{K}(H - H^*)^2 \leq 0.
\end{aligned}$$

Thus, orbits converge towards the line $H = H^*$. But along this line we only have the equilibrium E^* , thus E^* attracts all orbits.

We see here that the Lotka-Volterra system is not structurally stable, because a perturbation of it (adding the logistic term, which has little effect for H small) prevents the existence of a limit cycle.

7 Rosenzweig-MacArthur model

This lecture is based on [ip14, Section 6.4]

7.1 Rosenzweig-MacArthur model

We see that a perturbation of the Lotka-Volterra model, obtained by adding the logistic term, destroys the structural stability of the original model: no limit cycles are present.

In 1963, Rosenzweig and MacArthur proposed a new model, of Gause-type, with a Holling type II predation. Again, this can be seen as a small modification of the original system, thus we shall expect a similar dynamic for a wide range of parameters. The model reads as follows:

$$\begin{cases} H' = rH(1 - H/K) - a\frac{HP}{1 + aTH}, \\ P' = -\mu P + \gamma a\frac{HP}{1 + aTH}. \end{cases}$$

All parameters are positive. The new parameter $T > 0$ is the *handling time*, and the maximum killing rate is $1/T$.

7.1.1 Non-dimensionalization

We have 6 parameters in total, but we can reduce them by rescaling the quantities:

$$u = \frac{H}{K}, \quad v = \frac{P}{\gamma K}, \quad \tau = \mu t.$$

The rescaled equations are:

$$\begin{cases} \dot{u} = \rho u(1 - u) - \alpha \frac{\delta uv}{1 + \delta u}, \\ \dot{v} = -v + \alpha \frac{\delta uv}{1 + \delta u}, \end{cases}$$

where we defined the following quantities:

$$\rho = \frac{r}{\mu}, \quad \alpha = \frac{\gamma}{T\mu}, \quad \delta = aTK.$$

The total number of parameters is now only 3.

7.1.2 Equilibria

The nullclines are as follows:

$$\begin{cases} 0 = \rho u(1-u) - \alpha \frac{\delta uv}{1+\delta u}, \\ 0 = -v + \alpha \frac{\delta uv}{1+\delta u}, \end{cases}$$

Let us rewrite the system as follows:

$$\begin{cases} 0 = f(u)(\Phi(u) - v), \\ 0 = v(f(u) - 1), \end{cases}$$

where we defined:

$$f(u) = \frac{\alpha \delta u}{1 + \delta u}, \quad \Phi(u) = \frac{\rho}{\alpha \delta} (1 - u)(1 + \delta u).$$

Note that $f(0) = 0$, $0 \leq f(u) < \alpha$, and $f'(u) > 0$. The equilibria are:

- $E_0 = (0, 0)$, and is always present for any choice of the parameters.
- $E_1 = (1, 0)$, because $\Phi(1) = 0$. This equilibrium is also always present.
- $E^* = (u^*, v^*)$ where $f(u^*) = 1$ and $v^* = \Phi(u^*)$. The existence depends on the parameters.
 1. $\alpha > 1$, otherwise there exists no $u^* \geq 0$ such that $f(u^*) = 1$. In this case, we have $u^* = \frac{1}{\delta(\alpha-1)}$.
 2. $v^* = \Phi(u^*) \geq 0$, which implies $u^* \leq 1$. In terms of parameters, $\delta > \delta^* = \frac{1}{\alpha-1}$. (We define δ^* because is a bifurcation point, as we shall see.)

Thus we can have 2 or 3 equilibria. Note that $\alpha > 1$ means, in terms of the original parameters,

$$\frac{\gamma}{T} = \lim_{H \rightarrow 0} \omega(H) > \mu,$$

consistently with the hypothesis on $\omega(H)$ for general Gause-type models: for large H , predator population should grow.

7.1.3 Local stability

The Jacobian is

$$J(u, v) = \begin{pmatrix} f'(u)(\Phi(u) - v) + f(u)\Phi'(u) & -f(u) \\ v f'(u) & f(u) - 1 \end{pmatrix}.$$

We have:

$$\begin{aligned} J(0, 0) &= \begin{pmatrix} f'(0)\Phi(0) & 0 \\ 0 & -1 \end{pmatrix}, \\ J(1, 0) &= \begin{pmatrix} f(1)\Phi'(1) & -f(1) \\ 0 & f(1) - 1 \end{pmatrix}, \\ J(u^*, v^*) &= \begin{pmatrix} \Phi'(u^*) & -1 \\ v^* f'(u^*) & 0 \end{pmatrix}, \end{aligned}$$

where we used the definition of the equilibria, when they exist. Therefore:

- $E_0 = (0, 0)$ has eigenvalues $\lambda_1 = f'(0)\Phi(0) > 0$ and $\lambda_2 = -1 < 0$, so is a *saddle* (always).
- $E_1 = (1, 0)$ has eigenvalues $\lambda_1 = f(1)\Phi'(1) < 0$, because $\Phi'(1) < 0$, and $\lambda_2 = f(1) - 1$, which depends on the parameters. In particular,
 - if $\delta < \delta^*$ then $f(1) - 1 < 0$, so the equilibrium E_1 is asymptotically stable.
 - if $\delta > \delta^*$, E_1 is a saddle. Note that in this case we have the existence of E^* .

So, at $\delta = \delta^*$ (or $u^* = 1$) we have a bifurcation (transcritical).

- $E^* = (u^*, v^*)$ has $\det J^* = v^* f'(u^*) > 0$ and $\text{tr } J^* = \Phi'(u^*)$. Since $\Phi(u)$ is a concave parabola with a maximum for $u = u_{\max} = \frac{\delta-1}{2\delta}$, we have that:
 - if $u^* > u_{\max}$ then $\Phi'(u^*) < 0$, thus E^* is asymptotically stable.
 - if $u^* < u_{\max}$ then if $\Phi'(u^*) > 0$, thus E^* is unstable.

There is a second bifurcation point for $u^* = u_{\max}$, because E^* loses stability. In terms of parameters, this happens for $\delta^{**} = \frac{\alpha+1}{\alpha-1}$.

Additionally, the discriminant $\Delta = (\text{tr } J^*)^2 - 4 \det J^*$ tells us whether E^* is a spiral or not. Since $\Delta > 0$ for $u^* = 1$ and $\Delta < 0$ for $u^* = u_{\max}$, it changes sign at some point.

In summary, assuming $\alpha > 1$, we have 3 cases:

1. $\delta \in (0, \delta^*)$: 2 equilibria, E_0 saddle and E_1 asymptotically stable;
2. $\delta \in (\delta^*, \delta^{**})$: 3 equilibria, E_0 and E_1 are saddles, and E^* is asymptotically stable (and spiral at some point).
3. $\delta > \delta^{**}$: 3 equilibria, non of which is stable.

7.1.4 Limit cycles

For $\delta < \delta^{**}$ we also have no limit cycles. This can be shown by using the Dulac's criterium with integrating factor

$$h(u, v) = \frac{1}{2\rho\delta} \frac{uv^{q-1}}{(1 + \delta u)}, \quad q = \rho u^* |u_{\max}|.$$

Computing the divergence we have:

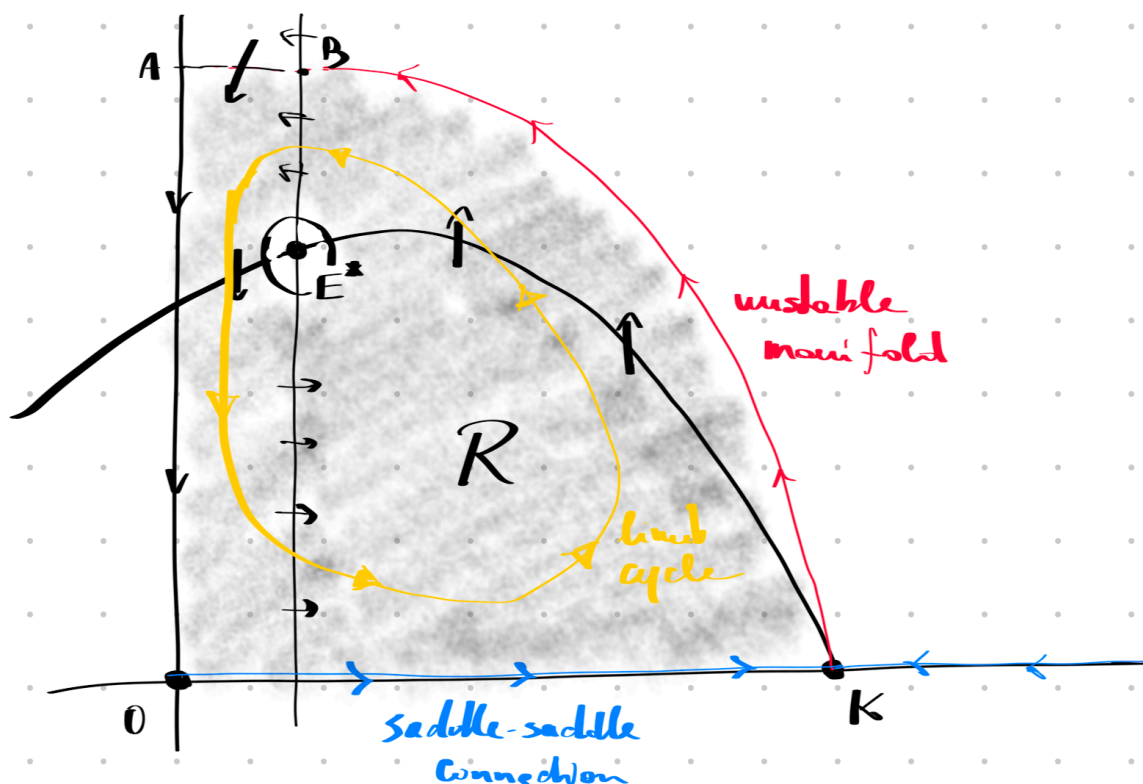
$$\operatorname{div}(h(u, v)\mathbf{f}(u, v)) = -uv^{q-1} \left(u^2 + (u_{\max} + |u_{\max}|)u - u^*u_{\max} \right),$$

so if $u_{\max} \leq 0$ the divergence is always negative. If $u_{\max} > 0$ and $u_{\max} \leq u^*$:

$$\begin{aligned} \operatorname{div}(h(u, v)\mathbf{f}(u, v)) &= -uv^{q-1} \left(u^2 + 2u_{\max}u - u^*u_{\max} \right) \\ &\leq -uv^{q-1} \left(u^2 + 2u_{\max}u - u_{\max}^2 \right) \\ &= -uv^{q-1} (u - u_{\max})^2, \end{aligned}$$

which is negative for $u \neq u^*$. So we cannot have limit cycles when the maximum of $\Phi(u)$ is on the negative side of, if present in the positive quadrant, when $u^* \geq u_{\max}$. This is equivalent to $\Phi'(u^*) > 0$, as seen above.

On the other hand, for $u^* \leq u_{\max}$ we cannot conclude anything from the Dulac criterium. Actually, we have a stable limit cycle orbiting around E^* in this case. To prove it, we use the Poincaré theorem, defining a trapping region for the orbits the does not contain any stable equilibrium. The region is the following:



The region $\mathcal{R}$ is delimited by the axes, the unstable manifold of E_K , the segment $\overline{AB}$, and a small hole around E^* . Trajectories are trapped, because the segment $\overline{AB}$ and the boundary of the hole are *inflow* for the vector field. On the boundary of the region there are 2 saddles, but trajectories *inside* of $\mathcal{R}$ cannot go there, otherwise they would be part of their stable manifold. We can also exclude the saddle-saddle connection, which is a heteroclinic orbit, because trajectories cannot go there. So, there must exist at least one limit cycle inside $\mathcal{R}$. It is also unique, but this is more difficult to prove.

7.1.5 Bifurcations

For $\delta = \delta^*$ we have a *transcritical bifurcation* and for $\delta = \delta^{**}$ we have a *Hopf bifurcation*. Let's see what we mean by this.

Consider a system of the form

$$\mathbf{y}' = \mathbf{f}(\mathbf{y}, p),$$

for some parameter $p \in \mathbb{R}$ (in our case, δ), and suppose that $\bar{\mathbf{y}}$ is an equilibrium for $p = \bar{p}$. Then, assuming that $\det J(\bar{\mathbf{y}}, \bar{p}) \neq 0$, we can locally define a curve of equilibria $\bar{\mathbf{y}} = \bar{\mathbf{y}}(p)$ such

that $\mathbf{f}(\bar{\mathbf{y}}(p), p) = 0$ for $p \in \mathcal{B}_\varepsilon(\bar{p})$. Moreover, the eigenvalues are continuous functions of p , so stability is not affected.

If one real eigenvalue becomes zero, then we have a bifurcation and a change of stability. We consider 3 cases:

- **Tangent bifurcation.** $J(\bar{\mathbf{y}}, \bar{p})$ has eigenvalue 0 and $\partial_p \mathbf{f}(\bar{\mathbf{y}}, \bar{p}) \neq 0$.
- **Transcritical bifurcation.** $J(\bar{\mathbf{y}}, \bar{p})$ has eigenvalue 0 and $\partial_p \mathbf{f}(\bar{\mathbf{y}}, \bar{p}) = 0$.
- **Hopf bifurcation.** $J(\bar{\mathbf{y}}, \bar{p})$ has pure imaginary eigenvalues $\pm i\omega$.

For $\delta = \delta^*$ the equilibria E_1 and E^* coincide and one eigenvalue is 0. Since E_1 does not depend on δ , $\partial_\delta \mathbf{f}(E_1, \delta) = 0$: we have a transcritical bifurcation. Graphically, the curve of equilibria of E_1 and E^* crosses each other and exchange their stability.

For $\delta = \delta^{**}$ we have a *Hopf bifurcation*: the equilibrium E^* becomes unstable and an infinitesimally small and stable limit cycle appears around it. We can detect a Hopf bifurcation from the eigenvalues of J^* : they are complex conjugate of the form:

$$\lambda(\delta) = \sigma(\delta) \pm i\omega(\delta).$$

We have that $\sigma(\delta^{**}) = 0$ but $\omega(\delta^{**}) = \sqrt{\rho/\delta^{**}} \neq 0$. Moreover, when eigenvalues are imaginary, the real part is twice the trace, so $\sigma(\delta) = \Phi'(u^*(\delta))$, and we can conclude that $\sigma'(\delta^{**}) > 0$. The eigenvalues are crossing the imaginary axis on the complex plane from the negative to the positive real part side, as δ crosses the bifurcation. Since the new limit cycle is stable, we have a *supercritical* Hopf bifurcation. When the limit cycle is unstable, we have a *subcritical* Hopf bifurcation.

7.1.6 Summary

In conclusion, we have the following picture:

- $\delta < \delta^*$: E_1 globally attractive.
- $\delta = \delta^*$: transcritical bifurcation.
- $\delta^* < \delta < \delta^{**}$: E^* globally attractive.
- $\delta = \delta^{**}$: Hopf bifurcation.
- $\delta > \delta^{**}$: limit cycle globally attractive.

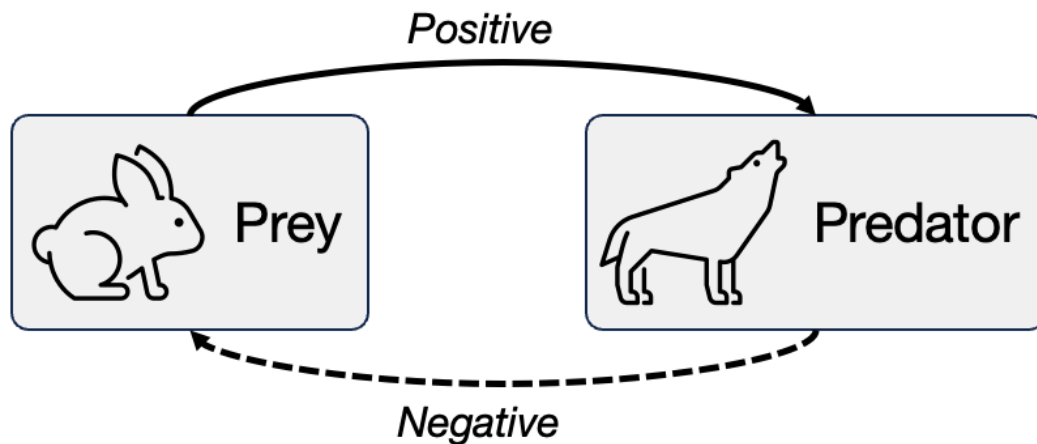
The global stability of E_1 and E^* follows from the same argument above, using a trapping region (the same in the case of E^* , because we know there cannot be a limit cycle and E^* is stable.)

8 Michaelis-Menten law

This lecture is based on [ks08_vol1, Section 1.1, 1.4.1-2]

8.1 Feedback control

The prey-predator model is an example of system with a *feedback*. The prey population has a *positive* effect on the predator population, whereas the predator population has a *negative* effect on the prey population.



Specifically, in this case we say that the system has a *negative feedback loop*. In a negative feedback, a positive value of a variable yields a decrease in that variable and, vice versa, a negative value leads to an increase. An important feat of negative loops is that they can lead to *oscillations*, as per prey-predator models.

We can find many good examples of negative loops in real life (try yourselves!) and in biology too:

- **Gene regulation.** Many genes inhibit their own transcription. An example is the protein *Hes1*, involved in the early stages of development of the embryo. *Hes1* oscillates over time, because the protein inhibits its own transcription.
- **Circadian rhythm.** Our body has an internal clock, synchronized with the Sun, located in the hypothalamus. The protein *PER* inhibits its own transcription, thus creating oscillations with a period of 24 hours.
- **Insuline-glucose oscillations.** An intake of glucose (from a meal) causes the pancreas to secrete more insulin, which eventually lowers the level of glucose in the body.
- **Hormones.** Many hormones like testosterone and progesterone are under negative feedback with the pituitary gland and the brain.

Positive feedback loops are important too. Using the analogy of ecology, consider the following (Kolmogorov) model:

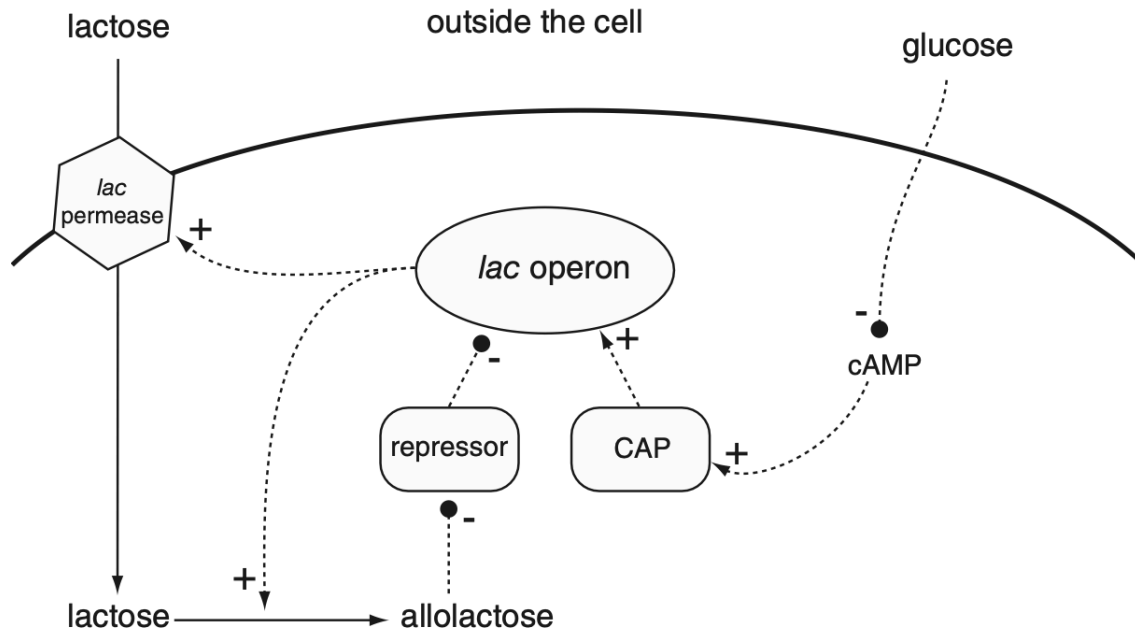
$$\begin{cases} N_1' = N_1 f(N_1, N_2), \\ N_2' = N_2 g(N_1, N_2), \end{cases}$$

We can have 3 situations:

1. $\frac{\partial f}{\partial N_2} < 0$ and $\frac{\partial g}{\partial N_1} > 0$: this is a prey-predator interaction, thus a negative feedback, because an increase of N_1 increases N_2' , which increases N_2 and that has a negative effect on N_1' itself.
2. $\frac{\partial f}{\partial N_2} < 0$ and $\frac{\partial g}{\partial N_1} < 0$: this is a *competitive* model, because an increase of N_1 (resp. N_2) has a negative effect of N_2 (resp. N_1). Here, there is no vicious or virtuous loop, possible equilibria are the extinction of either of the two species. (Coexistence is also possible but more difficult to achieve.)
3. $\frac{\partial f}{\partial N_2} > 0$ and $\frac{\partial g}{\partial N_1} > 0$: this is the case of *cooperation* or *mutualism* or *symbiosis*. Both species benefit from the presence of the other. This is a *positive feedback*, because as N_1 increases, N_2 increases too leading to an increase to N_1 itself.

Examples outside biology include **climate change**, for instance CO₂ emissions raise the global temperature, but at higher temperatures soil microbes have faster metabolism, thus breaking down organic matter faster, releasing more CO₂. Or methane release, which has much more greenhouse power than CO₂. Methane is trapped in permafrost, but its release increases temperatures which will melt permafrost and release more methane.

In general, life is an intricate network of positive and negative interactions. A positive effect is also called *activation*: a gene *activates* another gene, in the sense that its increase also increases the other one; a negative effect is called *inhibition*. An example is the *lac operon* shown below:



The *lac operon* is a feedback control loop present in the *E. coli*. When there is plenty of glucose, *E. coli* uses it for metabolism, even if other sugars are present. On the other hand, when glucose is absent, the bacterium activates other genes that are able to process lactose. This is an example of *genetic switch*.

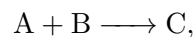
In this series of lectures we will explore more in depth feedback loops. We will start with enzymatic reactions, a good example of how to create complex and non-linear interactions out of “simple” chemistry.

8.2 Chemical kinetics

Before delving deeper into the matter, we need to refresh a bit our chemistry knowledge, specifically [chemical kinetics](#).

Chemical kinetics is the study of reaction rates. It is known that reaction rates depend on the concentration of reactants (and in some cases products) in characteristic ways that can be expressed in terms of differential equations known in chemistry as “rate laws”.

Consider the following reaction:



Here, the reaction involves 3 molecules, A, B, and C, and it simply states “the reaction of A and B gives C”. The formula tells us that exactly *one* molecule of A and B is needed to form

one molecule of C (stoichiometry of the reaction.) However, it does not give us information on the “rate” of product of C, for instance, or the consumption of A and B.

We measure the concentration of a species X at time $t \in \mathbb{R}$ by $[X](t)$. The unit of measure is mole per liter, that is $[X]$ is the molar concentration. (We assume that the volume of the system is constant.) A mole of X is exactly $N_A = 6.023 \times 10^{23}$ (Avogadro constant) molecules of X. So, the above reaction means that “one mole of A plus one mole of B gives exactly one mole of C”. This is important, since it does not work that well with molecular weights!

The rate of consumption or production of X is simply

$$\frac{d[X]}{dt},$$

with positive sign the case of production, and negative otherwise. From the stoichiometry of $A + B \longrightarrow C$ we have that

$$\frac{d[A]}{dt} = \frac{d[B]}{dt} = -\frac{d[C]}{dt}.$$

For the reaction $A + 2B \longrightarrow 3C + D$ we have:

$$\frac{d[D]}{dt} = \frac{1}{3} \frac{d[C]}{dt} = -\frac{d[A]}{dt} = -\frac{1}{2} \frac{d[B]}{dt}.$$

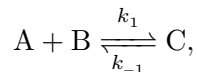
8.2.1 Law of mass action

The fundamental “law” of a chemical reaction is the law of mass action. This law describes the rate at which chemicals collide and interact to form different chemical combinations. For the reaction $A + B \longrightarrow C$, the law of mass action tells us that

$$\frac{d[C]}{dt} = k[A][B],$$

that is, the rate of production of $[C]$ is proportional to the concentration of A and B. The law is based on the idea that the bimolecular chemical reaction results from the *collision* of A and B, thus it must be proportional to the concentration of *both*. We encountered the same law for prey-predator systems. As for prey-predator system, therefore, the law is not inviolable, because in many situations also k depends on the concentrations. Deviations may occur at very large or very low concentrations of A or B, for instance.

For thermodynamical reasons, a chemical reaction always proceeds in both directions, that is:



where k_1 is the rate of the forward reaction $A + B \longrightarrow C$, and k_{-1} is the rate of the backward reaction $C \longrightarrow A + B$. Since the two reactions (forward and backward) are independent, we have that

$$\frac{d[C]}{dt} = k_1[A][B] - k_{-1}[C].$$

Note that we can apply the law of mass action also to the backward reaction: the rate is proportional to the concentration of the reactants, in this case only C.

The full system of equations results from the application of the law to each species:

$$\begin{cases} [A]' = -k_1[A][B] + k_{-1}[C], \\ [B]' = -k_1[A][B] + k_{-1}[C], \\ [C]' = k_1[A][B] - k_{-1}[C]. \end{cases}$$

There is some redundancy in the system, because of the conservation of mass. Indeed, $[A] + [C]$ and $[B] + [C]$ are constant. Another way to see this is noting that $[A]' + [C]' = 0$ from the above system.

At equilibrium, the production and consumption rate of C are the same. So:

$$\frac{d[C]}{dt} = 0, \quad \Rightarrow \quad k_1[A]_{\text{eq}}[B]_{\text{eq}} = k_{-1}[C]_{\text{eq}}.$$

The *equilibrium constant* of the reaction is:

$$K_{\text{eq}} = \frac{k_{-1}}{k_1} = \frac{[A]_{\text{eq}}[B]_{\text{eq}}}{[C]_{\text{eq}}}.$$

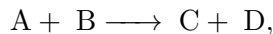
The equilibrium constant is very important. First, it provides a link to thermodynamics (see the end of the notes). Second, it can be used to understand the trend of the reaction: when K_{eq} is very small, then at equilibrium (so after waiting for long time) most of A and B are combined into C.

Quantitatively, since $[A] + [C]$ is constant, then $[A]_{\text{eq}} + [C]_{\text{eq}} = [A](0) + [C](0)$. Assuming that at the beginning $[C](0) = 0$ and $[A](0) = [A]_0 > 0$, then

$$[C]_{\text{eq}} = A_0 \frac{[B]_{\text{eq}}}{K_{\text{eq}} + [B]_{\text{eq}}}.$$

In other words, when $[B]_{\text{eq}} = K_{\text{eq}}$, half of A is combined into C.

Consider now the reaction



for some positive constants α , β , γ , and δ . We have that:

$$\frac{1}{\delta} \frac{d[D]}{dt} = \frac{1}{\gamma} \frac{d[C]}{dt} = -\frac{1}{\alpha} \frac{d[A]}{dt} = -\frac{1}{\beta} \frac{d[B]}{dt}.$$

For the law of mass action, we have:

$$\frac{1}{\alpha} \frac{d[A]}{dt} = -k_1[A]^\alpha[B]^\beta + k_{-1}[C]^\gamma[D]^\delta,$$

because we need α molecules of $[A]$ and β molecules of $[B]$ for the forward reaction (and similar for the backward reaction). The equilibrium constant is:

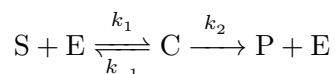
$$K_{\text{eq}} = \frac{[A]_{\text{eq}}^\alpha [B]_{\text{eq}}^\beta}{[C]_{\text{eq}}^\gamma [D]_{\text{eq}}^\delta}.$$

8.3 Michaelis-Menten law

A fundamental property of biochemical reactions in living cells is the fact that the rate at which they occur is regulated by the presence of other molecules. An enzyme is a protein that makes it possible that a reaction (that by itself would be extremely slow) occurs over a short time-scale, as necessary for the cell functioning, but is apparently not involved in the reaction itself, in the sense that its concentration at the end is the same as at the beginning.

Ezymes are catalysts that converts *substrates* into *products*. The speed-up of the reaction obtained thanks to the enzyme can be of several order of magnitudes, like millions of times or more. Enzymes are also *highly specific*. More importantly, enzymes are the building blocks of regulatory mechanisms present in every living body.

We consider now the archetypal example of enzymatic reaction. The diagram is the following:



In the diagram S refers to the “substrate” the molecule that is converted into P , the “product” of the reaction, while E refers to enzyme molecules. Looking only at the left and right end-points of the diagram, one sees that a molecule of S is converted into P , while one is left with same molecule of E present in the beginning. However, one sees that E is involved in intermediate reactions (the complex), necessary to understand why the presence of the enzyme is necessary for the reaction to occur.

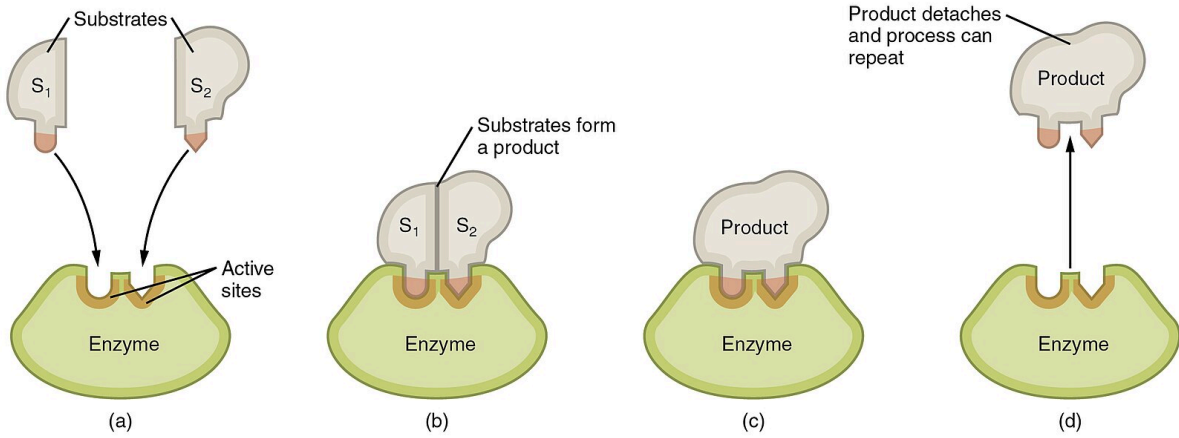


Figure 8.1: (From Wikimedia)

The letters over each arrow in the diagram represent the rate constants at which each reaction (represented by an arrow) occurs. We assume that all reactions occur following the *law of mass action*: when a reaction involves two substances reacting, the rate at which the reaction occurs is proportional to the product of the concentrations of the substances involved; the rate constant is the proportionality constant. On the other hand, when a reaction involves a single substance, the reaction rate is the product of the rate constant times the concentrations of that substance.

It is easy transforming the diagram into a set of differential equations, using the principle that the concentration of each substance will increase at the sum of the rates of all reactions producing it, and decrease at those destroying it. Namely setting

$$\begin{aligned} [S](t) &= \text{concentration of substrate } S \text{ at time } t, \\ [E](t) &= \text{concentration of enzyme } E \text{ at time } t, \\ [C](t) &= \text{concentration of complex } C \text{ at time } t, \\ [P](t) &= \text{concentration of product } P \text{ at time } t. \end{aligned}$$

we obtain,

$$\begin{cases} [S]' = k_{-1}[C] - k_1[S][E], \\ [E]' = (k_{-1} + k_1)[C] - k_1[S][E], \\ [C]' = k_1[S][E] - (k_{-1} + k_2)[C], \\ [P]' = k_2[C]. \end{cases}$$

This appears to be a system of 4 differential equations with 4 unknowns ($[S](t)$, $[E](t)$, $[C](t)$, $[P](t)$). Typical initial conditions will be $[S](0) = s_0 > 0$, $[E](0) = e_0 > 0$, $[C](0) = [P](0) = 0$ (at time $t = 0$, beginning of the reaction, there is neither product nor complex).

One can immediately see that the first 3 equations do not depend on $[P](t)$; we can then solve the first 3 equations, then compute $[P](t)$ (the quantity presumably more relevant) as:

$$[P](t) = k_2 \int_0^t [C](s) \, ds.$$

Moreover, one may note that

$$\frac{d}{dt} [C](t) + [E](t) = 0, \quad \Rightarrow \quad [C](t) + [E](t) = [C](0) + [E](0) = e_0.$$

We can therefore remove one equation and substitute $[E](t) = e_0 - [C](t)$. We end up with a system of 2 equations:

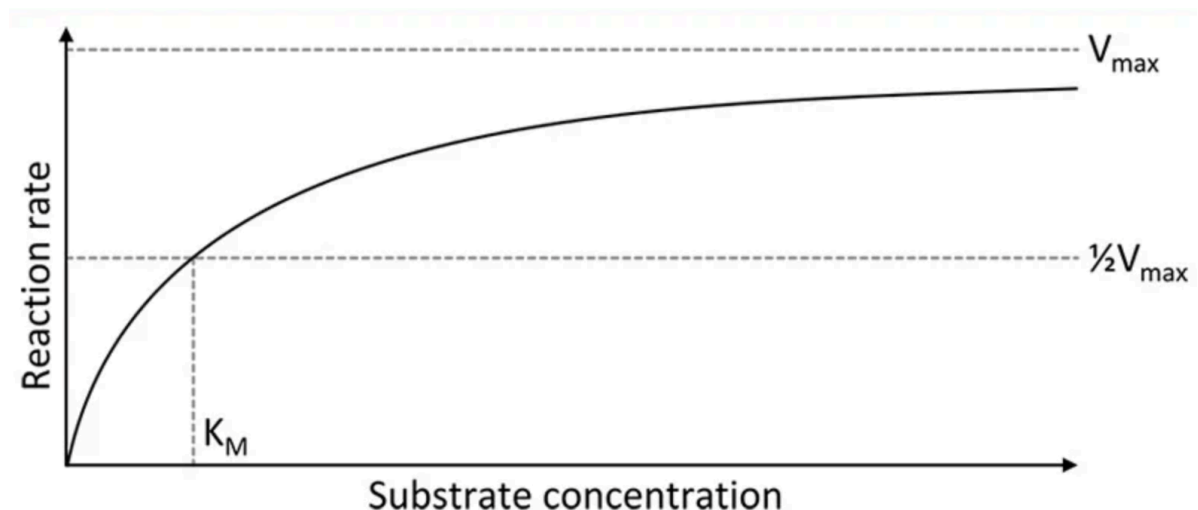
$$\begin{cases} [S]' = k_{-1}[C] - k_1[S](e_0 - [C]), \\ [C]' = k_1[S](e_0 - [C]) - (k_{-1} + k_2)[C], \end{cases}$$

with initial conditions $[S](0) = s_0 > 0$ and $[C](0) = 0$.

It is certainly possible analyse the behavior of system as it is. For instance, we would like to know how $[C]$, and hence $[P]'$ vary over time and depending on the concentration of the substrate. This type of analysis leads to the *Michaelis-Menten law*, which states that:

$$[P]' = V = \frac{V_{\max}[S]}{K_m + [S]},$$

for $V_{\max} > 0$ being the *maximum reaction rate*, that is $V \rightarrow V_{\max}$ for $[S] \rightarrow \infty$, and K_m is the *half-saturation rate*, that is when $[S] = K_m$ then $V = \frac{1}{2}V_{\max}$. Note that this law is exactly the Holling type II law from prey-predator models.



8.3.1 Equilibrium and phase portrait

As usual, we may have a look at the equilibria and their stability. Here, however, this analysis is not very insightful.

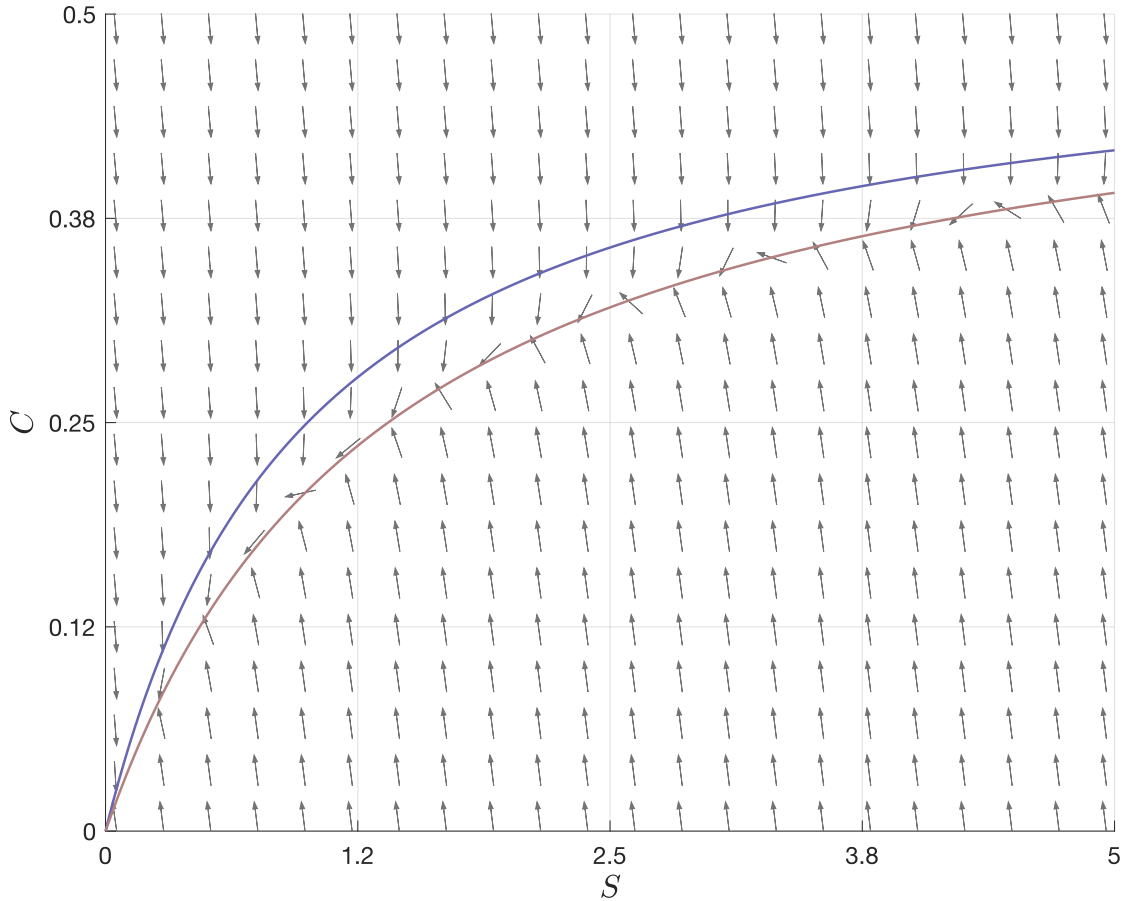
The only equilibrium is the origin. It is also locally asymptotically stable, since:

$$J(0,0) = \begin{pmatrix} -e_0 k_1 & k_{-1} \\ e_0 k_1 & -k_2 - k_{-1} \end{pmatrix}$$

so the trace is negative and $\det J = e_0 k_1 k_2 > 0$.

The nullclines are:

$$\begin{aligned} [S]' = 0 &\Leftrightarrow [C] = \frac{e_0 k_1 [S]}{k_{-1} + k_1 [S]}, \\ [S]' = 0 &\Leftrightarrow [C] = \frac{e_0 k_1 [S]}{k_{-1} + k_2 + k_1 [S]}. \end{aligned}$$



The nullclines are similar: they have the same horizontal asymptote e_0 , they intersect at the origin, but the second one is slightly shifted to the right, because they reach $\frac{1}{2}e_0$ (half of the maximum) at different values: for $[S] = \frac{k_{-1}}{k_1}$ in the first case, and $[S] = \frac{k_{-1}+k_2}{k_1}$ in the second case.

From the plot we see that once a trajectory enters into the corridor between the nullclines, it remains trapped. Since we only have the origin as equilibrium, all trajectories must converge there: the equilibrium is globally stable.

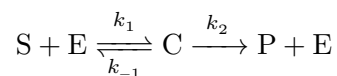
The corridor is shaped like a Michaelis-Menten law, in the sense that in the gap $[C]$ varies with respect to $[S]$ like in the Michaelis-Menten law. But which curve is the correct one? We will see it tomorrow.

9 Fast-Slow systems

This lecture is based on [ks08_vol1, Section 1.6.3]

9.1 Michaelis-Menten law

We consider again the reaction:



The system is:

$$\begin{cases} [S]' = k_{-1}[C] - k_1[S][E], \\ [E]' = (k_{-1} + k_1)[C] - k_1[S][E], \\ [C]' = k_1[S][E] - (k_{-1} + k_2)[C], \\ [P]' = k_2[C]. \end{cases}$$

We can remove one equation and substitute $[E](t) = e_0 - [C](t)$. We end up with a system of 2 equations:

$$\begin{cases} [S]' = k_{-1}[C] - k_1[S](e_0 - [C]), \\ [C]' = k_1[S](e_0 - [C]) - (k_{-1} + k_2)[C], \end{cases}$$

with initial conditions $[S](0) = s_0 > 0$ and $[C](0) = 0$.

Here, we would like to show that the production rate follows the Michaelis-Menten law:

$$[P]' = V = \frac{V_{\max}[S]}{K_m + [S]},$$

9.1.1 The quasi-equilibrium approximation

One way to derive the law is with the quasi-equilibrium assumption. This approach was proposed by Michaelis and Menten in 1913. They assumed that the first reaction $S + E \longrightarrow C$ is approximately at equilibrium, thus:

$$K_1 = \frac{k_{-1}}{k_1} = \frac{[S][E]}{[C]}.$$

Since $[C] + [E] = e_0$, we conclude that:

$$K_1 = \frac{[S](e_0 - [C])}{[C]}, \quad \Rightarrow \quad [C] = \frac{e_0[S]}{K_1 + [S]}.$$

But $[P]' = V = k_2[C]$, so:

$$V = \frac{k_2 e_0 [S]}{K_1 + [S]},$$

which is Michaelis-Menten law with $K_m = K_1$ and $V_{\max} = k_2 e_0$.

We remark that this derivation is an *approximation*. In fact, the only true equilibrium of the system is the origin, which is quite useless as an approximation. Here, we can *cannot* conclude that $[S]' = 0$, in spite of the assumption. What we are actually saying is that, as we use S in the reaction, the concentration of the complex approximately follows the concentration of S according to the Michaelis-Menten law.

We can derive a law for $[S]$ by noting that (from the system)

$$[S]' + [C]' = -k_2[C],$$

so

$$\frac{d}{dt}([S] + [C]) = \frac{d}{dt}\left([S] + \frac{e_0[S]}{K_1 + [S]}\right) = -k_2 \frac{e_0[S]}{K_1 + [S]},$$

and after rearranging we have

$$\frac{d[S]}{dt} \left(1 + \frac{K_1 e_0}{(K_1 + [S])^2}\right) = -\frac{k_2 e_0 [S]}{K_1 + [S]},$$

which gives a differential equation for $[S](t)$.

The quasi-equilibrium approximation is very easy, powerful, and extensively applied in enzymatic reaction. It is possible to prove this approximation in a more mathematical way by assuming $k_{-1} \gg k_2$, but we will follow a different approach below.

9.1.2 The quasi-steady-state approximation

First, let us rescale the equations by the transformations:

$$x = \frac{[S]}{s_0}, \quad y = \frac{[C]}{e_0}, \quad \tau = k_1 e_0 t.$$

After changing the variables, we get:

$$\begin{cases} \frac{dx}{d\tau} = \frac{k_{-1}}{s_0 k_1} y - x(1 - y), \\ \frac{dy}{d\tau} = \frac{s_0}{e_0} x(1 - y) - \frac{k_{-1} + k_2}{k_1 e_0} y, \end{cases}$$

that we can rearrange in the following form:

$$\begin{cases} \frac{dx}{d\tau} = -x + y(x + \alpha), \\ \varepsilon \frac{dy}{d\tau} = x - y(x + \beta), \end{cases}$$

where:

$$\alpha = \frac{k_{-1}}{s_0 k_1}, \quad \beta = \frac{k_{-1} + k_2}{k_1 s_0}, \quad \varepsilon = \frac{e_0}{s_0}.$$

Next, suppose that $\varepsilon \ll 1$, that is the initial concentration of the enzyme is, at the start, very low compared to the concentration of the substrate. Thus, we could assume that $\varepsilon y' \approx 0$:

$$\begin{cases} \frac{dx}{d\tau} = -x + y(x + \alpha), \\ 0 = x - y(x + \beta). \end{cases}$$

This approach is called *quasi-steady-state approximation*. From the second equation we conclude that

$$y = \frac{x}{\beta + x}, \quad \Rightarrow \quad [C] = \frac{[e_0 S]}{\frac{k_{-1} + k_2}{k_1} + [S]}.$$

We conclude that with the QSS approximation the Michaelis-Menten law has the same form with $V_{\max} = k_2 e_0$ (as before) and $K_m = \frac{k_{-1} + k_2}{k_1}$ (different, before was $K_m = K_1$). The difference is minimal: in the quasi-equilibrium approximation, we have $k_2 \ll k_{-1}$, so $K_m \approx K_1$, whereas for the QSS approximation we assume $\varepsilon \ll 1$.

The Michaelis-Menten law is quite general and often evoked without deriving it (similarly to Holling type predations.) For us, it is the main ingredient for more complex reactions.

9.2 Singular perturbation theory

Let dive a bit deeper on the QSS approximation. The idea is the following: since ε is small, we could try to approximate the solution of the IVP

$$\begin{cases} \frac{dx_\varepsilon}{d\tau} = -x_\varepsilon + y_\varepsilon(x_\varepsilon + \alpha), \\ \varepsilon \frac{dy_\varepsilon}{d\tau} = x_\varepsilon - y_\varepsilon(x_\varepsilon + \beta), \\ x_\varepsilon(0) = 1, \\ y_\varepsilon(0) = 0, \end{cases}$$

with *another* IVP obtained from the limit $\varepsilon \rightarrow 0$. Ideally, the solution of such IVP, say $(x(t), y(t))$, should be the limit of $(x_\varepsilon(t), y_\varepsilon(t))$ for $\varepsilon \rightarrow 0$. This approach is called *perturbation argument*.

Very often it is enough to take the limit in the ODE, as we did before. So we could approximate the problem with

$$\begin{cases} \frac{dx}{d\tau} = -x + y(x + \alpha), \\ 0 = x - y(x + \beta), \\ x(0) = 1, \\ y(0) = 0. \end{cases}$$

However, in this case it does not work, because the problem is not well-posed. In fact, we have 2 initial conditions, but only one ODE! The second ODE became an algebraic equation (this system is called Differential Algebraic Equation). We can only satisfy one initial condition. From the second equation we have:

$$y = \frac{x}{x + \beta},$$

for $x(0) = 1$, $y(0) = \frac{1}{1+\beta}$, whereas for $y(0) = 0$ we have $x(0) = 0$. The combination $x(0) = 1$ and $y(0) = 0$ does not work. (We say that the initial condition is not *compatible*.) The perturbation argument is *singular*.

We need a better strategy. Let us rescale the time as follows:

$$\eta = \frac{\tau}{\varepsilon}, \quad x(\tau) = x(\varepsilon\eta) = \tilde{x}(\eta), \quad y(\tau) = y(\varepsilon\eta) = \tilde{y}(\eta).$$

The scale η is *fast*, because one unit of η time corresponds to ε units of τ time. The new system reads:

$$\begin{cases} \frac{d\tilde{x}_\varepsilon}{d\eta} = \varepsilon(-\tilde{x}_\varepsilon + \tilde{y}_\varepsilon(\tilde{x}_\varepsilon + \alpha)), \\ \frac{d\tilde{y}_\varepsilon}{d\eta} = \tilde{x}_\varepsilon - \tilde{y}_\varepsilon(\tilde{x}_\varepsilon + \beta), \\ \tilde{x}_\varepsilon(0) = 1, \\ \tilde{y}_\varepsilon(0) = 0. \end{cases}$$

As before, we formally take $\varepsilon = 0$, and obtain:

$$\begin{cases} \frac{d\tilde{x}}{d\eta} = 0, \\ \frac{d\tilde{y}}{d\eta} = \tilde{x} - \tilde{y}(\tilde{x} + \beta), \\ \tilde{x}(0) = 1, \\ \tilde{y}(0) = 0. \end{cases}$$

The perturbation in this case is not singular, we can solve it and find:

$$\begin{aligned} \tilde{x}(\eta) &= 1, \\ \tilde{y}(\eta) &= \frac{1 - e^{-(1+\beta)\eta}}{1 + \beta}. \end{aligned}$$

The second equation follows from the integration of the second ODE, which is linear. The solution of the *fast problem* (in the variable η) represents the dynamic from the initial condition up to... something we will see. For $\eta \rightarrow \infty$, thus for long time in the metric of the fast system (but not so long time for the slow system), we have:

$$\begin{aligned} \tilde{x}(\eta) &\rightarrow 1, \\ \tilde{y}(\eta) &\rightarrow \frac{1}{1 + \beta}. \end{aligned}$$

Interestingly, for long η the fast solution matches the *slow solution* $(x(\tau), y(\tau))$ for $\tau = 0$! In other words, the equilibrium of the fast system provides a *compatible* initial condition for the slow system.

We are now in condition to approximate the solution as $\varepsilon \rightarrow 0$:

1. For $\tau \in [0, \varepsilon)$, we have the fast solution

$$\begin{aligned} x(\tau) &= 1, \\ y(\tau) &= \frac{1 - e^{-\frac{1+\beta}{\varepsilon}\tau}}{1 + \beta}. \end{aligned}$$

2. For $\tau > \varepsilon$, we have the slow solution:

$$\begin{aligned}\frac{dx}{d\tau} &= -x + \frac{x(x + \alpha)}{x + \beta}, \\ y(\tau) &= \frac{x(\tau)}{x(\tau) + \beta}.\end{aligned}$$

Note that in the limit we “forget” the fast dynamic.

9.2.1 The Tichonov theorem

In general, we can consider the following problem:

$$\begin{cases} \varepsilon \frac{dz_\varepsilon}{dt} = F(z_\varepsilon, y_\varepsilon, \varepsilon), \\ \frac{dy_\varepsilon}{dt} = f(z_\varepsilon, y_\varepsilon, \varepsilon), \\ z_\varepsilon(t_0) = y_0, \\ y_\varepsilon(t_0) = z_0. \end{cases}$$

Suppose that we satisfy all the hypotheses for well-posedness.

When $\varepsilon \ll 1$, we identify $z(t)$ as the *fast* variable, and $y(t)$ as the *slow* variable.

Let $z = \phi(y)$ be a root of the equation $F(z, y, 0) = 0$ defined in some closed and bounded domain $D \subset \mathbb{R}^m$, that is, $F(\phi(y), y, 0) = 0$ for $y \in D$. Consider the degenerate (slow) system:

$$\begin{cases} \frac{dy}{dt} = f(\phi(y), y, 0), \\ y(t_0) = y_0, \end{cases}$$

where we denote by $y(t)$ its solution. We would like to show that the solution $(\phi(y(t)), y(t))$ is the limit of the original solution $(z_\varepsilon(t), y_\varepsilon(t))$ as $\varepsilon \rightarrow 0$.

We have the following result.

Theorem 9.1. *Suppose that*

1. *$z = \phi(y)$ is an isolated root in D , positively stable uniformly in $y \in D$ with respect to the adjointed system*

$$\frac{dz}{d\tau} = F(z, y, 0), \quad \tau = \frac{t}{\varepsilon}.$$

2. *the initial point (z_0, y_0) is in the basin of attraction of the equilibrium $\phi(y)$,*

3. the solution $y(t)$ of the degerate system belongs to D for all $t \in [t_0, T]$.

Then the solution $(z_\varepsilon(t), y_\varepsilon(t))$ of the overall system tends to the degenerate solution $(\phi(y(t)), y(t))$ as $\varepsilon \rightarrow 0$, in the sense that, for any $T_0 \in (t_0, T)$,

$$\lim_{\varepsilon \rightarrow 0} (z_\varepsilon(t), y_\varepsilon(t)) = (\phi(y(t)), y(t)),$$

for $t_0 < t < T_0 \leq T$.

The proof of the theorem is technical, but intuitive. At the beginning, we have the fast dynamics, that will bring the initial point (z_0, y_0) at equilibrium for the fast system. This happens for y fixed and $z(\tau)$ converging towards the equilibrium $z = \phi(y)$. This dynamic occurs in $\mathcal{O}(\varepsilon)$ units of time. So, as $\varepsilon \rightarrow 0$ we can ignore it and suppose that we are already at equilibrium and starting from $(\phi(y_0), y_0)$. Once at equilibrium, we have the slow dynamics in $y(t)$. Note that we cannot deviate from the manifold $z = \phi(y)$, because it is asymptotically stable and we converge quickly towards it.

What if we move *off* the manifold? For instance, consider the spruce-budworm system,

$$\begin{cases} \varepsilon u' = \rho u(1 - u/q) - \frac{u^2}{1 + u^2}, \\ \rho' = f(u, \rho, q), \\ q' = g(u, \rho, q), \end{cases}$$

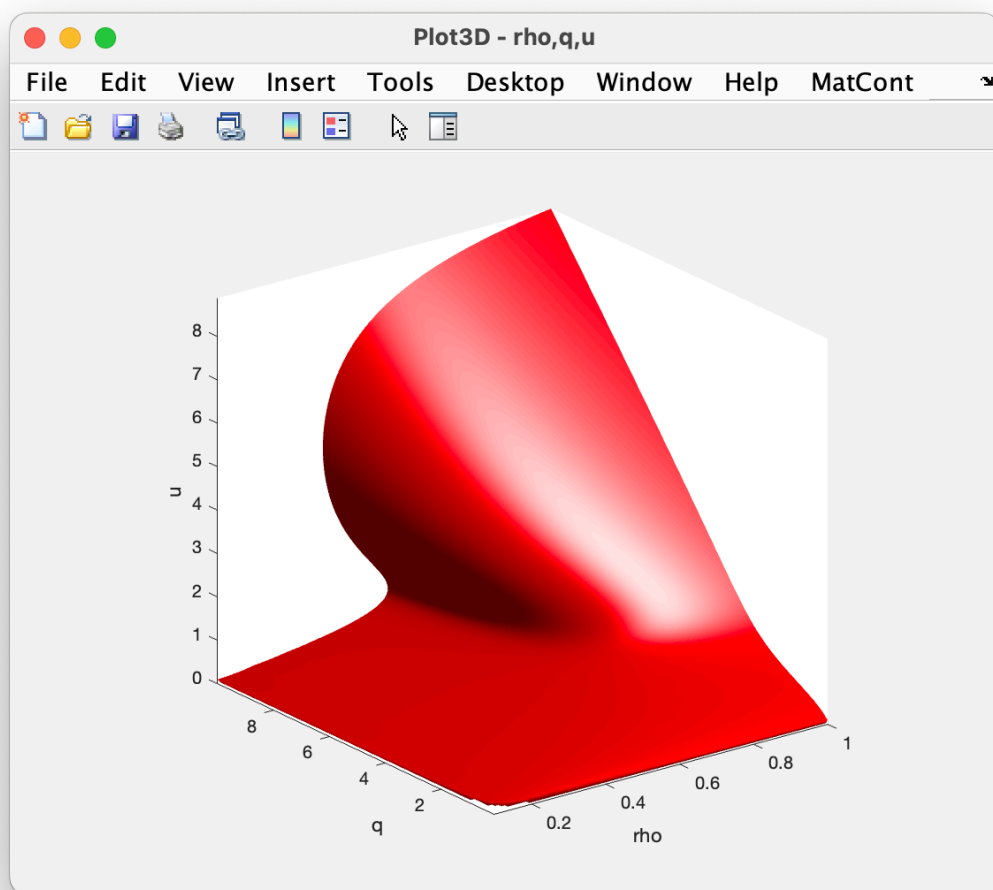
where $u(t)$ is *fast*, and $\rho(t)$ and $q(t)$ are *slow*. We know that

$$\psi(u, \rho, q) = \rho(1 - u/q) - \frac{u}{1 + u^2} = 0,$$

defines a surface in $\mathbb{R}^3$ of equilibria for the fast dynamics. However, we cannot represent this as a function:

$$u = \phi(\rho, q),$$

or at least not always. The function ϕ would be multi-valued (for some choice of the parameters we can have up to 3 equilibria).



We can define multiple surfaces:

$$\begin{aligned} u &= \phi_1(\rho, q), & (\rho, q) &\in D_1, \\ u &= \phi_2(\rho, q), & (\rho, q) &\in D_2, \\ u &= \phi_3(\rho, q), & (\rho, q) &\in D_3, \end{aligned}$$

one for each equilibrium. The domains overlap: in the bistable region, all functions are well-defined. Note also that ϕ_2 is always unstable, so we will never go there.

Now we can apply the Theorem, very some care concerning the initial conditions. Depending on where we start, we may go to a different branch.

Once on the surface, we have the slow dynamics:

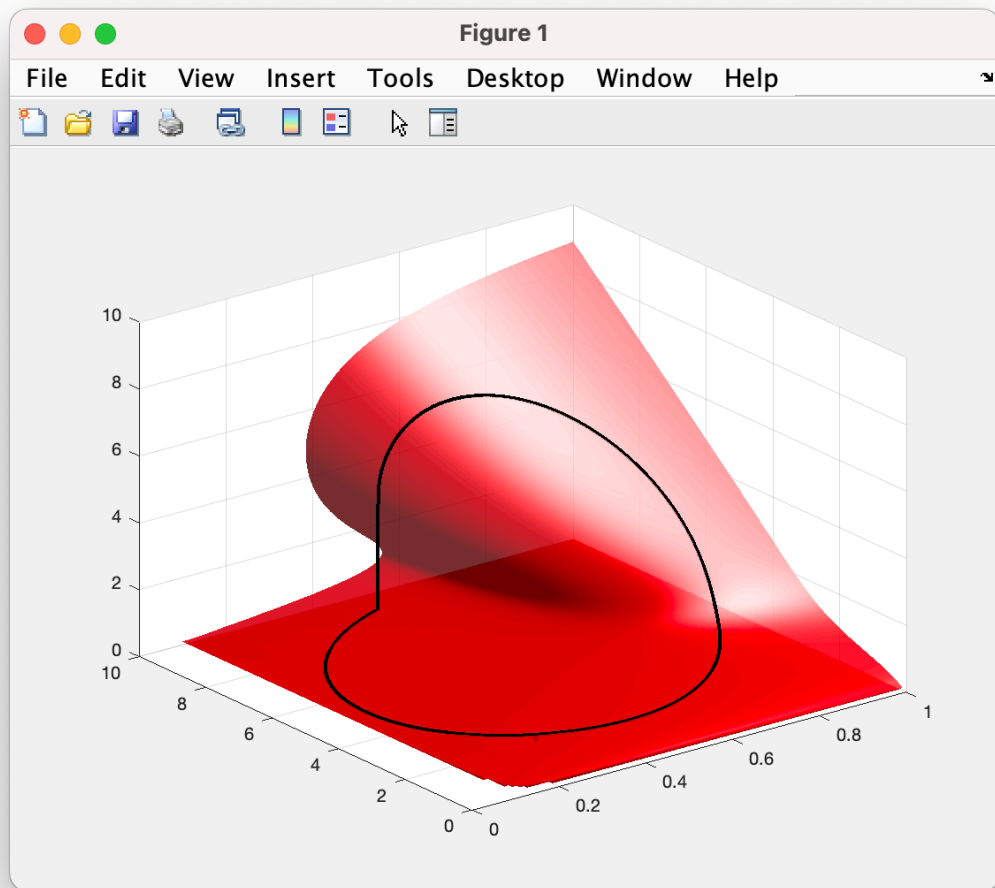
$$\begin{cases} \rho' = f(\phi_i(\rho, q), \rho, q), \\ q' = g(\phi_i(\rho, q), \rho, q), \end{cases}$$

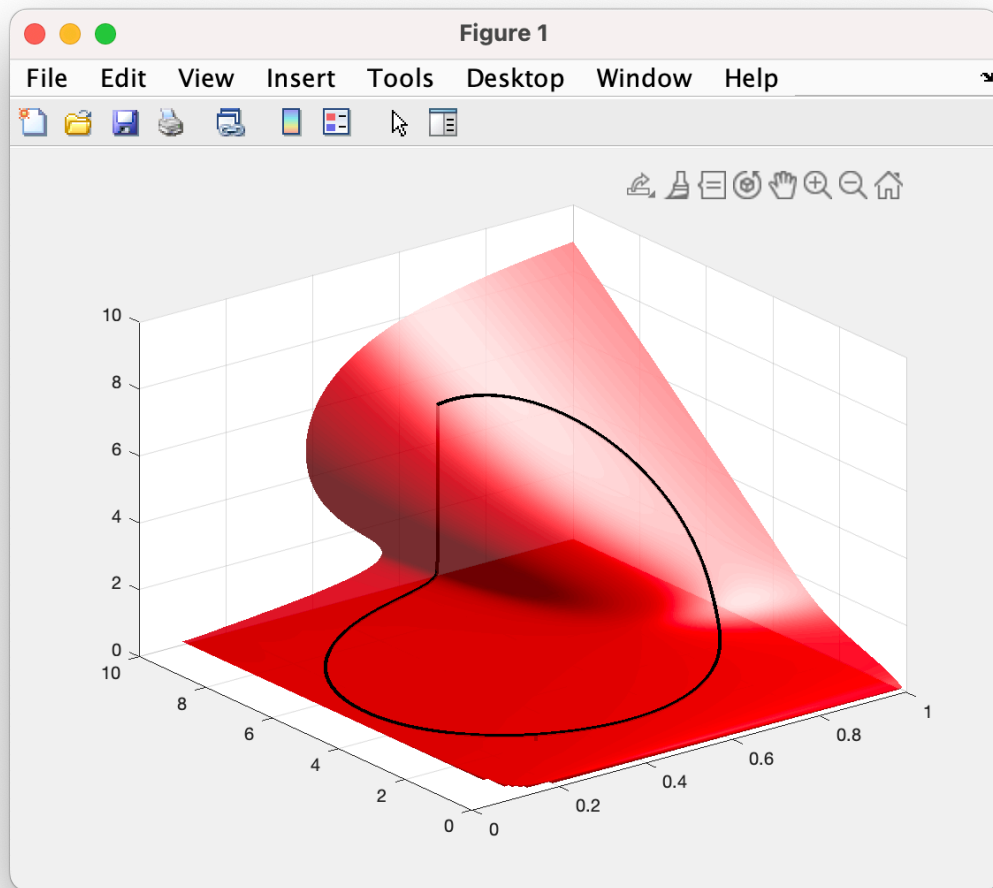
with $i = 1, 3$ depending on the stable branch. Actually, if the initial condition $(\rho_0, q_0) \in D_2$, the bistable region, then $i = 1$ if $u_0 < \phi_2(\rho_0, q_0)$ otherwise $i = 3$. If $(\rho_0, q_0) \in D_i \setminus D_2$, that is away from the bistable region, then the branch is ϕ_i .

Until $(\rho(t), q(t)) \in D_i$, the dynamic proceeds with no issues. However, if $(\rho(t), q(t))$ reaches the boundary of D_i , then the slow dynamic is over. Say that this happens at some time $T_0 > t_0$.

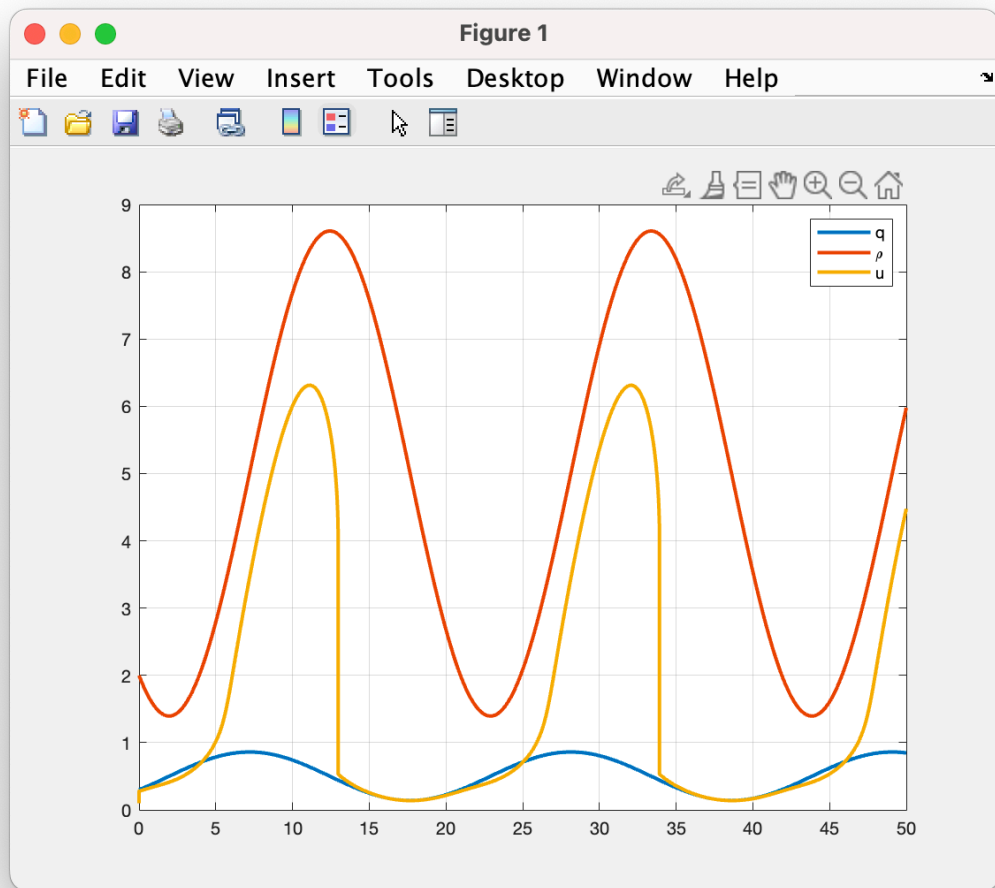
We can continue, and apply again the Tichonov theorem with initial condition slightly off the “rim” of the surface, and look for another stable branch of equilibria. Then we go there and we will have a new slow dynamic.

Below 2 examples: in the first case we move counterclockwise in the (ρ, q) plane. In the other, we move clockwise.





Note also the temporal dynamic, where u proceeds smoothly until a sudden drop (around $t = 12$ and $t = 34$). This is when we have the fast dynamic.



10 Inhibition

This lecture is based on [ks08_vol1, Section 1.4.3-4]

10.1 Competitive inhibition

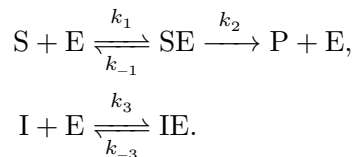
We have seen that a basic reaction involving enzymes proceeds according to the Michaelis-Menten law:

$$V = \frac{V_{\max}[S]}{K_m + [S]}.$$

where V is the velocity of production.

A very important aspect in the regulation of molecular networks comes from the fact that the presence of an inhibitory molecule may decrease the rate at which an enzymatic reaction occurs. Very schematically, we can think of two main mechanisms: competitive inhibition, or allosteric inhibition. In the first case the inhibiting molecule binds to the enzyme in the same site where it binds to the substrate, thus preventing the formation of SE complexes; in the latter case, the inhibiting molecule binds to the enzyme in a different site, but this changes the conformation of the enzyme reducing its affinity to the substrate.

We present now a very simple model of the first case. Letting I represent the inhibitory molecule, can be drawn as



where SE is the first complex, and IE is a new complex combining the inhibitor and the enzyme.

Let us define $s = [S]$, $e = [E]$, $i = [I]$, $c_1 = [SE]$, $c_2 = [IE]$, and $p = [P]$. The corresponding system is:

$$\begin{cases} s' = -k_1 se + k_{-1}c_1, \\ e' = -k_1 se + (k_{-1} + k_2)c_1 - k_3 ie + k_{-3}c_2, \\ c_1' = k_1 se - (k_{-1} + k_2)c_1, \\ i' = -k_3 ie + k_{-3}c_2, \\ c_2' = k_3 ie - k_{-3}c_2, \\ p' = k_2 c_1. \end{cases}$$

We can simplify the system by noting that:

$$i' + c_2' = 0, \quad \text{and} \quad e' + c_1' + c_2' = 0.$$

Since at the beginning $c_1(0) = c_2(0) = 0$, we have $e = e_0 - c_1 - c_2$ and $i = i_0 - c_2$. We obtain:

$$\begin{cases} s' = -k_1 s(e_0 - c_1 - c_2) + k_{-1}c_1, \\ c_1' = k_1 s(e_0 - c_1 - c_2) - (k_{-1} + k_2)c_1, \\ c_2' = k_3(i_0 - c_2)(e_0 - c_1 - c_2) - k_{-3}c_2. \end{cases}$$

Now we proceed with a rescaling. Here we use:

$$x = \frac{s}{s_0}, \quad y_1 = \frac{c_1}{e_0}, \quad y_2 = \frac{c_2}{e_0}, \quad \tau = k_1 e_0 t.$$

The system in the new variables is:

$$\begin{cases} \dot{x} = -x(1 - y_1 - y_2) + \frac{k_{-1}}{k_1 s_0} y_1, \\ \frac{e_0}{s_0} \dot{y}_1 = x(1 - y_1 - y_2) - \frac{(k_{-1} + k_2)}{k_1 s_0} y_1, \\ \frac{e_0}{s_0} \dot{y}_2 = \frac{k_3}{k_1 s_0} \left(\frac{i_0}{s_0} - \frac{e_0}{s_0} y_2 \right) (1 - y_1 - y_2) - \frac{k_{-3}}{k_1 s_0} y_2. \end{cases}$$

We define several constants:

$$\alpha_1 = \frac{k_{-1}}{k_1 s_0}, \quad \alpha_2 = \frac{(k_{-1} + k_2)}{k_1 s_0}, \quad \alpha_3 = \frac{k_{-3}}{k_1 s_0}, \quad \varepsilon = \frac{e_0}{s_0},$$

so the system reduces to:

$$\begin{cases} \dot{x} = -x(1 - y_1 - y_2) + \alpha_1 y_1, \\ \varepsilon \dot{y}_1 = x(1 - y_1 - y_2) - \alpha_2 y_1, \\ \varepsilon \dot{y}_2 = \frac{k_3}{k_1} \left(\frac{i_0}{s_0} - \varepsilon y_2 \right) (1 - y_1 - y_2) - \alpha_3 y_2. \end{cases}$$

We assume that $\varepsilon \ll 1$, so y_1 and y_2 are *fast*. Thus, with the quasi-steady-state approximation we get:

$$\begin{cases} 0 = x(1 - y_1 - y_2) - \alpha_2 y_1, \\ 0 = \beta(1 - y_1 - y_2) - \alpha_3 y_2, \end{cases}$$

where $\beta = \frac{k_3}{k_1} \frac{i_0}{s_0}$. We can solve with respect to c_1 and c_2 :

$$\begin{cases} y_1 = \frac{x}{x + \alpha_2 + \frac{\alpha_2}{\alpha_3} \beta}, \\ y_2 = \frac{\alpha_2}{\frac{\alpha_3}{\alpha_2} x + \alpha_3 + \beta}. \end{cases}$$

This equilibrium for the fast system is also asymptotically stable, because the Jacobian:

$$J = \begin{pmatrix} -x - \alpha_2 & -x \\ -\beta & -\alpha_3 - \beta \end{pmatrix}$$

has negative trace and positive determinant. Thus, according to Tichonov theorem 9.1, the system quickly converges to the stable equilibrium, and then the slow dynamics kicks in, and its given by the equation for x .

Since the second complex has no product, we are only interested in y_1 . Thus:

$$V = [P]' = k_2 c_1 = k_2 e_0 y_1 = \frac{k_2 e_0 x}{x + \alpha_2 + \frac{\alpha_2}{\alpha_3} \beta}.$$

Replacing the constants we have:

$$V = \frac{k_2 e_0 [S]}{[S] + K_m (1 + K_m \frac{i_0}{K_3})}$$

Going back to the production, we have that

$$V = k_2 e_0 y_1 = \frac{V_{\max} s}{s + K_m (1 + i_0/K_3)},$$

where $K_m = \frac{k_{-1} + k_2}{k_1}$ is the half-saturation rate. The effect of the competitive inhibition is to *increase* the half-saturation rate by a factor $(1 + i_0/K_3) > 1$. The Michaelis-Menten curves shifts to the right.

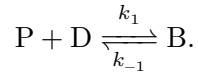
Note that $i_0 = i + c_2 = i + e_0 y_2$, so $i_0/s_0 = i/s_0 + \varepsilon y_2$ and hence $i_0 \approx i$. The variable i is also *slow*, like s , we could therefore replace i_0 with $i = [I]$ in the equation:

$$V = \frac{V_{\max} [S]}{[S] + K_m (1 + [I]/K_3)}.$$

An alternative derivation is to keep the equation for i , rescale it with respect to i_0 and then observe that $e_0/i_0 \ll 1$. In this case, the system has 2 slow variables, i and s , and two fast variables, y_1 and y_2 .

10.1.1 Example: TTX

Tetrodotoxin (TTX) is a powerful toxin that can quickly kill an adult even at very small dosage. It is a blocker of Na^+ channels, which are fundamental for impulse transmission in nerves and muscles. The effect of the drug is quite simple: suppose that $[P]$ is the number of available channels for correct conduction, $[B]$ is the number of blocked channels (thus not transmitting the signal), and $[D]$ is the concentration of the drug. Then:



At equilibrium we have:

$$[P][D] = K_1[B],$$

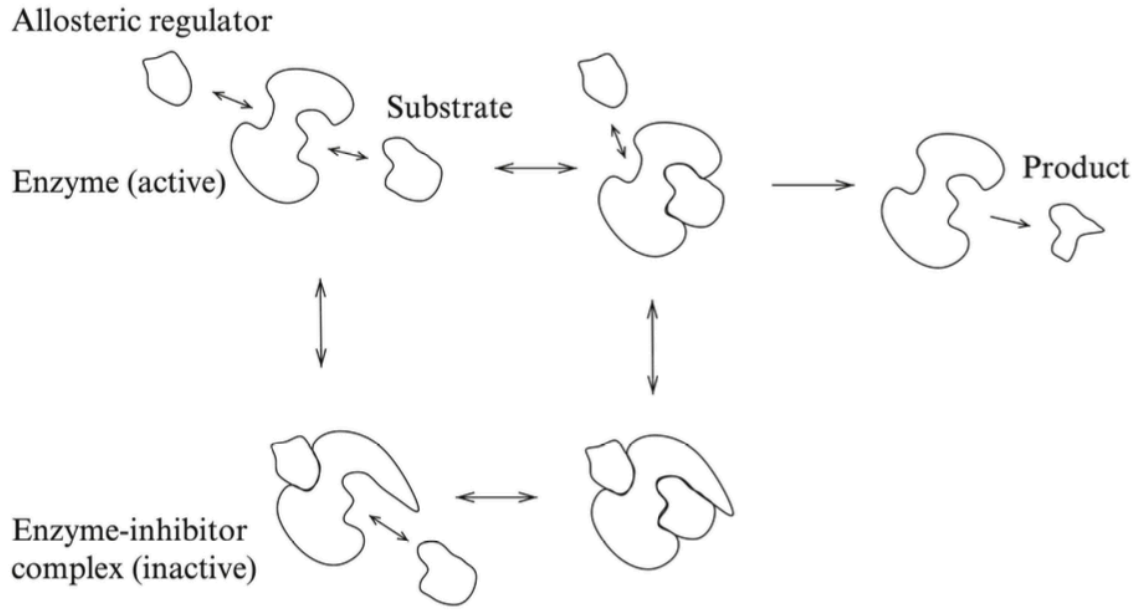
and, since $[P] + [B] = P_0$, the total number of channels, we have:

$$[P] = \frac{P_0}{[D]/K_1 + 1},$$

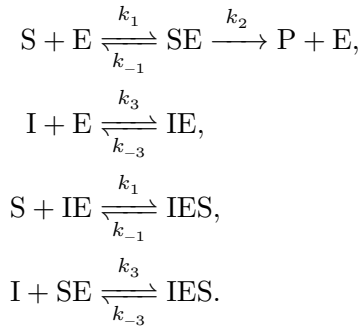
so the presence of the drug reduces by a factor $([D]/K_1 + 1)$ the number of available channels. For TTX, $K_1 \approx 1\text{-}5 \text{ nM}$, so a very small quantity can block most of the channels.

10.2 Allosteric inhibition

Allosteric inhibition is when the inhibitor binds in another site, but can still inhibit the production. The picture is the following



We have several equations to write, and the analysis with QSS approximation can be tedious. We opt here for the quasi-equilibrium approximation. The reactions involved are



Assuming all at equilibrium, and taking $s = [S]$, $e = [E]$, $i = [I]$, $c_1 = [SE]$, $c_2 = [IE]$, and $c_3 = [IES]$ we have:

$$\begin{cases} k_1 s e = k_{-1} c_1, \\ k_3 i e = k_{-3} c_2, \\ k_1 s c_2 = k_{-1} c_3, \\ k_3 i c_1 = k_{-3} c_3. \end{cases}$$

Note that $K_1 = \frac{k_{-1}}{k_1}$, $K_3 = \frac{k_{-3}}{k_3}$, and $e = e_0 - c_1 - c_2 - c_3$. The system becomes:

$$\begin{cases} s(e_0 - c_1 - c_2 - c_3) - K_1 c_1 = 0, \\ i(e_0 - c_1 - c_2 - c_3) - K_3 c_2 = 0, \\ sc_2 - K_1 c_3 = 0, \\ ic_1 - K_3 c_3 = 0. \end{cases}$$

Remember that s and i are fixed, and the unknowns are c_1 , c_2 , and c_3 . The system is *linear*:

$$\begin{pmatrix} 1 + K_1/s & 1 & 1 \\ 1 & 1 + K_3/i & 1 \\ 0 & 1 & -K_1/s \\ 1 & 0 & -K_3/i \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} e_0 \\ e_0 \\ 0 \\ 0 \end{pmatrix}$$

and of rank 3, although it is not immediate to see. So we have a unique solution. Note that we are only interested in c_1 , and this is easy to find since from the last two equations:

$$c_2 = \frac{K_1}{s} c_3, \quad c_3 = \frac{i}{K_3} c_1,$$

which can be substituted into the first equation to get:

$$(s + K_1)(1 + i/K_3)c_1 = e_0 s,$$

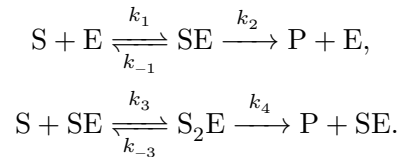
or, in terms of production rate: In this case we obtain:

$$V = k_2 c_1 = \frac{V_{\max}}{1 + [I]/K_3} \frac{[S]}{K_1 + [S]},$$

which $V_{\max} = k_2 e_0$. Again, we find the Michaelis-Menten law, but this time we have a reduction in $V_{\max}$ by a factor of $(1 + [I]/K_3) > 1$.

10.3 Cooperation

Consider the following reaction:



The enzyme has 2 (different) sites for binding. When the first complex is formed, there is still room for another substrate, but the rate is different (could be higher, for instance.)

Let us define $s = [S]$, $e = [E]$, $c_1 = [SE]$, $c_2 = [S_2E]$, and $p = [P]$. The system is:

$$\begin{cases} s' = -k_1se + k_{-1}c_1 - k_3sc_1 + k_{-3}c_2, \\ c_1' = k_1se - (k_{-1} + k_2)c_1 - k_3sc_1 + (k_4 + k_{-3})c_2, \\ c_2' = k_3sc_1 - (k_4 + k_{-3})c_2, \\ p' = k_2c_1 + k_4c_2. \end{cases}$$

As usual, we have $e + c_1 + c_2 = e_0$, so we ignore the equation for e . Note that the equation for p is different from what we are used to, but still independent from the rest of the system. We have 3 ODEs:

$$\begin{cases} s' = -k_1s(e_0 - c_1 - c_2) + k_{-1}c_1 - k_3sc_1 + k_{-3}c_2, \\ c_1' = k_1s(e_0 - c_1 - c_2) - (k_{-1} + k_2)c_1 - k_3sc_1 + (k_4 + k_{-3})c_2, \\ c_2' = k_3sc_1 - (k_4 + k_{-3})c_2. \end{cases}$$

We consider the usual non-dimensionalization:

$$x = \frac{s}{s_0}, \quad y_1 = \frac{c_1}{e_0}, \quad y_2 = \frac{c_2}{e_0}, \quad \tau = k_1e_0t,$$

and obtain the following system:

$$\begin{cases} \dot{x} = -x(1 - y_1 - y_2) + \frac{k_{-1}}{k_1s_0}y_1 - \frac{k_3}{k_1}xy_1 + \frac{k_{-3}}{k_1s_0}y_2, \\ \frac{e_0}{s_0}\dot{y}_1 = x(1 - y_1 - y_2) - \frac{k_{-1} + k_2}{k_1s_0}y_1 - \frac{k_3}{k_1}xy_1 + \frac{k_4 + k_{-3}}{k_1s_0}y_2, \\ \frac{e_0}{s_0}\dot{y}_2 = \frac{k_3}{k_1}xy_1 - \frac{k_4 + k_{-3}}{k_1s_0}y_2. \end{cases}$$

Defining as usual $\varepsilon = \frac{e_0}{s_0} \ll 1$, we recognise the fast dynamic in (y_1, y_2) and the slow dynamic in x . The fast system on the $\eta = \tau/\varepsilon$ time scale reads:

$$\begin{cases} \frac{dy_1}{d\eta} = x(1 - y_1 - y_2) - \alpha_1y_1 - \beta xy_1 + \alpha_2y_2, \\ \frac{dy_2}{d\eta} = \beta xy_1 - \alpha_2y_2. \end{cases}$$

where we introduces some constants, $\alpha_1 = \frac{k_{-1}+k_2}{k_1s_0}$, $\alpha_2 = \frac{k_{-3}+k_2}{k_1s_0}$, and $\beta = \frac{k_3}{k_1}$. The equilibrium is:

$$\begin{cases} y_1 = \frac{\alpha_2x}{\beta x^2 + \alpha_1\alpha_2 + x\alpha_2}, \\ y_2 = \frac{\beta x^2}{\beta x^2 + \alpha_1\alpha_2 + x\alpha_2}. \end{cases}$$

Going back to the original variables, we have:

$$\begin{cases} c_1 = \frac{K_{m_2} e_0 s}{K_{m_1} K_{m_2} + K_{m_2} s + s^2}, \\ c_2 = \frac{e_0 s^2}{K_{m_1} K_{m_2} + K_{m_2} s + s^2}, \end{cases}$$

where $K_{m_1} = \frac{k_{-1} + k_2}{k_1}$ and $K_{m_2} = \frac{k_{-3} + k_4}{k_3}$. The equilibrium is also asymptotically stable in the fast dynamic, since:

$$J = \begin{pmatrix} -(1 + \beta)x - \alpha_1 & -x - \alpha_2 \\ \beta x & -\alpha_2 \end{pmatrix}.$$

The Jacobian has negative trace, and the determinant is positive. We can apply Tichonov theorem, so the QSS approximation is valid. The total production rate is:

$$V = k_2 c_1 + k_2 c_2 = \frac{(k_2 K_{m_2} + k_4 s) e_0 s^2}{K_{m_1} K_{m_2} + K_{m_2} s + s^2}.$$

We can have different cases:

- **Case 1:** Identical, independent sites, so $k_1 = 2k_3$, $k_{-3} = 2k_{-1}$, and $k_4 = 2k_2$ so:

$$V = 2 \frac{k_2 e_0 s}{K_{m_1} + s}.$$

This is expected: the sites are independent, so they can work in parallel. The maximum rate is just twice. For n independent sites, we have the same formula with $V_{\max} = n \cdot k_2 e_0$.

- **Case 2:** We assume that the first reaction is very slow, but the second is fast. So, $k_1 \rightarrow 0$, $k_3 \rightarrow \infty$ with $k_1 k_3$ bounded. Therefore, $K_{m_1} \rightarrow \infty$, $K_{m_2} \rightarrow 0$, and $K_{m_1} K_{m_2}$ bounded. The production rate is:

$$V = \frac{k_4 e_0 s^2}{K_{m_1} K_{m_2} + s^2} = \frac{V_{\max} s^2}{K_m^2 + s^2},$$

where $K_m^2 = K_{m_1} K_{m_2}$. This function is *sigmoidal*, very much like Holling type III.

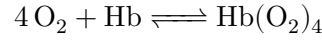
For n sites we can prove that:

$$V = \frac{V_{\max} s^n}{K_m^n + s^n}.$$

where $K_m = \prod_{i=1}^n K_{m_i}$. This is an example of a switch-like behavior: at low concentration of s , say $s \ll K_m$, the production rate is low, otherwise is very high.

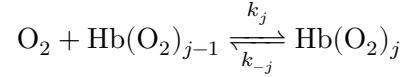
10.3.1 Example: hemoglobin

An example is the hemoglobin, which binds oxygen within the blood. The reaction is:



Hemoglobin has 4 sites, but they are not identical, in these sense that after the first oxygen molecule is binded, the second one binds easier, and so more the third and fourth ones.

We can rewrite the reaction as:



where $\text{Hb}(\text{O}_2)_j$ is hemoglobin with j oxygen molecules attached. Hence, at equilibrium we have:

$$[\text{Hb}(\text{O}_2)_j] = \frac{[\text{Hb}(\text{O}_2)_{j-1}][\text{O}_2]}{K_j} = \dots = \frac{[\text{Hb}][\text{O}_2]^j}{K_1 \dots K_j}$$

The saturation curve Y is the amount of hemoglobin bounded to oxygen over the total available one, and it reads:

$$Y = \frac{\sum_{j=0}^4 j[\text{Hb}(\text{O}_2)_j]}{4 \sum_{j=0}^4 [\text{Hb}(\text{O}_2)_j]}.$$

It is possible to fit K_j for $j = 1, \dots, 4$ so to match real data, and it is observed that K_j increases with j .

11 Feedback loops

This lecture is based on [tyson2003sniffers]

Actual molecular networks consist of a large number of molecular species with very complex patterns of interaction among them. We have seen the *lac operon* for instance.

Wishing to go beyond the simple reactions analyzed in the previous lectures, the natural next step is to consider reaction loops, i.e. networks with two molecular species, each acting on the rates, either of synthesis or of decay, of the other. Since the overall action may be positive (favouring the synthesis or inhibiting the decay) or negative (favouring the decay or inhibiting the formation), in principle there are 3 possible cases:

1. *mutual activation*: each molecular species has a positive effect on the other;
2. *mutual inhibition*: each has a negative effect on the other;
3. *activation-inhibition*: one has a positive effect on the other, which has a negative on the first one.

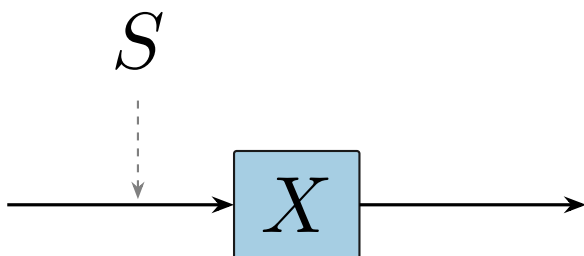
This matches exactly the Kolmogorov formalism in ecology: *mutualism*, *competition*, *prey-predator*.

We distinguish two kinds of feedback loops: *positive* and *negative*. Each can be realized in several different reaction schemes and each reaction may be modeled as linear or Michaelis-Menten-type or cooperative. The dynamical properties will then depend on the details of reactions; we will show however (by way of examples) that there are some common properties of positive vs. negative feedback loops.

As an initial point, we discuss how the concentration of a substance responds to a stimulus that increase its rate of synthesis in some simple reaction schemes. This is inspired to the article of Tyson, Chen, and Novak (2003) that classified the schemes according to the type of the signal-response curve, and used these as building blocks for more complex reaction networks.

11.1 Linear response

A simple scheme is following, where S (that may be the concentration of some substance favouring the synthesis of X) is the signal and the equilibrium concentration of X is the response:



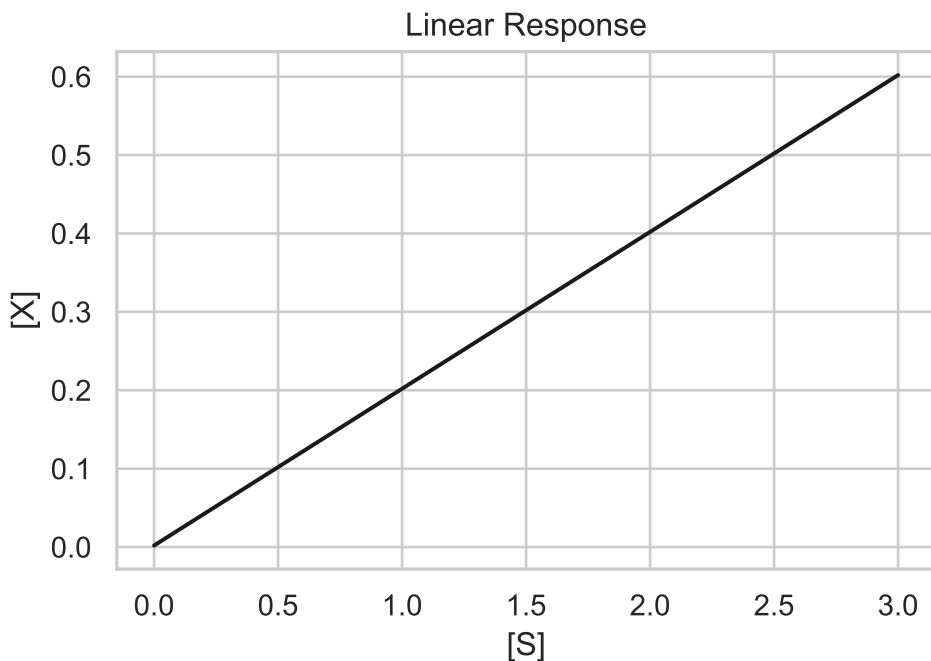
The corresponding equation is:

$$[X]' = (k_1 + k_1'[S]) - k_2[X],$$

hence the equilibrium is:

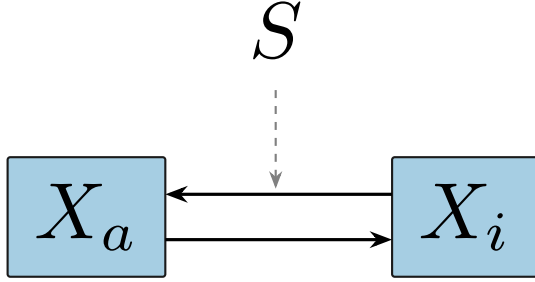
$$[X]_{\text{eq}} = \frac{k_1 + k_1'[S]}{k_2}.$$

The equilibrium is *linear* in $[S]$. Note that we allow for a production of $[X]$ at constant rate k_1 even in the absence of $[S]$.



11.2 Hyperbolic response

We assume that now the molecule X can be in active X_a or inactive X_i form, and that S favours the reaction that activates X :



The corresponding system is:

$$\begin{cases} [X_a]' = -k_{-1}[X_a] + k_1[S][X_i], \\ [X_i]' = k_{-1}[X_a] - k_1[S][X_i]. \end{cases}$$

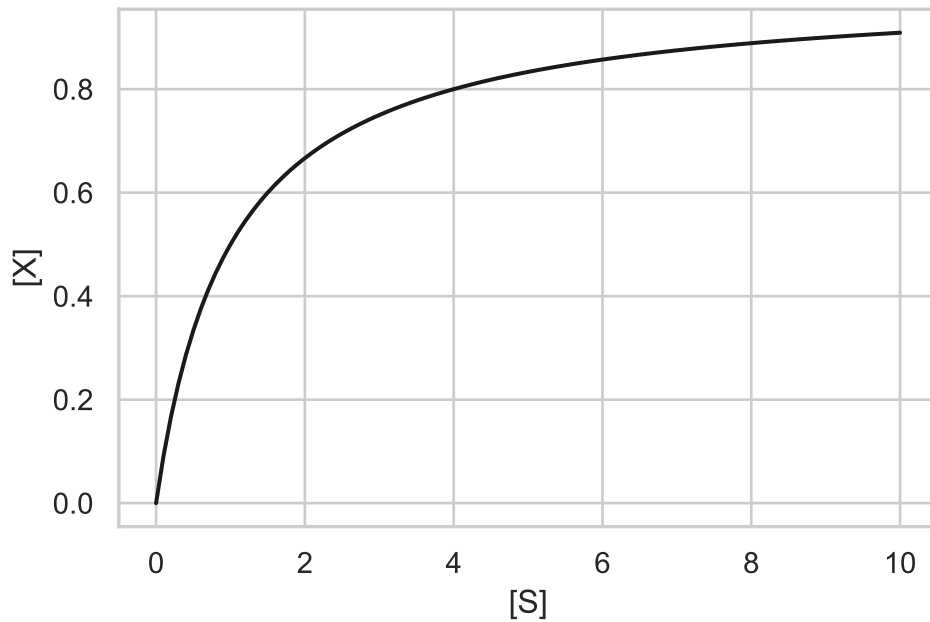
Since the total sum is conserved, say $X_{\text{tot}} = [X_a] + [X_i]$, we drop the second equation and keep the following

$$[X_a]' = -k_{-1}[X_a] + k_1[S](X_{\text{tot}} - [X_a]).$$

The equilibrium is globally attractive and equal to:

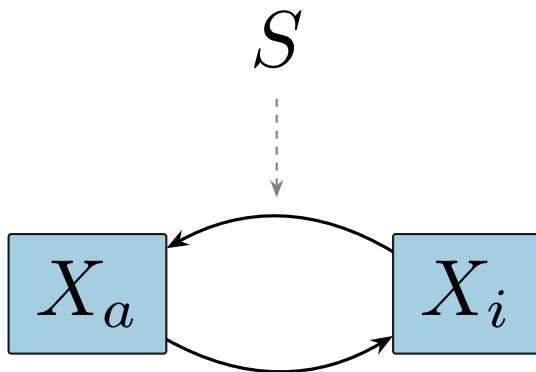
$$\frac{[X_a]_{\text{eq}}}{X_{\text{tot}}} = \frac{S}{S + K_1}.$$

where $K_1 = \frac{k_{-1}}{k_1}$. This yields a hyperbolic response to the signal, as in the figure. Note that this one is the Michaelis-Menten law (with quasi-equilibrium approximation).



11.3 Sigmoidal response

The scheme is like the previous one, but we assume that the activation and inactivation rates follow a Michaelis-Menten law (the curved edge is a symbol for this).



An example is phosphorylation, where we can have two enzymes acting on X_a and X_i : the first enzyme phosphorylate X_a into X_i , and the second one does the opposite. This “game” can reach an equilibrium.

We just need one equation, say for X_a , because as before $X_{\text{tot}} = [X_a] + [X_i]$:

$$[X_a]' = -\frac{V_2[X_a]}{K_{m_2} + [X_a]} + \frac{V_1[S](X_{\text{tot}} - [X_a])}{K_{m_1} + X_{\text{tot}} - [X_a]},$$

where K_{m_2} and K_{m_1} are the half-saturation constants of the two reactions.

The activated fraction at equilibrium $x = [X_a]/X_{\text{tot}}$ is then a solution of

$$\frac{x(L_1 + 1 - x)}{(1 - x)(L_2 + x)} = \frac{\nu_1}{\nu_2},$$

where:

$$L_1 = \frac{K_{m_1}}{X_{\text{tot}}}, \quad L_2 = \frac{K_{m_2}}{X_{\text{tot}}}, \quad \nu_1 = V_1[S], \quad \nu_2 = V_2.$$

First, we show that there exists a unique solution x of the above equation. Indeed, the problem is equivalent to:

$$Q(x) = \frac{\nu_1}{\nu_2}(1 - x)(L_2 + x) - x(L_1 + 1 - x) = 0$$

where $Q(x)$ is a quadratic function. Note that $Q(0) = \frac{\nu_1}{\nu_2} > 0$ and $Q(1) = -L_1 < 0$, so there must be a unique $x \in (0, 1)$ such that $Q(x) = 0$. We call this solution as a function of the parameters as *Goldbeter-Koshland function* G :

$$G(\nu_1, \nu_2, L_1, L_2) \quad \text{such that} \quad Q(G(\nu_1, \nu_2, L_1, L_2)) = 0.$$

We can explicitly write the solution using the quadratic formula:

$$G(\nu_1, \nu_2, L_1, L_2) = \frac{\beta - \sqrt{\beta^2 - 4\alpha\gamma}}{2\alpha},$$

where the parameters are:

$$\alpha = \nu_2 - \nu_1, \quad \beta = \nu_1(1 - L_1) - (1 + L_2)\nu_2, \quad \gamma = L_2\nu_1.$$

Note that $\nu_1 = V_1[S]$, so the response of the system is:

$$[X_a] = X_{\text{tot}}G(V_1[S], \nu_2, L_1, L_2),$$

which is not very informative. We proceed differently. Note that the Goldbeter-Koshland function satisfies following equation:

$$g_{L_1, L_2}(x) = \frac{x(L_1 + 1 - x)}{(1 - x)(L_2 + x)} = \frac{\nu_1}{\nu_2},$$

with $x = G(\nu_1, \nu_2, L_1, L_2)$. So, we can study the function $g_{L_1, L_2}(x)$ with L_1 and L_2 as parameters, and then notice that g is the inverse function of G , because

$$g_{L_1, L_2}(G(\nu_1, \nu_2, L_1, L_2)) = \frac{\nu_1}{\nu_2}.$$

Graphically, the inverse is obtained by switching the axes. The function g_{L_1, L_2} is defined for $x \in [0, 1)$:

$$\begin{aligned} g_{L_1, L_2}(0) &= 0, \\ g_{L_1, L_2}(1^-) &= +\infty, \\ g'_{L_1, L_2}(x) &= \frac{L_2(x-1)^2 + L_1(L_2 + x^2)}{(x-1)^2(L_2 + x)^2} > 0, \\ g''_{L_1, L_2}(x) &= \frac{2}{1+L_2} \frac{L_1(L_2 + x)^3 - L_2(1+L_1+L_2)(1-x)^3}{(1-x)^3(L_2 + x)^3}, \end{aligned}$$

The function is monotonically increasing with a vertical asymptote in $x = 1$. The second derivative may change sign depending on L_1 and L_2 . However, the numerator of $g''_{L_1, L_2}(x)$ is an increasing function of x , so if $g''_{L_1, L_2}(0) \geq 0$ then the function is convex in $x \in [0, 1)$. On the other hand, if $g''_{L_1, L_2}(0) < 0$, then there is a change of concavity for some $\bar{x} \in (0, 1)$, because $g''_{L_1, L_2}(x) \rightarrow +\infty$ as $x \rightarrow 1^-$. Specifically:

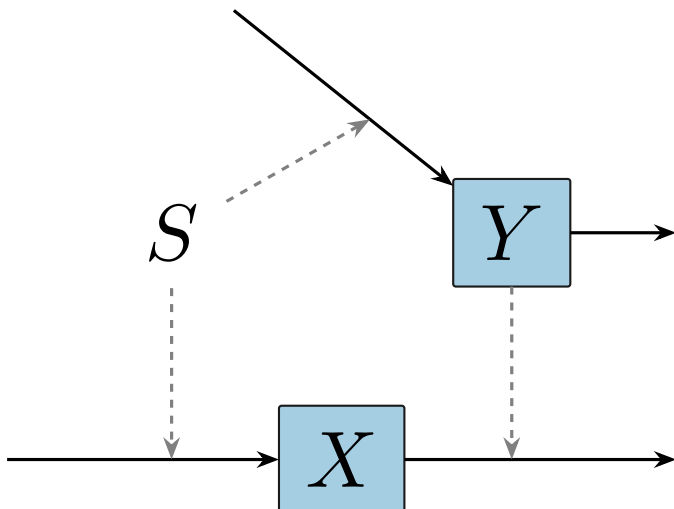
$$g''_{L_1, L_2}(0) = -2 \frac{1 + L_1 - L_1 L_2}{L_2^2},$$

hence when L_1 or L_2 are small, the function has a concave-convex shape; when they are large, it is a convex function. Since $G(\nu_1, \nu_2, L_1, L_2)$ as a function of ν_1/ν_2 is the inverse function of $g_{L_1, L_2}(x)$, it turns out that when $1 + L_1 - L_1 L_2 > 0$, G is a sigmoidal function, when $1 + L_1 - L_1 L_2 < 0$ it is a concave saturating function.

The Goldbeter-Koshland function is quite important in feedback networks. It can create a sigmoidal response with respect to the signal [S]. It is a switch-like behavior, different from cooperativity. It is also called zero-order ultrasensitivity.

11.4 Perfectly adapted response

There are several physiological systems organized so as to give a constant response, independent of the strength of the signal. Tyson and Novak propose a simple reaction scheme with this property; they call it “sniffer” as receptors get adapted to a smell. The scheme is the following



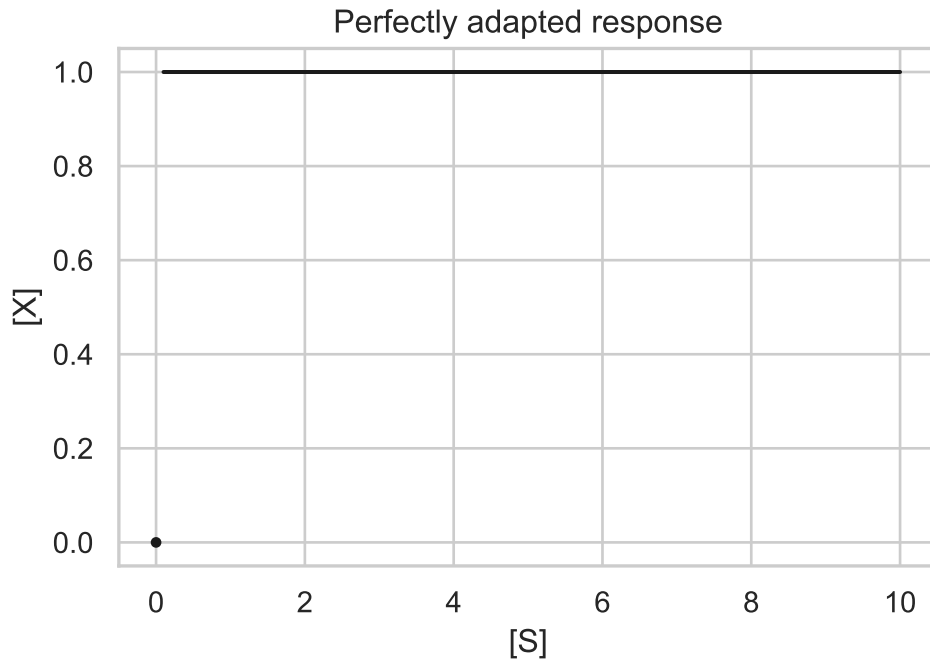
The corresponding differential equations are

$$\begin{cases} [X]' = k_1[S] - k_2[X][Y], \\ [Y]' = k_3[S] - k_4[Y]. \end{cases}$$

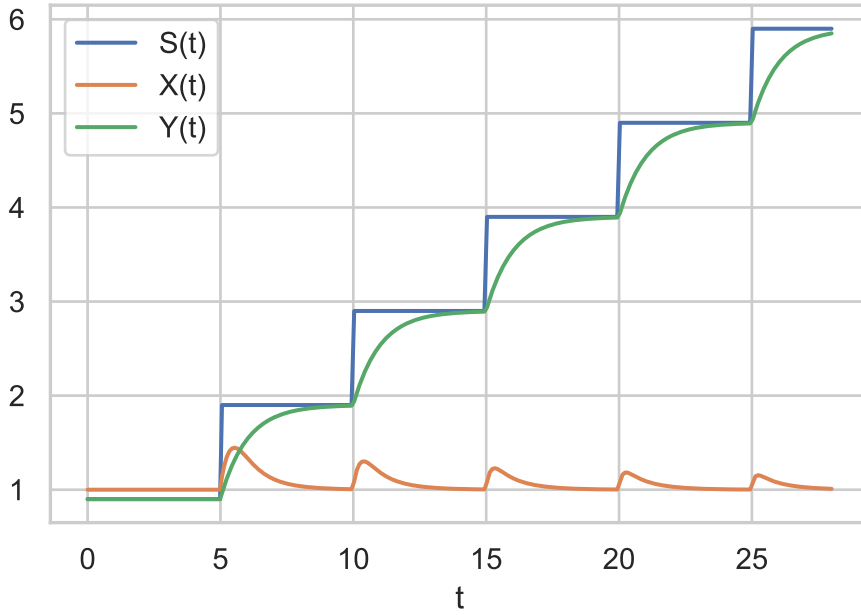
At equilibrium

$$[Y]_{\text{eq}} = \frac{k_3}{k_4}[S], \quad [X]_{\text{eq}} = \frac{k_1[S]}{k_2[Y]_{\text{eq}}} = \frac{k_1 k_4}{k_2 k_3},$$

as long as $[S] = 0$, otherwise the unique equilibrium is $(0,0)$. Considering X as the response to the signal S , one sees that the circuit in the figure (often named as a [feed-forward control](#), since the signal S acts on both Y and the molecule X that is controlled by Y) has the property that the response is independent of the signal, as long as it is not 0.



Beyond the response to a constant signal S , it is also interesting seeing how X responds to an increasing signal. Below we see how X and Y change, assuming that $[S](t)$ is a piecewise constant function increasing every 5 time unit. It can be seen that after each increase of S , there is a transient increase of X that then goes back towards the equilibrium, the exact pattern depending on the parameter values.



11.5 Positive feedback: mutual inhibition

The following is an example of feedback loop, of positive type.

Indeed, it assumed that the enzyme E in the unphosphorylated form favours the decay of X, hence decreases its concentration; on the other hand, X favours the phosphorylation of E. Hence an increase in the concentration of X decreases the concentration of E; this decrease decreases the decay rate of X, thus increasing its concentration. This is why this is a positive feedback loop.

The corresponding equations are

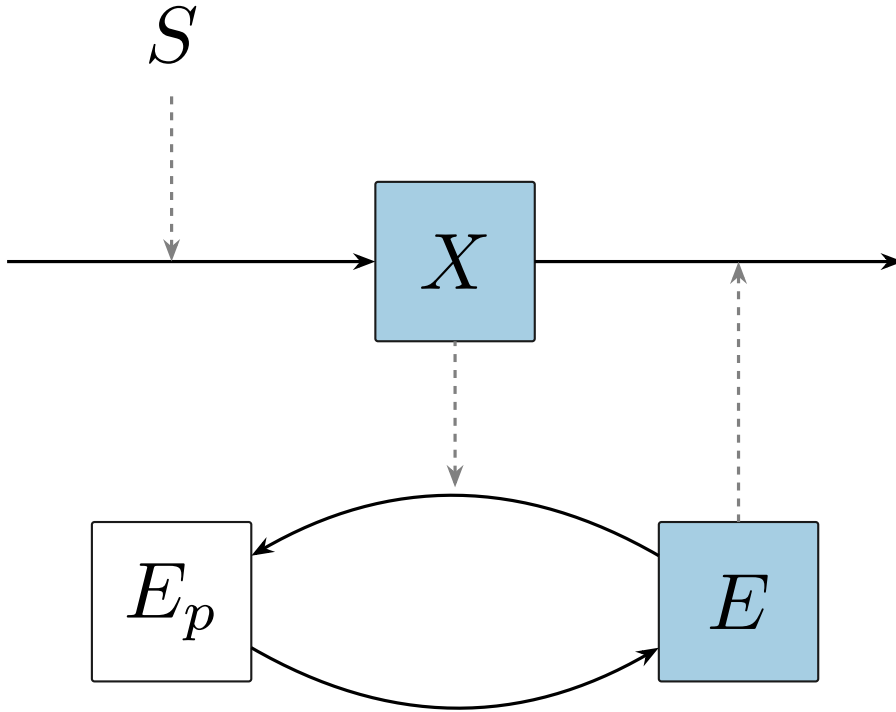
$$\begin{cases} [X]' = k_0 + k_1[S] - (k_2 + k'_2[E])[X], \\ [E]' = \frac{V_1(E_{\text{tot}} - [E])}{K_{m_1} + E_{\text{tot}} - [E]} - \frac{V_2[E][X]}{K_{m_2} + [E]}. \end{cases}$$

We rescale the equations with:

$$y = \frac{[E]}{E_{\text{tot}}}, \quad L_1 = \frac{K_{m_1}}{E_{\text{tot}}}, \quad L_2 = \frac{K_{m_2}}{E_{\text{tot}}}, \quad \nu_1 = \frac{V_1}{E_{\text{tot}}}, \quad \nu_2 = \frac{V_2}{E_{\text{tot}}}.$$

so we obtain:

$$\begin{cases} [X]' = k_0 + k_1[S] - (k_2 + k'_2[E]_{\text{tot}}y)[X], \\ y' = \frac{\nu_1(1 - y)}{L_1 + 1 - y} - \frac{\nu_2 y[X]}{L_2 + y}. \end{cases}$$

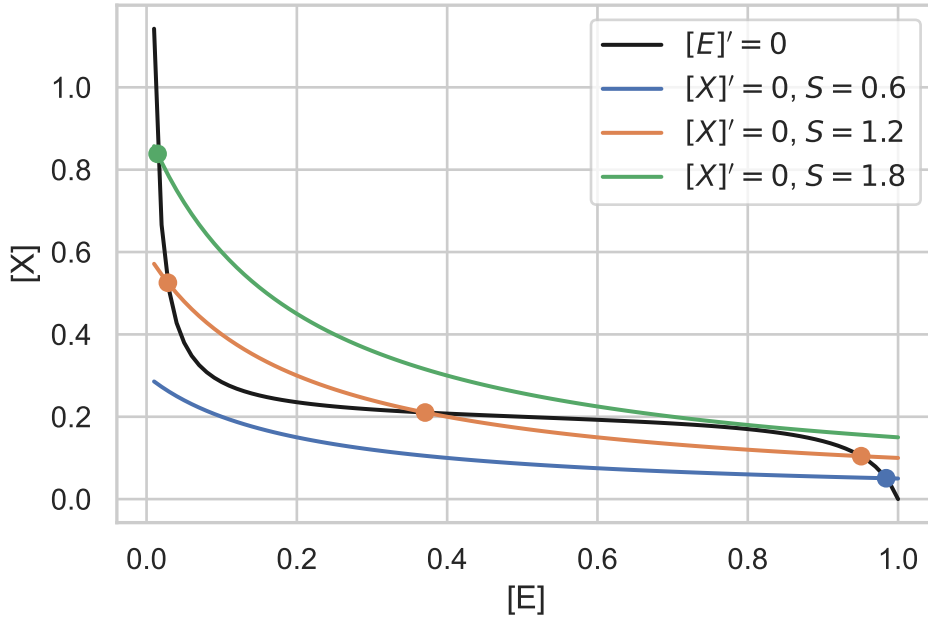


At equilibrium we have:

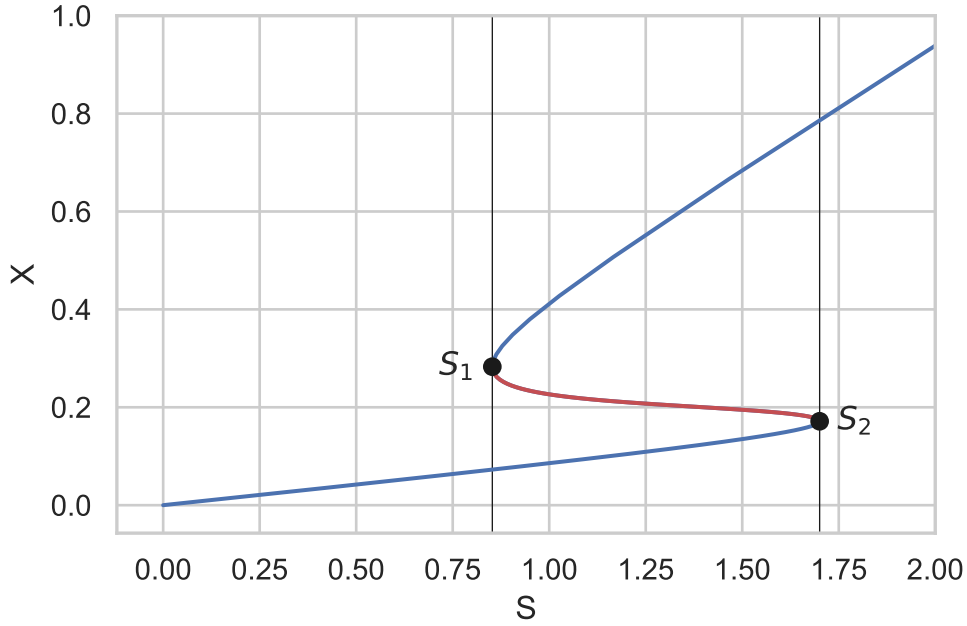
$$[X] = f(y) = \frac{k_0 + k_1[S]}{k_2 + k'_2[E]_{\text{tot}}y},$$

$$[X] = \frac{\nu_1}{\nu_2} \frac{(1-y)(L_1 + y)}{y(L_2 + 1-y)} = \frac{\nu_1}{\nu_2} g_{L_1, L_2}(1-y).$$

If L_1 and L_2 are very small, the function $g_{L_1, L_2}(1-y)$ has a very sharp concave-convex shape. Hence a hyperbolic function like $f(y)$ can cross it either once or three times, depending on parameter values.



In the figure below, we show how the equilibrium values of X and y depend on S . By studying the vector field when there are three equilibria, it is easy to see that the intermediate equilibrium is unstable, while the exterior ones are asymptotically stable. The scheme is called “switch” because, depending on the value of S , can give rise to a response in one of two possible stable states. In the generic model for the cell cycle it is shown how this scheme can be embedded in a more complex reaction network to give rise to repeated switches from one state to the other, and vice versa. We have hysteresis: an example is the *lac operon*.



11.6 Negative feedback: homeostatis

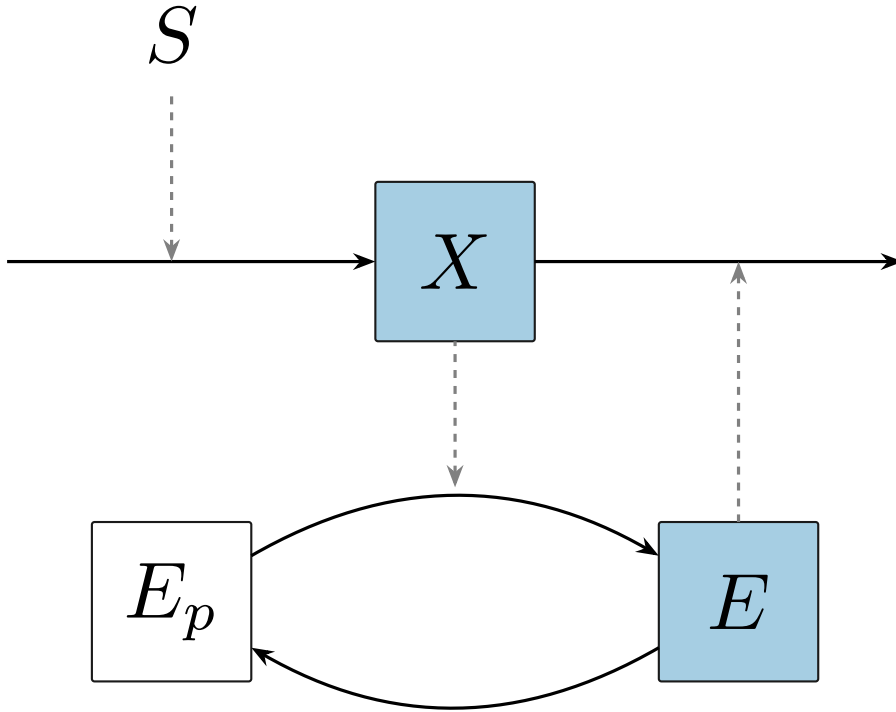
A simple variation on the previous loop provides an example of a negative feedback loop. Now X increases the rate at which E becomes active (de-phosphorylated in this scheme) at increasing the decay rate of X ; hence if the concentration of X is increased, the same will occur for the concentration of E that will then increase the decay of X , tending to reduce the concentration of X . Mechanisms of this kind are at the basis of homeostasis, the property of many biological systems to maintain some quantities as constant as possible.

The equations corresponding to this scheme are:

$$\begin{cases} [X]' = k_0 + k_1[S] - (k_2 + k'_2[E])[X], \\ [E]' = \frac{V_1[X](E_{\text{tot}} - [E])}{K_{m_1} + E_{\text{tot}} - [E]} - \frac{V_2[E]}{K_{m_2} + [E]}. \end{cases}$$

We proceed as before, with the same rescaling, and the equilibrium is such that:

$$\begin{cases} [X] = f(y) = \frac{k_0 + k_1[S]}{k_2 + k'_2[E]_{\text{tot}}y}, \\ [X] = \frac{\nu_2}{\nu_1} \frac{y(L_1 + 1 - y)}{(1 - y)(L_2 + y)} = \frac{\nu_2}{\nu_1} g_{L_1, L_2}(y). \end{cases}$$



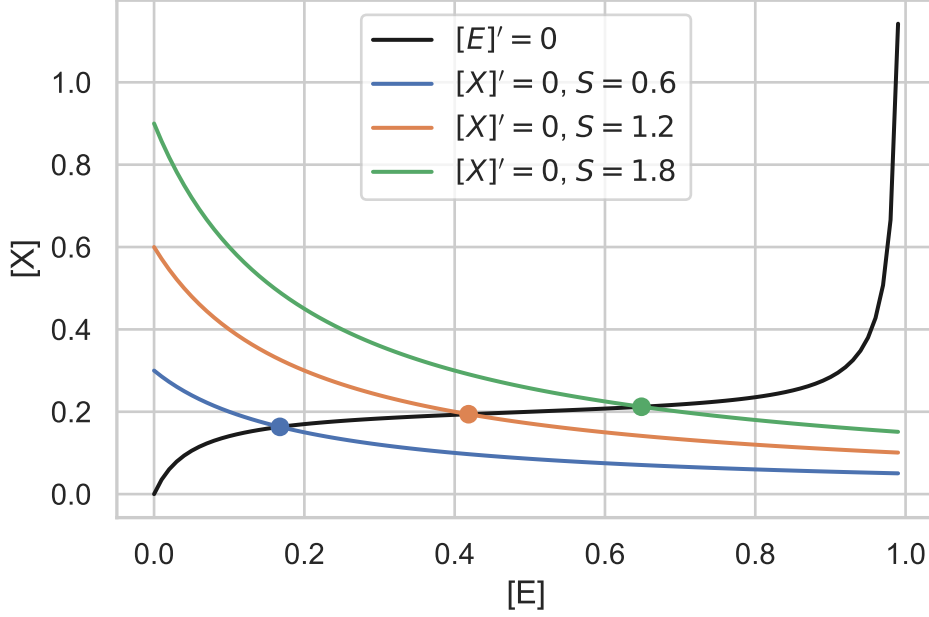
We observe that $g_{L_1, L_2}(y)$ is an increasing function of y , and $f(y)$ is a decreasing function. Moreover,

$$f(0) = \frac{k_0 + k_1[S]}{k_2} > 0,$$

$$g_{L_1, L_2}(0) = 0,$$

$$f(1) < \lim_{y \rightarrow 1^-} g_{L_1, L_2}(y) = \infty,$$

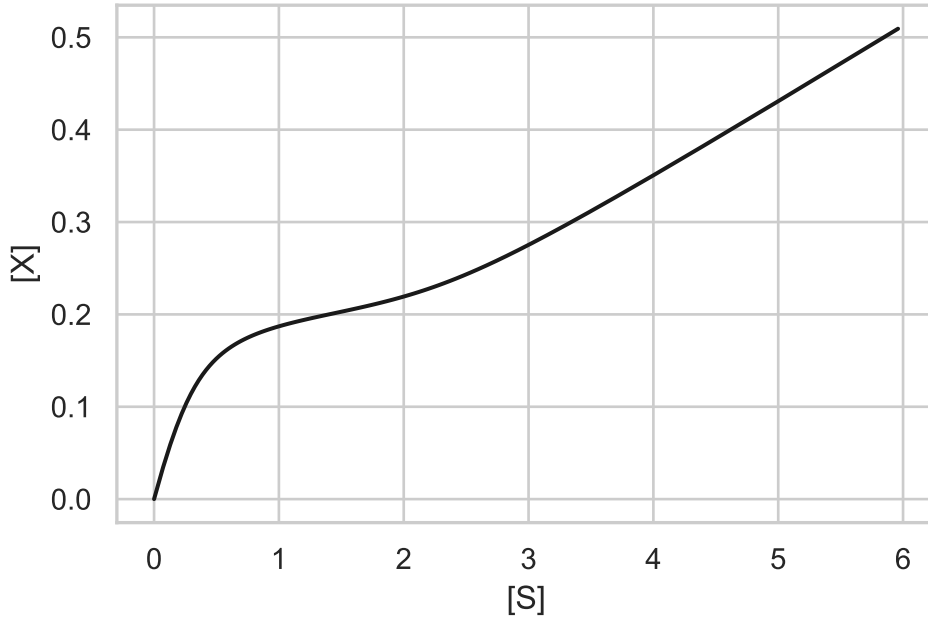
it follows that there always exists a unique equilibrium $([X]^*, y^*)$ with $[X]^* > 0$ and $0 < y < 1$.



Furthermore, it is easy to see that the equilibrium is asymptotically stable. To this purpose, it is enough computing the Jacobian at equilibrium

$$J([X], y) = \begin{pmatrix} -(k_2 + k'_2 E_{\text{tot}} y) & -k'_2 E_{\text{tot}} [X] \\ \frac{\nu_1 (1-y)}{L_1 + 1 - y} & -\frac{\nu_1 L_1 [X]}{(1 + L_1 - y)^2} - \frac{\nu_2 L_2}{(L_2 + y)^2} \end{pmatrix} = \begin{pmatrix} - & - \\ + & - \end{pmatrix}.$$

Because of the signs of the matrix entries, it is immediate to see that the trace is negative and the determinant is positive, implying asymptotic stability. This implies that after perturbations, the system is taken back to the equilibrium; furthermore, it can be observed through the Figure that the equilibrium values depend only weakly on the exact value of S in an intermediate range of S .



11.7 Negative feedback: oscillations

A system with a negative feedback loop may oscillate around the equilibrium (it can be seen by computing the vector field), but we have seen that in a feedback loop of length 2, the oscillations are damped as the equilibrium is asymptotically stable.

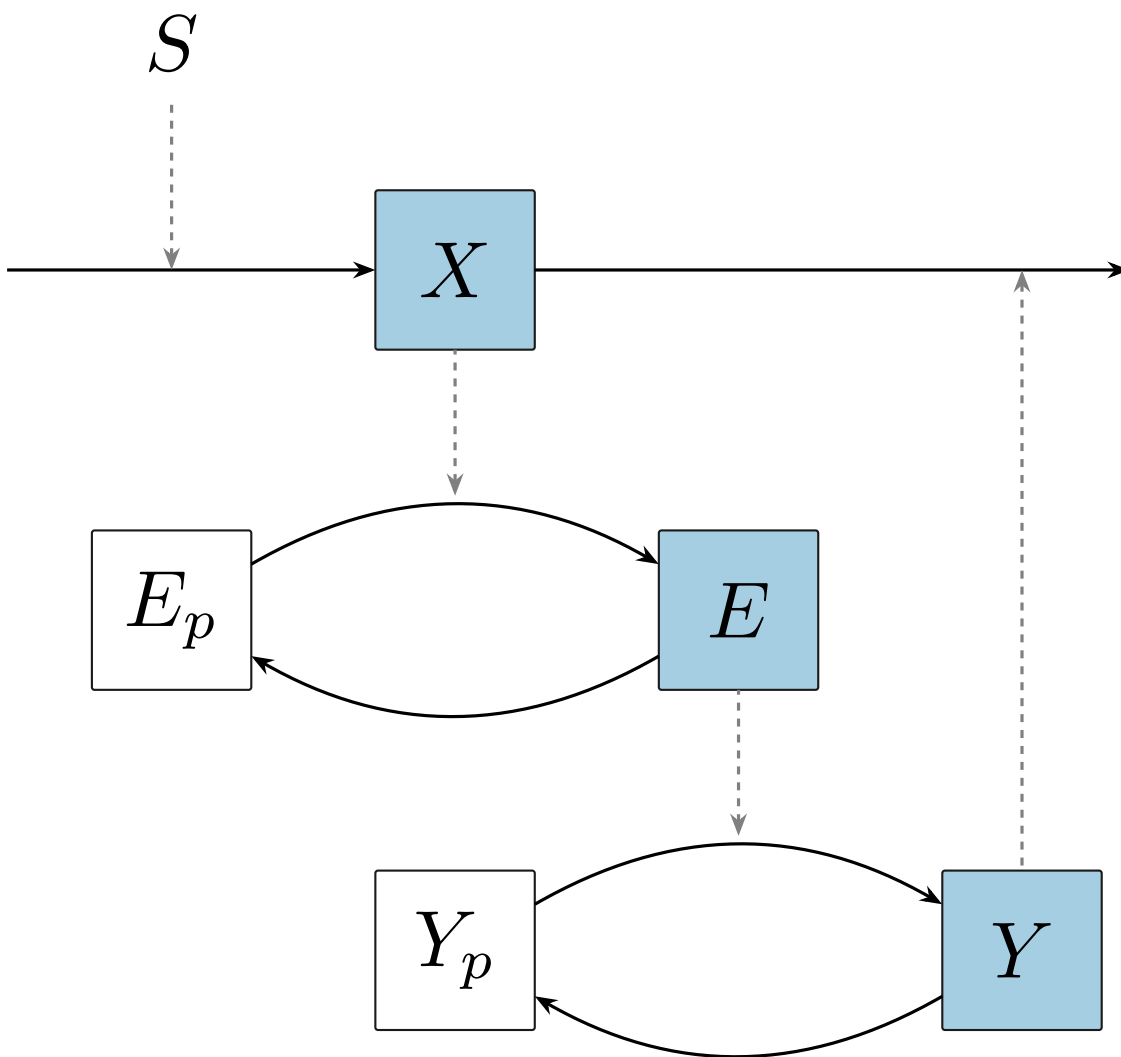
It then becomes natural asking whether a negative feedback loop of length 3 can give rise to sustained oscillations. A positive answer has been given by Goldbeter in 1991. The scheme proposed was the following (identical to the previous one, except that E instead of having a direct effect on the decay rate of X activates Y that in turn increases the decay rate of X):

Letting $e = [E]/[E]_{\text{tot}}$ and $y = [Y]/[Y]_{\text{tot}}$ the resulting equations are

$$\begin{cases} [X]' = k_0 + k_1[S] - (k_2 + k'_2[Y]_{\text{tot}}y)[X], \\ e' = \frac{\nu_1[X](1-e)}{L_1 + 1 - e} - \frac{\nu_2 e}{L_2 + e}, \\ y' = \frac{\nu_3 e(1-y)}{L_3 + 1 - y} - \frac{\nu_4 y}{L_4 + y}. \end{cases}$$

Setting the third equation equal to 0, we find that

$$e = \frac{\nu_4}{\nu_3} g_{L_3, L_4}(y),$$



hence an increasing function of y . Similarly, setting the second equation to 0, we find that

$$[X] = \frac{\nu_2}{\nu_1} g_{L_1, L_2}(e),$$

hence, setting second and third equation equal to 0, we see that $[X]$ is given by an increasing function of y ; on the other hand, setting the first equation to 0, we obtain

$$[X] = f(y) = \frac{k_0 + k_1[S]}{k_2 + k'_2[Y]_{\text{tot}}y},$$

a decreasing function of y . Necessarily there is unique y at which the two functions are equal, yielding a unique equilibrium $([X]^*, e^*, y^*)$.

In order to study its stability, consider the Jacobian:

$$J([X], e, y) = \begin{pmatrix} -(k_2 + k'_2 Y_{\text{tot}} y) & 0 & -k'_2 Y_{\text{tot}} [X] \\ \frac{\nu_1(1-e)}{L_1+1-e} & -\frac{\nu_1 L_1 [X]}{(1+L_1-e)^2} - \frac{\nu_2 L_2}{(L_2+e)^2} & 0 \\ 0 & \frac{\nu_3(1-y)}{L_3+1-y} & -\frac{\nu_3 L_3 [X]}{(1+L_3-y)^2} - \frac{\nu_4 L_4}{(L_4+y)^2} \end{pmatrix}$$

$$= \begin{pmatrix} - & 0 & - \\ + & - & 0 \\ 0 & + & - \end{pmatrix}$$

In order to study the stability, we can use the Routh-Hurwitz criterion. We recall that the invariants of a 3×3 matrix are:

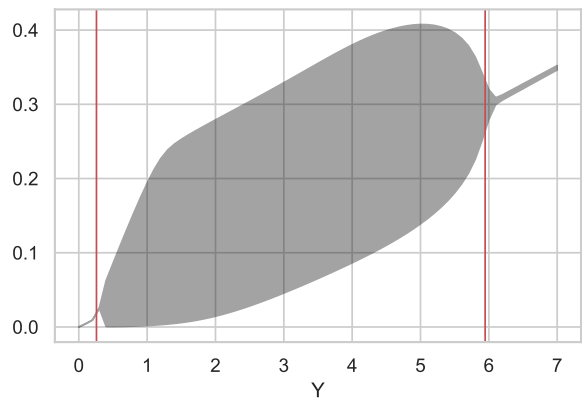
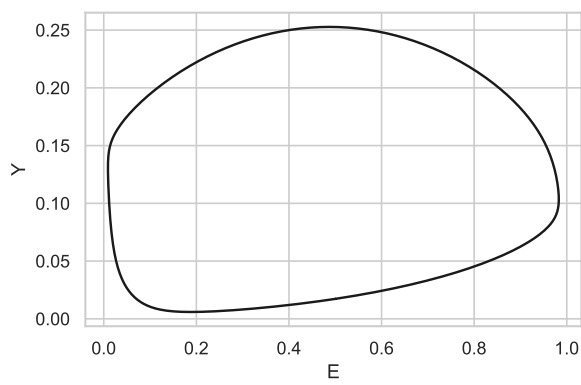
$$\begin{aligned} I_1 &= -a_1 = \text{tr } J, \\ I_2 &= +a_2 = \frac{1}{2}((\text{tr } J)^2 - \text{tr } J^2), \\ I_3 &= -a_3 = \det J, \end{aligned}$$

and they are such that:

$$\mathcal{P}_J(\lambda) = \lambda^3 - I_1 \lambda^2 + I_2 \lambda - I_3 = \lambda^3 + a_1 \lambda^2 + a_2 \lambda + a_3.$$

The Routh-Hurwitz requires that $a_1 > 0$, $a_2 > 0$, $a_3 > 0$, and $a_1 a_2 - a_3$. Using the signs, $\text{tr } J < 0$ and $\det J < 0$, so $a_1 > 0$ and $a_3 > 0$. For a_2 is slightly more complicated, but we have $a_2 > 0$. However, $a_1 a_2 - a_3$ may change sign. For some choice of the parameters, the equilibrium may lose stability through a Hopf bifurcation, giving rise to periodic solutions.

In the Figure above, we can see an example of a limit cycle for $S = 1.5$. On the right, the bifurcation plot, which shows the appearance of a limit cycle between two Hopf bifurcations for $S_{\text{crit}_1} \approx 0.26$ and $S_{\text{crit}_2} \approx 5.95$. The value of the parameters is in the code.



12 Bioelectricity

This lecture is based on [aks08__vol1, chapter 4]

Bioelectricity explores electrical phenomena within biological systems, primarily focusing on how cells and organs generate, transmit, and utilize electrical signals. Key examples of bioelectric organs include the brain, heart, and muscles, all of which rely on electrical activity to function effectively. This electrical signaling governs processes as varied as neural communication, heartbeat regulation, and muscle contraction.

In biological systems, electricity arises from the movement of ions, charged atoms or molecules such as sodium Na^+ , potassium K^+ , and calcium Ca^{2+} , across cell membranes. Each cell is structured with distinct intracellular (inside) and extracellular (outside) spaces, divided by a cell membrane. This membrane selectively controls ion flow, establishing a difference in ion concentration between the interior and exterior of the cell. This ionic imbalance creates an electric potential across the membrane, known as the transmembrane potential, defined as:

$$V_m := \phi_i - \phi_e,$$

where ϕ_i represents the intracellular potential and ϕ_e the extracellular potential.

A subset of cells, known as excitable cells, including neurons, cardiac cells, and muscle cells, can rapidly alter their transmembrane potential in response to stimuli. This shift, called an action potential, enables fast communication within and between cells and is fundamental to processes like nerve impulses and heartbeats.

The groundbreaking model of cellular excitability was developed by Hodgkin and Huxley in the 1950s, providing a mathematical framework to describe the ionic mechanisms underlying action potentials in neurons and inspiring later advances in bioelectrical research.

Hodgkin and Huxley conducted a series of seminal experiments on the giant axon of the squid (distinct from the axon of the giant squid!). This giant axon, unusually large in diameter, allowed for precise experimental manipulation and measurement, making it an ideal model for studying the mechanisms of action potential generation. In their experiments, Hodgkin and Huxley modeled the axon as a long, cylindrical structure through which ions flow, enabling them to develop a quantitative model of how nerve impulses propagate. By systematically

analyzing ionic currents across the axonal membrane, they identified key variables influencing excitability, including the roles of sodium and potassium ions and the voltage-gated ion channels that control their movement. Their work ultimately led to the formulation of the Hodgkin-Huxley model, a mathematical framework that describes the dynamics of action potentials and remains foundational in neurobiology.

12.1 The membrane as a leaky capacitor

In cellular bioelectricity, the cell membrane can be understood as a *leaky capacitor*. A capacitor typically stores electrical charge, and in a similar way, the cell membrane can maintain a separation of charges between the intracellular and extracellular environments. However, unlike an ideal capacitor, the cell membrane isn't completely insulating—it allows certain ions to “leak” across through ion channels. This controlled leakage plays a crucial role in setting the cell's resting membrane potential and modulating electrical signals.

The *capacitance* is the measure of how much charge Q we need to move from one side to another of a conductor in order to have a potential difference V_m :

$$C := \frac{Q}{V_m}.$$

The unit of measure is the Farad (symbol [F]). In other words, the membrane stores a potential energy in a fully reversible manner. The capacitance is essentially a geometrical measure. For a membrane, we can use the formula:

$$C = \frac{\varepsilon \varepsilon_0 A}{d},$$

where $d \approx 2.3 \text{ nm}$ is the thickness of the membrane, $\varepsilon_0 \approx 8.85 \cdot 10^{-12} \text{ Fm}^{-1}$ is the vacuum permittivity, $\varepsilon \approx 2.1$ is the dielectric constant of a lipidic chain, and A is the area. For a cell, the area may vary significantly, so it is more convenient to work per unit area. The specific membrane capacitance is

$$C_m = \frac{C}{A}.$$

For a typical cell, $C_m \approx 1 \mu\text{F}/\text{cm}^2$.

The membrane has also ion channels, allowing for current flows. Therefore, the membrane has a specific resistance R_m (resistance per unit area), whose value highly depends on the density of channels and their state (open or close).

Overall, the membrane can be approximated with an RC circuit, where the total transmembrane current is the sum of the current through the channels I_R and the capacitor I_C :

$$I_{\text{tot}} = I_C + I_R.$$

For the capacitor, as we move charges, we have that:

$$\frac{dV_m}{dt} = \frac{I_C}{C_m},$$

while for the resistance, we can use Ohm's law:

$$I_R = \frac{V_m}{R_m}.$$

Assuming an isolated membrane, we have $I_{\text{tot}} = 0$ so:

$$C_m \frac{dV_m}{dt} + \frac{V_m}{R_m} = 0.$$

The equilibrium of the ODE is $V_m = 0$ that is, no potential difference between the ends of the membrane. With an initial potential of $V_m(0) = V_0$, the initial value problem reads:

$$\begin{cases} \frac{dV_m}{dt} + \frac{V_m}{R_m C_m} = 0, \\ V_m(0) = V_0, \end{cases}$$

where the solution is:

$$V_m(t) = V_0 e^{-t/\tau}, \quad (12.1)$$

where $\tau = R_m C_m$ is the membrane time constant. The constant τ measures how quickly the membrane charges up (or discharges): when $t = \tau$, $V(\tau) = V_0 e^{-1} \approx 0.37 V_0$. There is a lot of variability from cell to cell: usually from $1 \mu\text{s}$ to 1s . Since C_m is mostly constant across cells (it is always the same material), it is R_m to vary a lot. Therefore, ion channels density and their state is highly variable.

12.2 Membrane response to current stimulus

When injecting a transmembrane current, we have the following model:

$$C_m \frac{dV_m}{dt} + \frac{V_m}{R_m} = I_s(t),$$

where $I_s(t)$ is a stimulation current. The general solution is:

$$V_m(t) = V_0 e^{-t/\tau} + \frac{1}{C_m} \int_0^t e^{(-t-s)/\tau} I(s) ds.$$

Suppose that $I_s(t)$ is zero for $t \leq 0$, and equal to I for $t > 0$. Assuming $V_m(0) = 0$, the solution is:

$$V_m(t) = \begin{cases} 0, & t \leq 0, \\ R_m I (1 - e^{-t/\tau}), & t > 0. \end{cases}$$

Therefore, the system tends to the equilibrium $R_m I$.

An important example is when we have a square pulse, that is we switch off the current after some time T . When we switch off the current, the solution is just Equation 12.1. Therefore we have:

$$V_m(t) = \begin{cases} 0, & t \leq 0, \\ R_m I (1 - e^{-t/\tau}), & 0 < t \leq T, \\ R_m I (e^{-T/\tau} - 1) e^{-t/\tau}, & t > T. \end{cases}$$

The peak value is $R_m I (1 - e^{-T/\tau})$. Notice that this type of cell cannot be excitable, because the peak value is fully determined by the current. There must be another mechanism for triggering the action potential.

12.3 Ion channels

The main mechanism is based on the gating dynamics of ion channels. We have the following observations:

1. Ion channels are very specific, in the sense that they allow the passage of a specific ion: there are sodium channels, potassium channels, and so on.
2. The current through a channel cannot directly follow the Ohm's law, because even in absence of an electric field across the membrane, we can have a flow of ions (and thus, a current), simply because of difference of concentration. In fact, Hodgkin and Huxley observed that sodium concentration is higher outside while potassium concentration is higher inside (see Table 12.1).
3. Ion channels can change their conductance over time, in response to voltage (voltage-gated ion channels). Since the cell can have several channels per unit area, the overall current for a specific ion varies because some channels are closed and other are open.

For point 1., we deduce the following model:

$$C_m \frac{dV_m}{dt} + I_{Na}(V_m, t) + I_K(V_m, t) + I_{leak}(V_m, t) = I_s(t), \quad (12.2)$$

where the currents I_{Na} , I_K , and I_{leak} are respectively for sodium ions Na^+ , potassium K^+ , and minor ions like chloride Cl^- . Now we need to characterize the form of each current.

12.3.1 Nernst potential

The functional form of each ion results from the Nernst equation. In fact, in view of point 2., the flow of ions through a channel is determined by a balance of diffusion and electric field. At equilibrium, the total flow is zero, hence:

$$0 = J_{\text{el}} + J_{\text{diff}}.$$

Assuming a 1-D channel of length L , with x in the longitudinal direction ($x = 0$ is the intracellular side of the membrane, $x = L$ is the extracellular one), we model the concentration of an ion along the channel with $c(x)$, and the electric potential with $\phi(x)$. For the diffusion flux we have the Fick's law:

$$J_{\text{diff}} = -D \frac{dc}{dx},$$

where $D > 0$ is the diffusion coefficient, whereas for the electric flux we have (see Section 12.5.4):

$$J_{\text{E}} = c \operatorname{sign}(z) u E,$$

where z is the valence of the ion ($z = +1$ for sodium and potassium, $z = +2$ for calcium, $z = -1$ for chloride), u is the electrical mobility, and $E = E(x)$ is the electric field. At equilibrium, the sum of the two fluxes should be zero:

$$J_{\text{diff}} + J_{\text{E}} = -D \frac{dc}{dx} - c \operatorname{sign}(z) u \frac{d\phi}{dx} = 0.$$

where we introduced the electric potential $\phi(x)$, so that $E = -\phi'$.

Using the Einstein-Smoluchowski relationship Section 12.5.5, that is $u = \frac{|z|q_E}{k_B T} D$, we have that:

$$J_{\text{E}} = -c \operatorname{sign}(z) u \frac{d\phi}{dx} = -D c \operatorname{sign}(z) \frac{|z|q_E}{k_B T} \frac{d\phi}{dx} = -D c \frac{Fz}{RT} \frac{d\phi}{dx},$$

where we introduced the constant of gases $R = N_A k_B \approx 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$ and the Farad constant $F = N_A q_E$. Now the equation reads:

$$J_{\text{diff}} + J_{\text{E}} = -D \left(\frac{dc}{dx} + c \frac{Fz}{RT} \frac{d\phi}{dx} \right) = 0.$$

We can now solve the equation. We have that:

$$\frac{dc}{c} = -\frac{Fz}{RT} d\phi$$

We want to improve the following boundary conditions:

$$\begin{aligned}\phi(0) &= \phi_i, & c(0) &= c_i, \\ \phi(L) &= \phi_e, & c(L) &= c_e.\end{aligned}$$

Therefore we have:

$$\int_{c_i}^{c_e} \frac{dc}{c} = -\frac{Fz}{RT} \int_{\phi_i}^{\phi_e} d\phi.$$

After integration, we get

$$\ln \frac{c_e}{c_i} = -\frac{Fz}{RT} (\phi_e - \phi_i).$$

Solving for *Nernst equilibrium* $V_{\text{eq}} = \phi_i - \phi_e$, we have:

$$V_{\text{eq}} = \frac{RT}{Fz} \ln \frac{c_e}{c_i}.$$

Wrapping up, given a concentration difference, the Nernst equilibrium tells up the potential difference that we need to maintain so that there is no flux (and so no current) through the channel. This potential is clearly zero when $c_i = c_e$. For example, the Nernst potential V_K of potassium is:

$$V_K = \frac{RT}{zF} \ln \frac{[K^+]_e}{[K^+]_i}.$$

With $[K^+]_e = 4 \text{ mM}$, $[K^+]_i = 155 \text{ mM}$, $R = 8.314 \text{ JK}^{-1}\text{M}^{-1}$, $z = 1$, $F = 9.65 \cdot 10^4 \text{ C mol}^{-1}$ we obtain $V_K \approx -98 \text{ mV}$. In Table 12.1 we report the Nernst potential for some ions in the axon.

Table 12.1: Nernst potentials for the squid giant axon at 27°C.

Ion	Extracellular	Intracellular	Nernst potential
Na^+	437 mM	50 mM	56.1 mV
K^+	20 mM	397 mM	−77.7 mV
Cl^-	556 mM	40 mM	68.1 mV

12.3.2 Current-voltage relations

Deviations in potential from the Nernst equilibrium yield a current. The simplest form is a linear one. For an ion S , we have:

$$I_S = g_S(V - V_S),$$

where g_S is the conductance. This relationship was introduced by Hodgkin and Huxley as a generalization of the Ohm's law. Note that the current is zero for $V = V_S$, as expected by the Nernst potential.

A more general current-voltage relation is given by the Goldman-Hodgkin-Katz (GHK) formula:

$$I_S = P_S \frac{z^2 F^2}{RT} V \frac{[S]_i - [S]_e e^{-\frac{VFz}{RT}}}{1 - e^{-\frac{VFz}{RT}}}.$$

The GHK current-voltage curve is non-linear, and corresponds to the generalized Ohm's law only for $V = V_S$, where they are both zero. The parameter P_S can be selected so that I_S for $V = 0$ is exactly $-g_S V_S$, as predicted by the Ohm's law.

i Derivation of the GHK formula

The GHK formula is based again on the balance of fluxes, but away from equilibrium:

$$J_{\text{diff}} + J_E = -D \left(\frac{dc}{dx} + c \frac{Fz}{RT} \frac{d\phi}{dx} \right) = J_{\text{tot}} \neq 0,$$

where J_{tot} is a constant. We also assume, consistently with boundary conditions, that $\phi(x) = \phi_i + (\phi_e - \phi_i)x/L$, so $\phi' = (\phi_e - \phi_i)/L = -V/L$. We have the differential equation for $c(x)$:

$$-D \left(\frac{dc}{dx} - c \frac{FzV}{RTL} \right) = J_{\text{tot}},$$

for $x \in (0, L)$ and with boundary conditions $c(0) = c_i$, $c(L) = c_e$. We can rewrite the equation as:

$$\frac{dc}{dx} = \frac{FzV}{RTL} \left(c - \frac{J_{\text{tot}} RTL}{DFzV} \right) = 0.$$

The ODE is linear and easy to integrate:

$$c(x) = c_i e^{\frac{VFz}{RTL}x} + \frac{J_{\text{tot}} RTL}{DFzV} (1 - e^{\frac{VFz}{RTL}x}).$$

Now, we know that $c(L) = c_e$. So, we can impose the condition and solve for J_{tot} :

$$J_{\text{tot}} = \frac{DFz}{LRT} V \frac{c_i - c_e e^{-\frac{VFz}{RT}}}{1 - e^{-\frac{VFz}{RT}}}.$$

Remember that J_{tot} is the flux of molecules, not charge. To get the current density I_S (current per unit area), we multiply by zF , the total charge per mole of ions, to obtain

$$I_S = zF J_{\text{tot}} = \frac{D}{L} \frac{F^2 z^2}{RT} V \frac{c_i - c_e e^{-\frac{VFz}{RT}}}{1 - e^{-\frac{VFz}{RT}}},$$

which corresponds to the above relation with $P_S = D/L$.

12.3.3 Reversal potential

Using the generalized Ohm's law in Equation 12.2 we have:

$$C_m \frac{dV_m}{dt} + g_{\text{Na}}(V - V_{\text{Na}}) + g_{\text{K}}(V - V_{\text{K}}) + g_{\text{leak}}(V - V_{\text{leak}}) = 0.$$

The *reversal potential* is the potential V such that the total ionic current is zero. In other words, it is the equilibrium value for the ODE. In this case we have that:

$$C_m \frac{dV_m}{dt} = (g_{\text{Na}} + g_{\text{K}} + g_{\text{leak}}) \left(V - \frac{g_{\text{Na}} V_{\text{Na}} + g_{\text{K}} V_{\text{K}} + g_{\text{leak}} V_{\text{leak}}}{g_{\text{Na}} + g_{\text{K}} + g_{\text{leak}}} \right)$$

so the reversal potential is:

$$V_r = \frac{g_{\text{Na}} V_{\text{Na}} + g_{\text{K}} V_{\text{K}} + g_{\text{leak}} V_{\text{leak}}}{g_{\text{Na}} + g_{\text{K}} + g_{\text{leak}}},$$

and the membrane time constant is:

$$\tau = \frac{C_m}{g_{\text{Na}} + g_{\text{K}} + g_{\text{leak}}}.$$

12.3.4 Gating mechanism

The last point to address concerns the fact that the conductances of the channels may change over time:

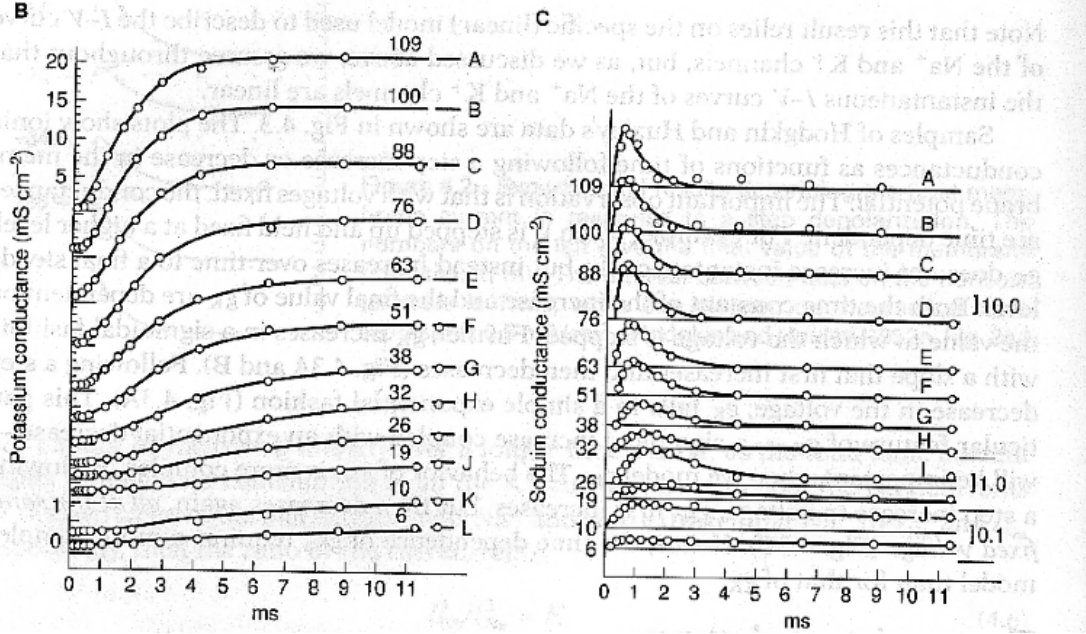
$$I_S = g_S(t)(V - V_S).$$

Hodgkin and Huxley deduced the time course of $g_S(t)$ for sodium and potassium thanks to a couple of techniques:

1. voltage clamping, a method they invented to hold the potential V to a given value. In this way, they could measure the total ionic current over time;

2. by rendering inactive the sodium channels in a separate set of measures, so that they could deduce from the total current the individual potassium and sodium currents.

From the currents, they obtained the variation over time of the conductances g_{Na} and g_K for different values of V .



In the Figure, we see the conductance curves fitted to experimental data, at various levels of potential (with respect to the resting value). We see that the potassium current always increases, with a sigmoidal shape, and with a limit value for large times that increases as well with the potential. For the sodium, the conductance has a peak value and then goes to some lower limit value.

Hodgkin and Huxley speculated that ion channels could be in at least two states: open and close; the observed conductance is the sum of all open channels:

$$g_S(t) = \bar{g}_S \cdot p(t),$$

where $\bar{g}_S$ is the maximum conductance per unit area of all S -channels, and $p(t)$ is the probability that any S -channel is open.

The simplest model for $p(t)$ is a 2-state Markov model. Suppose that $Q(t) \in \{\mathcal{C}, \mathcal{O}\}$ is the state of the channel at time t , then:

$$p(t) = \mathbb{P}[Q(t) = \mathcal{O}].$$

The Markov model is as follows:

$$\mathcal{C} \xrightleftharpoons[\beta]{\alpha} \mathcal{O},$$

where α and β are the transition rates. The corresponding equation for $p(t)$ is:

$$\frac{dp}{dt} = \alpha(1 - p) - \beta p. \quad (12.3)$$

Note that we can interpret $p(t)$ as the “concentration” of open channels, thus we can formally apply the laws of chemical reactions.

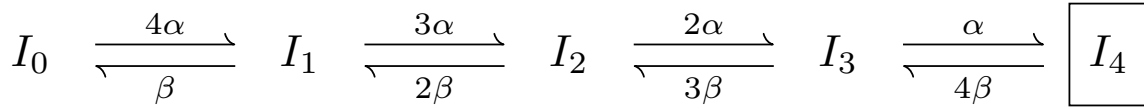
The ODE in Equation 12.3 is linear and can be rewritten as:

$$p' = -(\alpha + \beta)\left(p - \frac{\alpha}{\alpha + \beta}\right) = -\frac{1}{\tau_p}(p - p_\infty),$$

where we introduced the characteristic time τ_p and the limit value p_∞ , that is $p \rightarrow p_\infty$ for $t \rightarrow \infty$. The general solution is:

$$p(t) = p_\infty - e^{-t/\tau_p}(p_\infty - p(0)).$$

This function captures the time transient of the conductance, but without the correct shape. For instance, in order to have a sigmoidal shape for the potassium current we need to allow for multiple states: for the potassium, the model is as follows:



where the state I_4 is open, that is $p(t) = \mathbb{P}[Q(t) = I_4]$. We can imagine the potassium channel as made of 4 identical units, each possibly being in either open or closed state. The channel is open when *all* the 4 units are open. Then, I_k represents the state with k open units. The rates of transition are the same for all units: if the rate of single unit from closed to open is α , then the rate from I_0 to I_1 is exactly 4α , because we have 4 closed units that could become open. A similar argument applies to the other rates.

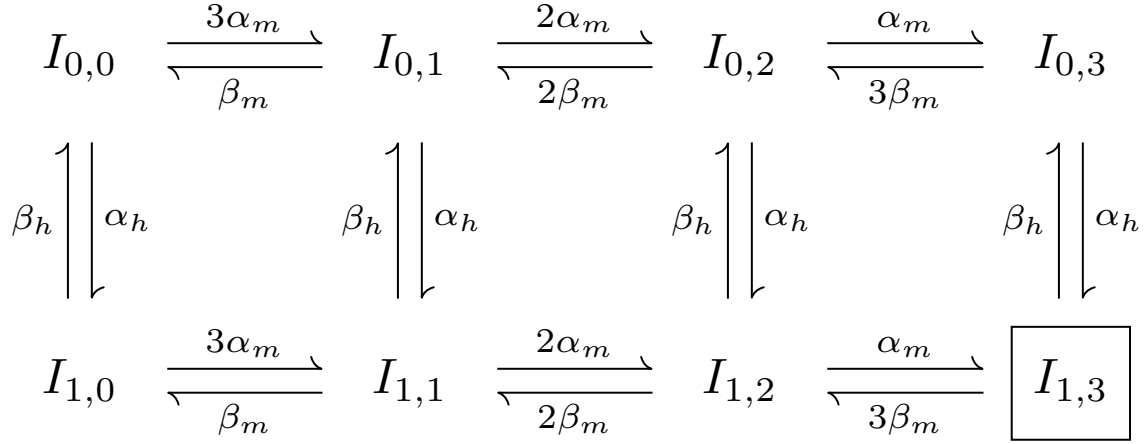
It is possible to derive a full set of ODEs for this Markov model, but we can anticipate that, if $n(t)$ is the probability that a single unit is open, then $p(t) = n^4(t)$. The equation for $n(t)$ is just

$$\frac{dn}{dt} = \alpha_n(1 - n) - \beta_n n.$$

For the sodium channel, the mechanism is a bit more complicated:

Here we have a Markov model with 8 possible states, and we have $p(t) = m^3(t)h(t)$ where $m(t)$ and $h(t)$ satisfies:

$$\begin{aligned} \frac{dm}{dt} &= \alpha_m(1 - m) - \beta_m m, \\ \frac{dh}{dt} &= \alpha_h(1 - h) - \beta_h h. \end{aligned}$$



In this way, the time course of the conductances can be fully characterized.

The very last ingredient is the voltage sensitivity of the conductances. Here, Hodgkin and Huxley assumed that all the rates for m , h , and n , are explicit functions of V , fitted to the data.

i Alternative derivation of the gating equation

First note that

$$\mathbb{P}[S(t) = \mathcal{C}] = 1 - \mathbb{P}[S(t) = \mathcal{O}] = 1 - p(t).$$

The probability $p(t + dt)$ writes as follows:

$$\begin{aligned}
p(t + dt) &= \mathbb{P}[S(t + dt) = \mathcal{O}] \\
&= \mathbb{P}[S(t + dt) = \mathcal{O} | S(t) = \mathcal{C}] \mathbb{P}[S(t) = \mathcal{C}] \\
&\quad + \mathbb{P}[S(t + dt) = \mathcal{O} | S(t) = \mathcal{O}] \mathbb{P}[S(t) = \mathcal{O}] \\
&= \mathbb{P}[S(t + dt) = \mathcal{O} | S(t) = \mathcal{C}] (1 - p(t)) \\
&\quad + (1 - \mathbb{P}[S(t + dt) = \mathcal{C} | S(t) = \mathcal{O}]) p(t) \\
&= \alpha dt (1 - p(t)) + p(t) - \beta dt p(t).
\end{aligned}$$

where we used the rates:

$$\begin{aligned}
\mathbb{P}[S(t + dt) = \mathcal{C} | S(t) = \mathcal{O}] &= \alpha dt \\
\mathbb{P}[S(t + dt) = \mathcal{O} | S(t) = \mathcal{C}] &= \beta dt
\end{aligned}$$

Rearranging the terms we have:

$$\frac{p(t + dt) - p(t)}{dt} = \alpha(1 - p) - \beta p.$$

12.4 The Hodgkin-Huxley model

Putting all the ingredients together, we arrive that the final form of the Hodgkin-Huxley model for the excitability of the axon:

$$\begin{cases} C_m \frac{dV_m}{dt} + \bar{g}_{Na} m^3 h (V - V_{Na}) + \bar{g}_K n^4 (V - V_K) + \bar{g}_{leak} (V - V_{leak}) = I_{stim}(t), \\ \frac{dn}{dt} = \alpha_n (1 - n) - \beta_n n, \\ \frac{dm}{dt} = \alpha_m (1 - m) - \beta_m m, \\ \frac{dh}{dt} = \alpha_h (1 - h) - \beta_h h. \end{cases}$$

The model, supplemented with initial conditions for the variables $V_m(t)$, $m(t)$, $h(t)$, and $n(t)$, can quantitatively reproduce the action potential of the squid axon.

12.5 Appendix: Electrochemistry

We review here some basic concepts of electrochemistry.

12.5.1 Conductivity

Consider a volume (a cube, for instance) with some ions inside, for instance, a volume of water with salt inside, which dissociated into sodium Na^+ and chloride Cl^- .

If we apply a uniform electric field E across two sides of the cube, with two electrodes of area A , *cations* (positive ions) will move towards the *cathode* (negative electrode) and *anions* (negative ions) will move towards the *anode* (positive electrode.) This movement of ions will close the circuit and give rise to a current I .

The current is proportional to the area A of the electrodes, so we define the *current density* J as follows:

$$J = \frac{I}{A}.$$

Experimentally, we observe that the current is proportional to the potential difference V between the electrode:

$$I = gV,$$

where g is the conductance of the solution (unit: Siemens [S]), that is the inverse of the resistance. Overall, the conductance may depend on several factors:

- the concentration of ions (more ions, more current),
- the mobility of ions in the solution (faster ions, more current),
- the distance between the electrodes,
- the viscosity of the solution (similar to mobility),
- the temperature,
- ...

In first approximation, we can take $E = \frac{V}{d}$, where d is the distance between the electrodes. We end up with:

$$J = \frac{gd}{A}E = \sigma E,$$

where σ is the *conductivity*. The conductivity is an intrinsic property of the solution. For instance, sea water has $\sigma \approx 50\text{mS/cm}$, and for the body is $\sigma \approx 1.5\text{mS/cm}$.

12.5.2 Molar conductivity

The conductivity σ , in turn, depends on the concentration of ions. The conductivity per unit mole is the *molar conductivity* $\Lambda_m = \frac{\sigma}{c}$. In first approximation, we assume that Λ_m is independent of c .

The molar conductivity of a solution with multiple ions involved is given by the [Kohlrausch law of independent migration of ions](#):

$$\Lambda_m = \sum |z_i| \lambda_m$$

where z_i is the valence of the i -th ion and λ_i its molar conductivity. For instance, $\lambda_{\text{Na}} = 5\text{mS m}^2/\text{mol}$ and $\lambda_{\text{Cl}} = 7.6\text{mS m}^2/\text{mol}$. So, the molar conductivity of salt NaCl in water is $\Lambda_{\text{NaCl}} = 12.6\text{mS m}^2/\text{mol}$.

12.5.3 Mobility of ions

In a solution, each ion will experience an electric force according to the Coulomb's law:

$$f_E = q_E z E,$$

where $q_E \approx 1.602 \cdot 10^{-19}\text{ C}$ is the elementary charge, z is the valence, and E the electric field. On the other hand, the ion will also experience a viscous friction, due to the molecules of the solution. The balance of forces gives the dynamics:

$$m\dot{v} = q_E z E - f v,$$

where m is the mass, v is the velocity, and f is the friction. At equilibrium, the limit velocity v_∞ is

$$v_\infty = \frac{zq_E}{f}E = uE,$$

where $u = \frac{zq_E}{f}$ is the electrical mobility. More generally, when $z < 0$ also the friction should be negative, so:

$$u = \frac{|z|e_E}{f}.$$

Again, the mobility is an intrinsic property of the solution. Specifically, we can consider the [Stokes law](#) for the friction, $f = 6\pi a\eta$ where a is the radius of the ion and η is the viscosity of the solution. Therefore,

$$u = \frac{zq_E}{6\pi a\eta}.$$

For instance, Na^+ mobility in water at 25°C is $5.19 \cdot 10^{-8} \text{ m}^2/(\text{s V})$. (Clearly, the viscosity depends on the temperature.)

12.5.4 Mobility and conductivity

There is a relationship between mobility and conductivity, because moving ions yields a current. Assuming $v_\infty = uE$ as the speed of ions, in the time dt the ions cover a distance of $dx = v_\infty dt$. The total volume of ions that moves across an area A is

$$A \cdot dx \cdot c \cdot N_A = AuEcN_A dt,$$

where $N_A \approx 6.022 \cdot 10^{23} \text{ mol}^{-1}$ is the Avogadro constant. The total charge is the above quantity multiplied by $|z|q_E$. Hence, the current density is:

$$J = \frac{|z|q_E AuEcN_A dt}{A dt} = Fcu|z| \cdot E = \sigma E,$$

with $F = q_EN_A \approx 9.65 \cdot 10^4 \text{ C mol}^{-1}$ is the Farad constant. So we arrive at the expression:

$$\sigma = Fcu|z|, \quad \Rightarrow \quad \Lambda_m = \frac{\sigma}{c} = F|z|u.$$

Note that we take the absolute value $|z|$ because we should separate cations (+) and anions (-) in the analysis. Cations will follow the direction of E , while anions go in the opposite direction. Since the latters carry an negative charge, the virtually move a positive charge in the direction of E .

Given the total current as $J = F|z|ucE$, we can go back to the flux of ions J_E (as a change of concentration) to obtain the Planck's equation:

$$J_E = \frac{J}{Fz} = \frac{F|z|uc}{Fz}E = uc \text{ sign}(z)E.$$

12.5.5 Einstein-Smoluchowski equation

The connection between friction and diffusion has been provided by Einstein in 1905 and independently by others around the same period (Sutherland in 1904, Smoluchowski in 1906), in a seminal work on Brownian motion. Briefly, friction dissipates kinetic energy into heat, whereas Brownian motion-the rapid fluctuation of molecules at microscopic scale due to thermal agitation-converts heat into kinetic energy. The duality between friction and Brownian motion translates into the general relationship

$$D = \frac{k_B T}{f}$$

where k_B is the Boltzmann constant, D is the diffusion coefficient and T is the absolute temperature. Given the definition of electrical mobility, we have that:

$$D = \frac{k_B T}{f} = \frac{k_B T}{|z|q_E} u,$$

which provides a link between D and u .

13 Hodgkin-Huxley model

This lecture is based on [aks08_vol1, chapter 4]

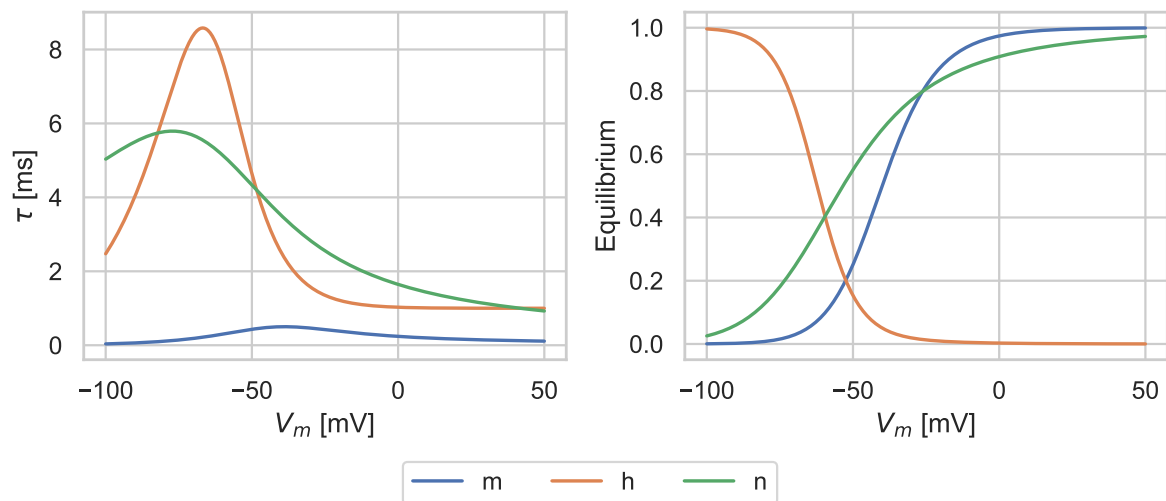
13.1 The model

The complete form of the Hodgkin-Huxley model is the following:

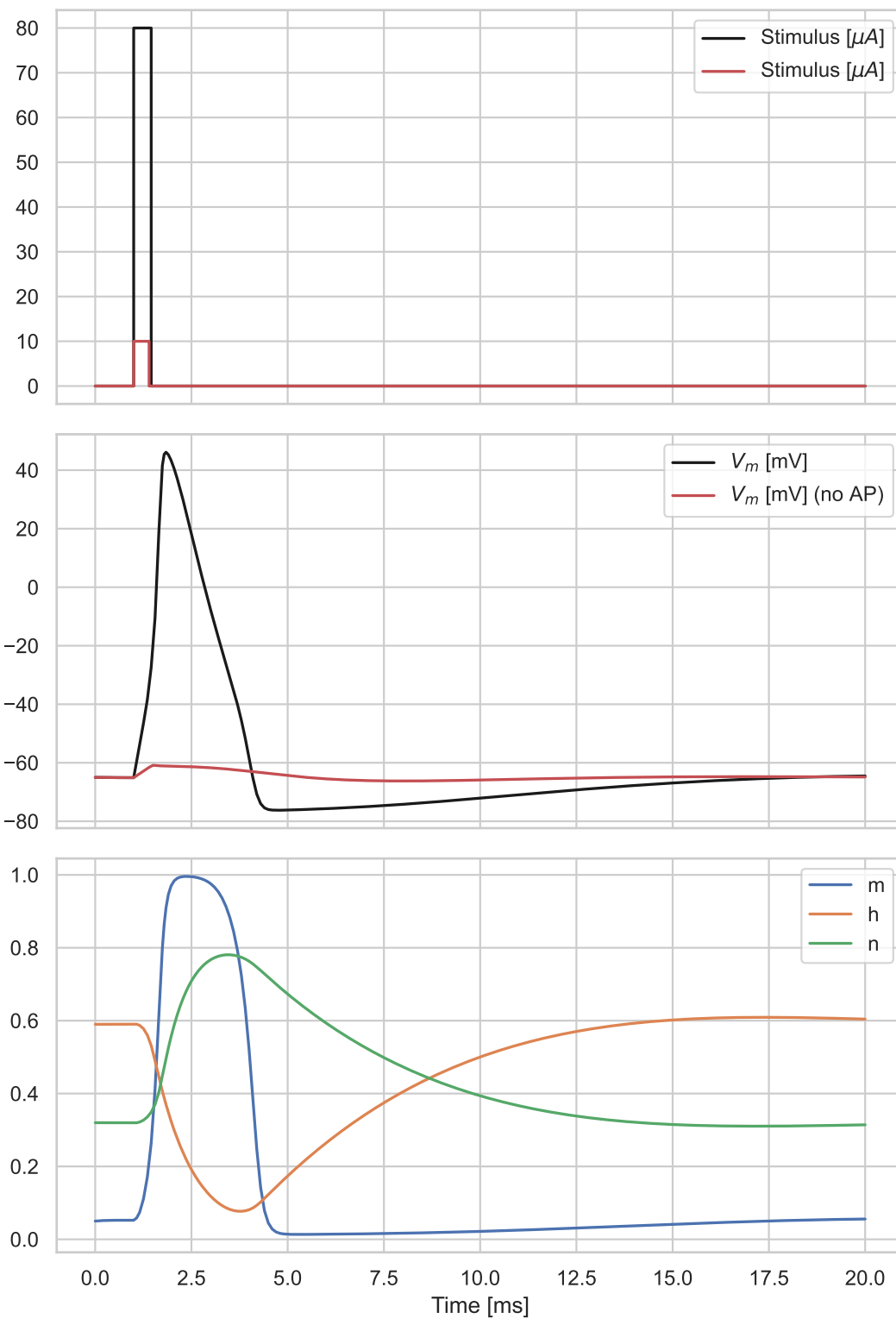
$$\left\{ \begin{array}{l} -C_m \frac{dV_m}{dt} = \bar{g}_{Na} m^3 h (V - V_{Na}) + \bar{g}_K n^4 (V - V_K) + \bar{g}_{leak} (V - V_{leak}) - I_{stim}(t), \\ \frac{dn}{dt} = \alpha_n(V_m)(1 - n) - \beta_n(V_m)n, \\ \frac{dm}{dt} = \alpha_m(V_m)(1 - m) - \beta_m(V_m)m, \\ \frac{dh}{dt} = \alpha_h(V_m)(1 - h) - \beta_h(V_m)h. \end{array} \right. \quad (13.1)$$

The parameters of the model per 1 cm^2 are as follows: $C_m = 1 \mu\text{F}$, $g_{Na} = 120 \text{ mS}$, $g_K = 36 \text{ mS}$, $g_{leak} = 0.3 \text{ mS}$, $V_{Na} = 55 \text{ mV}$, $V_K = -77 \text{ mV}$, $V_{leak} = -54.4 \text{ mV}$.

Concerning the transition rates, we have the following for $\tau_{\{m,h,n\}}(V_m)$ and $\{m, h, n\}_\infty$:



If we apply a square stimulus, we obtain the following solution:



Notice that we can trigger an action potential only for a sufficiently large stimulus. This is a manifestation of *excitability*: when the stimulus is large enough, the potential will raise automatically to an excited value, and recover afterwards. Otherwise, when the stimulus is small (or too short), we essentially have a passive behavior, similar to an RC circuit.

13.2 Fast-fast analysis

The model is too complex for an explicit computation, but we can get some insights into excitability with the fast-slow analysis.

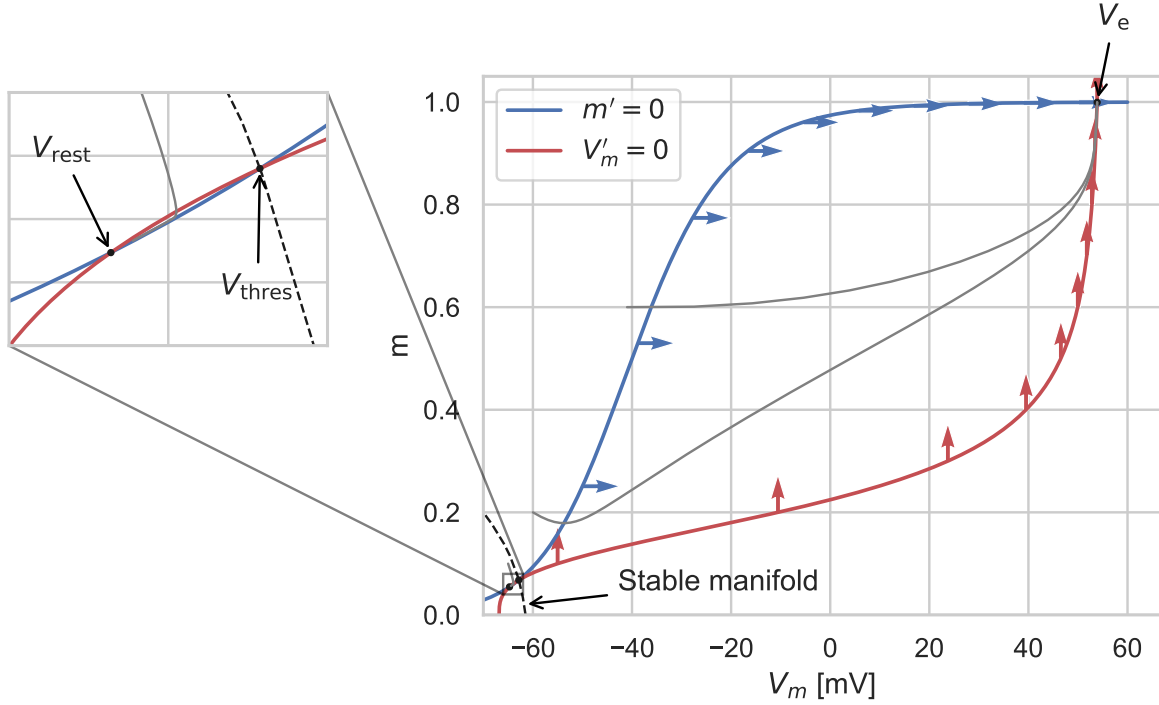
In the 1960s, FitzHugh observed that the model has a fast-slow structure, were:

- $m(t)$ and $V_m(t)$ are fast,
- $n(t)$ and $h(t)$ are slow.

In fact, $\tau_m \ll \tau_n, \tau_h$. The same we observe for V_m due to the large conductivity of the sodium. Therefore, we can study the fast dynamics by fixing $n(t) = n_0$ and $h(t) = h_0$:

$$\begin{cases} -C_m \frac{dV_m}{dt} = \bar{g}_{Na} m^3 h_0 (V - V_{Na}) + \bar{g}_K n_0^4 (V - V_K) + \bar{g}_{leak} (V - V_{leak}), \\ \tau_m(V) \frac{dm}{dt} = m_\infty(V) - m. \end{cases}$$

In the (V_m, m) phase space, we have:

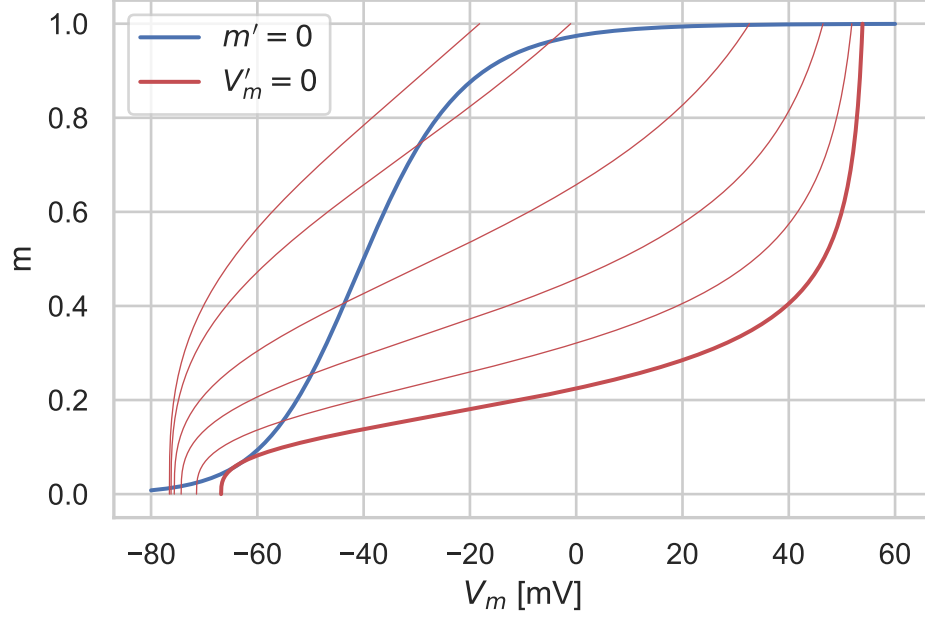


We have 3 equilibria:

1. $V_m = V_{\text{rest}} \approx -64.72$ mV, which is stable,
2. $V_m = V_{\text{thres}} \approx -62.85$ mV, a saddle,
3. $V_m = V_e \approx 53.87$ mV, stable.

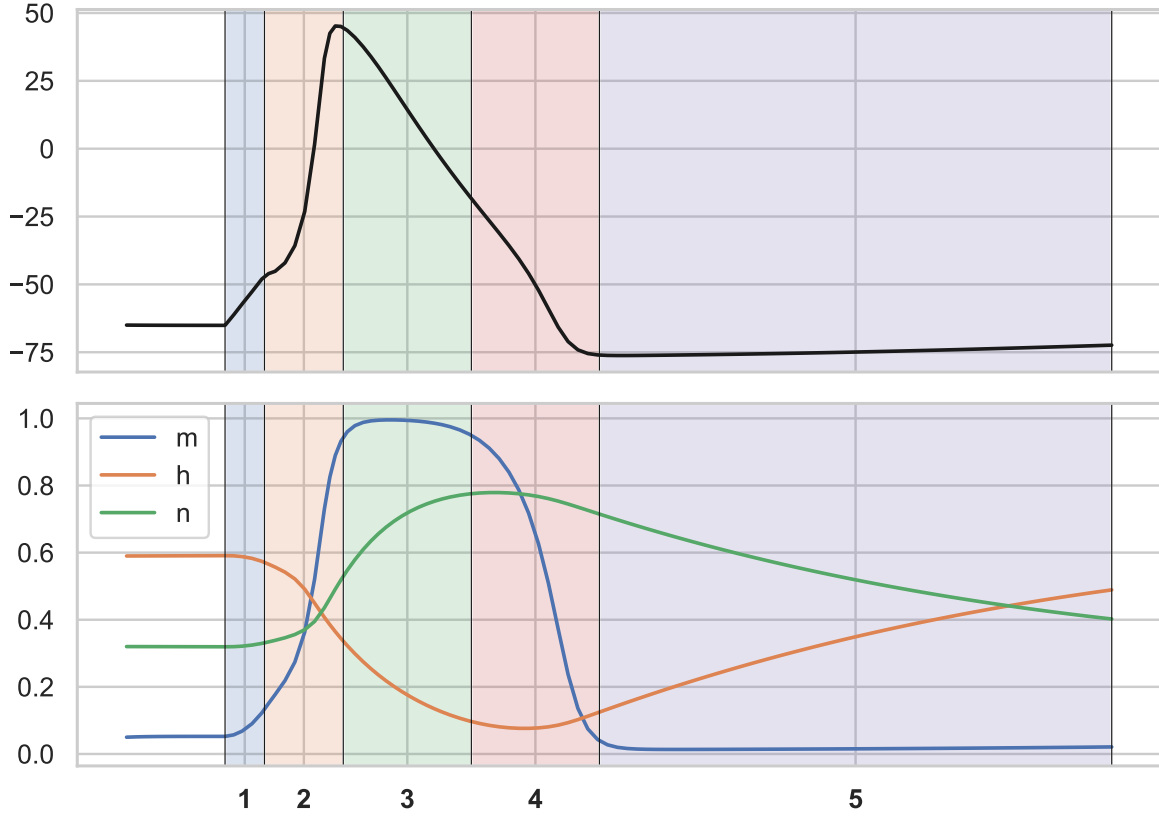
The saddle has a stable manifold, indicated in the above figure, that separates the phase space in two parts: if we start from the right of it, we converge towards the resting equilibrium, otherwise we go towards the excited state. The bistable behavior explains the excitability of the system: starting from the resting membrane potential, if we apply a sufficiently large current we can move the state to the right of the manifold, and obtain an action potential.

Therefore, the fast dynamics brings the system to the excited state (the upstroke phase of the action potential). Once we reach the equilibrium, the slow dynamics enters, moving n and h away from their resting value. In particular, n will increase while h will decrease. As we do so, the nullcline for V_m , in the fast system, will shift:



We notice that equilibria shift as the nullcline changes. Let us summarize the dynamics:

1. (Fast) Fixed $n = n_0$, $h = h_0$, a stimulus excite the system from the resting state $(V_{\text{rest}}, m_{\infty}(V_{\text{rest}}))$ to the excited $(V_e, m_{\infty}(V_e))$;
2. (Slow) The fast system is now at equilibrium. We can track V_m and m with respect to n and h the surface of equilibria defined by $V_e(h, n)$ and $m(h, n) = m_{\infty}(V_e(h, n))$. On this surface, we have the slow dynamics given by $n' = \dots$ and $h' = \dots$. Since V_m is large, and $n_0 > n_{\infty}$, n will start to increase while h will do the opposite. As these values changes, the nullcline for V_m in the fast system will also shift, as in the figure.
3. (Tipping point) The process will continue until the excited equilibrium and the saddle collide in a tangent bifurcation. This is a tipping point.
4. (Fast) Since the encountered a tipping point, we cannot continue the slow dynamics and we need to integrate the fast system. After the bifurcation, there is only one equilibrium left, the resting state, so the system will converge there (quickly).
5. (Slow) Once back at the resting state, the slow system will continue on the lower branch of equilibria, defined by $V_{\text{rest}}(h, n)$ and $m(h, n) = m_{\infty}(V_{\text{rest}}(h, n))$. Here, n will decrease and h will increase, moving back the system to the original equilibrium.

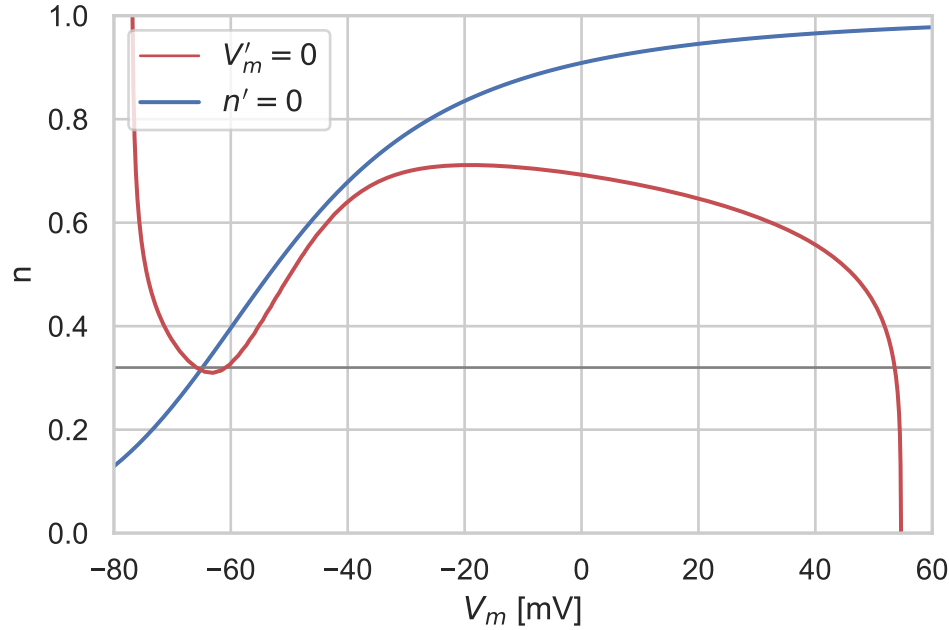


13.3 Fast-slow analysis

In the 1960s, FitzHugh observed that $m(t)$ is actually *very fast* compared to other variables, so we could always assume $m = m_\infty(V)$. A second observation is that $n + h \approx 0.8$, so we could take $h \approx 0.8 - n$. In this way, we end up with a system of 2 equations:

$$\begin{cases} -C_m \frac{dV_m}{dt} = \bar{g}_{Na} m_\infty(V_m)^3 (0.8 - n)(V - V_{Na}) \\ \quad + \bar{g}_K n^4 (V - V_K) + \bar{g}_{leak} (V - V_{leak}), \\ \frac{dn}{dt} = \alpha_n(V_m)(1 - n) - \beta_n(V_m)n. \end{cases}$$

In the phase plane, nullclines are $n = n_\infty(V_m)$ and another function that looks like a cubic function:



In the above plot, we have one equilibrium. Furthermore, the variable V_m is fast compared to n , so we can again use the fast-slow analysis:

- The fast system has up to 3 equilibria, depending on the value of n . Initially, for $n = n_0$, we have 3 equilibria: 2 stable and one unstable (in the middle).
- The slow system proceeds on the branches of solutions $V_-(n)$ or $V_+(n)$, both stable. The branch in the middle, $V_0(n)$, is unstable and acts as a threshold.

After applying a stimulus, the potential quickly goes beyond $V_0(n_0)$, and the fast dynamics continues until $V_m \rightarrow V_+(n_0)$. Here, the slow dynamics starts, where n increase along the branch $V_+(n)$, up to the tipping point. After that, there is a fast dynamics up to the other branch, $V_-(n)$, until the system goes back to equilibrium.

13.4 The FitzHugh-Nagumo model

A further simplification of the above model is to assume $n_\infty(V)$ linear and the cubic-like function a cubic with three zeros (and shift-scale everything):

$$\begin{cases} \varepsilon \frac{dV}{dt} = f(V) - w + I(t), \\ \frac{dw}{dt} = V - \gamma w, \end{cases}$$

where w is the *recovery variable* (formerly n), $f(V) = V(1 - V)(V - \alpha)$ and $\varepsilon \ll 1$. Typical values are $\varepsilon = 0.01$, $\gamma = 0.5$, $\alpha = 0.1$.

This model is sufficiently simple to be studied in the phase plane. Suppose $I(t) = 0$. Since $\varepsilon \ll 1$, V is fast and w is slow.

13.4.1 Fast dynamics

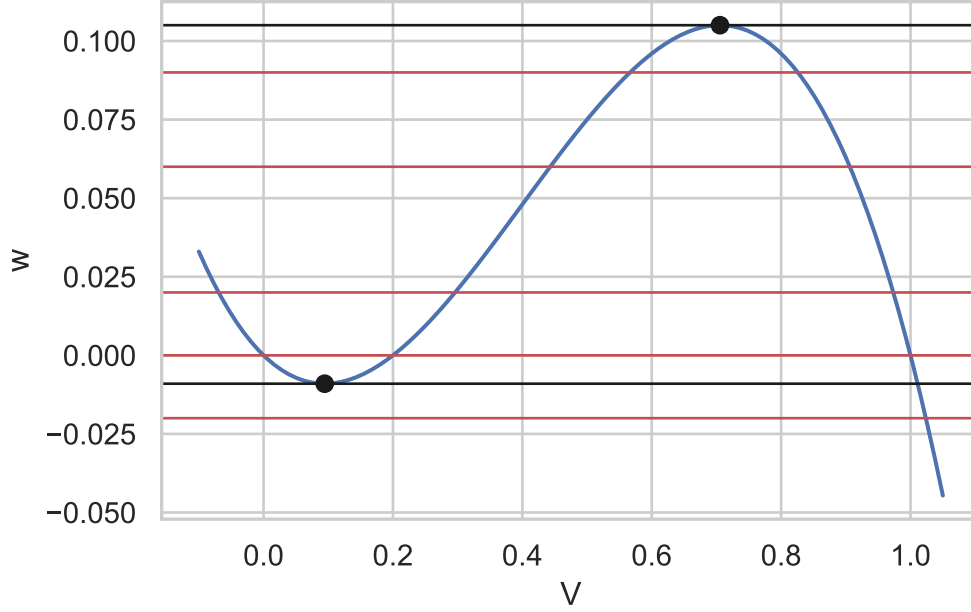
We set $\eta = \frac{t}{\varepsilon}$ and

$$\begin{cases} \dot{V} = f(V) - w, \\ \dot{w} = \varepsilon(V - \gamma w), \end{cases}$$

so for $\varepsilon = 0$ we have $w = w_0$ and

$$\dot{V} = f(V) - w_0.$$

Depending on w_0 , we can have a different number of equilibria:



Given w_* and w^* respectively the minimum and the maximum of the cubic, we have three equilibria

$$V_-(w) \leq V_0(w) \leq V_+(w)$$

that exists for $w_* < w < w^*$. Otherwise, we only have either $V_-(w)$ or $V_+(w)$. The equilibria $V_{\pm}(w)$ are always stable, while $V_0(w)$ is unstable (the threshold).

13.4.2 Slow dynamics

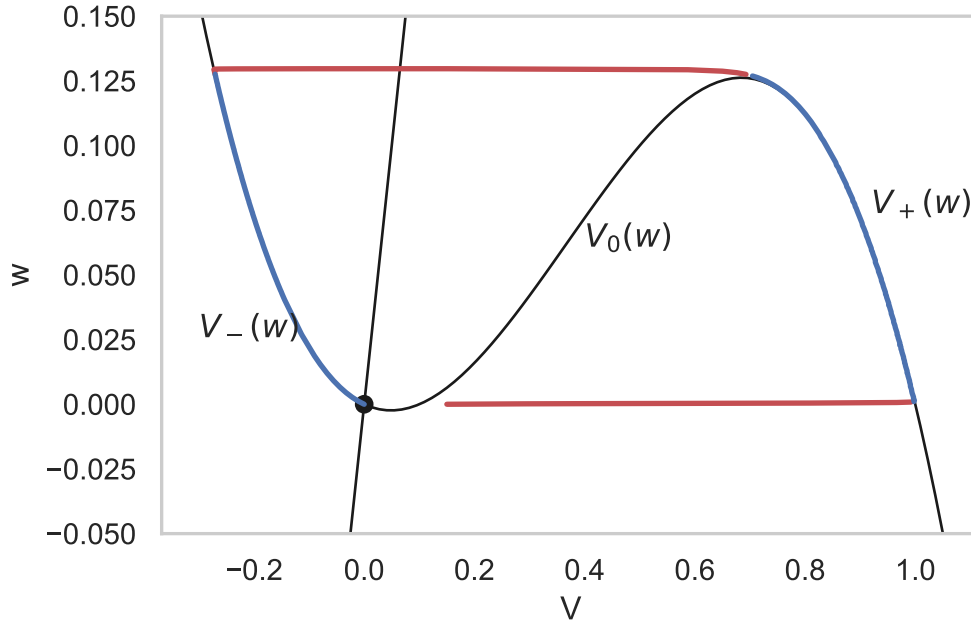
In the original system, we set $\varepsilon = 0$ to obtain:

$$\begin{cases} 0 = f(V) - w, \\ \frac{dw}{dt} = V - \gamma w, \end{cases}$$

The first equation can be replaced by one branch of solution $V_{\pm}(w)$. Thus we are left with

$$\frac{dw}{dt} = V_{\pm}(w) - \gamma w = G_{\pm}(w),$$

which dictated the slow dynamics.



From the Figure, we can follow the fast-slow dynamics (red is fast, blue is slow). Notice that we have one tipping point.

We can also compute the “excited” time T_e , that is the time the system is on the excited state:

$$\dot{w} = V_+(w) - \gamma w, \quad \Rightarrow \quad T_e = \int_{w_0}^{w^*} \frac{dw}{V_+(w) - \gamma w}.$$

During the excited interval, the system cannot be further excited (it is *refractory*). The recovery period, on the other hand, is

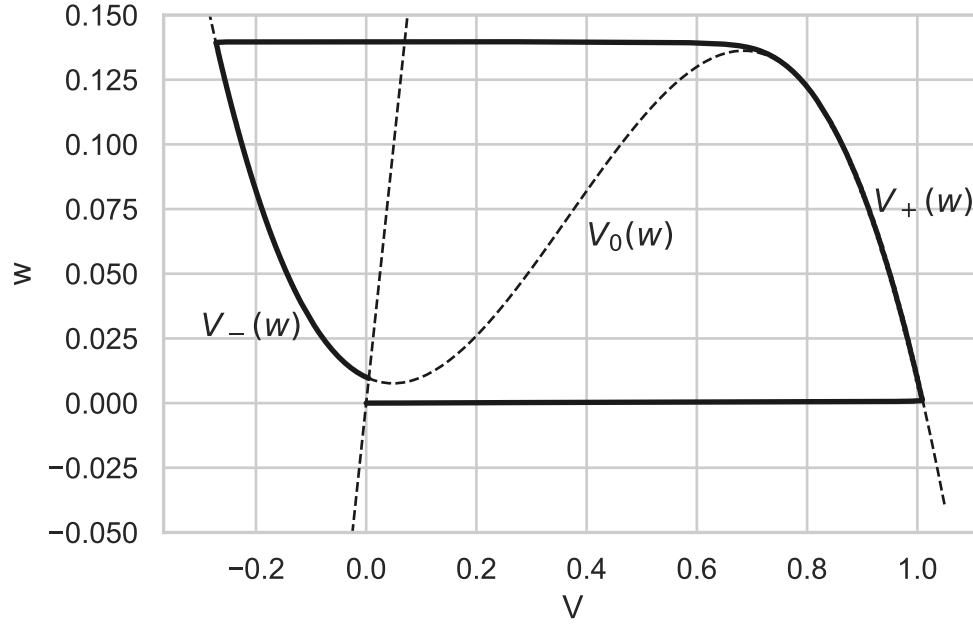
$$T_r = \int_{w^*}^0 \frac{dw}{V_-(w) - \gamma w},$$

and during this period the system is excitable.

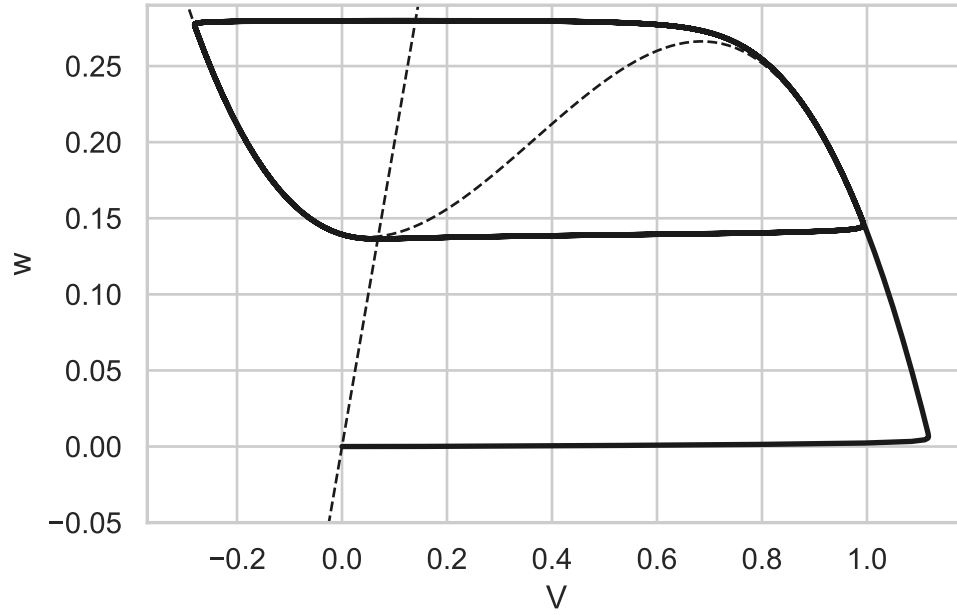
13.4.3 Oscillations

Suppose now that we apply a constant current $I > 0$. The analysis is as before with a minor difference: the cubic is shifted vertically. The equilibrium is also non-zero now.

Suppose we start for the origin at time $t = 0$, and then we apply a current $I > 0$. The cubic will shift upwards, and the state is not at equilibrium anymore. The fast-slow dynamics is the same as before.



However, if we increase I , we end up in the following situation:



We have stable oscillations in a limit cycle. In fact, for I sufficiently large, there is a second tipping point at $w = w_*$: if the equilibrium is on the right of the tipping point, the solution will jump from $V_-(w)$ to $V_+(w)$, and continue indefinitely.

We can also approximate the period of the limit cycle:

$$T \approx T_e + T_r = \int_{w_*}^{w^*} \left(\frac{1}{V_+(w) - \gamma w} - \frac{1}{V_-(w) - \gamma w} \right) dw.$$

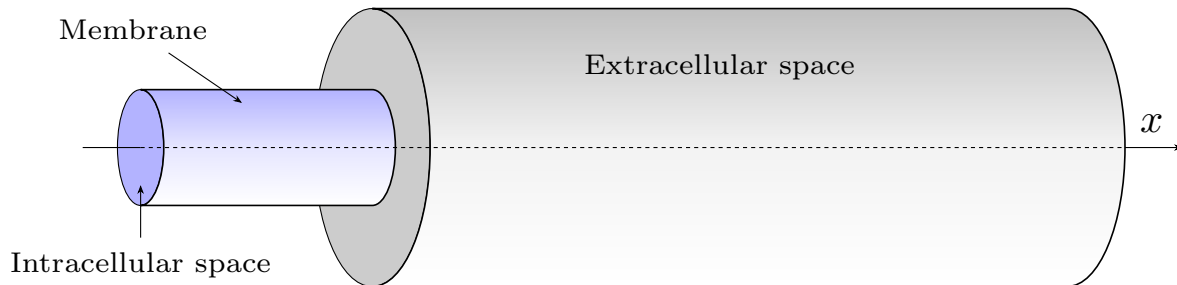
14 The cable equation

This lecture is based on [aks08_vol1, chapter 5]

14.1 Derivation of the cable equation

The axon is a tubular structure through which ions and currents diffuse primarily along its main axis. As a result, a localized stimulus propagates spatially, though its amplitude diminishes with increasing distance from the point of stimulation. In excitable membranes, the diffusive currents associated with an action potential can be strong enough to trigger additional action potentials in neighboring regions. This cascade effect leads to the propagation of action potentials along the axon.

We can approximate the axon with a cylinder:



The axon is immersed in an extracellular medium (the bath solution) that we also suppose cylindrical in first approximation. Usually, the extracellular space is much larger in volume than the intracellular one. The cellular membrane, where ion channels are, is the boundary between the extra- and intra-cellular regions. We also suppose that all quantities (currents, potential) vary only in the longitudinal direction.

Consider a intra-cellular (resp. extra-cellular) cross-section of area A_i (resp. A_e) and length dx . We denote the intra- and extra-cellular electrical potential along the longitudinal direction with

$\phi_i(x, t)$ and $\phi_e(x, t)$. For the [Ohm's law](#), we have that the intra- and extra-cellular currents are:

$$I_i(x, t) = -\frac{1}{R_i} \cdot (\phi_i(x + dx) - \phi_i(x)),$$

$$I_e(x, t) = -\frac{1}{R_e} \cdot (\phi_e(x + dx) - \phi_e(x)),$$

where $I_i(x, t)$ and $I_e(x, t)$ are the currents and R_e , R_i are the resistances of the cross-sections. Remember that the resistance is:

$$R = \frac{r\ell}{A},$$

where r is the resistivity, A the cross-section, and ℓ the length. Therefore:

$$I_i(x, t) = -\frac{A_i}{r_i} \frac{\phi_i(x + dx) - \phi_i(x)}{dx} \xrightarrow{dx \rightarrow 0} -\frac{A_i}{r_i} \frac{\partial \phi_i}{\partial x},$$

$$I_e(x, t) = -\frac{A_e}{r_e} \frac{\phi_e(x + dx) - \phi_e(x)}{dx} \xrightarrow{dx \rightarrow 0} -\frac{A_e}{r_e} \frac{\partial \phi_e}{\partial x}.$$

Now we do a balance of the currents. For the intracellular section, we have 3 surfaces for the cylinder: the two ends, at x and $x+dx$, and the membrane. By convention, the transmembrane current density, denoted by $J_m(x, t)$, flows from the intracellular to the extracellular space. Summing up we have:

$$I_i(x + dx, t) = I_i(x, t) - \int_x^{x+dx} p J_m(z, t) dz,$$

where p is the perimeter of the axon. Rearranging, dividing by dx , and taking the limit assuming that J_m is regular, we have:

$$\frac{\partial I_i}{\partial x} = -p J_m(x, t).$$

For the extracellular space, the hollow cylinder has 4 faces: the ends, the membrane, and the outer surface. For this last one, we suppose that the bath is insulated, so there is no current flow through it. For the rest is as above:

$$I_e(x + dx, t) = I_e(x, t) + \int_x^{x+dx} p J_m(z, t) dz,$$

$$\rightsquigarrow \frac{\partial I_e}{\partial x} = p J_m(x, t).$$

The transmembrane current density (current per unit area) depends on the transmembrane voltage, according to the Hodgkin-Huxley model, in [Equation 13.1](#):

$$J_m(x, t) = C_m \frac{\partial V_m}{\partial t} + I_{\text{ion}}(V_m),$$

where $V_m(x, t) = \phi_i(x, t) - \phi_e(x, t)$ is the transmembrane potential and I_{ion} is the sum of ion currents. Also the gating variables, $m(x, t)$, $n(x, t)$, $h(x, t)$, are defined at each point of the membrane, that is they vary both in space and time.

We can now put all together and obtain the following system:

$$\begin{cases} \frac{\partial}{\partial x} \left(\frac{A_i}{r_i} \frac{\partial \phi_i}{\partial x} \right) = p \left(C_m \frac{\partial V_m}{\partial t} + I_{\text{ion}}(V_m) \right), \\ \frac{\partial}{\partial x} \left(\frac{A_e}{r_e} \frac{\partial \phi_e}{\partial x} \right) = -p \left(C_m \frac{\partial V_m}{\partial t} + I_{\text{ion}}(V_m) \right), \\ V_m = \phi_i - \phi_e. \end{cases} \quad (14.1)$$

The above system can be further simplified by exploiting some symmetries. Summing up the first 2 equations we have:

$$\frac{\partial}{\partial x} \left(\frac{A_i}{r_i} \frac{\partial \phi_i}{\partial x} + \frac{A_e}{r_e} \frac{\partial \phi_e}{\partial x} \right) = 0,$$

hence:

$$\frac{A_i}{r_i} \frac{\partial \phi_i}{\partial x} + \frac{A_e}{r_e} \frac{\partial \phi_e}{\partial x} = I_{\text{tot}}(t),$$

where $I_{\text{tot}}(t)$ is the total current. Now we replace ϕ_i , since $\phi_i = \phi_e + V_m$:

$$\frac{A_i}{r_i} \frac{\partial V_m}{\partial x} + \left(\frac{A_i}{r_i} + \frac{A_e}{r_e} \right) \frac{\partial \phi_e}{\partial x} = I_{\text{tot}}(t),$$

and solve with respect to ϕ_e :

$$\frac{\partial \phi_e}{\partial x} = \frac{r_i r_e}{A_i r_e + A_e r_i} I_{\text{tot}}(t) - \frac{A_i r_e}{A_i r_e + A_e r_i} \frac{\partial V_m}{\partial x}.$$

Now we replace this expression into the second equation of Equation 14.1. We have that:

$$\frac{\partial}{\partial x} \left(\frac{A_e}{r_e} \frac{\partial \phi_e}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{A_e r_i}{A_i r_e + A_e r_i} \right) I_{\text{tot}}(t) - \frac{\partial}{\partial x} \left(\frac{A_i A_e}{A_i r_e + A_e r_i} \frac{\partial V_m}{\partial x} \right).$$

Assuming that the cross-section A is constant, as well as the resistivities, the first term cancels and we end up with the following equation *cable equation*:

$$p \left(C_m \frac{\partial V_m}{\partial t} + I_{\text{ion}}(V_m) \right) = \frac{A_i A_e}{A_i r_e + A_e r_i} \frac{\partial^2 V_m}{\partial x^2}. \quad (14.2)$$

Note that this equation has only V_m as variable, and not ϕ_i and ϕ_e . In this way, we can approximate the transmembrane potential without caring of the potential in the whole space.

14.2 Non-dimensionalization

We rescale the time and space as follows:

$$\tilde{x} = \frac{x}{\lambda_m}, \quad \tilde{t} = \frac{t}{\tau_m},$$

where λ_m and τ_m are respectively the space and time membrane constants. We also consider the membrane specific resistance at rest as:

$$R_m = \left. \frac{dI_{\text{ion}}}{dV_m} \right|_{\text{equilibrium}}$$

and define the $f = -R_m I_{\text{ion}}$ as the rescaled current. Substituting into the equation we have:

$$\frac{C_m R_m}{\tau_m} \frac{\partial V_m}{\partial \tilde{t}} - f(V_m) = \frac{A_i A_e R_m}{p(A_i r_e + A_e r_i)} \frac{1}{\lambda_m^2} \frac{\partial^2 V_m}{\partial \tilde{x}^2}.$$

Taking:

$$\begin{aligned} \tau_m &= C_m R_m, \\ \lambda_m^2 &= \frac{A_i A_e R_m}{p(A_i r_e + A_e r_i)}, \end{aligned}$$

we reduce the equation to:

$$\frac{\partial \tilde{V}_m}{\partial \tilde{t}} - f(\tilde{V}_m) = \frac{\partial^2 \tilde{V}_m}{\partial \tilde{x}^2}, \quad (14.3)$$

which is a non-linear reaction-diffusion equation in the variable V_m and the gating variables. The equation is a partial differential equation, because we have derivatives with respect to both x and t .

14.3 Travelling waves

A key characteristic of some reaction-diffusion systems are travelling waves, that is solutions of the form:

$$V_m(x, t) = U(x + ct) = U(\xi),$$

for some function $U(\xi)$ (the wave profile) and velocity c . In space-time, $V_m(x, t)$ will look the same along the characteristics $\xi = x + ct$. The variable ξ is also called travelling coordinate. Note that both U and c are still unknowns (unless we solve the cable equation), but at least we reduced the dimensionality of the problem from (x, t) to only ξ .

Given a travelling wave for the cable equation in Equation 14.3, we have:

$$\tilde{V}_m(\tilde{x}, \tilde{t}) = \tilde{U}(\tilde{x} + s\tilde{t}) = \tilde{U}\left(\frac{x}{\lambda_m} + s\frac{t}{\tau_m}\right) = \tilde{U}\left(\frac{x + s\frac{\lambda_m}{\tau_m}t}{\lambda_m}\right) = U(x + ct),$$

where $\xi = \lambda_m \tilde{\xi}$ and

$$c = s \frac{\lambda_m}{\tau_m},$$

is the velocity in terms of a parameter s (the velocity in the rescaled equations) and the time and space constants. The advantage of this formula is that s is a fixed parameter, only depending on f (thus, the ionic dynamics). Replacing the true values we have:

$$c = s \frac{1}{R_m C_m} \sqrt{\frac{A_i A_e R_m}{p(A_i r_e + A_e r_i)}}.$$

For a cylindrical axon of diameter d and $p = \pi d$, $A_i = \pi d^2/4$. We also take $A_i \rightarrow \infty$, because we assume that the bath is very large compared to the size of the axon. Thus:

$$c = \frac{s}{C_m} \sqrt{\frac{d}{4R_m r_i}}.$$

With this expression we see that the velocity of propagation increases when d increases, or when the intracellular (or cytoplasmic) resistivity decreases, or when the membrane conductivity R_m^{-1} increases. In the squid, the axon is *giant* because d is large, and this ensures a fast propagation in the body. Alternatively, we can reduce (or even block) the propagation by reducing sodium channels, because $R_m^{-1} \approx g_{\text{Na}}$ at rest, or increase the speed by increasing the density of channels per unit area of the membrane.

14.4 FitzHugh-Nagumo model with diffusion

We still need to prove that travelling waves actually exist for the cable equation. We can tackle the problem by studying the FitzHugh-Nagumo approximation of the Hodgkin-Huxley model:

$$\begin{cases} \varepsilon \frac{\partial V}{\partial t} = \varepsilon^2 \frac{\partial^2 V}{\partial x^2} + f(V) - w, \\ \frac{\partial w}{\partial t} = v - \gamma w = g(v, w), \end{cases}$$

with $\varepsilon \ll 1$. Note that we are assuming $\tau_m = \mathcal{O}(\varepsilon)$ and $\lambda_m = \mathcal{O}(\varepsilon)$, so that $c = \mathcal{O}(1)$, that is the velocity is finite in the limit. (Unless s goes to infinity with ε , but this will not be the case.)

We already studied the FitzHugh-Nagumo system in the singular limit $\varepsilon \rightarrow 0$. We can do the same here, but we need some care.

14.5 The bistable approximation

For the sake of simplicity, we ignore the repolarization, that is we set $g(v, w) = 0$ and $w = 0$. That is, once excited, the membrane remains so, and never recovers. This is not realistic, but it considerably simplifies the analysis. Therefore, we study the so-called *bistable* or *Nagumo* equation:

$$\varepsilon \frac{\partial V}{\partial t} = \varepsilon^2 \frac{\partial^2 V}{\partial x^2} + f(V). \quad (14.4)$$

14.5.1 The slow system

We substitute:

$$V(x, t) = U(\xi) = U(x + st),$$

and obtain:

$$s\varepsilon U' = \varepsilon^2 U'' + f(U),$$

where the derivatives are with respect to ξ . Taking $\varepsilon = 0$ we have:

$$0 = f(U).$$

Remember that $f(U)$ is a cubic function,

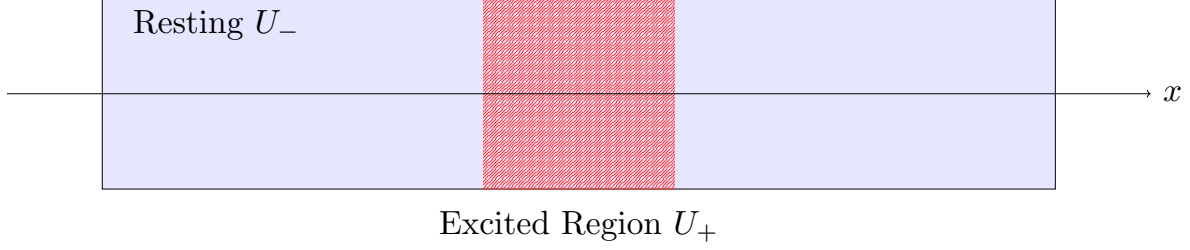
$$f(U) = U(1 - U)(U - \alpha).$$

The equation $f(U) = 0$ has 3 solutions, or branches, see Section 13.4.2. That is, we denote the solutions as

$$U_- \leq U_0 \leq U_+,$$

where $U_- = 0$, $U_0 = \alpha$, $U_+ = 1$ for this particular choice of $f(U)$.

Therefore, on the slow scale, the axon can only be in 3 states: resting $U = U_-$, excited $U = U_+$, or the threshold state $U = U_0$. However, as we are going to see, the threshold state is unstable, so the axon can only be excited or resting. Note that this yields a partition of the travelling domain ξ into regions where the axon is either excited or resting. In the physical space (x, t) , such partitions remain, and they move with constant speed s . We still don't know the velocity (even its sign), though.



14.5.2 The fast system

To recover the speed, we must “zoom in” around the interfaces between excited and resting regions. Therefore, we set:

$$V(x, t) = U\left(\frac{x + st}{\varepsilon}\right).$$

After substitution we obtain:

$$sU' = U'' + f(U), \quad (14.5)$$

for $\xi \in \mathbb{R}$. This is a second-order, non-linear ODE. In order to solve it, we need some boundary conditions. The most interesting cases are when we are at the interface between excited and resting regions, otherwise the solution is constant and the equation is trivially satisfied. We can have 2 cases:

- $U(-\infty) = U_-$ and $U(+\infty) = U_+$, like the left boundary of the figure above, or
- $U(-\infty) = U_+$ and $U(+\infty) = U_-$, like the right boundary of the figure above.

In general, we expect that the front moves from the excited region to the resting region, because we excite the neighbor tissue. We can prove this fact more precisely.

We multiply Equation 14.5 by U' and then integrate:

$$s \int_{\mathbb{R}} (U')^2 d\xi = \int_{\mathbb{R}} U' U'' d\xi + \int_{\mathbb{R}} f(U) U' d\xi.$$

Now, $U'' U' = (1/2(U')^2)'$, so we can integrate the second term, and $U' d\xi = dU$, to simplify the third integral:

$$s \int_{\mathbb{R}} (U')^2 d\xi = \frac{1}{2} \left[(U')^2 \right]_{-\infty}^{+\infty} + \int_{U(-\infty)}^{U(+\infty)} f(U) dU.$$

We need $U'(\xi) \rightarrow 0$ for $\xi \rightarrow \pm\infty$, otherwise the first integral in $(U')^2$ would be unbounded. Hence, the boundary terms cancel and we get:

$$s = \frac{\int_{U(-\infty)}^{U(+\infty)} f(U) dU}{\int_{\mathbb{R}} (U')^2 d\xi}.$$

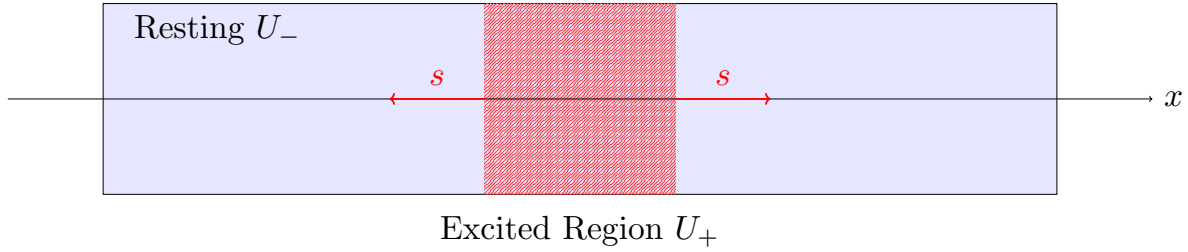
Using the boundary conditions for U at infinity, we have:

$$s = \frac{\int_{U_-}^{U_+} f(U) dU}{\int_{\mathbb{R}} (U')^2 d\xi}.$$

The denominator is always strictly positive (when zero, the solution is constant.) Hence, the sign of s is determined by the signed area of $f(U)$ between U_- and U_+ . For the cubic $f(U)$, this will depend on the position of α : when $\alpha < 1/2$, then $s > 0$, otherwise $s < 0$. We henceforth always assume $\alpha < 1/2$ (the other case is symmetric.) Note that for $\alpha = 1/2$ we have $s = 0$, so no propagation (standing wave).

In summary,

- $s > 0$ for the left boundary (U_- to U_+), so the wave moves to the left,
- $s < 0$ for the right boundary (U_+ to U_-), so the wave moves to the right.



The above formula does not give us a clue of the actual value of s (just the sign). For the cubic f , we can actually perform the full computation and find s (and U) exactly. For a general cubic-like f , it is possible to prove the existence and uniqueness of the travelling wave.

We go back to Equation 14.5, and recast it as a system:

$$\begin{cases} U' = Z, \\ Z' = sZ - f(U). \end{cases}$$

We focus on connecting U_- to U_+ (the left boundary). Therefore, we look for a trajectory of the system that connects $(0, 0)$ to $(1, 0)$, that is $U(\xi) \rightarrow U_- = 0$ for $\xi \rightarrow -\infty$, and $U(\xi) \rightarrow U_+ = 1$ for $\xi \rightarrow \infty$. Of course, $Z(\xi) = U'(\xi) \rightarrow 0$ for $\xi \rightarrow \pm\infty$.

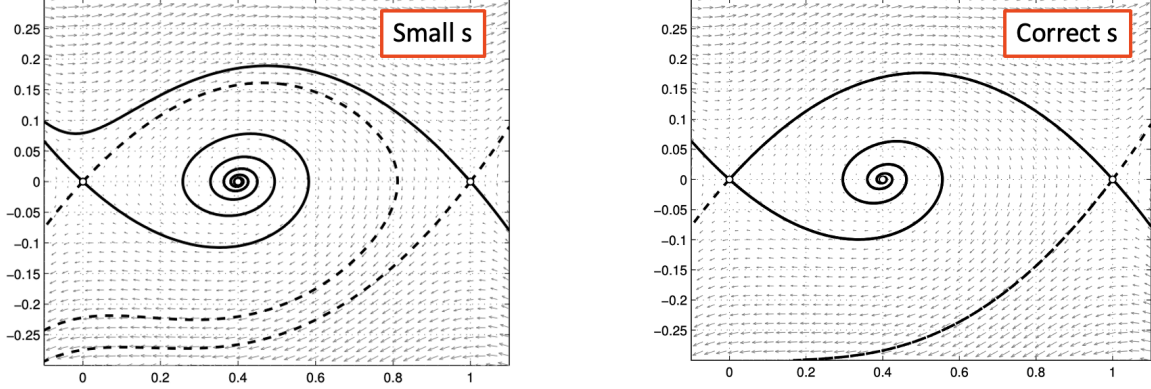
The system has 3 equilibria: $E_0 = (0, 0)$, $E_1 = (1, 0)$ and $E_\alpha = (\alpha, 0)$. For their stability, we look at the Jacobian:

$$J = \begin{pmatrix} 0 & 1 \\ -f'(U) & s \end{pmatrix}.$$

We have that $\text{tr } J = s > 0$, so no equilibrium is stable. Since $\det J = f'(U)$, we have that E_0 and E_1 are saddles, and E_α is unstable.

The objective is construct an orbit that connects E_0 to E_1 . Such orbit is called *heteroclinic*, and it only exists for a very specific value of s : for this value, we have a heteroclinic bifurcation. Since the orbit must start from E_0 , we can follow the unstable manifold of E_0 . Likewise, since we need to converge into E_1 , we can follow the stable manifold of E_1 . The bifurcation occurs when the two manifolds collide and merge.

To understand what is happening, we need the phase portrait:



We see that for small s , the unstable manifold of E_0 falls too short, it does not reach E_1 and instead diverges to $-\infty$. For large values of s , the unstable manifold always stays high, and eventually diverges to $+\infty$ without reaching E_1 . With a bisection argument, since E_0 depends continuously on s , there must exist a value $\bar{s}$ such that the unstable manifold reaches exactly E_1 . This value gives the heteroclinic orbit.

Luckily, for $s = \bar{s}$, the orbit is a parabola, so there must be:

$$Z = h(U) = bU(1 - U),$$

for some value of $b > 0$. Therefore:

$$Z' = h(U)' = h'(U)U' = b(1 - 2U)Z = b^2U(1 - U)(1 - 2U).$$

Substituting into the equation for Z' :

$$b^2U(1 - U)(1 - 2U) = sbU(1 - U) - U(1 - U)(U - \alpha).$$

We can remove some terms and collect:

$$bs - b^2 + \alpha + (1 - 2b^2)U = 0.$$

This expression must be zero for every $\xi \in \mathbb{R}$, thus the coefficients must be zero. We have:

$$b = \pm \frac{\sqrt{2}}{2}, \quad s = \pm \frac{\sqrt{2}}{2}(1 - 2\alpha).$$

We take the solution with the plus, because $s > 0$ and $b > 0$. The other solution is also possible, but connects E_1 to E_0 . So we conclude that the propagation velocity is:

$$s = \frac{\sqrt{2}}{2}(1 - 2\alpha). \quad (14.6)$$

We can also compute the front shape $U(\xi)$. Since $U' = Z = h(U)$, we need to solve:

$$U' = bU(1 - U),$$

which is the logistic equation! The general solution is:

$$U(\xi) = \frac{1}{1 + Ce^{-b\xi}}.$$

The constant is arbitrary, and we cannot determine it from the boundary conditions, because $U(\xi) \rightarrow 1$ for $\xi \rightarrow \infty$ and $U(\xi) \rightarrow 0$ for $\xi \rightarrow -\infty$, for any $C > 0$. The arbitrariness is expected, because the heteroclinic orbit is defined up to a phase shift (if we put $\hat{\xi} = \xi + \xi_0$ the solution is the same.) Therefore, we set $C = e^{-b\xi_0}$ or more simply $C = 1$.

We also note that:

$$\tanh \xi = \frac{e^{2\xi} - 1}{e^{2\xi} + 1},$$

so,

$$U(\xi) = \frac{1}{1 + e^{-b\xi}} = \frac{1}{2}(\tanh(b\xi/2) + 1) = \frac{1}{2} \left(\tanh\left(\frac{\xi}{2\sqrt{2}}\right) + 1 \right).$$

For the other case, when we connect U_+ to U_- , the solution is exactly the opposite.

In conclusion, the solution is:

$$V(x, t) = U\left(\frac{x + st}{\varepsilon}\right) = \frac{1}{2} \left(\tanh\left(\frac{x + st}{2\varepsilon\sqrt{2}}\right) + 1 \right),$$

and s as in Equation 14.6. The solution is actually *exact* for Equation 14.4, we don't even need the fast-slow analysis. In the limit $\varepsilon \rightarrow 0$, we recover the slow dynamics: the solution converges to a Heaviside function and the front moves with velocity s .

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Part II

Labs

Lab 01: Numerical integration

The purpose of this lab session is to introduce some basic numerical schemes for the following initial value problem (IVP)

$$\begin{cases} \mathbf{y}' = \mathbf{f}(t, \mathbf{y}), \\ \mathbf{y}(t_0) = \mathbf{y}_0. \end{cases}$$

The simplest method is *Forward Euler* method or *Explicit Euler*.

In the Forward Euler method, the numerical solution $\mathbf{u}$ is obtained from the following algorithm: given $h > 0$ (timestep), we set $\mathbf{u}_0 = \mathbf{y}_0$ and we iterate for $n \geq 0$:

$$u_{n+1} = u_n + hf(t_n, u_n)$$

until $nh > T$, being T the final time.

Alternatively to Forward Euler, we can have the following scheme, called *Backward Euler*:

$$u_{n+1} = u_n + hf(t_{n+1}, u_{n+1}).$$

Please note that the unknown u_{n+1} is implicitly defined. If f is nonlinear, then it might not be trivial to obtain it! In our example, the ODE is linear and the solution is easily found:

$$u_{n+1} = u_n + h\lambda u_{n+1} \quad \Rightarrow \quad u_{n+1} = (1 - h\lambda)^{-1}u_n.$$

In the function below we directly use λ and not f .

Stability and convergence

Consider the IVP (in numerics, this is also called *test problem*):

$$\begin{cases} y' = \lambda y, \\ y(t_0) = y_0. \end{cases}$$

1. Write down the forward Euler scheme and solve for the equation for various values of h , with $\lambda = -1$. Is it always true that $y_n \rightarrow 0$ for $n \rightarrow \infty$? Use $t_0 = 0$ and $y_0 = 1$.

2. Write down the backward Euler scheme and repeat the previous step.
3. Solve the equation with the forward Euler scheme and compare the solution to the exact one. How does the absolute error decrease with respect to h ? Plot the error in the logarithmic scale. Set $\lambda = 1$ and $y(0) = 1$.
4. Repeat the previous step with the Heun scheme:

$$y_{n+1} = y_n + \frac{h}{2}(f(y_n) + f(y_n + hf(y_n))).$$

Nonlinear equations

Consider the Nagumo or bistable model:

$$u' = au(1-u)(u-\alpha),$$

with $\alpha = 0.2$ and $a = 10$. Implement and solve the model in $t = 0, \dots, 2$ with forward Euler for $h = 10^{-3}$ and various values of $u(0)$.

Van Der Pol equation

Here a more complicate example of a system. We solve the Van Der Pol equation:

$$y'' - \mu(1 - y^2)y' + y = 0.$$

1. Solve the van der Pol equation with forward Euler, using $\mu = 5$, $y(0) = 2$ and $y'(0) = 0$. How small h needs to be? Find the approximate stability limit by bisection.
2. Solve the van der Pol equation with backward Euler. Design a strategy to solve the non-linear equation. Is the scheme always stable?

Conservation of energy

Consider the 2nd-order ODE $y'' + \omega^2 y = 0$ (linear oscillator).

1. After recasting the equation to a first-order system, solve it with forward Euler using $\omega = 1$, $h = 0.01$, $y(0) = 1$, $y'(0) = 0$ and $t = 0, \dots, 100$. Does the solution match the expected behavior?
2. Try again with backward Euler. *Hint: the system is linear, you can use the backslash to solve it at each iteration.*

3. Compute the discrete energy of the system $E(t) = \frac{1}{2}((y'(t))^2 + \omega^2 y(t)^2)$ and inspect the problem.
4. Implement the following strategy: first, update the velocity, then the position with the newly compute velocity. (This is called the *symplectic Euler method*.) Does the method conserve the (discrete) energy?

Lab 02: Predation

Holling type predation

Holling type functional responses can be experimentally observed with a simple game. This is inspired to the MSc thesis from Pulley (2020), where the author proposes a similar game with marshmallows.

The experiment

We need the following tools:

- A desk or a clean area where to perform the experiment;
- Two stopwatches (there's an app in every phone);
- A bag of small objects to be collected or eaten. In the original proposal, they were marshmallow, which can be eaten. If you're concerned about your diet or teeth, you can use other objects (coins, beans, pens,...)
- A blindfold.
- A group of at least 4 students.



Figure 14.1: Desk with a 0.31 m^2 delimited area with $N = 16$ candies. The density here is $16/0.31 \approx 52 \text{ candies/m}^2$. You can measure the area with an app.

The game is as follows:

1. We *randomly* place on the desk a certain number of objects. We take note of this number N ;
2. Student A is blindfolded;
3. Student B says “Go!” and starts the stopwatch. In the meanwhile, student A starts searching on the desk for an object.
4. When Student A finds one object, he will collect/eat it.
5. Student C, with another stopwatch, will measure the total time spent eating/collecting. Thus, it starts the stopwatch when student A picks up the object, and stops it when student A starts searching again.
6. Student D will put a new object at random on the desk so to keep N constant.

The experiment can last 90 seconds. At the end of the experiment, we need to mark

- the total number of preys N ,
- the total number of preys captured C ,
- the total handling time T^* .

To avoid statistical fluctuations, we repeat the experiment 3 times. Data can be collected in a table. You can use the following link to copy the table: [CandyLab spreadsheet](#).

Number of candies	Number of captured prey				Total handling time [sec]				Handling time
	Trial 1	Trial 2	Trial 3	Avg	Trial 1	Trial 2	Trial 3	Avg	
2	1	1	1	1	2	2	2	2	2
4									
6									
15									
20									

Figure 14.2: Spreadsheet for data collection

Once done, we compute the handling time $T_h = \bar{T}^*/\bar{C}$.

The analysis

Suppose that T is the total experiment time (90 seconds for us). During this time, we spend T_s in searching for a prey, and T_h in handling a prey. Thus, we have

$$T = T_s + CT_h,$$

where the total handling time is multiplied by the number of prey captured. The assumption we make is that the total number of captured preys is proportional to the searching time and the total number of prey N . So:

$$C = aNT_s,$$

for some $a > 0$. Substituting T_s we obtain:

$$C = aNT_s = aN(T - CT_h),$$

that we can solve for C/T , that is the rate of predation (number of prey captured per unit time):

$$\frac{C}{T} = \frac{aN}{1 + aT_hN}$$

So, the functional response should be exactly Holling type II.

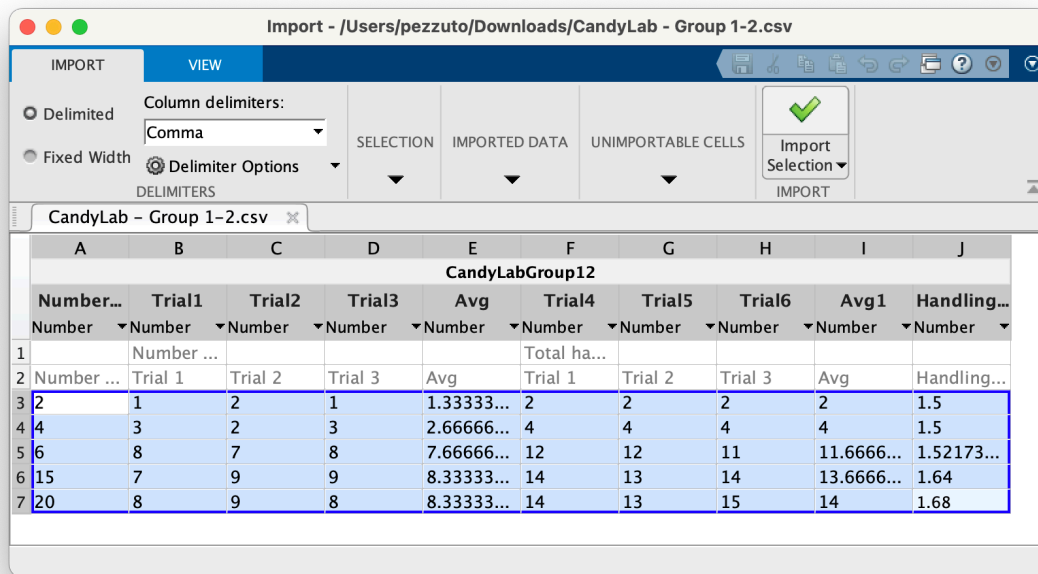
Import data

Before trying any fitting, we need to import the data into a software. For instance, this can be MATLAB or Python.

Save and download the spreadsheet as **csv** format.

14.5.1 MATLAB

Use the **Import Data** button, then select the columns of your spreadsheet. The variables will appear in your workspace.



You can access the variables with `CandyLabGroup.NumberOfCandies`, for instance. Just type `CandyLabGroup` to see all variables.

Plot the results.

14.5.2 Python

In Python there is no GUI for importing csv files, but you can use a module:

```
import csv

import csv
with open('CandyLab - Group 1.csv') as csvfile:
    reader = csv.reader(csvfile)
    lines = list(reader)

N = [int(l[0]) for l in lines[3:]]
```



```
C1 = [int(l[1]) for l in lines[3:]]
C2 = [int(l[2]) for l in lines[3:]]
C3 = [int(l[3]) for l in lines[3:]]
```

Fitting

The objective is to fit the function:

$$\frac{C}{T} = \frac{aN}{1 + aT_h N}$$

Since T is fixed, we can fit C versus N for various values, and find a and T_h .

There are at least 2 options you can try:

Option 1: Linear fit via Lineweaver–Burk plot

The [Lineweaver–Burk plot](#) uses a simple transformation to make the above equation linear. In fact:

$$y = \frac{T}{C} = \frac{1}{a} \frac{1}{N} + T_h = \alpha x + \beta.$$

So, the Holling type II relationship is *linear* with respect to the reciprocal. We can perform a linear regression to find the coefficients (use `polyfit`).

Option 2: Nonlinear regression

We can approach the problem as follows: find (a, T_h) such that

$$g(a, T_h) := \sum_{i=1}^3 \left(\frac{C}{T_i} - \frac{aN_i}{1 + aT_h N_i} \right)^2 \rightarrow \min.$$

This problem can be solved via optimization. In MATLAB, you can use `nlinfit` function, in Python `scipy.optimize.curve_fit`.

Otherwise, you can try to use a steepest descent approach: given an initial guess $(a^{(0)}, T_h^{(0)})$, we update with:

$$(a^{(k+1)}, T_h^{(k+1)}) = (a^{(k)}, T_h^{(k)}) - \eta \nabla g((a^{(k)}, T_h^{(k)})),$$

where $\eta > 0$ is the step length or learning rate.

Lab 03: Bifurcations

This exercise is based on [MatCont](#), a sophisticated software for computing bifurcation diagrams of systems of ODEs and discrete maps. It is based on Matlab.

Installation of MatCont

The installation is easy, but it requires some care on macOS and Windows. In Matlab, try the following:

```
unzip('https://gitlab.utwente.nl/m7686441/matcont/-/archive/main/matcont-main.zip')
cd matcont-main
matcont
```

If the last command fails, then follow the instruction below and try again with the command `init`, only after `mex -setup` returns success.

14.5.1 macOS

You need to install the [XCode](#), which is several GBs. After installation, open it once and follow the instruction to accept the license.

If you succeeded, then you should see in Matlab:

```
mex -setup
MEX configured to use 'Xcode with Clang' for C language compilation.
```

14.5.2 Windows

An easy option is to install the Matlab add-on [MiniGW](#) for Windows 10 or 11 64-bit. See the webpage. Try with `mex -setup` to check the compiler after installation.

Note: if you already have Microsoft Visual C++ it should work already without MiniGW.

14.5.3 GNU/Linux

It should work out of the box. However, the compiler provided by your distribution might be too recent to work well with MEX. In this case, you might get some warnings during the compilation. On widespread distributions like Ubuntu it is usually fine.

Spruce-budworm system

We start with a simple model to get familiar with **MatCont**. Open the program by typing `matcont` in the program folder. (only after running `init` at least once.) You should see

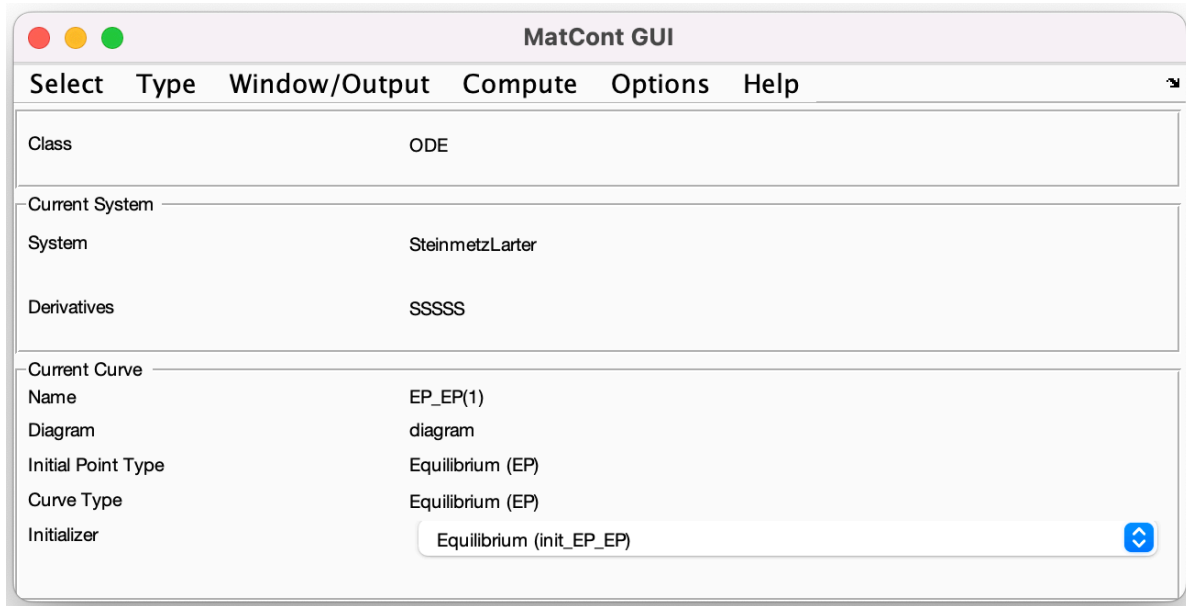


Figure 14.3: The MatCont interface.

plus some other windows that you can ignore for now.

In this exercise we want to study the bifurcation diagram of the Spruce-Budworm ODE

$$u' = f(u, \rho, q) = \rho u \left(1 - \frac{u}{q}\right) - \frac{u^2}{1 + u^2},$$

with respect to the parameters ρ and q . Remember that we rescaled the ODE assuming:

$$u(t) = \frac{N(t)}{\nu}, \quad \tau = \frac{\alpha P^*}{\nu} t, \quad \rho = \frac{r\nu}{\alpha P^*}, \quad q = \frac{K}{\nu},$$

where

- $N(t)$ is the population of budworms,
- r is the intrinsic growth rate,
- P^* is the number of predators of the budworms (e.g., birds),
- α is the attack rate,
- K is the carrying capacity,
- ν is the value of N yielding half of the maximum killing rate.

1. Create a new system **Select->System->New**. In the new window, compile as follows:

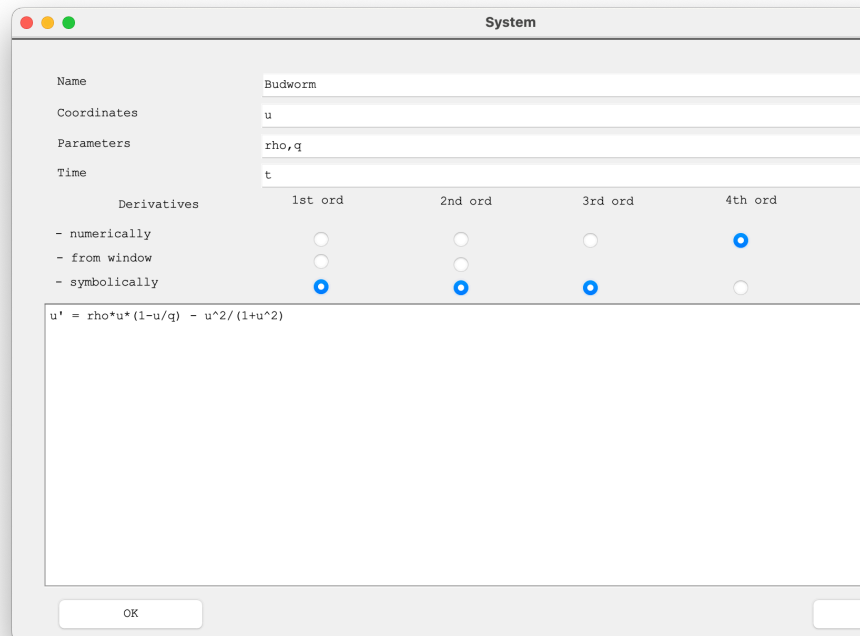
- **Name:** SpruceBudworm
- **Coordinates:** u
- **Parameters:** rho,q

Put the first 3 derivatives to **symbolically**. (If **symbolically** is disabled, install the Symbolic Toolbox.)

Write the equation in the textbox at the bottom, using a natural language:

$$u' = \text{rho} * u * (1 - u/q) - u^2 / (1 + u^2)$$

Press OK.



We should see something like this:

2. Now we compute an orbit. We start at $t = 0$ with $u(0) = 0.1$. The initial parameters are: $\rho = 1$ and $q = 8$.

We do as follows:

1. **Type**->**Initial Point**->**Point**
2. In **Starter**, we set the initial conditions and parameters
3. In **Methods**, reduce the tolerance for numerical integration
4. In the main window, **Compute**->**Forward**, then inspect **View Results**.

To visualize the solution,

1. **Window/Output**->**Graphic**->**2D Plot**,
2. Select in **Layout** time as ascissa and u as ordinate,
3. In the plot window, **MatCont**->**Redraw Curve**.

Clearly we are not yet at equilibrium. Go back and increase the integration time in **Methods**, then try again.

3. Once we have reached an equilibrium, we can try to continue it with respect to ρ .

Select as initial point the last integration point of the orbit: **Select**->**Initial Point**. You should notice that in the **Starter** window we now have different values.

To continue an equilibrium,

1. **Type**->**Initial Point**->**Equilibrium**
2. **Type**->**Curve**->**Equilibrium**
3. Select in **Starter** the parameter of interest, that is ρ for us
4. In **Continuer** window, reduce the **MaxStepsize** to 0.02 and the **MaxNumPoints** to 100. (*This is not strictly necessary, just to be safe.*)
5. In **Plot2D** (if not open, open it with **Window/Output**->**Graphic**->**2D Plot**) change the layout to parameter **rho** (range $[0, 2]$) in the ascissa and **u** (range $[0, 10]$) on the ordinate. Clear the plot.
6. In the main window, **Compute**->**Forward**.

What do you observe?

Try now with **Compute**->**Backward** and then **Compute**->**Extend**. You should get a *Limit Point* (LP in MatCont). Continue to extend (on the left) the curve until you get another LP and a *Branch Point* (BP). Stop here.

4. In order to visualize the stability, you need to change the **Plot Properties**, and set what you prefer for **EP if unstable**, e.g., a dashed line. Redraw the previous plot and check which branch of the equilibrium curve is unstable.

Alternatively, try to plot in 3-D the parameter, the coordinate and the real part of the eigenvalue (navigate the **Layout** panel).

5. Try again with different values of q , for instance $q = 1$, $q = 5$ and $q = 10$.

One way to compute the initial equilibrium point is to start with a new orbit (**Select**->**Curve** and then **New**). Then you can repeat the above process. Do not clear the plot: we can keep multiple curves.

Do you observe limit points in all cases?

6. It seems that for small q there are no limit points, whereas for large q we have 2 points. In particular, as we decrease q the limit points get closer, thus we expect a collision. Such collision is another type of bifurcation, called *cuspl bifurcation*, which has co-dimension 2. That is, it involves 2 parameters.

We can easily compute the cusp point in MatCont

1. Select an equilibrium curve with at least a limit point (**Select**->**Curve**)
2. Select the limit point (**Select**->**Initial Point**)
3. Select **Type**->**Initial Point**->**Limit Point**
4. Select **Type**->**Curve**->**Limit Point**
5. In the **Starter** window, we now have to select 2 parameters: select both **rho** and **q** (as active parameters, the box on the right.)

Now we can compute the curve with **Compute**->**Backward** or **Compute**->**Forward**. Extend it a few times if necessary. Find ρ and q corresponding to the cusp.

7. Try to visualize the plot in 3-D.

To get an idea of the function, consider $\rho \in [0, 1]$, $q \in [0, 10]$ and $u \in [0, 10]$. Find the equation that implicitly define the surface of equilibria (parametrized in ρ and q).

Use the command **isosurface** as follows:

```
[R Q U] = meshgrid(0.0:0.01:1,0.0:0.1:10,0:0.1:10);
V = ...; % <- EDIT!
p = patch(isosurface(R,Q,U,V,0.0));
isonormals(R,Q,U,V,p);
p.FaceColor = 'red';
p.EdgeColor = 'none';
daspect([1 10 10])
view(3)
camlight; lighting phong
```

8. With the 3-D plot at hands, with a fixed value of q , explain the effect of hysteresis by imaging a “person moving on the landscape”. What does the ridge of limit points represent? And the cusp?

Rosenzweig-MacArthur model

A sufficiently complex prey-predator model is the following:

$$\begin{cases} u' = u\left(\rho(1-u) - \frac{\alpha\delta v}{1+\delta u}\right), \\ v' = v\left(-1 + \frac{\alpha\delta u}{1+\delta u}\right), \end{cases}$$

where $u(t)$ and $v(t)$ are the (non-dimensionalized) number of preys and predators, respectively.

1. Implement the model in MatCont and integrate a trajectory $\alpha = 3$, $\rho = 1$ and with initial conditions $u(0) = 0.9$ and $v(0) = 0.9$. As integration time put a large number, say $T = 100$. Try different values of $\delta \in [0, 5]$, and discuss the behaviour of the system. *Hint: you can plot multiple trajectories on the same graph. Try to visualize the phase portrait.*
2. Approximate the equilibrium E^* , intersection of the two non-trivial nullcline, for $\delta = 0.6$. Then, continue the equilibrium with respect to δ . How many bifurcation point do you observe? Visualize the diagram in the coordinates (δ, v^*) and discuss it.
3. Is the equilibrium E^* always a focus? *Hint: plot the imaginary part of one eigenvalue with respect to δ .*
4. At some value of δ (which one?) there is a *Hopf* (H) bifurcation. It means that from the equilibrium we should expect the emergence of a limit cycle. Select as initial point the Hopf point. Then do:
 1. Type->Initial Point->Hopf
 2. Type->Curve->Limit Cycle
 3. In **Starter**, select **delta** and **period** as continuation parameters
 4. Compute->Backward (or forward, it may vary)

Try to visualize the limit cycles in 3-D, using the coordinates (ρ, u^*, v^*) . What's the period of the cycle at the end of the branch?

5. Define a user function (**Select->System->Manage Userfunctions**) to monitor when $\delta = 2.5$ (for instance, **res=delta-2.5**). Once defined, activate it in the **Starter** window when continuing the Hopf bifurcation into a limit cycle. Find the period of the cycle for $\delta = 2.5$.

Lab 04: Chaos

Food chain

The standard prey-predator system only considers 2 species. In reality, many species interact and compete for survival. Ecological systems are typically very much interconnected. More species means more variables and parameters, thus more complexity.

Partial food web: North American grassland and forest

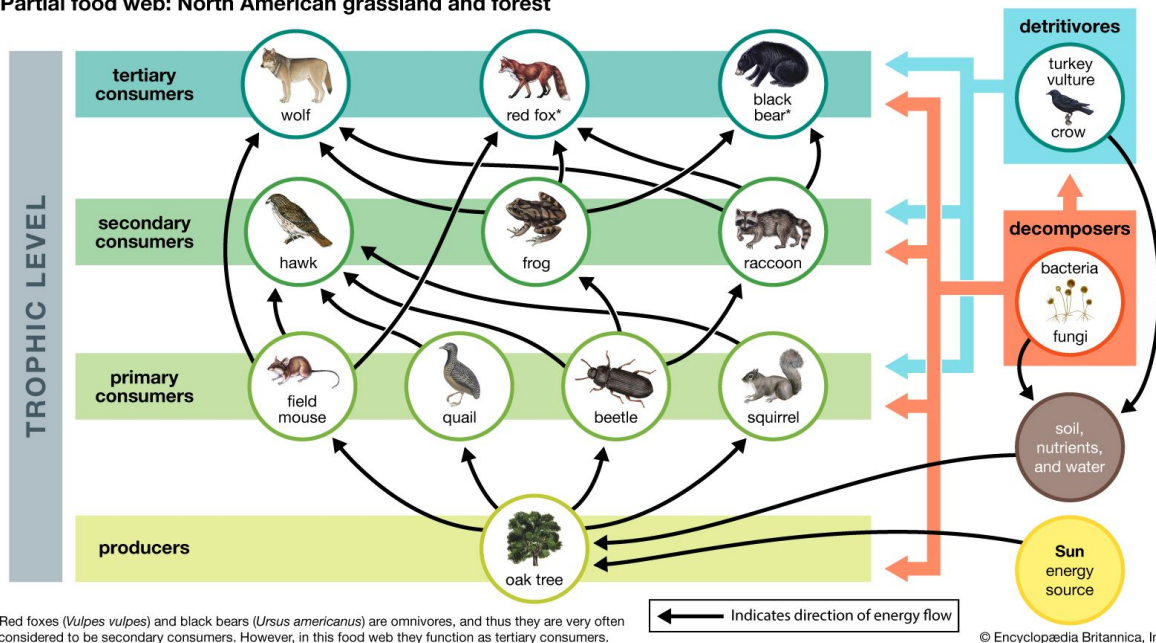


Figure 14.4: A partial food web illustrating the network of feeding relationships in a North American grassland and forest. (*Encyclopædia Britannica, Inc./Patrick O'Neill Riley*)

To better appreciate the jump in complexity, we analyze here a simple system of 3 species: plant (X), herbivore (Y), and carnivores (Z).

The model reads as follows (see Hastings and Powell (1991)):

$$\begin{cases} X' = X \left(r(1 - X/K) - \frac{a_1 Y}{1 + b_1 X} \right), \\ Y' = Y \left(-d_1 + \frac{c_1 a_1 X}{1 + b_1 X} - \frac{a_2 Z}{1 + b_2 Y} \right), \\ Z' = Z \left(-d_2 + \frac{c_2 a_2 Y}{1 + b_2 Y} \right). \end{cases}$$

We implement the model in a Matlab function `FoodChain` that can take b_1 as free parameter.

```
function dy = FoodChain(~,y,b1)
    % parameters
    r=1; K=1; c1=1; c2=1;
    a1=5; a2=0.1; b2=2; d1=0.4; d2=0.01;
    % default for b1
    if (nargin < 3), b1 = 2.0; end
    % rhs
    X = y(1);
    Y = y(2);
    Z = y(3);
    dX = r*X.*(1-X/K)-a1*X./(1+b1*X).*Y;
    dY = c1*a1*X./(1+b1*X).*Y-d1*Y-a2*Y./(1+b2*Y).*Z;
    dZ = c2*a2*Y./(1+b2*Y).*Z-d2*Z;
    dy = [dX;dY;dZ];
end
```

The parameters are set as follows:

Parameter	Value
r	1
K	1
a_1	5
a_2	0.1
b_1	$2.0 \div 3.0$
b_2	2
c_1	1
c_2	1
d_1	0.4
d_2	0.01

1. Simulate the model with `ode45` for $b_1 = 2$, initial conditions $X(0) = Y(0) = Z(0) = 1$, and for a final time $T = 1000$. Plot the solution both as a time series and in a phase portrait. Does the system converge towards a limit cycle? *Hint: try to extend the final time.*

To use `ode45`, please do as follows:

```
y0 = [1,1,1]; % initial condition
TT = 1000.0;

% we define a function with fixed b1
f = @(t,y) FoodChain(t,y,2.0);
opts = odeset('AbsTol',1e-10,'RelTol',1e-8);
[T,Y] = ode45(f,[0,TT],y0,opts);

disp(Y(end,:));

fig=figure;
plot(T,Y,'LineWidth',1.0);
title('Time series');
legend('X','Y','Z'); grid on;

fig=figure;
plot3(Y(:,1),Y(:,2),Y(:,3));
title('Phase portrait');
grid on;
xlabel('X'); ylabel('Y'); zlabel('Z');
```

2. Now simulate the system with $b_1 = 2.2$. Do you observe a limit cycle? Report the amplitude of the oscillations for each species.
3. Visualize the Poincaré map (see Section A.7.4) for the limit cycle. For constructing it, consider a plane orthogonal to the coordinate Z and placed so that passes through the midpoint of the cycle. Hopefully, the limit cycle will intersect it at some point. MATLAB ODE integrator offers a way to estimate “events”. Events are conditions computed along the trajectory that, once met, are returned by the integrator. For us, the condition is “check if the trajectory intersects the Poincaré plane”. If this happens, the integrator will record the position and time of the event.

Using the `ode45` function (but it works also for other integrators), we need to define an event function as follows:

```
function [position,isterminal,direction] = PoincareMap(~,y,c,n)
    % C is center of the plane
    % N is normal to the plane
```

```

% when the orthogonal projection is zero, we are on the plane
position = dot(y(:)-c(:),n);
isterminal = 0; % do not halt integration at the event
direction = -1; % only intersection in direction n
end

```

Use this code to visualize the Poincaré map. Try to estimate the period.

```

b1 = 2.2;
f = @(t,y) FoodChain(t,y,b1);
y0 = [1,1,1];

% we integrate for some long time
opts = odeset('AbsTol',1e-10,'RelTol',1e-8);
[~,Y] = ode45(f,[0,1e5],y0,opts);
% the last point should be on the LC
yc = Y(end,:);
% we integrate more to get the LC
[T,Y] = ode45(f,[0,1e3],yc,opts);

% the Poincare' map is a plane orthogonal to nn
% and centered in pc
nn = [0;0;1];
pc = trapz(T,Y,1) / T(end);

% we can use the nullspace to get tangent vectors
V = null(nn');

% this function is called every time 'position' is
% zero in the right 'direction'
% we measure the distance from the plane
event_fun = @(t,y) PoincareMap(t,y,pc,nn);
opts_event = odeset(opts,'Events',event_fun);

% integration with events
% YP is the solution; te,ye are the events
[~,YP,te,ye,~] = ode45(f,[0,1e3],yc,opts_event);

fig = figure;
plot3(YP(:,1),YP(:,2),YP(:,3),'LineWidth',2.0);
hold on; grid on;
plot3(ye(:,1),ye(:,2),ye(:,3),'r.','MarkerSize',16);
title(sprintf('Phase portrait (b_1 = %g)',b1));

```

```

xlabel('X'); ylabel('Y'); zlabel('Z');

% plot the Poincare plane
rr = 0.25;
pp = [pc'+rr*V,pc'-rr*V];
p = patch(pp(1,:),pp(2,:),pp(3:),'r');
p.FaceAlpha = 0.4;

% in-picture visualization of the Poincaré plot
ye_proj = (ye-pc) - ((ye-pc) * nn) * nn';
axes('Position',[.6 .6 .3 .3],'XTickLabel',[],...
     'YTickLabel',[],'nextplot','add');
plot(ye_proj(:,1),ye_proj(:,2),'r.',...
     ye_proj(end,1),ye_proj(end,2),'b+','MarkerSize',16);
xlim([-rr,rr]); ylim([-rr,rr]);
grid on; box on;
title('Poincare' map');

```

4. (Optional) A limit cycle is stable if, when perturbed, attracts the orbits. Test this in the Poincaré map, by perturbing the equilibrium within the plane, with a perturbation of $\varepsilon = 10^{-2}$:

```

% ye is a point of the LC on the plane
y0_new = ye(end,:) + 1e-1*(V * (2*rand(2,1)-1));

```

Adapt the previous code to see the effect of the perturbation.

5. Slightly increase the value of b_1 . Try with $b_1 = 2.3$, $b_1 = 2.39$, $b_1 = 2.392$. What do you observe?
6. (Optional) One key feature of chaotic systems is the sensitivity to initial conditions. One measure of sensitivity is the *Lyapunov characteristic exponent*. Given two initial conditions $\mathbf{y}_0$ and $\tilde{\mathbf{y}}_0 = \mathbf{y}_0 + \varepsilon \mathbf{z}$, we visualize the distance between the trajectories over time:

$$d(t) = \|\tilde{\mathbf{y}}(t) - \mathbf{y}(t)\| \approx e^{Lt}d(0).$$

We call L the Lyapunov exponent. Try to visualize $d(t)$ in the case of an equilibrium ($b_1 = 2$), a limit cycle ($b_1 = 2.2$), or a strange attractor ($b_1 = 3$). Use the `semilogy` command for visualizing the distance.

The Lyapunov exponent gives a measure *stretching* of the initial data. In this system there are 3 exponents: the dominant one we have estimated above (not exactly, but we got an idea about it). The others show up only with a careful selection of $\mathbf{y}_0$. In general we have:

Attractor	L_1	L_2	L_3
Equilibrium	—	—	—
Cycle	0	—	—
Torus	0	0	—
Strange attractor	+	0	—

Discuss the results. (There is no torus in this system.)

7. A positive dominant Lyapunov exponent means that the strange attractor is “stretching apart” orbits. Take several points as initial conditions by perturbing a single point from the strange attractor, similarly as done before. Plot the cloud of points in the phase portrait for various time instants, to highlight the mixing phenomenon.

The code samples 1000 points, then shows their position along with the attractor over time.

```

b1 = 3.0;
f = @(t,y) FoodChain(t,y,b1);
y0 = [1,1,1];

% we integrate for some long time
opts = odeset('AbsTol',1e-10,'RelTol',1e-8);
[~,Y] = ode45(f,[0,1e5],y0,opts);
yc = Y(end,:);
% we integrate more to get the attractor
[T,Y] = ode45(f,[0,1e4],yc,opts);

% plot the attractor
f1 = figure;
ll = plot3(Y(:,1),Y(:,2),Y(:,3));
ll.Color = [0 0.4470 0.7410 0.2];
grid on; hold on;
title(sprintf('Phase portrait (b_1 = %g)',b1))

% we take a cloud of points around the initial condition
Np = 1000; % number of points
ep = 1e-3; % perturbation
y00 = yc + ep*randn(Np,3);
[T,Y] = ode45(f,0:1:4e3,y00',opts); % vectorized version

Ysol = shiftdim(reshape(Y,[],3,Np),1);

```

```

pp = plot3(Ysol(1,:,1),Ysol(2,:,1),Ysol(3,:,1),'.',...
           'MarkerSize',16.0,'Color',[1,0,0,0.2]);
pause;

for tt=1:length(T)
    pp.XData = Ysol(1,:,tt);
    pp.YData = Ysol(2,:,tt);
    pp.ZData = Ysol(3,:,tt);
    pause(0.01);
end
hold off;

```

Transition to chaos

A prototypical example of a chaotic system is the discrete logistic map:

$$y^{(k+1)} = f(y^{(k)}) = \alpha y^{(k)}(1 - y^{(k)}),$$

for some $\alpha > 0$. Note that this is a discrete in time equation, there is no time derivative! Still, it is a dynamical system, and we can discuss equilibria, cycles, and chaos. Surprisingly, the dynamic can be very complex even in dimension 1, whereas for ODEs we need at least 3 dimensions for chaos. The reason is that orbits in the discrete case can intersect and jump one over the other one.

1. Set $\alpha \in [0, 1)$. Compute the trajectory of the system for $k \geq 1$ with $y^{(0)} \in [0, 1]$. Orbits should converge toward the fixed point $y^* = 0$. Note that fixed points (or equilibria) satisfy the equation

$$y^* = \alpha y^*(1 - y^*).$$

2. Try with $\alpha \in (1, 3]$. Now you should see a new equilibrium, while zero becomes unstable.
3. Now set $\alpha \in (3, 1 + \sqrt{6})$. A stable 2-cycle should appear, while the non-zero equilibrium is unstable. Note, a 2-cycle is a pair of points $\{y_1, y_2\}$ such that $f(y_1) = y_2$ and $f(y_2) = y_1$, where $f(y) = \alpha y(1 - y)$.
4. The last part is to further increase α up to 4. We use the following Theorem (Fatou): for a fixed point iteration with $f(y) = ay^2 + by + c$ with a cycle, the point $-b/2a$ is attracted by such orbit. So, we can fix the initial condition to $-b/2a$, and then iterate until we reach the cycle. Try with various values of α .

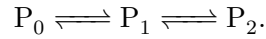
Lab 05: Feedback loops

Circadian Clock

The circadian clock is an internal physiological mechanism that shows oscillations of 24 hours. The existence of such internal cycle is apparent when we travel or right after daylight saving time switch.

An established mechanism for producing limit cycles in molecular networks is *negative feedback*. For instance, we can have an enzyme X_1 that produces X_2 with some rate, then X_2 produces X_3 , and so on until X_n , but then X_n has a negative influence of X_1 . The chain of reaction is necessary to create a delay.

We consider here a model for the *Drosophila*. We have the protein PER (for *period*), denoted by P_0 , that is phosphorylated into P_1 and then P_2 :



The protein P_2 is then transported into the nucleus, where it inhibits the production of PER mRNA (and thus the protein PER).

After a few steps and simplifications, Tyson et al. (1999) showed that the model can be reduced to

$$\begin{cases} M' = \frac{\nu_m}{1 + \left(\frac{P(1-q)}{2P_{\text{crit}}}\right)^2} - k_m M, \\ P' = \nu_p M - \frac{k_{p_1} P q + k_{p_2} P}{J_p + P} - k_{p_3} P, \\ q = \frac{2}{1 + \sqrt{1 + 8K_{\text{eq}} P}}, \end{cases}$$

where P is the total PER protein and M is the mRNA concentration. The parameters are:

Parameter	Value
ν_m	1
k_m	0.1
ν_p	0.5

Parameter	Value
k_{p_1}	10
k_{p_2}	0.03
k_{p_3}	0.1
K_{eq}	200
P_{crit}	0.1
J_p	0.05

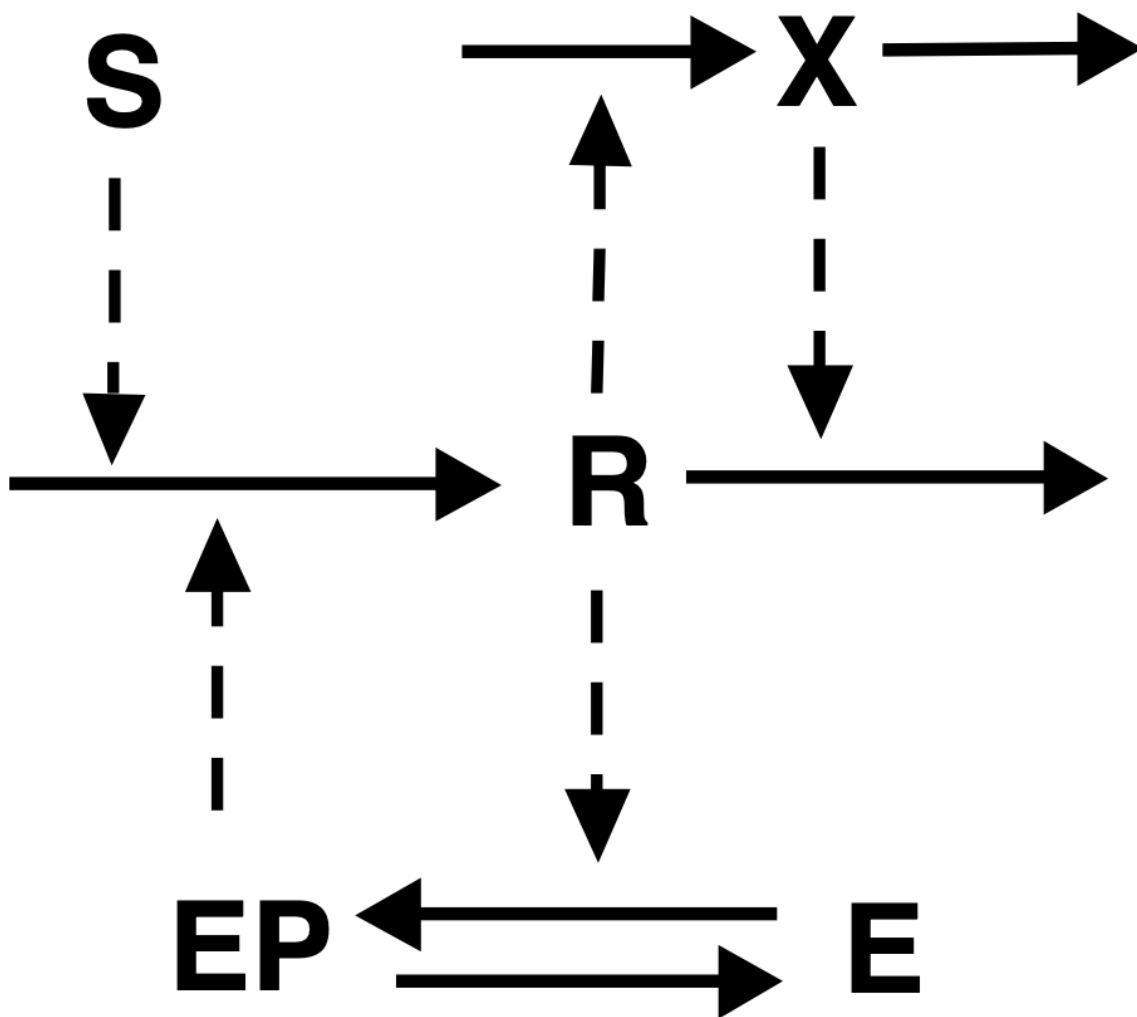
1. Implement the model in MatCont and simulate an orbit. (The time unit is *hours*.) The initial condition is $M(0) = 0.1$ and $P(0) = 0.1$. You can copy and paste this code for the equations:

```
q=(2/(1+sqrt(1+8*Keq*PP)))
MM'=vm/(1+(PP*(1-q)/(2*Pcrit))^2) - km*MM
PP'=vp*MM-(kp1*PP*q+kp2*PP)/(Jp+PP)-kp3*PP
```

2. Manually find the period. To do so, start from any point on the cycle and then integrate for enough time so to close the cycle. To get the exact value, do **Select**→**Initial Point** and select **LC** **Select Cycle**. It will ask for tolerance and number of discretization points.
3. A limit cycle should contain an (unstable) equilibrium or another unstable limit cycle (why?) Integrate the system backward in time to approximate it.
4. Continue the unstable equilibrium with respect to k_{p_1} , both backward and forward. Report all bifurcations.
5. Continue the first Hopf point with respect to both k_{p_1} and K_{eq} , and find a region in the (k_{p_1}, K_{eq}) plane that allows for a limit cycle to exist. *Hint: if it is too slow, increase Maxstepsize in the Continuer.*
6. Go back to the original Hopf point, and now continue it with respect to the period and k_{p_1} . Plot how the period varies with respect to k_{p_1} .
7. Select the original LC of 24 hours. Then, try to continue it with respect to k_{p_1} and K_{eq} . The curve should represent all possible combinations of the parameters giving a cycle of period 24 hours. Notice that you will see some limit point of cycles (LPC).

Activator-inhibitor oscillations

Consider the following network loop from Tyson, Chen, and Novak (2003):



(The activation for the enzyme E are of Michaelis-Menten type.)

The corresponding system of equations is:

$$\begin{aligned}
 [R]' &= k_0[E_p] + k_1[S] - k_2[R] - k'_2[X][R], \\
 [X]' &= k_5[R] - k_6[X], \\
 [E_p]' &= \frac{k_3[R](1 - [E_p])}{J_3 + 1 - [E_p]} - \frac{k_4[E_p]}{J_4 + [E_p]}.
 \end{aligned}$$

1. Simulate an orbit for $[R](0) = [X](0) = [E_p](0) = 0.1$ and parameters:

Parameter	Value
$[S]$	0.2
k_0	4
k_1, k_2, k'_2, k_3, k_4	1
k_5	0.1
k_6	0.075
J_3, J_4	0.3

2. Try to approximate the period of the limit cycle.
3. For larger $[S]$, say $[S] = 0.5$, there is no limit cycle. Approximate the equilibrium (it should be a spiral), then continue the curve backward with respect to $[S]$. Stop at $[S] \approx 0$. Do you observe bifurcations?
4. Select the Hopf bifurcation, and continue the limit cycle with respect to $[S]$. You should obtain a limit point of cycle. Find the corresponding value of it.
5. Plot in 3D the limit cycle family with respect to $[S]$. Notice that for a range of $[S]$ (which one?) the system is bistable. The leftmost Hopf is subcritical, whereas the other one is supercritical. Which one is a catastrophic bifurcation?

Lab 06: Electrophysiology

The Hodgkin-Huxley model

The Hodgkin-Huxley model for the action potential is a system of 4 ODEs. The first ODE is the balance of transmembrane currents:

$$C_m \frac{dV_m}{dt} + J_{\text{ion}}(V, m, h, n) = I_{\text{stim}}(t),$$

where $J_{\text{ion}}(V, m, h, n)$ is the current through the ion channels, which depends on the gating variables m , h , and n . In the Hodgkin-Huxley model we have:

$$J_{\text{ion}}(V, m, h, n) = g_{\text{Na}} m^3 h (V_m - V_{\text{Na}}) + g_{\text{K}} n^4 (V_m - V_{\text{K}}) + g_{\ell} (V_m - V_{\ell}),$$

where g_{Na} and g_{K} are the *maximal conductivities* for sodium and potassium, respectively. The gating variables $m(t)$, $n(t)$, and $h(t)$ all satisfy the same ODE (with different coefficients), as follows

$$\begin{aligned}\frac{dm}{dt} &= \alpha_m(V_m)(1 - m) - \beta_m(V_m)m, \\ \frac{dh}{dt} &= \alpha_h(V_m)(1 - h) - \beta_h(V_m)h, \\ \frac{dn}{dt} &= \alpha_n(V_m)(1 - n) - \beta_n(V_m)n,\end{aligned}$$

where the coefficients are (experimentally fitted) functions of V_m .

Use the following code for the model.

Action potential simulation

1. Compute the resting membrane potential and the equilibrium values for the gating variables.
2. Try to simulate an action potential by raising the initial resting potential. Find the threshold potential, that is minimal deviation from the resting potential so to obtain an action potential.

3. Implement a square-pulse for the stimulus current, and find the combination of height (intensity) and width (duration) that yields an action potential. As initial condition, use the resting state from point 1., and set $T_{\text{end}} = 20.0$ ms.
4. Visualize the currents using the extended output, that is

```
[~,currents] = fun(sol.x,sol.y);

fig = figure();
subplot(121);
plot(sol.x,sol.y(1,:), 'LineWidth',2.0);
xlabel('Time [ms]');
ylabel('Potential [mV]');
grid on; ylim([-80,60]);
subplot(122);
plot(sol.x,currents', 'LineWidth',2.0);
xlabel('Time [ms]');
ylabel('Currents [\mu A]');
legend('I_{Na}', 'I_K', 'I_{leak}', 'I_{stim}');
grid on;
```

5. Take $t_{\text{dur}} = 2$ ms and $I_{\text{max}} = 5 \mu\text{A}$. Implement a periodic stimulation protocol with a fixed value of cycle length T_{CL} . Start from $T_{\text{CL}} = 100$ ms and reduce it until it not possible to provoke an action potential anymore.
6. Implement a constant-in-time current $I_{\text{stim}}(t) \equiv I_{\text{max}}$. Increase the current from zero and show that in a certain range the system has a limit cycle. Is the transition smooth or catastrophic?

Voltage-clamp

Hodgkin and Huxley invented the [voltage-clamp technique](#), a key tool for measuring the transmembrane currents and establishing the gating mechanism. The voltage-clamp approach is a feedback control that can inject a current in the axon so to keep the transmembrane potential at some given level, V_{clamp} .

The simplest model is as follows:

$$C_m \frac{dV_m}{dt} + \mathcal{J}_{\text{ion}}(V) = -\kappa(V - V_{\text{clamp}}),$$

for some large $\kappa > 0$. In fact, for $\kappa \gg 1$, the system has a stable equilibrium $V = V_{\text{clamp}}$. Implement the voltage clamp. Note that the function `hhmodel` allows `Istim` to be a function of time *and* voltage.

Hyperkalemia

Hyperkalemia is an increase of potassium in the body. High level of potassium are dangerous for the body because the excitability of cells is affected. By using Nernst equilibrium equation,

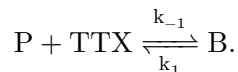
$$V_K = \frac{RT}{F} \log\left(\frac{[K]_e}{[K]_i}\right),$$

show that as $[K]_e$ increases from 20 mM self-sustained oscillations appear. Find the values of bifurcations, as above. For the constants, use $R = 8.314 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ (gas constant), $F = 9.648 \cdot 10^4 \text{ C} \cdot \text{mol}^{-1}$ (Farad constant), $T = 26 + 273.15 \text{ K}$, $[K]_i = 397 \text{ mM}$.

Tetrodotoxin

Tetrodotoxin (TTX) is a powerful neurotoxin, famous for the puffer fish. TTX acts by blocking the sodium channels, hence making them unavailable for action potential formation.

Here is how we can model it. If P_0 represents the total number of Na^+ channels, and we denote by P and B respectively the number of non-blocked and blocked channels, then the reaction is



In order words, TTX is an inhibitor. The equilibrium is

$$\frac{P}{P_0} = \frac{1}{1 + \text{TTX}/K_1}.$$

Since $K_1 \approx 2.5 \text{ nM}$, a very small quantity of TTX is sufficient to block several channels. (At K_1 , half of the channels are blocked.)

Next, in the Hodgkin-Huxley model the sodium conductance is proportional to the number of open channels. Hence, $g_{\text{Na}} = 120 \text{ mS}$ at $P = 1$, so:

$$g_{\text{Na}}(\text{TTX}) = \frac{120}{1 + \text{TTX}/K_1}.$$

Find the quantity of TTX that blocks axonal propagation.

The cable equation

i Note

This section is just for your information. We will see reaction-diffusion equations in the second part of the course.

The cable equation is a 1-D reaction-diffusion model for axonal propagation. The equation reads as follows:

$$p \left(C_m \frac{\partial V_m}{\partial t} + J_{\text{ion}}(V, m, h, n) - I_{\text{stim}}(x, t) \right) = \frac{1}{r_i + r_e} \frac{\partial^2 V_m}{\partial x^2},$$

for $V_m(x, t)$, $x \in [0, L]$, $t > 0$ is the transmembrane potential. Note that V_m is a function of the coordinate x , which represents the longitudinal direction of the cable.

The additional parameters are p , the perimeter of the axon, and r_i , r_e , respectively the intra- and extra-cellular electrical resistivity. We also assume homogeneous Neumann boundary conditions for V_m , that is:

$$\frac{\partial V_m}{\partial x}(0, t) = \frac{\partial V_m}{\partial x}(L, t) = 0.$$

We can readily use the previous code, `hhmodel.m`, to simulate the reaction-diffusion model. We will consider a discretization in space as follows: given a grid $x_i = h \cdot i$ of N nodes, with $0 \leq i \leq N - 1$ and $h = \frac{L}{N-1}$, we approximate the Laplacian of V_m at x_i with the formula:

$$\frac{\partial^2 V_m(x_i, t)}{\partial x^2} \approx \frac{V_m(x_{i-1}, t) - 2V_m(x_i, t) + V_m(x_{i+1}, t)}{h^2}.$$

Therefore, we end up with a system of ODEs of $4 \cdot N$ equations in the unknowns $V_m(x_i, t)$, $m(x_i, t)$, $h(x_i, t)$, $n(x_i, t)$.

The system can still be solved with `ode15s`, with some care for the dimensions of the variables and by providing some information on the sparsity of the Jacobian matrix.

Listing 14.1 hhmodel.m

```
function [dy,varargout] = hhmodel(t,y,varargin)
% Hodgkin-Huxley model for nerves

% Create input parser object
p = inputParser;

% Add required inputs
addRequired(p, 't');
addRequired(p, 'y');

% Define optional parameters with their default values
addParameter(p, 'Istim', @(t) 0*t); % no stimulation current
addParameter(p, 'g_Na', 120); % sodium conductance [mS]
addParameter(p, 'g_K', 36); % potassium conductance [mS]
addParameter(p, 'g_l', 0.3); % leakage conductance [mS]
addParameter(p, 'V_K', -77); % Nernst potential K [mV]
addParameter(p, 'V_Na', 55); % Nernst potential Na [mV]
addParameter(p, 'V_l', -54.4); % Nernst potential leakage [mV]
addParameter(p, 'Cm', 1.0); % membrane capacitance [uF]

% Parse input arguments
parse(p, t, y, varargin{:});

% Assign parameters to variables
Istim = p.Results.Istim;
g_Na = p.Results.g_Na;
g_K = p.Results.g_K;
g_l = p.Results.g_l;
V_K = p.Results.V_K;
V_Na = p.Results.V_Na;
V_l = p.Results.V_l;
Cm = p.Results.Cm;

% State variables
%
sy = size(y);
disp(t);
y = reshape(y,4,[]);
V = y(1,:);
m = y(2,:);
h = y(3,:);
n = y(4,:);

% gate rates [ms-1]
%
alpha_m = -0.1*(40+V)./(exp(-(40+V)/10)-1);
beta_m = 4*exp(-(V+65)/18);
alpha_h = 0.07*exp(-(V+65)/20);
beta_h = 1./(exp(-(35+V)/10)+1);
alpha_n = -0.01*(55+V)./(exp(-(55+V)/10)-1);
beta_n = 0.125*exp(-(V+65)/80);
```

Listing 14.2 cable_hhmodel.m

```
%% 1-D ode15s

Cm = 1.0;          % membrane capacitance [uF/cm^2]
L = 10;           % length [cm]
Te = 15;          % simulation time [ms]
a = 0.0238;       % axon radius [cm]
p = 2*pi*a;       % perimeter [cm]
Rc = 35.4;        % cytoplasm resistance [Ohm cm]
Ai = pi*a^2;      % cross section area [cm^2]
ri = Rc/Ai;       % intracellular resistivity [Ohm/cm]
re = 0.0;         % extracellular resistivity

D = 1/(p*(ri+re)) * 1e3; % diffusion [mS]

dx = 0.02; % spatial resolution [cm]
x = 0:dx:L;
Nx = length(x);

% diffusion operator in 1-D
e = ones(Nx,1);
A = spdiags([e -2*e e],-1:1,Nx,Nx);
A(1,2) = 2; A(end,end-1) = 2; % non-symmetric version
A = D/dx^2*A;
J = spdiags([e e e],-1:1,Nx,Nx);
I = speye(Nx,Nx);
O = sparse(Nx,Nx);

% initial conditions
m = 0.0532;
h = 0.5945;
n = 0.3184;
Vr = -64.9538;

% solution vector
Y0 = repmat([Vr,m,h,n]',1,Nx);
I_diff = @(~,V) reshape(A*reshape(V,Nx,[]),1,[]);
I_stim = @(t,y) 80*(t<2)*(t>1)*(repmat(x,1,size(y,2)/Nx)<L/20);

fun = @(t,y) hhmodel(t,y,'Istim',@(t,y) I_stim(t,y) + I_diff(t,y));

Jpattern = sparse(4*Nx,4*Nx);
ii = reshape(1:4*Nx,4,[]);
Jpattern(ii(:),ii(:)) = [J, I, I, I; ...
                        I, I, 0, 0; ...
                        I, 0, I, 0; ...
                        I, 0, 0, I];

opts = odeset('Vectorized','on','JPattern',Jpattern);
sol = ode15s(fun, [0,Te], Y0(:), opts);

Y = reshape(sol.y, 4, Nx, []);
```


Lab 07: Parameter fitting

Holling type II

We continue here the analysis of the experimental data from the “CandyLab”. Data is here: [CandyLab spreadsheet](#). You can download the file as `.xlsx`.

The objective is to fit the following function:

$$\frac{C}{T} = \frac{aN}{1 + aT_h N}$$

where $T = 90$ sec, C is the number of “captured” candies, and N is the number of candies on the table (the density of prey). The unknown parameters are:

- $a > 0$, the predation rate,
- $T_h > 0$, the handling time.

Import data

Before trying any fitting, we need to import the data into a software. Save and download the spreadsheet as `.xlsx` format.

In MATLAB, you can use the **Import Data** button. Otherwise, this should work:

```
sheet_num = 3; % <- select yours
data = readtable('CandyLab.xlsx', 'Sheet', sheet_num);
```

You can access the variables with `data.Candies`, for instance. Just type `data` to see all variables.

Plot the results C/T versus N . Be careful in filtering NaNs (use `isnan`) and recompute the average and standard deviation.

Fitting via Lineweaver–Burk plot

The [Lineweaver–Burk plot](#) uses a simple transformation to make the above equation linear. In fact:

$$y = \frac{T}{C} = \frac{1}{a} \frac{1}{N} + T_h = \alpha x + \beta.$$

So, the Holling type II relationship is *linear* with respect to the reciprocal. Perform a linear regression to find the coefficients (use `polyfit`, `fit`, or **Tools->Basic Fitting** in the menu bar of the plot window). Compare the estimated value of T_h with the one measured by you. Explain possible differences.

Nonlinear least squares

We can approach the problem as follows: find (a, T_h) such that

$$g(a, T_h) := \sum_{i=1}^3 \left(\frac{C_i}{T} - \frac{aN_i}{1 + aT_h N_i} \right)^2 \rightarrow \min.$$

This problem can be solved via optimization. In MATLAB, you can use `lsqnonlin` function, in Python `scipy.optimize.curve_fit`. Check the documentation.

Loss function

Try to visualize the function $g(a, T_h)$. You can use the function `arrayfun` to vectorize the residual (or just do a for loop). Use the log scale to plot the contour of g .

Gradient descent

The function `lsqnonlin` is implementing an optimization procedure. Here, we would like to implement our own, based on gradient descent.

Given an initial guess $(a^{(0)}, T_h^{(0)})$, we update with:

$$(a^{(k+1)}, T_h^{(k+1)}) = (a^{(k)}, T_h^{(k)}) - \eta \nabla g((a^{(k)}, T_h^{(k)})),$$

where $\eta > 0$ is the step length or learning rate and ∇g is the gradient with respect to a and T_h .

Implement the method and check the converge for various values of η . What could be a reasonable stopping criterium?

Uncertainty quantification

How sure we are of the fitting? Here, we fitted the data to the mean, and the mean will vary if we'd perform more experiments. In turn, also the fitting will change. What is the probability distribution of the parameters, given the probability distribution of the noise?

A possibility is to use a Bayesian viewpoint. We assume some prior knowledge on the parameters, we check the predictions through the likelihood, and then compute the posterior distribution according to the Bayes' Theorem. With the simplest model of noise, we have:

$$C_i/T - \frac{aN_i}{1 + aT_h N_i} = \varepsilon \sim \mathcal{N}(0, \sigma^2),$$

assuming that measurements are i.i.d. The likelihood is:

$$\mathcal{L}_D(a, T_h) = \prod_{i=1}^K \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2} \left(C_i/T - \frac{aN_i}{1 + aT_h N_i} \right)^2}$$

where K is the number of measurements and $D = \{(C_i, N_i)\}_{i=1}^K$ is the data. Bayes' formula gives us the posterior:

$$\rho_{\text{post}}(a, T_h | D) = \frac{\mathcal{L}_D(a, T_h) \rho_{\text{prior}}(a, T_h)}{\rho(D)},$$

where $\rho(\cdot)$ is the probability density function. Now we can maximize the posterior distribution to find the *Maximum A-posteriori Estimator (MAP)*:

$$a^{\text{MAP}}, T_h^{\text{MAP}} = \arg \max \rho_{\text{post}}(a, T_h | D).$$

There is a simple case. If ρ_{prior} is *uninformative*, like a uniform distribution, it does not depend on a and T_h so we can ignore it. We can simply maximize the likelihood, leading to the so-called *Maximum Likelihood Estimator (MLE)*. Equivalently, we can minimize $-\log \mathcal{L}_D$, that is:

$$-\log \mathcal{L}_D = \text{constant} + \frac{1}{2\sigma^2} \sum_{i=1}^K \left(C_i/T - \frac{aN_i}{1 + aT_h N_i} \right)^2 = c + \frac{1}{2\sigma^2} g(a, T_h).$$

Thus, we already know the MLE, because it is provided by the minimum of g .

We can further simplify the model with the Laplace approximation. Let us define $\theta = (a, T_h)$. Setting $\theta^{\text{MLE}} = (a^{\text{MLE}}, T_h^{\text{MLE}})$ we have:

$$g(\theta) \approx g(\theta^{\text{MLE}}) + \underbrace{\nabla g(\theta^{\text{MLE}})}_{=0} \cdot (\theta - \theta^{\text{MLE}}) + \frac{1}{2} (\theta - \theta^{\text{MLE}}) \cdot \nabla^2 g(\theta^{\text{MLE}}) (\theta - \theta^{\text{MLE}}).$$

The Hessian $\nabla^2 g(\theta^{\text{MLE}})$ is positive definite, because θ^{MLE} is a minimum. Thus, $g(\theta)$ is approximately a quadratic function in θ . Going back to the likelihood:

$$\mathcal{L}_D(\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}g(\theta)} = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}g(\theta^{\text{MLE}})} e^{-\frac{1}{2\sigma^2}(\theta-\theta^{\text{MLE}}) \cdot \nabla^2 g(\theta^{\text{MLE}})(\theta-\theta^{\text{MLE}})}$$

The posterior distribution is:

$$\rho_{\text{post}}(\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(\theta-\theta^{\text{MLE}}) \cdot \nabla^2 g(\theta^{\text{MLE}})(\theta-\theta^{\text{MLE}})}$$

If we define $H = \frac{1}{\sigma^2} \nabla^2 g(\theta^{\text{MLE}})$ (the Fisher information), then the distribution of the posterior is

$$\theta \sim \mathcal{N}(\theta^{\text{MLE}}, H^{-1}).$$

Compute the covariance matrix. Sample from the posterior the parameters and plot the curves.

Part III

Assignments

Solving ODEs

The logistic model

Consider the following model for a population:

$$N' = \beta(N)N - \mu(N)N,$$

where $\beta(N)$ is the *fertility rate* and $\mu(N)$ is the *mortality rate*. We assume here both are function of the population size N as follows:

$$\begin{aligned}\beta(N) &= \beta_0 - \tilde{\beta}N, \\ \mu(N) &= \mu_0 + \tilde{\mu}N.\end{aligned}$$

1. Show that the model can be written as

$$N' = r\left(1 - \frac{N}{K}\right)N. \tag{1}$$

2. Prove that the ODE, supplemented with an initial condition $N(0) = N_0$, has a unique local solution. *Hint: verify the hypotheses of the Cauchy-Lipschitz theorem.*
3. Consider $u(t) = 1/N(t)$. Show that $u(t)$ satisfies a linear ODE. Solve it, then use the solution to solve the original ODE in $N(t)$ with the initial condition $N(0) = N_0 > 0$.
4. Make the change of variables as follows: $\tau = rt$ and $y = N/K$. Find the corresponding equation in $y(\tau)$.

Blow-up of solutions

It may be considered reasonable that, for a sexual species, births are proportional to the number of encounters, hence, disregarding the mortality process, we have the equation

$$N'(t) = \beta N^2(t).$$

1. This equation can be solved by the method of separation of variables. Show that the solutions of this equation with $N(0) > 0$ tend to infinity in a finite time. (Blow-up of solution.)

2. Let us correct the equation, introducing deaths:

$$N'(t) = \beta N^2(t) - \mu N(t).$$

Discuss how the dynamics changes. In particular, does the equation still have the problem of solutions going to infinity in a finite time? *Hint: do not solve analytically the equation.*

3. Solve the above equation analytically and confirm the qualitative analysis of the previous point.

mRNA

[Zeisel et al.](#) considered the following system for the concentration of mRNA. The variables are $M(t)$, the concentration of mature mRNA, and $P(t)$, the concentration of precursor mRNA:

$$\begin{cases} P'(t) = b(t) - \alpha_1 P(t), \\ M'(t) = \alpha_1 P(t) - \alpha_2(t) M(t). \end{cases}$$

where $b(t)$ is the (gene-specific and time-dependent) production rate, α_1 is the conversion (splicing) rate of pre-mRNA to mRNA, $\alpha_2(t)$ the (time-dependent) degradation rate of mRNA.

1. Assume $b(t) \equiv b$ and $\alpha_2(t) \equiv \alpha_2$ are constant. Find the explicit solution given $P(0) = P_0$ and $M(0) = M_0$. *Hint: use the method of variation of constants.*
2. Assume now that $\alpha_2(t) \equiv \alpha_2$ and

$$b(t) = \begin{cases} \bar{b} & t < \bar{t}, \\ 0, & t > \bar{t}. \end{cases}$$

Show that $M(t)$ tends to 0 as $t \rightarrow \infty$. At which rate does it decay?

Thorium-Uranium dating

The thorium-uranium method for dating rocks is based on the fact that Uranium-234 decays into Thorium-230 which in turn decays into other elements. Set $t = 0$ the rock formation time and denoting $U(t)$ (resp. $T(t)$) the amount of Uranium-234 (resp. Thorium-230) in the rock at time t (measured in years), the following differential equation system is written:

$$\begin{cases} U'(t) = -aU(t), \\ T'(t) = aU(t) - bT(t), \\ U(0) = U_0, \\ T(0) = 0, \end{cases}$$

where $a \approx 5.9 \cdot 10^{-6} \text{ years}^{-1}$, $b \approx 1.9 \cdot 10^{-5} \text{ years}^{-1}$, U_0 represents the initial (generally unknown) amount of Uranium-234. Note that, based on geological principles, it is believed that there was no thorium at the time of rock formation.

1. Solve the equation for $U(t)$.
2. How can the quantities of a and b be interpreted? From the data provided can we infer the half-life of Uranium-234 and Thorium-230?
3. Calculate $T(t)$, solution of the second differential equation.
4. Compute

$$\lim_{t \rightarrow \infty} \frac{T(t)}{U(t)}.$$

5. Explain why it is possible to estimate the rock age from the knowledge of T/U at current time, but it is not possible from the knowledge of T alone. *Hint: study the function $T(t)/U(t)$.*

Population dynamics

Periodic solutions

Consider the time dependent Malthus model

$$\begin{cases} N'(t) = r(t)N(t), \\ N(0) = N_0. \end{cases} \quad (14.7)$$

with periodic Malthus parameter: $r(t+T) = r(t)$ and denote by $\bar{r}$ the average over one period:

$$\bar{r} = \frac{1}{T} \int_0^T r(s) \, ds. \quad (14.8)$$

1. After showing that the function

$$\pi(t) = \int_0^t r(s) \, ds - \bar{r}t$$

is periodic with period T , prove that

$$\bar{r} = \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t r(s) \, ds.$$

2. Show that the solution of the ODE is

$$N(t) = e^{\bar{r}t} N_\pi(t),$$

where $N_\pi(t)$ is a periodic function.

3. Modify the ODE by adding a time-dependent migration

$$N'(t) = r(t)N(t) + m(t), \quad (14.9)$$

assuming both $r(t)$ and $m(t)$ continuous and periodic with period T . Using Equation 14.8, show that, if $\bar{r} < 0$, then the function

$$N_\infty(t) = \int_{-\infty}^t e^{\int_s^t r(\sigma) \, d\sigma} m(s) \, ds$$

is well-defined, is a solution to the ODE, and is periodic with period T .

4. Show that if $\bar{r} < 0$, then the solution to Equation 14.9 with initial condition $N(0) = N_0$ is

$$N(t) = e^{\int_0^t r(\sigma) d\sigma} N_0 + \int_0^t e^{\int_s^t r(\sigma) d\sigma} m(s) ds$$

and is such that $\lim_{t \rightarrow \infty} (N(t) - N_\infty(t)) = 0$.

Logistic model

Consider the logistic equation

$$N'(t) = r \left(1 - \frac{N}{K} \right) N.$$

1. Find the equilibria of the ODE and study their stability.
2. Study the global stability of the equilibria.
3. Would it make sense to assume $r < 0$ and $K > 0$?

Global well-posedness

Consider the following ODE model:

$$N'(t) = (\beta N^2 - \mu N)(1 - \gamma N).$$

Find all non-negative equilibria and discuss their stability. From the direction field conclude that all solutions are bounded, and hence solutions are globally defined (no blow-up in finite time).

Predation

Fishery model

(From Strogatz, 2015, exercise 3.7.4)

A fishery model with harvesting has the form

$$N' = r\left(1 - \frac{N}{K}\right)N - E\frac{N}{A + N},$$

where all parameters are positive. The parameter E is the harvest effort. Note that harvest rate increases with N because it is harder to catch fish when the population size is small.

1. Give a biological interpretation of A .
2. Show that the system can be rewritten in dimensionless form as

$$\dot{u} = u(1 - u) - h\frac{u}{a + u}$$

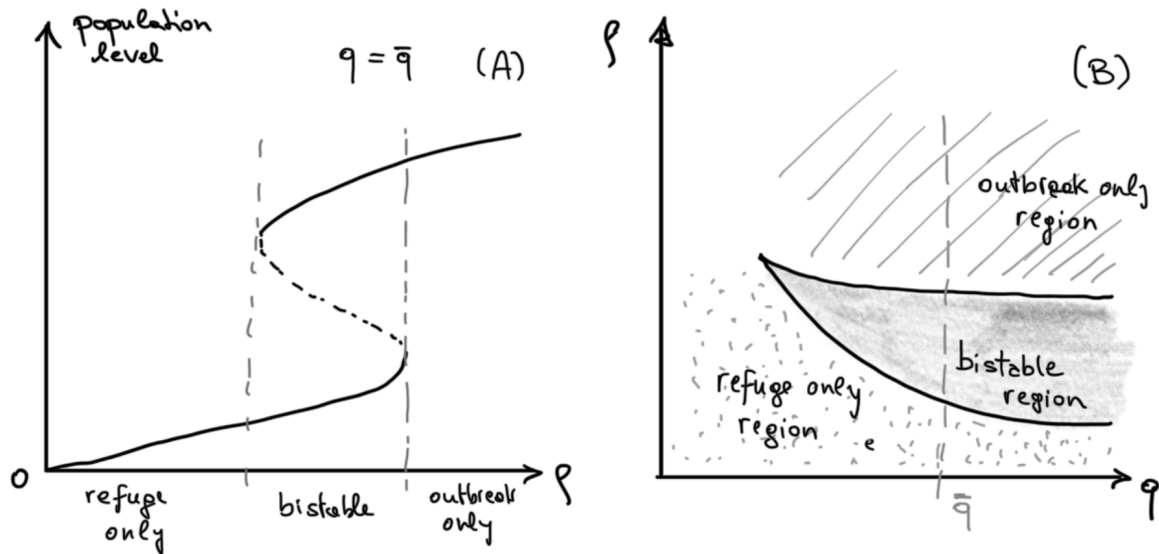
for a suitable choice of quantities.

3. Show that the system can have one, two, or three equilibria, depending on the values of a and h . Classify the stability in each case.
4. Plot the bifurcation diagram with respect to h , for a fixed value of a . Determine all bifurcation points and whether they are catastrophic or not. Is there a bistable region?
5. Suppose that we slowly increase the fishing effort from zero, assuming that the fish population is at the carrying capacity at the beginning. Show the dynamics.
6. Plot the stability diagram of the system in (a, h) parameter space. Can hysteresis occur in any of the stability regions?

Spruce-budworm model

The spruce-budworm model describes the population of budworms over time, for a given initial population. There are only 2 relevant parameters: ρ , the intrinsic growth rate of budworms; and q , the carrying capacity. An *outbreak* is when the stable population is very large, whereas a *refuge* is when it is very low.

Depending on ρ , we may have various stable population levels, according to the following diagrams:



In diagram (A) we see the effect of changing ρ with a fixed value of $q = \bar{q}$. In diagram (B) we visualize the region in the parameters space where 3 equilibria are present (bistable region).

We would like to analyze various strategies to efficiently go from an outbreak to a refuge.

1. Strategy (1): we reduce the population by reducing ρ (for instance, by preventing mating). Does it work? Use the diagrams to show how the population varies as we reduce ρ .
2. Strategy (2): we reduce the population by using an insecticide. Does it work? Use diagram (A). Observe that if we kill worms, we move the population away from the equilibrium, hence it will quickly rebound to the closest stable equilibrium.
3. Strategy (3): we reduce q , for instance by spraying a defoliant. Does it work? Use diagram (B).
4. Strategy (1+2+3): we reduce q and ρ simultaneously, and then use the insecticide. Does it work?

5. After the application of Strategy (1), ρ starts to increase again back to its original value (large). Explain how the population varies. Does it increase in the exactly same matter as it decreased?

Logistic with resources

Assume that the growth rate of a population depends on available resources $\rho(t)$, according to a general law $r(t) = G(\rho(t))$.

Assume for the moment that the total amount of resources is a fixed constant C , but they can be free (hence available, denoted by $\rho(t)$) or used by the population, denoted by $H(N(t))$, where dependence on the population is shown. In other words

$$\rho(t) = C - H(N(t)), \quad (14.10)$$

and the resulting model

$$N' = G(C - H(N))N, \quad (14.11)$$

will be specified when the functions $G(\cdot)$ and $H(\cdot)$ are given.

1. Explain why reasonable assumptions are that both G and H are increasing functions with $G(0) < 0$, $H(0) = 0$.
2. Choose a linear form for G and H and show that the growth rate has the form $r(N) = r - \alpha N$.
3. With the previous assumptions, does Equation 14.11 always have a positive equilibrium? If not, find the conditions under which it does and find the expressions for the intrinsic rate of growth r and the carrying capacity K .
4. Using generic functions G and H satisfying the assumptions in 1., find under which conditions the equation (2) has a unique positive equilibrium. When is it asymptotically stable?
5. What could be other reasonable assumptions for ρ instead of Equation 14.10?
6. Modify the equation to

$$\begin{cases} N' = G(R)N, \\ R' = \rho_0 - H(R, N), \end{cases}$$

where

$$G(R) = m \frac{R-a}{R+1} - \mu_0, \quad H(R, N) = (c + bN) \frac{R-a}{R+1} N - R.$$

Here $N(t)$ represents population density, and $R(t)$ available resources, and all constants are supposed to be positive.

With the help of [Phase Plane](#) in MATLAB, explore the solutions of the problem using parameter values such that:

$$\rho_0 > \frac{\mu_0 + ma}{m - \mu_0}.$$

System of ODEs

Linear dynamics

Consider the linear system $\mathbf{v}' = \mathbf{A}\mathbf{v}$ where

$$\mathbf{A} = \begin{pmatrix} -3 & 2 \\ -2 & 2 \end{pmatrix}.$$

1. Find the eigenvalues and eigenvectors of $\mathbf{A}$.
2. Compute the matrix exponential of $t\mathbf{A}$ and the transition matrix.
3. How can we approximate the behaviour of solutions for different initial conditions?
4. Find the equilibrium of the ODE and compute the stable and unstable manifold. Draw them on the phase portrait. Note, by definition the stable and unstable manifolds are

$$\mathcal{W}^s(\mathbf{y}^*) = \{\mathbf{y}_0 \in \mathbb{R}^2 : \lim_{t \rightarrow +\infty} \phi(t, \mathbf{y}_0) = \mathbf{y}^*\},$$
$$\mathcal{W}^u(\mathbf{y}^*) = \{\mathbf{y}_0 \in \mathbb{R}^2 : \lim_{t \rightarrow -\infty} \phi(t, \mathbf{y}_0) = \mathbf{y}^*\}.$$

5. Qualitatively describe or draw the orbits.

Phase portraits

Take a look at the following set of phase portraits:

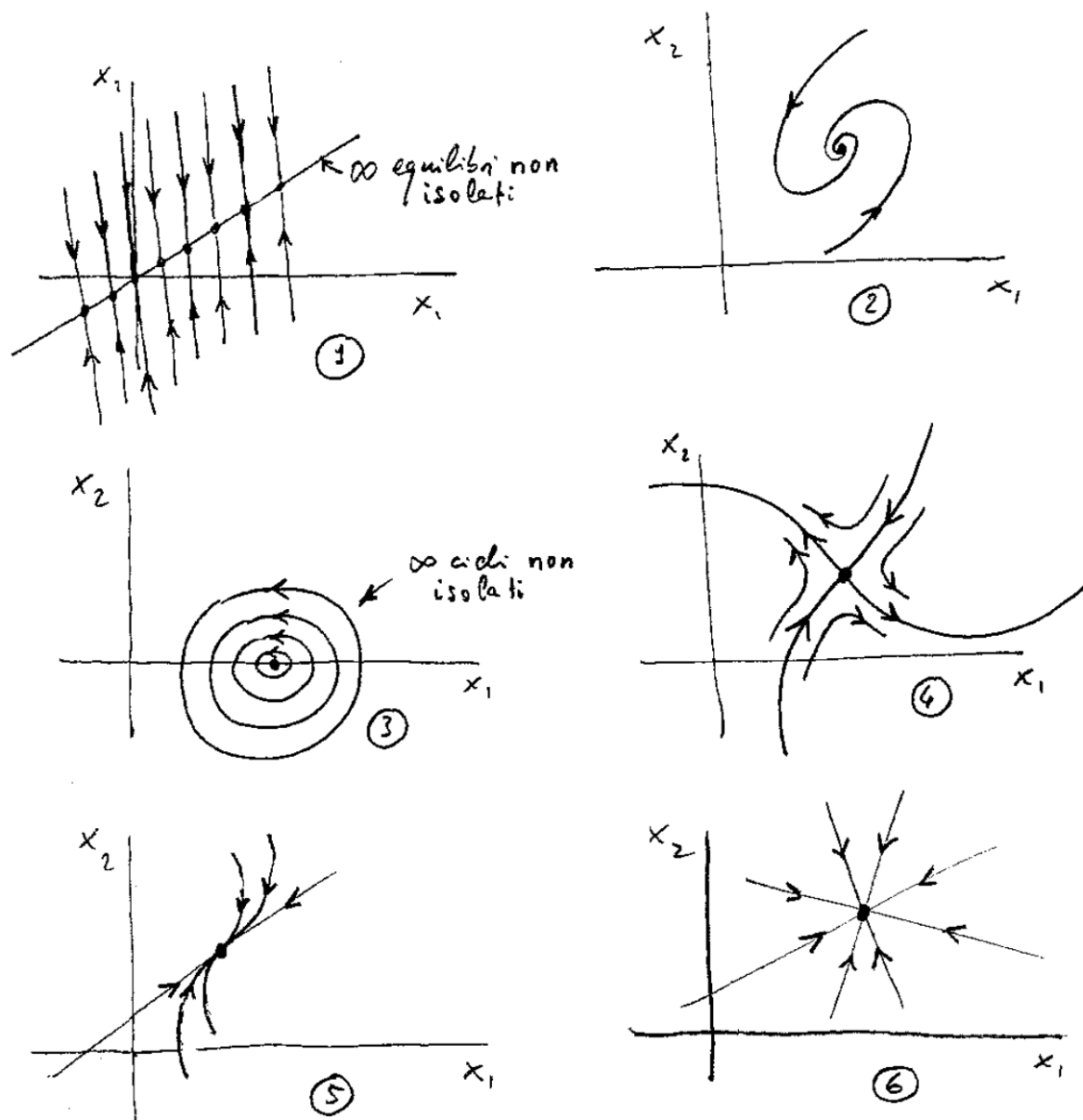


Figure 14.5: Phase portraits

1. Find the phase portrait that *cannot* be from a linear ODE.
2. For all the others, qualitatively describe the set of eigenvectors, eigenvalues, equilibria, and stability.

Double-well potential

Consider the following ODE

$$y'' = F(y) = y - y^3.$$

1. Recast the problem as a system of ODEs.
2. Find the Hamiltonian of the system. *Hint: use the potential energy.*
3. Graphically represent the orbits in the phase plane for various choices of the initial energy. Are all orbits closed?
4. Show that the system has 3 equilibria, two stable (not asymptotically) and one unstable.
5. The system has 2 *homoclinic orbits*, one on the left and one on the right of the origin. Homoclinic orbits look like periodic orbits, but with an infinite period. Try to describe what happens if you start close to the origin, and then you take the initial condition that tends to the origin. What does it happen to the solution?

Gone with the Wind

Based on exercise 7.2.19 from Strogatz (2015).

Rinaldi *et al.* (2013) used mathematical modeling to describe the stormy love affair between Scarlett O'Hara and Rhett Butler. The system reads:

$$\begin{cases} R' = -R + A_S + kSe^{-S}, \\ S' = -S + A_R + kRe^{-R}, \end{cases}$$

where $R(t)$ is Rhett's love for Scarlett, and $S(t)$ is Scarlett's love for Rhett. All parameters are positive.

Note: "Frankly, my dear, I don't give a damn" is not a valid answer.

1. Interpret the three terms on the right hand side of each equation. What do they mean, romantically speaking? In particular, what does the functional form of the exponentials terms signify about how Rhett and Scarlett react to each other's endearments?
2. Show that all trajectories that begin in the first quadrant $R, S > 0$ stay in the first quadrant forever, and interpret that result psychologically.
3. Using Dulac's criterion, prove that the model has no periodic solutions. *Hint: the simplest $h(R, S)$ you can think of will work.*

4. Using Phase Plane in Matlab, plot the phase portrait for the system, assuming parameter values $A_S = 1.2$, $A_R = 1$, and $k = 15$. Assuming that Rhett and Scarlett are indifferent when they meet, so that $R(0) = S(0) = 0$, plot the predicted trajectory for what happens in the first stage of their relationship. *Hint: don't use the default range for the phase plane, increase it. If you spot a heart , it's not on purpose.*

Prey-predator systems

Existence of cycles

Consider the following competition model:

$$\begin{cases} N_1' = r_1 N_1 (1 - N_1/K_1) - b_1 N_1 N_2, \\ N_2' = r_2 N_2 (1 - N_2/K_2) - b_2 N_1 N_2, \end{cases}$$

where $N_1(t)$ and $N_2(t)$ are two species. (The modeling details are unimportant now.) Show that the system has no periodic orbits. All parameters are positive. *Hint: use the function $h(N_1, N_2) = 1/(N_1 N_2)$ in the Dulac's criterium.*

Leslie model

Leslie (1948) proposed the following system as a model for the predator-prey dynamics:

$$\begin{cases} H' = H(a - bH - cP), \\ P' = P(r - sP/H). \end{cases}$$

In other words, the predators P have a logistic type dynamics with carrying capacity proportional to the density of preys H .

1. State clearly, and in case criticize, the assumptions of the model.
2. Draw the isoclines in the phase plane, and find the positive equilibrium.
3. Study its stability.

Optimal hunting strategy

Consider the general Gause model with logistic growth rate $r(H)$ and numerical response $\omega(H)$ proportional to the functional response $\pi(H)$, that is $\omega(H) = \gamma\pi(H)$.

1. Write down the model.
2. Consider $\pi(H)$ linear, with attack rate $a \geq 0$. Compute the stable equilibrium population size of the predators.
3. Now suppose that the predator species evolves over several generations, by adapting the value of a , so to maximize its equilibrium size. Find the optimal value of a .
4. Repeat the previous point by assuming a Holling type 2 response.

Juvenile predators

One feature not very realistic of standard predator-prey models is that predators start catching prey as soon as they are born. One way to circumvent this problem is to introduce a class of juvenile predators, say $J(t)$, that do not catch prey; juveniles become adults at constant rate, say ρ .

Taking all other assumption from the first model that has been presented, and assuming that death rate is the same for juvenile and adult predators, we arrive at the system

$$\begin{cases} H' = r(1 - \frac{H}{K})H - aHP, \\ J' = \gamma aHP - \rho J - \mu J, \\ P' = \rho J - \mu P. \end{cases}$$

1. Read from the model, which is the probability that a newly born juvenile will still be a juvenile at time t . And which is the probability that a newly born juvenile will still be a juvenile at time t , conditional on being still alive at time t ? From these computations, interpret the parameter ρ .
2. On the basis of the previous point discuss the realism of the assumption that juveniles become adults at constant rate (assumption necessary to frame the model in terms of ordinary differential equations). Suggest possible changes to the assumption.
3. Show that using appropriate changes of variables system can be written as

$$\begin{cases} x' = \sigma x(1 - x) - \alpha xz, \\ y' = \alpha xz - (\nu + 1)y, \\ z' = \nu y - z. \end{cases}$$

4. Show that the new system has always the equilibria $E_0 = (0, 0, 0)$ and $E_1 = (1, 0, 0)$. Find the conditions under which it has also a positive equilibrium $E^+ = (x^+, y^+, z^+)$ and compute its coordinates.
5. Find the conditions for the stability of E_1 and of E^+ . Note that there are parameter values such that all equilibria are unstable; if possible, use a computer to explore the behaviour of solutions in such a case. *Hint: use the Routh-Hurwitz criteria Theorem A.7*.

Green pools

It is often observed that, in the pools where fish have been added, water has a greenish colour, indicating a high algal biomass. Try to explain this using a prey-predator system, assuming that algae (prey)-zooplankton (predators) follow a Holling-type 2 predator-prey system, where prey grow according to the logistic equation and predators have mortality rate given by $d + bF$ where F is fish density (assumed to be constant).

1. Write down the model.
2. Find how equilibrium algal biomass and equilibrium stability vary with fish density F , and comment the results.
3. Assume now that the zooplankton mortality rate is $d + \frac{bF}{1+cZ}$ where Z is zooplankton density (Holling-type 2 functional response). Study the equilibria of the system.

Holling type 3 response

Consider the following prey-predator system:

$$\begin{cases} H' = H \left(r(1 - H/K) - \frac{aHP}{H^2 + \beta^2} \right), \\ P' = P \left(-c + \frac{\gamma a H^2}{H^2 + \beta^2} \right). \end{cases}$$

where all parameters are positive.

1. Give a biological interpretation to these equations. Consider especially the functional response used in this model.
2. Find all equilibria (in the first quadrant) and study their stability for $a\gamma < c$.

3. Assume $a\gamma > c$. Find the conditions on K that make unstable the equilibrium without predators (but with a positive number of preys).
4. Show that a positive equilibrium (H^*, P^*) , if it exists, is stable if

$$\frac{r}{K} + aP^*\phi'(H^*) > 0, \quad \phi(H) = \frac{H}{H^2 + \beta^2}.$$

Hint: to show this, it is convenient to rewrite the system using the function ϕ , and compute the positive equilibrium and the Jacobian through $\phi(H^)$ and $\phi'(H^*)$, without explicitly computing these.*

5. Show that the previous conditions is equivalent to $\Phi'(H^*) < 0$, where $\Phi(H) = \frac{r(1-H/K)(H^2+\beta^2)}{aH}$ is the function for the nullcline $H' = 0$, when $P = \Phi(H)$.
6. Scale the system by setting $x = H/\beta$, $y = P/(\gamma\beta)$, $\tau = ct$. The resulting system should have only 3 parameters, that can be chosen as $\eta = \beta/K$, $\alpha = a\gamma/c$, $\rho = r/c$.

Reactions

Oscillating reaction

Lengyel, Rabai, and Epstein (1990) have proposed and analyzed a particularly elegant model of an oscillating chemical reaction, the chlorine dioxide-iodine-malonic acid reaction. (You can check out on [some YouTube videos](#) of oscillating chemical reactions.)

After suitable non-dimensionalization, the model becomes

$$\begin{cases} x' = a - x - \frac{4xy}{1+x^2}, \\ y' = bx - \frac{bxy}{1+x^2}, \end{cases}$$

where $x = [\text{I}^-]$ and $y = [\text{ClO}_2^-]$, and a, b are positive parameters.

1. Find the equilibrium of the system, and prove that it is never a saddle point.
2. In the (a, b) plane, find the region $\mathcal{S}$ such that the equilibrium is asymptotically stable.
3. Show that there exists a limit cycle for any choice of the parameters (a, b) in the complement of $\mathcal{S}$. *Hint: apply the Poincaré-Bendixson theorem, by constructing an equilibrium-free region in the phase space that traps all the orbits.*
4. What does it happen when we are on the boundary of $\mathcal{S}$? *Hint: the eigenvalues of the linearized system are purely imaginary, thus the equilibrium is actually an infinitesimally small limit cycle.*
5. Find the period of the (infinitesimally small) limit cycle on the boundary as a function of a . Show that as $a \rightarrow \infty$, $T \rightarrow \frac{2\pi}{\sqrt{15}}$. *Hint: since eigenvalues are purely imaginary, locally the solution is of the form $e^{i\omega t}$ for some ω . Use this fact to obtain the period.*

Fast-slow systems

Consider the following system of ODEs:

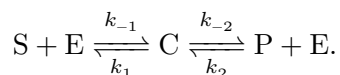
$$\begin{cases} y' = z, \\ z' = \frac{1}{\varepsilon}(z - z^3 - y). \end{cases}$$

with $\varepsilon \ll 1$.

1. Explain why the system has fast-slow dynamics, and identify the fast and the slow variable.
2. With the help of Tikhonov's theorem, qualitatively study the dynamics of the system.

Reversible reactions

In the derivation of Michaelis-Menten equation, it is assumed that the back-reaction is negligibly slow. Consider now the full system of reactions:

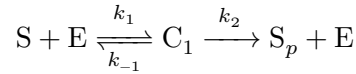


1. Write down the system of equations corresponding to this scheme. Assume that at the start, there is neither complex C nor product P .
2. Note that two quantities are conserved: $[E] + [C]$ and $[S] + [C] + [P]$. Use this to decrease the number of equations.
3. Make the equations non-dimensional, and in so doing introduce the parameter $\varepsilon = \frac{e_0}{s_0}$, the ratio of initial values of enzyme and substrate.
4. Go to the limit $\varepsilon \rightarrow 0^+$ which amounts to make the quasi-steady-state approximation.
5. Consider the resulting differential equation for $S(t)$ and show that it will converge to an equilibrium S_* where the concentrations of substrate and product will satisfy Haldane's relationship:

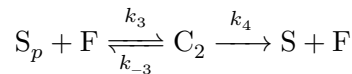
$$\frac{P_*}{S_*} = \frac{k_1 k_2}{k_{-1} k_{-2}}.$$

Phosphorylation

Assume that an enzyme E can bind to a substrate molecule S allowing it to become phosphorylated, S_p , as in the following scheme:



A different enzyme F can bind to the phosphorylated molecule S_p allowing it to be dephosphorylated, according to the scheme:

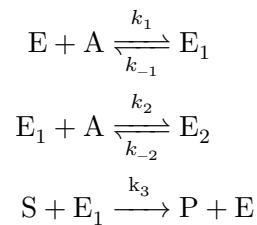


1. Write down the system of equations corresponding to this scheme. Assume that at the start, there are no complexes C_1 or C_2 .
2. Prove that two quantities are conserved: $[E] + [C_1]$ and $[F] + [C_2]$. Use this to decrease the number of equations.
3. Following the same approach used in the derivation of Michaelis-Menten equation, arrive at a single equation for the (appropriately scaled) variable $[S]$.

Chain of reactions

(Similar to Keener and Sneyd (2009), Exercise 1.5)

Consider an enzymatic reaction in which an enzyme E can be activated or inactivated by the same chemical substance A , as follows



Assume that the initial concentration of E is $e_0 \ll a_0$, where a_0 is the initial concentration of A . Further assume that E_1 , E_2 , and P are not present at $t = 0$.

1. Write down the system of equations corresponding to this scheme.

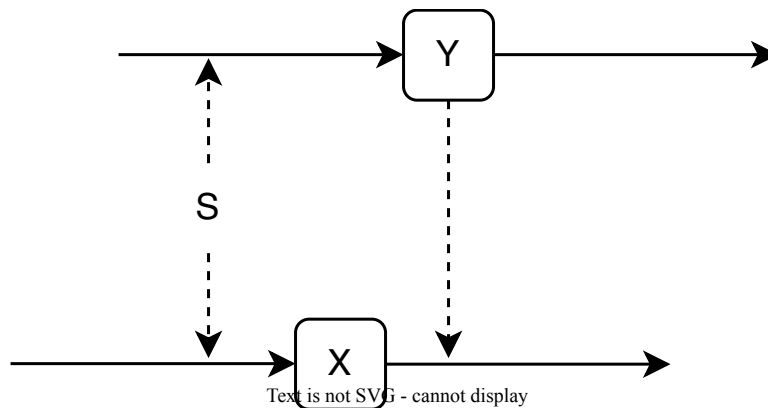
2. After having identified quantities that are conserved, rescale the variables with the initial concentrations of E or A . Using the assumption $e_0 \ll a_0$ (while the values of all rates are comparable), show that the equations result in a slow-fast system.
3. Using the quasi-steady-state analysis, show that $x = [A]/a_0$ satisfies an equation of the type:

$$x' = -\frac{\beta x s}{1 + \alpha x + \delta s + \gamma x^2}.$$

Feedbacks

Sniffer

There are several physiological systems organized to deliver a constant response, independent of the strength of the signal. Tyson and Novak propose a simple reaction scheme with this property; they call it a “sniffer”, as receptors get adapted to a smell. The scheme is the following

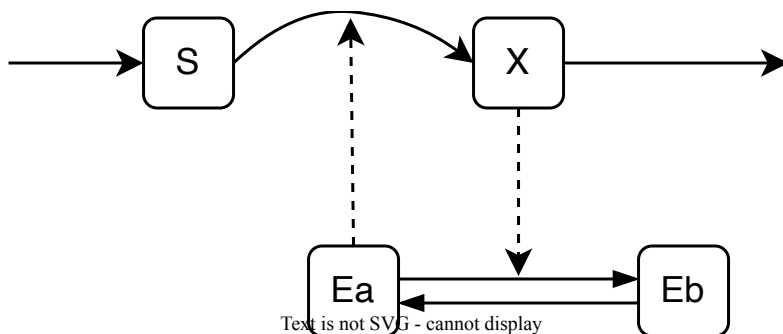


1. Write down a system of differential equations describing the scheme.
2. Show that the equilibrium value of $[X]$ is independent of $[S]$, and that the equilibrium is globally attractive.

A loop

A molecule X is synthesized from a substrate S through an enzymatic reaction of Michaelis-Menten kinetics, proportionally to the concentration of the enzyme E in the active configuration. The enzyme E moves from bound to active configuration at rate k_1 , while the presence of X helps the opposite transition that occurs at rate $k_{-1}[X]$.

These reactions can be summarised in the following scheme:

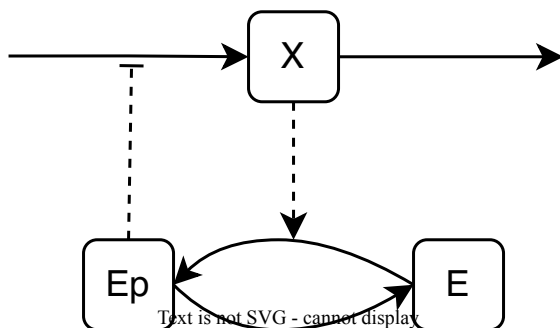


Finally, the substrate is produced at a constant rate, and X dissociates proportionally to its concentration.

1. Is there positive or negative feedback on X synthesis?
2. Write down a system of differential equations corresponding to the assumptions.
3. Compute the (unique) equilibrium of the system, finding conditions for its feasibility.
Hint: consider the equation for the sum $[S] + [X]$.
4. Suggest what might be the behavior of the system, when the conditions for the feasibility of the equilibrium are not satisfied.

Inhibition

Consider the following 2-molecule reaction scheme



where the edge with a bar ending means *inhibition* of the reaction. In particular, we have seen 2 types of inhibition: competitive and allosteric. For competitive inhibition, the half-saturation constant of the Michaelin-Menten law is reduced by a factor $(1 + I/K_i)$; for allosteric inhibition, the same factor reduces $V_{\max}$. For this exercise, we assume that the rate of the inhibited reaction is of the type:

$$k = \frac{k_1}{k_2 + [E_p]}.$$

Note that an increase of $[E_p]$ *inhibits* the production of $[X]$. More generally, we can have the law:

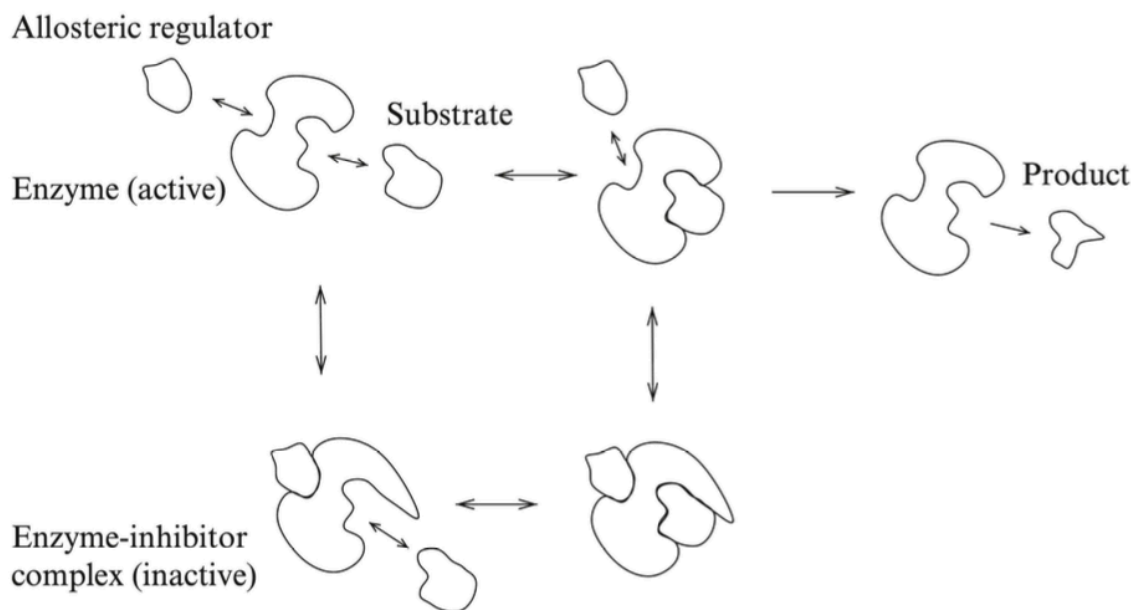
$$k' = \frac{k_1}{k_2^n + [E_p]^n},$$

for some $n \geq 1$.

1. Does the scheme model a negative or positive feedback loop?
2. Write down a system of equations representing the scheme.
3. Find the equilibria, and study their stability.

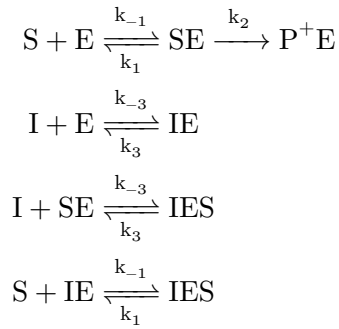
Allosteric inhibition

Allosteric inhibition of an enzyme occurs is a mechanism briefly sketched in the following figure:



When the allosteric regulator binds to the enzyme, a conformation change occurs and it is more difficult for the substrate to bind and/or for the reaction substrate-to-product to occur. In the Figure it is assumed that when the enzyme is bound to the inhibitor (inactive state) the reaction substrate-to-product does not occur at all.

We translate this picture into a reaction scheme as follows:



1. Write down the equations corresponding to this scheme.
2. Assume that each of the reversible reactions in the scheme is at equilibrium (i.e., the rates in both directions are equal) and find the values of $[SE]$, $[IE]$ and $[IES]$ as a function of the concentrations of $[S]$ and $[I]$.
3. Find out the rate of product production as a function of $[S]$ and $[I]$ using the relations found at the previous point. Interpret the result as a modified Michaelis-Menten equation.
4. Show the steps leading instead to the quasi-equilibrium approximation, assuming that the concentration of enzyme is much lower than that of the substrate and that the concentration of the allosteric inhibitor is constant. Identify the fast and the slow variables in the system. *Hint: It is enough to arrive at a system of equations representing the quasi-steady-state approximation without solving these.*
5. In the reaction scheme it has been assumed, for the sake of simplicity, that the rates at which the regulator binds and detach are the same, whether the substrate is bound or not to the enzyme, and similarly for the rate at which the substrate binds. How would you change that if it assumed that the inactive conformation of the enzyme makes it more difficult for the substrate to bind to the enzyme?

Electrophysiology

FitzHugh-Nagumo model

Consider the FHN system:

$$\begin{cases} \varepsilon v' = f(v) - w + I_{\text{app}}, \\ w' = v - \gamma w, \end{cases}$$

with $f(v) = v(1-v)(v-\alpha)$ and $I_{\text{app}} \in \mathbb{R}$. We assume that $\varepsilon \ll 1$, so that we can always apply the singular perturbation analysis.

1. Show that for $I_{\text{app}} > 0$ constant and sufficiently large (but not too large), the system has a (singular) limit cycle. Provide a formula for the period.
2. Try to sketch the bifurcation diagram with respect to I_{app} , mentioning all bifurcations.
3. *Anode break excitation* is when the potential is slowly decreased (thus, I_{app} is negative rather than positive) and then switched off. Can we trigger an action potential in this way? Does it work also when the application time of the current is short?

Roger-McCulloch model

Consider the Roger-McCulloch system:

$$\begin{cases} \varepsilon v' = -kv(v-\alpha)(v-1) - vw + I_{\text{app}}, \\ w' = v - \gamma w, \end{cases}$$

Suppose $\varepsilon \ll 1$. The parameters have the same meaning as in the FitzHugh-Nagumo system.

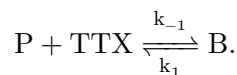
1. Consider $I_{\text{app}} = 0$. Study the phase portrait and show how an action potential may occur.
2. What does it happen when $I_{\text{app}} \neq 0$? Is the situation similar to the previous exercise?

Hodgkin-Huxley model

When using the voltage-clamp technique, the total transmembrane current can be accurately measured. However, Hodgkin and Huxley were interested in the *individual* sodium I_{Na} and potassium I_{K} currents. How can we estimate them from $I_{\text{tot}} = I_{\text{Na}} + I_{\text{K}}$? This exercise addresses the question.

1. Are there potential levels (fixed by clamping) at which we can measure a pure I_{Na} or I_{K} current?
2. Suppose 90% of the Na^+ in the bath of a squid axon is replaced by inert choline, preserving electroneutrality. What happens to the equilibrium potentials?
3. Explain why replacing the extracellular Na^+ with choline has little effect on the resting potential of an axon. Calculate the new resting potential with 90% of the extracellular Na^+ removed. Why is the same not true if K^+ is replaced? Assume $m = 0.0532$, $n = 0.3184$, $h = 0.5945$ (constants), $\bar{g}_{\text{Na}} = 120 \text{ mS}$, $\bar{g}_{\text{K}} = 36 \text{ mS}$, $\bar{g}_{\text{leak}} = 0.3 \text{ mS}$, $V_{\text{leak}} = -54.4 \text{ mV}$ and $V_{\text{K}} = -77.3 \text{ mV}$.
4. Suppose we know $K = I_{\text{Na}}^1 / I_{\text{Na}}^2$, where I_{Na}^1 is the normal sodium current and I_{Na}^2 is the one obtained with 90% of extracellular sodium replaced by choline. Suppose also that K does not change over time and $I_{\text{K}}^1 = I_{\text{K}}^2$. Find I_{Na}^1 and I_{Na}^2 from I_{tot}^1 and I_{tot}^2 . (A possible way to measure K is to look at I_{tot} immediately after the voltage clamp, because most of the initial current is carried by sodium.)
5. Nowadays a simpler way to block channels is with drugs. Tetrodotoxin (TTX) is a powerful neurotoxin, famous for the puffer fish. TTX acts by blocking the sodium channels, hence making them unavailable for action potential formation.

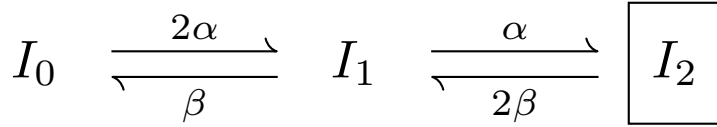
Here is how we can model it. If P_0 represents the total number of Na^+ channels, and we denote by P and B respectively the number of non-blocked and blocked channels, then the reaction is



Modify the sodium conductance to account for the presence of TTX.

Gating mechanism

A ion channel can be modelled with a Markov model. For instance, for a channel we have the following state diagram:



Here we want to show that the probability of the state I_2 is exactly $n^2(t)$, where:

$$\frac{dn}{dt} = \alpha_n(1 - n) - \beta_n n.$$

1. Write down the equations of the system, where $s_k(t) = \mathbb{P}[\text{state is } I_k]$, with $k = 0, 1, 2$. Observe that $s_0(t) + s_1(t) + s_2(t) = 1$.
2. Set $s_0(t) = (1 - n)^2$, $s_1(t) = 2n(1 - n)$, $s_2(t) = n^2$, with $n(t)$ solution of the above ODE. Show that this choice provides a solution of the full system.
3. Try to generalize to a system with $k + 1$ states, showing that the ODE is always the same with the probability of I_k being $n^k(t)$.

Multidimensional fronts

In excitable media like the cardiac tissue, the cable equation generalises to its multidimensional version:

$$\varepsilon \frac{\partial V}{\partial t} = \varepsilon^2 \Delta V + f(V).$$

where Δ is the Laplacian, that is for $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ we have:

$$\Delta V = \frac{\partial^2 V}{\partial x_1^2} + \frac{\partial^2 V}{\partial x_2^2} + \dots + \frac{\partial^2 V}{\partial x_n^2}.$$

1. Show that a planar wave, that is $V(x, t) = U(\frac{x \cdot \vec{k} + st}{\varepsilon})$ with $\vec{k}$ the unit vector pointing in the propagation direction, one has exactly the 1-D result for the velocity:

$$s = \frac{1}{\sqrt{2}}(2\alpha - 1).$$

2. Consider now a spherical front, that is $V(x, t) = U(\frac{|x| + st}{\varepsilon}) = U(r + st)$ where $r = |x|$ is the radius of the sphere. Using the fast system, show that the speed is as follows:

$$s = s_0 - \kappa \varepsilon,$$

where s_0 is the speed of a planar front and $\kappa = \frac{n-1}{r}$ is the *curvature* of the front. Show that, compared to a planar front, the velocity of a concave front is larger and for a convex front is lower.

Extra exercises

Seasons

Let us consider a population growing according to the logistic model, but with periodic carrying capacity K , e.g.,

$$K(t) = K_0(1 + \varepsilon \cos(\omega t)),$$

with $0 < \varepsilon < 1$. Precisely, population size $N(t)$ satisfies the equation

$$N'(t) = r \left(1 - \frac{N}{K(t)} \right) N,$$

where $r > 0$.

1. Show that, using the same trick used for solving the logistic equation, the solutions of can be written explicitly, although involving integrals that may be impossible to evaluate analytically.
2. Draw the direction field for this equation. Try to draw one solution of the equation, and convince yourself that it will converge to a positive periodic solution.

Mutualism

It is assumed that the dynamics of the density of two mutualistic species follows, for each species, the logistic equation where however the carrying capacity increases due to the presence of the other species

$$\begin{cases} N_1'(t) = r_1 N_1 \left(1 - \frac{N_1}{K_1 + a N_2} \right), \\ N_2'(t) = r_2 N_2 \left(1 - \frac{N_2}{K_2 + b N_1} \right). \end{cases}$$

1. Explain why, according to this model, the interaction of the two species is of mutualistic type.
2. Assuming $0 < ab < 1$, draw the isoclines of the systems and the vector field.

3. Can we understand from the vector-field the long-time behaviour of the solutions?
4. What happens if $ab > 1$?

Plant-herbivore system

The following system has been proposed to study a plant-herbivore system: q represents plant quality (low q means that the plant is toxic because of chemicals released, while high q means that the plant is a good food for the herbivores). We assume that quality decreases (toxic compounds are synthesised) as a result of high herbivory). The density of herbivores is $H(t)$ and their growth rate depends on the quality of food they consume. The model equations are

$$\begin{cases} q' = k_1 - k_2 q H (H - H_0), \\ H' = k_3 H (1 - k_4 H / q), \end{cases}$$

1. Explain why the equations correspond to the biological assumptions, and suggest possible meanings for its parameters.
2. Show that the equations can be written in dimensionless form as follows

$$\begin{cases} x' = 1 - kxy(y - 1), \\ y' = \alpha y(1 - y/x), \end{cases}$$

Determine k and α in terms of the original parameters.

3. Show (graphically) that there is only one positive equilibrium.
4. Determine its stability from a qualitative analysis of the isoclines.

Mooses and wolves

In all Canada moose are preyed by packs of wolves that cause a relevant mortality. The ecologist Messier (1994) has collected data in different areas in the country and found that the following relation (approximately) holds between the density W (# of wolves per km²) of wolves and the density M (# of moose per km²) of moose:

$$W = \frac{0.0587(M - 0.3)}{0.76 + M}. \quad (1)$$

1. Let us interpret this relation as the isocline of predators, i.e. the equilibrium number of wolves for each constant density M of moose. Is it possible to obtain such an isocline from a prey-predator system of Gause-Rosenzweig-McArthur type?

2. Suppose instead that the numerical response of wolves has the Holling-type 2 form

$$\pi(M) = \frac{aM}{1 + aT_h M},$$

and that mortality of wolves is equal to $\mu_0 + \mu_1 W$. Find the constants a/μ_1 , aT_h and μ_0/μ_1 such that the predator isocline can be written as (1). *Hint: we cannot determine all 4 parameters (a, T_h, μ_0, μ_1) by this relation, but only 3 combinations of them.*

3. Assume that, in absence of wolves, the growth of moose is logistic with intrinsic growth rate equal to 0.51 years^{-1} and carrying capacity equal to $0.5 \text{ moose per km}^2$.

Furthermore, assume that the functional response is proportional to the numerical response (the one found at the previous point).

Draw the prey and predator isoclines and show that there exists a unique positive equilibrium of the system.

4. Study the stability of the positive equilibrium. *Hint: it is enough computing the signs of the terms of the Jacobian at the equilibrium; the computations are easier using the function $\Phi(M)$ defined for the isocline $W = \Phi(M)$.*

Fishing

Consider the predator-prey model with logistic prey growth rate and bilinear predation rate

$$\begin{cases} H'(t) = r\left(1 - \frac{H}{K}\right)H - aHP, \\ P'(t) = -\mu P + \gamma aHP. \end{cases}$$

Assume that the parameters are such that a positive equilibrium exists, and add fishing to the equations of prey and predators, following Volterra's assumptions.

1. Show that, if the fishing effort becomes large, the predators will get extinct; if it becomes even larger, the prey also will get extinct.
2. Show that, with lower fishing efforts, the equilibrium predator density decreases and the equilibrium prey density increases.
3. Change the time unit to $\tau = rt$ and the variable H to $u = H/K$ and write down the resulting system.
4. Propose a change of variable also for P such to further decrease the number of parameters of the resulting system.

Gause-Rosenzweig-McArthur model

Gause-Rosenzweig-McArthur predator-prey models with logistic prey growth rate have the following structure

$$\begin{cases} H'(t) = r\left(1 - \frac{H}{K}\right)H - \pi(H)P, \\ P'(t) = -\mu P + \gamma\pi(H)P. \end{cases}$$

where $\pi(H)$ is a function, named *functional response*, representing the per capita predation rate, i.e., the number of prey captured by each predator in unit time.

Assume that $\pi(H)$ is an increasing function such that:

$$p(0) = 0, \quad \lim_{H \rightarrow \infty} \pi(H) = \bar{p} < +\infty.$$

The prototypical example is Holling type-2 function

$$\pi_2(H) = \frac{aH}{1 + aT_h H}, \quad \lim_{H \rightarrow \infty} \pi_2(H) = \frac{1}{T_h} < +\infty.$$

1. Show graphically that, if $\gamma\bar{p} > \mu$, there exists a unique H_* such that $\gamma\pi(H_*) = \mu$. Compute H_* in the case of the Holling type-2 function.
2. Compute the isoclines of the system, showing that $H' = 0$ if and only if $H = 0$ or $P = \Phi(H)$ for an appropriate function Φ . Compute the function $\Phi(H)$ in the case of the Holling type-2 function.
3. Show that the system has a positive equilibrium E_* if and only if $K > H_*$.

Competition

Consider the following system describing competition between two species:

$$\begin{cases} N_1' = (122 - 3N_1 - N_2)N_1, \\ N_2' = (80 - 2N_1 - N_2)N_2. \end{cases}$$

1. Draw the isoclines of the system and find its equilibria.
2. Identify what can be defined the carrying capacity for each species.
3. Find out the behaviour of the solutions of the system (a graphical analysis may help).
4. Discuss the assumptions of the model.

Population dynamics

We model the population dynamics of a species of insects by the following system of differential equations:

$$\begin{cases} J' = \beta_0 \phi(A)A - (\rho + \mu_J)J, \\ A' = \rho J - \mu_A A. \end{cases}$$

where $J(t)$ represents the density of juveniles, and $A(t)$ represent the density of adults.

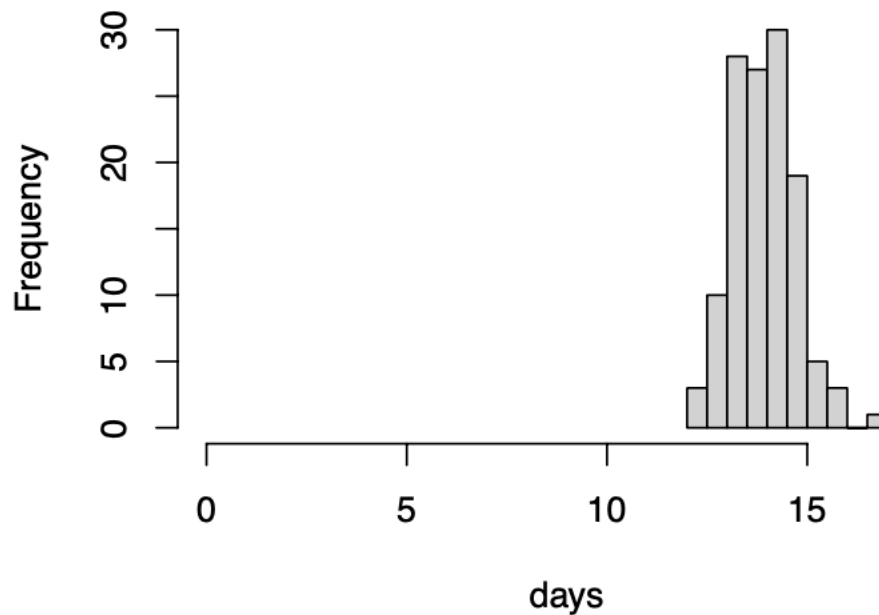
In order to fit the parameters of the model, we use some information obtained by raising the insects in the laboratory. We know that under those conditions

- around 70% of the larvae emerging from eggs reach the adult stage;
 - 14 days is the average age at which the adult stage is reached;
 - the average length of the adult stage is 30 days;
 - each (female) adult produces on the average 120 eggs during its whole life.
1. Use this information to choose all parameters of the model, except for the function ϕ .
 2. Laboratory experiments cannot yield reliable information on the function ϕ . Instead we assume that

$$\phi(x) = e^{-\gamma x}.$$

Estimate γ , using the information that the density of the juvenile stages of the insect in a cultivated area is around 0.7 per m^2 and that this must correspond to the equilibrium of model.

3. Analyse the stability of the equilibrium you just found of system.
4. The complete data on the age at which the adult stage is reached are shown in the histogram below. Do you think that these data are compatible with the assumptions implicit in model? If not, do you have suggestions on how to modify?



SEIR model

An extension of SIR models is to introduce a class E (Exposed) of individuals that have been infected, but are not yet infectious. Consider thus an SEIR model (with births and deaths) similar to the SIR model, assuming that individuals in E move to the class I at rate ν .

1. Find the equilibria of the system, discussing the conditions under which a positive equilibrium exists.
2. Study the stability of the disease-free equilibrium, and write down the condition for its stability as $R_0 < 1$ for an appropriately defined quantity R_0 .
3. Noting that R_0 will differ (slightly) from that obtained for the SIR model, interpret it anyway as the average number of new infections produced by a newly infected individual along her/his infectious period, assuming everybody else is susceptible.
4. Does the value of R_0 change if we consider instead an SEIRS model in which immunity decays?

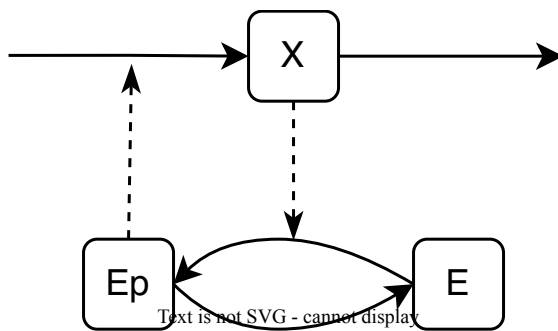
Asymptomatics

It is known that infections with the same pathogen strain can give to diseases of different severities according to the person infected. We model this by modifying the SEIR model of the previous problem; we assume that there may be infected individuals that are severely ill I and others that are asymptomatic A ; it is assumed that at the end of the incubation period, infected individuals become I with probability p and A with probability $1 - p$. Furthermore the infectiousness of asymptomatic individuals is equal to ϕ times the infectiousness of severely ill, and the mean duration of the A stage is (neglecting deaths) $1/\alpha$, while that of the I stage is $1/\gamma$.

1. Write down an ODE model for the classes of susceptibles (S), exposed (E), severely ill (I), asymptomatic (A) and removed (R) individuals consistent with this verbal description.
2. If empirical information is missing, would you expect ϕ to be greater or smaller than 1? And α smaller or greater than γ ? Discuss.
3. Find R_0 in an intuitive way, by looking at the possible paths of a newly infected individual, and the average number of infections caused along each path.
4. Show that the disease-free equilibrium is asymptotically stable (resp. unstable) if $R_0 < 1$ (resp. $R_0 > 1$).
5. Show that the system has a unique positive equilibrium if and only if $R_0 > 1$.

A feedback network

Consider a small molecular network as presented in the following scheme:



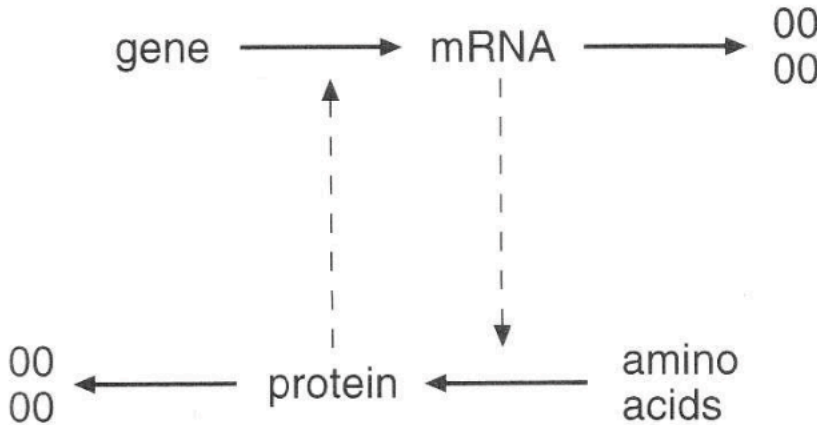
We assume that E gets phosphorylated and de-phosphorylated following Michaelis-Menten type reactions; the reaction rates for X are instead of linear type with respect to E_p (rate k'_1) and we assume that X is synthesized (at rate k_1) also in absence of E_p . The product X is removed at rate k_2 .

1. Does the scheme model a negative or a positive feedback loop?
2. Write down a system of equations representing the scheme.
3. Study the equations satisfied by equilibria; this will involve the Goldbeter-Koshland function.
4. Consider the bifurcation graph of equilibria with respect to the parameter k_1 ; show that, when the half-saturation constants of Michaelis-Menten reactions are small (this because of the shape of the Goldbeter-Koshland function), there is one equilibrium for small k_1 or large k_1 , and three equilibria for intermediate k_1 .

Gene Transcription

Note: this model has been proposed by JS Griffith in 1968 in two papers on [positive](#) and [negative](#) feedback loops.

Consider the simple case of a protein that activates transcription of its own gene, as in the following figure:



This mechanism is described by a pair of ordinary differential equations:

$$\begin{cases} [M]' = \nu_1 \frac{\varepsilon^2 + ([P]/K_\ell)^2}{1 + ([P]/K_\ell)^2} - k_2[M], \\ [P]' = k_3[M] - k_4[P]. \end{cases}$$

1. How must the variables be scaled to write the ODEs in dimensionless form:

$$\begin{cases} \frac{dx}{d\tau} = \frac{\varepsilon^2 + y^2}{1 + y^2} - x, \\ \frac{dy}{d\tau} = \kappa(\sigma x - y). \end{cases}$$

2. Assume $\varepsilon = 0.2$, $\kappa = 1$. Draw phase plane portraits for several values of σ .
3. Show graphically how the number of equilibria of the system changes with σ .
4. Plot the equilibrium value of y as a function of σ . *Hint: it is enough a picture showing the qualitative pattern.*

A Introduction to ODEs

A.1 Basic definitions

Let us start with some simple definitions about Ordinary Differential Equations (ODEs).

Definition A.1 (ODE). An scalar ODE is an equation (generally nonlinear) involving a function $y(t)$ and its derivatives. Let $f: D \rightarrow \mathbb{R}$, $D \subseteq \mathbb{R}^{n+1}$ open set not empty. Then an ODE of order n is an equation of the type:

$$y^{(n)} = f(t, y, y', \dots, y^{(n-1)}).$$

Let's have a look to some of the equations we are going to deal with during this course. We will analyze the equations for a purely mathematical point of view, with no biological insights: this will come later.

Example: Malthusian growth model

$$N'(t) = r N(t).$$

This is a simple model of population growth where $N(t)$ represents the population size at time t . We have that $f(t, N) = rN$. It is a *linear* ODE (f is linear with respect to N), with *constant coefficients* (r is a constant), *first-order*, and *autonomous* (f does not explicitly depend on t).

Example: Logistic Growth Model

$$N'(t) = rN \left(1 - \frac{N}{K}\right)$$

The logistic model modifies the Malthusian growth by introducing a carrying capacity. We have that $f(t, N) = rN(1 - N/K)$. It is a *nonlinear* ODE (f is non-linear with respect to N), *first-order*, and *autonomous*.

i Example: Damped Harmonic Oscillator

$$y''(t) + 2\beta y'(t) + \omega_0^2 y(t) = F(t).$$

This is a second-order ODE, linear, constant coefficients, non-autonomous (because of the presence of $F(t)$).

In general, an ODE does not tell anything about a specific solution; it is just the law of motion. We can have many solutions. As a rule of thumb, an n -order scalar ODE requires n constants for uniquely defining a solution. An **Initial Value Problem (IVP)** is an ODE supplemented with additional conditions. It is called “initial condition” because it sets the value of the solution at some initial time t_0 . (Please note that an IVP is also called “Cauchy problem” on some books.)

Definition A.2 (Initial Value Problem). An initial value problem (IVP) is a system of the form:

$$\begin{cases} y^{(n)} = f(t, y, y', \dots, y^{(n-1)}), \\ y(t_0) = y_0, \\ y'(t_0) = y_1, \\ \vdots \\ y^{(n-1)}(t_0) = y_{n-1}. \end{cases}$$

In all the above cases, the ODEs were scalar. A system of ODEs is the natural generalization to $\mathbb{R}^n$. We will only consider *first-order system of ODEs*, because their extension to higher-order ODEs is superfluous (we will see why in a moment.)

Consider the system of first-order ODEs:

$$\begin{cases} y'_1 = f_1(t, y_1, y_2, \dots, y_n), \\ y'_2 = f_2(t, y_1, y_2, \dots, y_n), \\ \dots \\ y'_n = f_n(t, y_1, y_2, \dots, y_n). \end{cases}$$

We set $y(t) = (y_1, \dots, y_n)^\top$ and $f(t, y) = (f_1, \dots, f_n)^\top$ so the system rewrites to

$$y' = f(t, \mathbf{y}).$$

(Please note that the time derivative of the vector is the vector of the derivatives.) In compact form, the IVP reads as:

$$\begin{cases} y' = f(t, \mathbf{y}), \\ y(t_0) = y_0. \end{cases} \quad (\text{A.1})$$

The *solution* $\phi \in \mathcal{C}^1(I, \mathbb{R}^n)$ of the system is a *curve* in $\mathbb{R}^n$ such that:

$$\phi'(t) = f(t, \phi(t)), \quad \text{for all } t \in I.$$

A.2 Separation of variables

Solving ODE analytically is not always possible. As for integration, sometimes we cannot find a solution in closed form. However, we have alternatives to rescue us:

- **Qualitative analysis:** we don't solve the ODE explicitly, we rather analyse the behaviour of the trajectories as we would do for a function study. Very often we proceed graphically (in 1-D and 2-D). It is very helpful for parametric studies.
- **Numerical integration:** we solve the IVP on a computer. We have no limits, but we always need to provide all parameters and initial values. Parametric studies are more difficult.

A combination of the above approaches is always wise. On the other hand, it is good to know how to integrate an ODE when it is possible.

A simple yet effective technique to solve IVPs is the method of *separation of variables*. It applies to IVPs of the form:

$$\begin{cases} y' = f(t)g(y), \\ y'(t_0) = y_0. \end{cases}$$

for some functions f and g . Using the Leibniz notation, we write the ODE as follows:

$$\frac{dy}{dt} = f(t)g(y),$$

which suggests to separate the variables t and y :

$$\frac{dy}{g(y)} = f(t) dt.$$

Here we consider y and t as independent variables. The advantage is that we can integrate both sides to get rid of differentials:

$$\int_{y_0}^y \frac{d\tilde{y}}{g(\tilde{y})} = \int_{t_0}^t f(\tilde{t}) d\tilde{t}.$$

Please note that amongst all anti-derivatives, we select those satisfying the initial condition $y(t_0) = y_0$. Suppose that $F(t)$ and $G(y)$ are respectively anti-derivatives of $f(t)$ and $\frac{1}{g(y)}$,

i.e. $F'(t) = f(t)$ and $G'(y) = \frac{1}{g(y)}$. Then, according to the Fundamental Theorem of Calculus, we have:

$$G(y) - G(y_0) = F(t) - F(t_0),$$

which gives an implicit expression for $y(t)$:

$$G(y(t)) = F(t) + G(y_0) - F(t_0).$$

Please note that the expression on the right hand side only contains the variable t . When G is invertible, the expression of $y(t)$ can be made explicit.

i Example (Malthusian growth model)

We aim at solving the following IVP:

$$\begin{cases} N' = rN, \\ N(t_0) = N_0. \end{cases} \quad (\text{A.2})$$

We apply the method of separation of variables. We have:

$$\int_{N_0}^N \frac{d\tilde{N}}{\tilde{N}} = \int_{t_0}^t r \, d\tilde{t}.$$

After integration (recall that $\ln'(y) = 1/y$):

$$\ln\left(\frac{N}{N_0}\right) = r(t - t_0),$$

or, by inverting the logarithm,

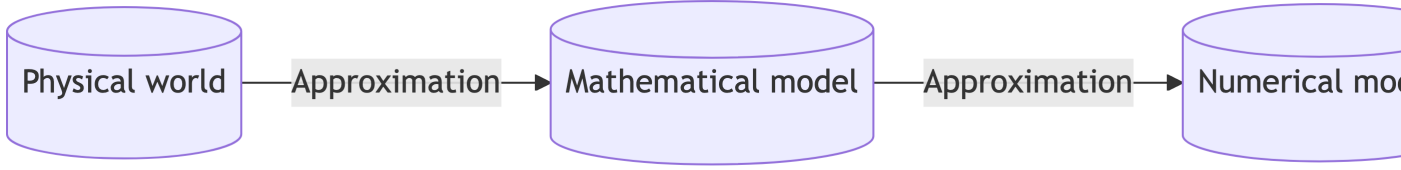
$$N(t) = N_0 e^{r(t-t_0)}.$$

! Exercise (Time-dependent Malthus equation)

Check that $N(t) = N_0 e^{r(t-t_0)}$ solves the problem (Equation A.2).

A.3 Well-posedness

Let us recall this (simplistic) view of mathematical modeling:



A good mathematical model shall approximate well the physical world. For instance, it is able to fit experimental data in a validation test. However, a validation test requires that we are able to *solve* the mathematical problem. For that, we require that the problem should satisfy **well-posedness properties**.

We are dealing with IVPs. Here, well-posedness translates into:

1. **Existence** of a solution $\phi \in \mathcal{C}^1([0, T], \mathbb{R}^n)$ for every choice of the initial data;
2. **Uniqueness** of the solution, in the sense that if $\phi_1(t)$ and $\phi_2(t)$ are both solutions of the IVP for the *same* initial data, then $\phi_1(t) = \phi_2(t)$ for all t .
3. **Stability** of the solution to perturbations, that is if $\tilde{\phi}(t)$ is the solution of the IVP with some perturbation, e.g., applied to the initial datum or to the right hand side, then $\|\tilde{\phi} - \phi\| \rightarrow 0$ as the perturbation goes to zero.

For IVPs, stability to perturbation is also called **zero stability**. The zero stability for initial data follows from the Cauchy-Lipschitz theorem, as shown below.

For the numerical realm, we have a similar definition of well-posedness. Additionally, we typically need to show that the numerical solution *converges* (in some sense) to the analytical solution. That is, the approximation error goes to zero as we refine the numerical problem. For an IVP, convergence means that the error between the numerical solution and the true solution goes to zero as the time step goes to zero.

We start with a definition of a solution for the IVP.

Definition A.3 (Classic solution). A classic solution of the IVP (Equation A.1) is a function $\phi \in \mathcal{C}^1(I; \mathbb{R}^n)$, where $I \subset \mathbb{R}$ is a closed interval, such that

1. $\phi'(t) = f(t, \phi(t))$ for all $t \in I$;
2. $\phi(t_0) = y_0$;
3. $(t, \phi(t)) \subset D$ for all $t \in I$, where $D \subset \mathbb{R}^{n+1}$ is the domain of f .

The first 2 conditions are simply the IVP (equation and initial condition). The third condition concern the domain of definition of the right hand side of the ODE.

i Example (lack of existence)

When the f is not continuous, we cannot expect existence in general. In fact, consider the IVP

$$\begin{cases} y' = \mathcal{H}(t) = \begin{cases} 0, & t \leq 0, \\ 1, & t > 0, \end{cases} \\ y(0) = y_0. \end{cases}$$

A possible and intuitive solution is

$$\phi(t) = y_0 + \max\{0, t\} = \begin{cases} y_0, & t \leq 0, \\ y_0 + t, & t > 0. \end{cases}$$

The function $\phi(t)$ is, however, not $\mathcal{C}^1(I)$ for any neighborhood I of the initial time $t_0 = 0$. Therefore, the function cannot be a solution, at least according to the above definition of solution. Note that with a different initial condition, say $y(1) = y_0$, we can find a $\mathcal{C}^1(I)$ solution, for I away from $t = 0$.

A.3.1 Local well-posedness

A general result for the local existence of a solution is due to [Giuseppe Peano](#):

Theorem A.1 (Peano). *Let $\mathbf{f}: D \rightarrow \mathbb{R}^n$, $D \subseteq \mathbb{R}^{n+1}$ open set, be continuous. Then, for all $(t_0, \mathbf{y}_0) \in D$, there exists a neighborhood of t_0 , denoted by $I_\delta = [t_0 - \delta, t_0 + \delta]$ with $\delta > 0$, on which is possible to construct a solution $\phi \in \mathcal{C}^1(I_\delta; \mathbb{R}^n)$ to the IVP.*

Peano's theorem ensures the existence but not the uniqueness. He also provided a simple counterexample to uniqueness.

i Peano's counterexample on uniqueness

Consider the ODE

$$\begin{cases} y' = \sqrt[3]{y}, \\ y(1) = 0. \end{cases}$$

The function $f(t, y) = \sqrt[3]{y}$ is continuous with respect to t and y (note that it does not explicitly depends on time), thus Peano's Theorem ensures the existence of at least one solution.

We can find a solution by separation of variables. We have (please check!):

$$\phi_1(t) = \begin{cases} (\frac{2}{3}(t-1))^{3/2}, & t \geq 1, \\ 0, & t < 1. \end{cases}$$

This solution is however not the only one. Also the following function is a solution (again, please check!)

$$\phi_2(t) = \begin{cases} -(\frac{2}{3}(t-1))^{3/2}, & t \geq 1, \\ 0, & t < 1. \end{cases}$$

Yet another solution is $\phi_3(t) \equiv 0$. Actually, the IVP is so subtle that we can build an *infinite* number of solutions with the power of continuum! In fact, for any $\alpha \geq 1$, we have that

$$\phi_{\alpha^\pm}(t) = \begin{cases} \pm(\frac{2}{3}(t-\alpha))^{3/2}, & t \geq \alpha, \\ 0, & t < \alpha, \end{cases}$$

are all solutions of the IVP. That's a lot to deal with.

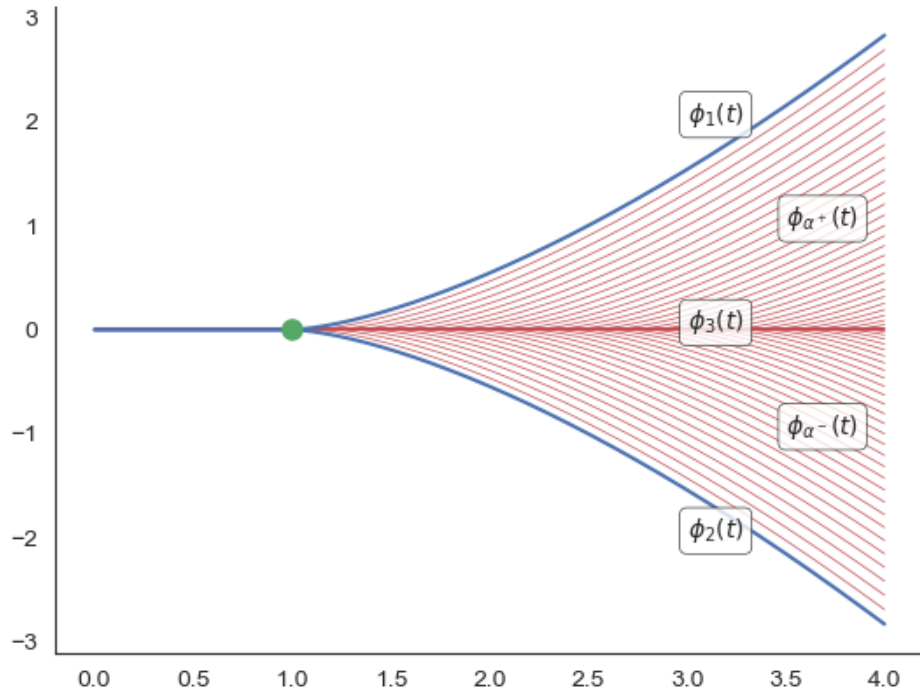


Figure A.1: Peano's brush

The example shows that the lack of uniqueness is somehow related to a lack of stability of the solution. Consider a particle located, at $t = 1$, in $y = 0$. The velocity at some generic time t is $y'(t) = \sqrt[3]{y(t)}$. Then, the trajectory of the particle is exactly the solution of Peano's example.

The example shows that we need some control on the growth rate of f . The correct concept is the Lipschitz continuity:

Definition A.4 (Lipschitz continuity). Let $f: \Omega \rightarrow \mathbb{R}^n$, $\Omega \subseteq \mathbb{R}^n$, a continuous function. We say that f is *Lipschitz continuous* if there exists a constant $L > 0$ such that

$$\|f(y) - f(z)\| \leq L\|y - z\|,$$

for all $y, z \in \Omega$.

Definition A.5 (Local Lipschitz continuity). Let $\mathbf{f}: \Omega \rightarrow \mathbb{R}^n$ a continuous function, with $\Omega \subseteq \mathbb{R}^n$ open domain. We say that f is *locally Lipschitz continuous* in Ω if for every $y_0 \in \Omega$ there exists a neighborhood of y_0 in which f is Lipschitz continuous, with a constant L possibly depending on the neighborhood.

! Exercise

Show that $\mathcal{C}^1$ functions are always locally Lipschitz. For simplicity consider functions on $\mathbb{R}$.

Theorem A.2 (Cauchy-Lipschitz). Let $f \in \mathcal{C}(D; \mathbb{R}^n)$, $D \subseteq \mathbb{R}^{n+1}$ open domain. If f is locally Lipschitz in D , with respect to y and uniformly in t , then for every $(t_0, y_0) \in D$ there exists an interval $I_\delta = [t_0 - \delta, t_0 + \delta]$, with $\delta > 0$, on which is possible to construct a solution $\phi \in \mathcal{C}^1(I_\delta; \mathbb{R}^n)$ to the IVP (Equation A.1). Such solution is also unique, in the sense that every other solution of the IVP coincides with $\phi(t)$ on I_δ .

The Cauchy-Lipschitz theorem has very important consequences on the study of ODEs. We mention here the following

Corollary A.1 (Zero stability). Consider the following IVPs:

$$\begin{cases} y' = f(t, \mathbf{y}), \\ y(t_0) = y_0, \end{cases} \quad \text{and} \quad \begin{cases} \tilde{y}' = f(t, \tilde{\mathbf{y}}), \\ \tilde{y}(\tilde{t}_0) = \tilde{y}_0. \end{cases}$$

Suppose we are under the hypotheses of the Cauchy-Lipschitz theorem. Then, we have the following stability estimate:

$$\|\tilde{y}(t) - y(t)\| \leq M|\tilde{t}_0 - t_0| + e^{L|t-t_0|}\|\tilde{y}_0 - y_0\|,$$

where L is the Lipschitz constant of f .

In other words, if we perturb the initial value of the IVP, the perturbed solution remains close to the unperturbed solution as long as the perturbation is small and small time. For large time, however, the initial perturbation may grow exponentially, with a rate proportional to the Lipschitz constant.

A.3.2 Global well-posedness

The solution provided by the Cauchy-Lipschitz Theorem is only defined in a local interval, that is up to $t = t_0 + \delta$, with $\delta > 0$. Unfortunately, δ can be small, thus there is no guarantee that we can integrate the IVP for an arbitrarily long time.

We can try to circumvent the problem as follows. Let us call $\phi_1(t)$ the solution and $\delta_1 = \delta$. Now we set up a new problem of the form:

$$\begin{cases} y' = f(t, y), \\ y(t_1) = y_1, \end{cases}$$

where $t_1 = t_0 + \delta_1$ and $y_1 = \phi_1(t_1)$. Since also the above problem is well-posed, we get a new solution $\phi_2(t)$ up to $t_2 = t_1 + \delta_2$, for some $\delta_2 > 0$.

So, we can define a new solution on the interval $[t_0, t_0 + \delta_1 + \delta_2]$ by gluing the solutions around t_1 :

$$\phi(t) = \begin{cases} \phi_1(t), & t \in [t_0, t_0 + \delta_1], \\ \phi_2(t), & t \in [t_0 + \delta_1, t_0 + \delta_1 + \delta_2]. \end{cases}$$

Please note that $\phi_1(t_1) = \phi_2(t_1)$ by construction, and the same applies to the derivative, so the full solution is still $\mathcal{C}^1([t_0, t_0 + \delta_1 + \delta_2])$.

By iterating the process, we obtain a sequence $\{\delta_n\}$ of extensions. The hope is that $\sum_n \delta_n \rightarrow \infty$, that is we can extend the solution indefinitely. However, it is certainly possible that $t_0 + \sum_n \delta_n \rightarrow T_{\max} < \infty$.

Definition A.6 (Maximal interval). The *maximal right interval* $[t_0, T_{\max})$ is simply the maximum time $T_{\max}$ up to which we can extend the solution to the right. In general the interval is open. Similarly, the *left maximal interval* is $(T_{\min}, t_0]$. Finally, the *maximal interval* is just $(T_{\min}, T_{\max})$. We denote the maximal interval of the solution with J_ϕ .

i Example (blow-up of solutions)

Consider the problem

$$\begin{cases} y' = y^2, \\ y(0) = 1. \end{cases}$$

We solve it by separation of variables. So,

$$\begin{aligned}
\int_1^y \frac{d\tilde{y}}{\tilde{y}^2} &= \int_0^t d\tilde{t} \\
\Rightarrow \left[-\frac{1}{\tilde{y}} \right]_1^y &= t \\
\Rightarrow -\frac{1}{y} + 1 &= t \\
\Rightarrow y(t) &= \frac{1}{1-t}.
\end{aligned}$$

We know that the problem is locally well-posed, because $f(t, y) = y^2$ is continuous and locally Lipschitz. Thus, the above solution is locally unique. However, the interval of integration cannot be arbitrarily large: starting at $t = 0$, we can go forward in time only up to $t = 1 - \varepsilon$, with $\varepsilon > 0$. That is, we cannot reach $t = 1$. The reason is clear: the solution exhibit a vertical asymptote at $t = 1$, thus the solution *blows up*. In mathematics, this event is indeed called *blow-up in finite time* of the solution.

Conversely, backward integration exhibit no problem. Therefore, the maximal interval of well-posedness is $J_\phi = (-\infty, 1)$.

Computing the maximal interval for a solution by hand is clearly not practical. We would like to estimate the interval without having to solve the problem analytically.

Theorem A.3. *Let $\|\phi(t)\| \leq M$ for all $t \in (a, b)$. Then $J_\phi = (a, b)$.*

In the theorem, we can also take $a = -\infty$ and $b = +\infty$. In this case the maximal interval would be $\mathbb{R}$.

If the right hand side of the IVP grows almost linearly, then we have global existence as well. Indeed, the following theorem holds.

Theorem A.4. *Let $f: (a, b) \times \mathbb{R}^n \rightarrow \mathbb{R}^n$. Suppose we are under the assumptions of the Cauchy-Lipschitz theorem. If there exist 2 non-negative constants k_1, k_2 such that*

$$\|f(t, y)\| \leq k_1 + k_2\|y\|$$

for every $(t, y) \in [a, b] \times \mathbb{R}^n$, then for all (t_0, y_0) the unique solution $\phi(t)$ is defined in $[a, b]$.

In general, we use *a priori* information on the solution to apply the above Theorems. For that, we may use energy estimates (common for problems in physics), or we can study the qualitative behavior of the solutions for various initial conditions, e.g., in the phase space. Let us try that with a simple example.

i Example (logistic equation)

Consider the ODE

$$y' = y(1 - y),$$

where $y(t)$ may represent, for instance, the density of a population. We would like to show that $J = \mathbb{R}$ for every choice of the initial data $y(0) = y_0$. Note that the right hand side $f(t, y) = y(1 - y)$ is quadratic: we cannot exclude a blow up as in the above example! We have the following cases:

- if $y_0 = 0$ or $y_0 = 1$, the solution is constant in time, $y(t) = y_0$. In fact, $f(t, y_0) = 0$, thus $y'(t) = 0$ and we have a constant solution. We are going to call this type of solutions *fixed points* or *equilibria* (see the next section.)
- if $y_0 \in (0, 1)$, then $y(t) \rightarrow 1^-$ as $t \rightarrow +\infty$ and $y(t) \rightarrow 0^+$ as $t \rightarrow -\infty$. Thus, $y(t) \in (0, 1)$ for all t . The limit follows from an analysis of the sign of $f(t, y)$. For $y \in (0, 1)$, $f(t, y) > 0$, thus $y' > 0$ for all t and $y(t)$ is monotonically increasing. But $y = 1$ is an equilibrium that cannot be crossed, and there are no other equilibria in $(0, 1)$. Thus, it must be that $y \rightarrow 1$ from below as $t \rightarrow \infty$.
- if $y_0 > 1$, then $y(t) \rightarrow 1^+$ as $t \rightarrow +\infty$. Again, the statement follows from $f(t, y) < 0$ for $y > 1$, thus $y' < 0$ and $y(t)$ is monotonically decreasing. Since it is bounded from below by $y(t) = 1$, we have $y(t) \rightarrow 1$. However, for $t \rightarrow -\infty$ we have a blow up in finite time, specifically for $t = \bar{t} < 0$ where (show this by explicit integration of the IVP!)

$$\bar{t} = \ln(1 - y_0) - \ln(y_0).$$

We conclude that the solution $y(t)$ is *bounded* in $[y_0, 0]$ for any choice of the initial condition $y_0 \geq 0$ and $t \geq 0$. (Why we exclude $y_0 < 0$?) Thus, we can apply the above theorem with $a = 0$ and $b = \infty$, concluding that $J = (0, \infty)$. For $t < 0$, the solution is bounded for $y_0 \in [0, 1]$, thus $J = \mathbb{R}$. However, for $y_0 > 1$, we have $J = (\bar{t}, \infty)$.

A.4 Linear ODEs

A general linear ODE is an equation of the form:

$$\mathbf{y}' = \mathbf{A}(t)\mathbf{y} + \mathbf{b}(t)$$

with

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad \mathbf{A}(t) = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ a_n(t) & a_{n-1}(t) & \cdots & a_2(t) & a_1(t) \end{pmatrix}, \quad \mathbf{b}(t) = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ b(t) \end{pmatrix}.$$

Note that $\mathbf{f}(t, \mathbf{y}) = \mathbf{A}(t)\mathbf{y} + \mathbf{b}(t)$ is only linear in $\mathbf{y}$, that is

$$\mathbf{f}(t, \alpha \mathbf{y} + \mathbf{z}) = \alpha \mathbf{f}(t, \mathbf{y}) + \mathbf{f}(t, \mathbf{z}),$$

for all choices of $\alpha \in \mathbb{R}$ and $\mathbf{y}, \mathbf{z} \in \mathbb{R}^n$.

A.4.1 Well-posedness

Theorem A.5. *Assuming that $\mathbf{A} \in \mathcal{C}^0([\alpha, \beta]; \mathbb{R}^{n \times n})$ and $\mathbf{b} \in \mathcal{C}^0([\alpha, \beta]; \mathbb{R}^n)$, that is all coefficients are continuous function of $t \in [\alpha, \beta]$, then we have a unique global solution $\phi \in \mathcal{C}^1((\alpha, \beta); \mathbb{R}^n)$ for each choice of the initial data.*

The proof of the local well-posedness follows for the Cauchy-Lipschitz theorem, because $\mathbf{f}(t, \mathbf{y})$ is locally Lipschitz uniformly in t . In fact, since $\mathbf{A}(t)$ is continuous on a closed interval, it is also bounded. Thus:

$$\|\mathbf{f}(t, \mathbf{y}) - \mathbf{f}(t, \mathbf{z})\| = \|\mathbf{A}(t)(\mathbf{y} - \mathbf{z})\| \leq \|\mathbf{A}(t)\| \|\mathbf{y} - \mathbf{z}\| \leq \underbrace{\max_{t \in [\alpha, \beta]} \|\mathbf{A}(t)\|}_L \|\mathbf{y} - \mathbf{z}\|.$$

The global existence follows from Theorem A.4.

$$\|\mathbf{f}(t, \mathbf{y})\| = \|\mathbf{A}(t)\mathbf{y} + \mathbf{b}(t)\| \leq \|\mathbf{A}(t)\| \|\mathbf{y}\| + \|\mathbf{b}(t)\| \leq \underbrace{\max_{t \in [\alpha, \beta]} \|\mathbf{A}(t)\|}_{k_1} \|\mathbf{y}\| + \underbrace{\max_{t \in [\alpha, \beta]} \|\mathbf{b}(t)\|}_{k_2}.$$

A.4.2 Solution of the homogeneous problem

An interesting aspect of linear ODEs is that we can represent explicitly all the solutions of the ODE. Obviously, a linear IVP has a unique solution. The ODE, on the other hand, can have many solutions. We start with the *homogeneous problem*:

$$\mathbf{y}' = \mathbf{A}(t)\mathbf{y}.$$

Theorem A.6. Let $\mathcal{U} = \{\phi \in \mathcal{C}^1((\alpha, \beta); \mathbb{R}^n) : \phi'(t) = \mathbf{A}(t)\phi(t) \ \forall t \in (\alpha, \beta)\}$ be the set of solutions of the linear ODE. Then $\dim \mathcal{U} = n$.

Therefore, there exists a basis $\{\phi_1(t), \dots, \phi_n(t)\}$ of $\mathcal{U}$ such that each solution $\phi(t) \in \mathcal{U}$ reads as follows:

$$\phi(t) = \sum_{i=1}^n c_i \phi_i(t),$$

for some choice of $\mathbf{c} = [c_1, \dots, c_n]^T \in \mathbb{R}^n$.

Definition A.7 (Wronskian matrix). We define the Wronskian matrix $\mathbf{W}(t)$ as the column-matrix of the basis of $\mathcal{U}$:

$$\mathbf{W}(t) = \begin{bmatrix} \phi_1(t) & | & \phi_2(t) & | & \dots & | & \phi_n(t) \end{bmatrix}.$$

Note that from its definition it follows that:

$$\mathbf{W}(t)' = \mathbf{A}(t)\mathbf{W}(t).$$

The *general solution* of the linear ODE is:

$$\phi(t) = \mathbf{W}(t)\mathbf{c},$$

for some $\mathbf{c} \in \mathbb{R}^n$. For an IVP with initial condition $\mathbf{y}(t_0) = \mathbf{y}_0$, we have that:

$$\phi(t_0) = \mathbf{W}(t_0)\mathbf{c} = \mathbf{y}_0, \quad \Rightarrow \quad \mathbf{c} = \mathbf{W}^{-1}(t_0)\mathbf{y}_0,$$

thus the unique solution is:

$$\phi(t) = \mathbf{W}(t)\mathbf{c} = \mathbf{W}(t)\mathbf{W}^{-1}(t_0)\mathbf{y}_0 = \mathbf{W}(t, t_0)\mathbf{y}_0,$$

where $\mathbf{W}(t, t_0)$ is called *transition matrix*. The transition matrix bears this name because it *transfers* the initial condition $\mathbf{y}_0$ at time t_0 to the solution at time t . It is a particular case of *flow* of an ODE. We have the following very useful properties, all straightforward to prove using the definition above:

1. $\mathbf{W}(t_0, t_0) = \mathbf{I}$,
2. $\mathbf{W}(t, s)\mathbf{W}(s, t_0) = \mathbf{W}(t, t_0)$,
3. $(\mathbf{W}(t, t_0))^{-1} = \mathbf{W}(t_0, t)$.

Note that we haven't proved the invertibility of $\mathbf{W}(t_0)$. We are going to do it later for the case of $\mathbf{A}$ with constant coefficients.

! Exercise

Prove the properties of the transfer matrix.

A.4.3 Solution of the general problem

The general solution is useful for building a *particular solution* for the non-homogeneous problem

$$\mathbf{y}' = \mathbf{A}(t)\mathbf{y} + \mathbf{b}(t). \quad (\text{A.3})$$

Starting from the solution $\phi(t) = \mathbf{W}(t)\mathbf{c}$, we can apply the method of variation of constants to find a solution of the form $\phi(t) = \mathbf{W}(t)\mathbf{c}(t)$, for some choice of $\mathbf{c}(t)$. It turns out that the *particular solution* is:

$$\phi(t) = \mathbf{W}(t) \int_{t_0}^t \mathbf{W}^{-1}(s)\mathbf{b}(s)ds = \int_{t_0}^t \mathbf{W}(t,s)\mathbf{b}(s)ds.$$

Supplemented with an initial condition $\mathbf{y}(t_0) = \mathbf{y}_0$, the solution of the IVP is

$$\phi(t) = \mathbf{W}(t, t_0)\mathbf{y}_0 + \int_{t_0}^t \mathbf{W}(t, s)\mathbf{b}(s)ds. \quad (\text{A.4})$$

! Exercise

Show that (Equation A.4) is the solution of the IVP (Equation A.3).

A.4.4 Matrix exponential

We focus now on the following case:

$$\begin{cases} \mathbf{y}' = \mathbf{A}\mathbf{y} + \mathbf{b}(t), \\ \mathbf{y}(t_0) = \mathbf{y}_0. \end{cases}$$

We need the Wronskian matrix. For computing it, we use the fact that $\mathbf{W}' = \mathbf{A}\mathbf{W}$ and $\mathbf{W}(0) = \mathbf{I}$ (identity matrix). Formally, notice that

$$\begin{aligned} \mathbf{W}(0) &= \mathbf{I}, \\ \mathbf{W}'(0) &= \mathbf{A}\mathbf{W} = \mathbf{A}, \\ \mathbf{W}''(0) &= \mathbf{A}\mathbf{W}' = \mathbf{A}^2, \\ &\dots \\ \mathbf{W}^{(k)}(0) &= \mathbf{A}^k. \end{aligned}$$

Thus we can construct the solution as a Taylor expansion:

$$\mathbf{W}(t) = \sum_{n=0}^{\infty} \frac{\mathbf{W}^{(n)}(0)}{n!} t^n = \sum_{n=0}^{\infty} \frac{\mathbf{A}^n}{n!} t^n = \mathbf{I} + t\mathbf{A} + \frac{t^2}{2}\mathbf{A}^2 + \dots$$

Definition A.8 (Matrix exponential). We define the *matrix exponential* as:

$$e^{\mathbf{A}} := \sum_{n=0}^{\infty} \frac{\mathbf{A}^n}{n!}.$$

Thus, the Wronskian is:

$$\mathbf{W}(t) = e^{t\mathbf{A}}.$$

The series converges for each $\mathbf{A} \in \mathbb{R}^{n \times n}$. Below some useful properties:

1. If $\mathbf{A} = \text{diag}\{a_1, \dots, a_n\}$, then $e^{\mathbf{A}} = \text{diag}\{e^{a_1}, \dots, e^{a_n}\}$. The matrix exponential of diagonal matrices is then straightforward to compute. The proof is simple: just plug the diagonal matrix in the definition and note that $\mathbf{A}^k = \text{diag}\{a_1^k, \dots, a_n^k\}$. Hence:

$$\sum_{k=0}^{\infty} \frac{\mathbf{A}^k}{k!} = \text{diag}\left\{\sum_{k=0}^{\infty} \frac{a_1^k}{k!}, \dots, \sum_{k=0}^{\infty} \frac{a_n^k}{k!}\right\} = \text{diag}\{e^{a_1}, \dots, e^{a_n}\}.$$

2. $e^{\mathbf{0}} = \mathbf{I}$. If $\mathbf{A}$ is the matrix with all zero entries $\mathbf{0}$, then the matrix exponential is the identity matrix $\mathbf{I}$. This is trivial because all the entries of the sum are zero except for $k = 0$.
3. $\det(e^{\mathbf{A}}) = e^{\text{tr} \mathbf{A}}$. This is reminiscent of the Wronskian (determinant of Wronskian matrix), and the proof is similar, based on the Taylor expansion $|\mathbf{I} + \varepsilon \mathbf{A}| = 1 + \varepsilon \text{tr} \mathbf{A} + \mathcal{O}(\varepsilon^2)$. We use the alternative definition of matrix exponential, and the fact that the determinant is a continuous function with respect to the coefficients of the matrix:

$$\begin{aligned} |e^{\mathbf{A}}| &= \left| \lim_{n \rightarrow \infty} \left(\mathbf{I} + \frac{1}{n} \mathbf{A} \right)^n \right| \\ (\text{continuity}) &= \lim_{n \rightarrow \infty} \left| \left(\mathbf{I} + \frac{1}{n} \mathbf{A} \right)^n \right| \\ (\text{det of product}) &= \lim_{n \rightarrow \infty} \left| \mathbf{I} + \frac{1}{n} \mathbf{A} \right|^n \\ (\text{det expansion}) &= \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \text{tr} \mathbf{A} + \mathcal{O}\left(\frac{1}{n^2}\right) \right)^n = e^{\text{tr} \mathbf{A}}. \end{aligned}$$

4. $e^{\mathbf{A}}$ is invertible. This is a consequence of the previous property, since the determinant is always strictly positive.

5. If $\mathbf{A}$ and $\mathbf{B}$ commute, that is $\mathbf{AB} = \mathbf{BA}$, then $e^{\mathbf{A}+\mathbf{B}} = e^{\mathbf{A}}e^{\mathbf{B}}$. We are not proving this fact.
6. $e^{0\mathbf{A}} = \mathbf{I}$. This follows immediately from property 2. of the matrix exponential.
7. $e^{t\mathbf{A}}e^{s\mathbf{A}} = e^{(t+s)\mathbf{A}}$. In this case we could use property 5., but instead we use the formula for the [product of two series](#) and [Newton's binomial formula](#):

$$\begin{aligned}
e^{t\mathbf{A}}e^{s\mathbf{A}} &= \left(\sum_{k=0}^{\infty} \frac{t^k \mathbf{A}^k}{k!} \right) \cdot \left(\sum_{j=0}^{\infty} \frac{s^j \mathbf{A}^j}{j!} \right) \\
&= \sum_{k=0}^{\infty} \sum_{l=0}^k \frac{t^l \mathbf{A}^l}{l!} \frac{s^{k-l} \mathbf{A}^{k-l}}{(k-l)!} \\
&= \sum_{k=0}^{\infty} \frac{\mathbf{A}^k}{k!} \sum_{l=0}^k \frac{k!}{l!(k-l)!} t^l s^{k-l} \\
&= \sum_{k=0}^{\infty} \frac{(t+s)^k \mathbf{A}^k}{k!} = e^{(t+s)\mathbf{A}}.
\end{aligned}$$

8. $\frac{d}{dt}e^{t\mathbf{A}} = \mathbf{A}e^{t\mathbf{A}}$. It follows from the definition:

$$\frac{d}{dt}e^{t\mathbf{A}} = \sum_{k=0}^{\infty} \frac{d}{dt} \frac{t^k \mathbf{A}^k}{k!} = \sum_{k=1}^{\infty} \frac{k t^{k-1} \mathbf{A}^k}{k!} = \mathbf{A} \sum_{k=1}^{\infty} \frac{t^{k-1} \mathbf{A}^{k-1}}{(k-1)!} = \mathbf{A}e^{t\mathbf{A}}.$$

This last property in particular shows rigorously that the matrix exponential is the Wronskian matrix of the ODE $\mathbf{y}' = \mathbf{A}\mathbf{y}$, while with the other properties we define the transition matrix.

A.4.5 Solution of the case with constant coefficients

The solution of

$$\begin{cases} \mathbf{y}' = \mathbf{A}\mathbf{y} + \mathbf{b}(t), \\ \mathbf{y}(t_0) = \mathbf{y}_0, \end{cases}$$

follows now trivially from the matrix exponential. We have:

$$\phi(t) = e^{(t-t_0)\mathbf{A}}\mathbf{y}_0 + \int_{t_0}^t e^{(t-s)\mathbf{A}}\mathbf{b}(s)ds.$$

i Example (scalar case)

When $n = 1$, with $\mathbf{A} = a \in \mathbb{R}$, the matrix exponential is the usual exponential function. The solution is

$$\phi(t) = e^{a(t-t_0)}y_0 + \int_{t_0}^t e^{a(t-s)}b(s)ds.$$

For instance, with $b = e^{\omega t}$, we have:

$$\int_{t_0}^t e^{a(t-s)}b(s)ds = \int_{t_0}^t e^{a(t-s)}e^{\omega s}ds = \frac{e^{at} - e^{\omega t}}{a - \omega}.$$

So the solution to the problem:

$$\begin{cases} y' = ay + e^{\omega t}, \\ y(t_0) = y_0, \end{cases}$$

is as follows:

$$\phi(t) = e^{a(t-t_0)}y_0 + \frac{e^{at} - e^{\omega t}}{a - \omega}.$$

A.5 Dynamical systems

Intuitively, a dynamical system is a state or a set of variables evolving over time according to some rule. The concept is very general. [Markov chains](#), [cellular automata](#) and [large language models](#) are examples of dynamical systems. Roughly speaking, a dynamical system is composed by

1. A *state space*, the set of all admissible states of the system, and
2. A *rule* for moving from the current state to the next one.

We can associate to a system of ODEs a dynamical system. The state space (called *phase space*) is the subset of $\mathbb{R}^n$ of all possible states. The update rule is the solution of the ODE (called *flow*). For ODEs, one is not interested in studying a specific instance (or trajectory) of the system, but rather the system as a whole, for all possible initial states.

Definition A.9. A dynamical system is well-posed autonomous ODE (*autonomous* means that $\mathbf{f}$ is not an explicit function of t). Specifically, given $\mathbf{f} \in \mathcal{C}^1(\Omega; \mathbb{R}^n)$, with $\Omega \neq \emptyset$ an open set of $\mathbb{R}^n$, we suppose that for every initial condition $\mathbf{y}_0 \in \mathbb{R}^n$, the IVP

$$\begin{cases} \mathbf{y}' = \mathbf{f}(\mathbf{y}), \\ \mathbf{y}(0) = \mathbf{y}_0, \end{cases}$$

has a unique solution $\phi(t)$ defined for $t \in [0, \infty)$, and such that $\phi(t) \in \Omega$ for all $t \geq 0$. We call $\mathbf{y}' = \mathbf{f}(\mathbf{y})$ *dynamical system* on the *phase space* Ω . Succintly, the dynamical system is fully qualified by the pair $(\mathbf{f}, \Omega)$.

Definition A.10. The *flow* of an ODE is a function $\Phi: \mathbb{R} \times \Omega \rightarrow \Omega$ such that

$$\Phi(t, \mathbf{y}_0) = \phi(t),$$

where $\phi(t)$ is the solution of the ODE with initial condition $\mathbf{y}_0$.

For example, for a linear dynamical system $\mathbf{y}' = \mathbf{A}\mathbf{y}$ the flow is given by the transition matrix:

$$\Phi(t, \mathbf{y}_0) = \mathbf{W}(t, 0)\mathbf{y}_0 = e^{t\mathbf{A}}\mathbf{y}_0.$$

Note. It is worth to mention that also the general, non-autonomous ODEs $\mathbf{y}' = \mathbf{f}(t, \mathbf{y})$ in $\mathbb{R}^n$ can be recast to the autonomous form $\mathbf{z}' = \mathbf{g}(\mathbf{z})$ in $\mathbb{R}^{n+1}$ by setting

$$\mathbf{z}(t) = \begin{bmatrix} \tau(t) \\ \mathbf{y}(t) \end{bmatrix}, \quad \mathbf{g}(\mathbf{z}) = \begin{bmatrix} 1 \\ \mathbf{f}(\tau, \mathbf{y}) \end{bmatrix},$$

in fact, $\tau(t) = t$.

A.5.1 Orbits and trajectories

Definition A.11. For a given solution (or trajectory) $\phi(t)$, we consider the associated *orbit* $\mathcal{O}(\phi)$ defined as follows

$$\mathcal{O}(\phi) = \bigcup_{t \geq 0} \phi(t),$$

that is, the orbit is the “shadow” of a trajectory onto the phase space Ω .

In practice, the study of a dynamical system is not limited to a particular solution (defined by some initial condition), but rather it covers the entire domain. In some sense, there is a strong geometrical interpretation: the r.h.s. of the ODE determines a vectorial field in the phase space Ω , so that a given orbit is tangent at every point to such field.

Proposition A.1. *If $\phi(t)$ is a trajectory of the dynamical system for all $t \geq 0$, also $\psi(t) = \phi(t + c)$ is a solution for all $t \geq -c$, with $c \in \mathbb{R}$.*

The proposition is a simple consequence of the fact that a dynamical system is an autonomous ODE, that is it $\mathbf{f}$ is not a function of time.

Proposition A.2. *Two distinct orbits of a dynamical system either coincide or they never intersect.*

This proposition is extremely important when analyzing the phase portrait of a system, because some orbits may act as “barriers” for other orbits. Please note that the statement refers to *orbits*, not simply to the trajectories.

To prove the proposition, suppose there are two distinct orbits associated to the trajectories ϕ and ψ , and such that $\phi(t_1) = \psi(t_2)$ for some $t_1, t_2 \geq 0$. That is, the orbits intersect. Now consider the function $\tilde{\psi}(t) = \psi(t + t_2 - t_1)$. This function is still a solution, and $\tilde{\psi}(t_1) = \psi(t_2) = \phi(t_1)$. Thus, $\tilde{\psi}$ and ϕ have the same initial condition, and they must coincide for all t , that is $\tilde{\psi}(t) = \phi(t)$. In other words, $\tilde{\psi}$ and ϕ have the same orbit. But also $\tilde{\psi}$ and ψ have the same orbit, because the system is autonomous. In conclusion, if two orbits intersect at a least one point, they must coincide at all points.

i Example (Van der Pol)

Below an example of the phase space for the equation $y'' - \mu(1 - y^2)y' + y = 0$, which models a stiff oscillating system.

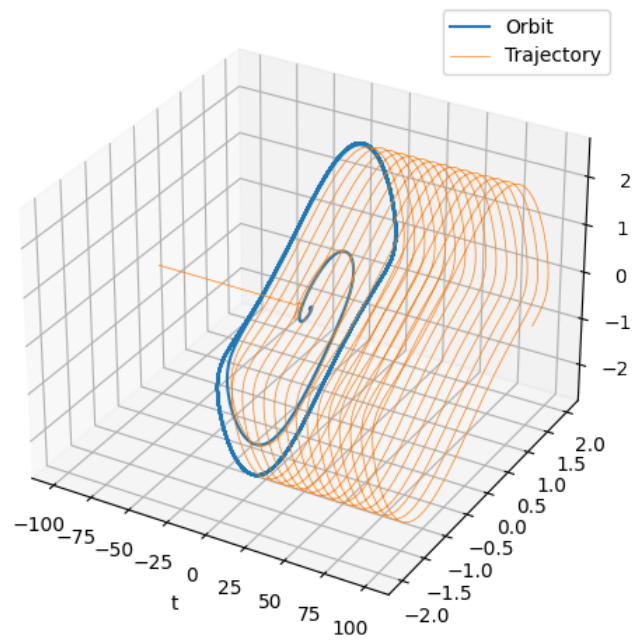


Figure A.2: Trajectory

In the figure, in blue we have the orbit associated to the trajectory in orange.

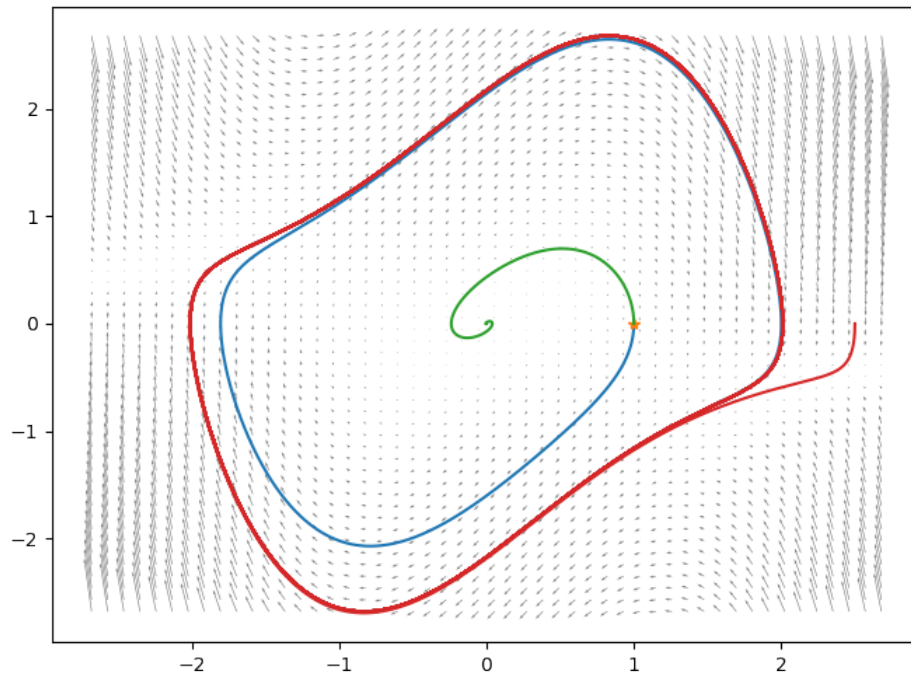


Figure A.3: Phase space

This second figure represents the phase portrait of the system, with the orbit and the limit cycle.

i Example (positive systems)

An example are dynamical systems of the form:

$$\begin{cases} y_1' = y_1 \cdot f(y_1, y_2), \\ y_2' = y_2 \cdot g(y_1, y_2). \end{cases}$$

In fact, the line $(y_1, y_2) = (0, y_2)$ (the ordinate) is an orbit of the system, because if we start from $(0, y_2)$ we never leave the line, since $y_1' = 0$. Similarly, the line $(y_1, y_2) = (y_1, 0)$ (the abscissa) is also an orbit. Therefore, any other orbit starting in the positive quadrant will stay in the positive quadrant for all times. That is, the two orbits are barriers that ensures the positivity of the system for all times.

A.5.2 Types of orbits

Some orbits of a dynamical system are special. Here, we only mention the two most important ones: equilibria and periodic orbits.

Definition A.12. We say that $\mathbf{y}_0 \in \Omega$ is an *equilibrium* of the system if $\mathbf{f}(\mathbf{y}_0) = 0$. In particular, the trajectory $\phi(t) = \mathbf{y}_0$ for all $t \in \mathbb{R}$ is called solution of the equilibrium. The orbit is simply the point $\{\mathbf{y}_0\}$.

What if an orbit self-intersect at some finite time? We have a *periodic orbit*.

Proposition A.3. *If there exist $\tau_1, \tau_2 \geq 0$ such that $\phi(\tau_1) = \phi(\tau_2)$, then $\phi(t)$ is a periodic solution.*

In fact, as above, consider the trajectory $\psi(t) = \phi(t - \tau_1 + \tau_2)$. Then $\psi(\tau_1) = \phi(\tau_2) = \phi(\tau_1)$. So we conclude that $\psi(t)$ and $\phi(t)$ coincide for all time. In particular, $\phi(t) = \phi(t + \tau_2 - \tau_1)$, that is, $\phi(t)$ is periodic with period $\tau_2 - \tau_1$.

The kind of orbits that we may find in a dynamical system is actually limited to

1. Orbits consisting of a single point (equilibria);
2. Orbits corresponding to closed regular curves (periodic solutions);
3. Orbits corresponding to open regular curves, with no self-intersections.

The dimension of the phase space plays a key role in the type of attractor. For $\Omega \subset \mathbb{R}^n$, we can only have equilibria. In fact:

Proposition A.4. *For $n = 1$ a dynamical system cannot have periodic solutions.*

For the proof, suppose that $\phi(t)$ is periodic with period $T > 0$. Then, we multiply by ϕ' both sides of the ODE and integrate in $[t, t + T]$:

$$\int_t^{t+T} (\phi'(s))^2 ds = \int_t^{t+T} f(\phi(s))\phi'(s) ds = \int_{\phi(t)}^{\phi(t+T)} f(u) du = 0,$$

where we applied the change of variables $u = \phi(s)$. The last integral is zero because $\phi(t) = \phi(t + T)$. But the first integral is strictly positive, so we have a contradiction.

Note. For $n = 1$, a more general non-autonomous ODE $y' = f(t, y)$ can have periodic solutions. But this is not a dynamical system, unless we recast it as a system, thus $n = 2$ and periodic solutions are possible.

i Example (test problem)

The dynamical system $y' = \lambda y$ has only one equilibrium, $y^* = 0$.

i Example (logistic equation)

The dynamical system $y' = y(1 - y)$ has two equilibria, $y_1 = 0$ and $y_2 = 1$.

i Example (linear system)

Given $\mathbf{A} \in \mathbb{R}^{n \times n}$, the dynamical system $\mathbf{y}' = \mathbf{A}\mathbf{y}$ has equilibria such that $\mathbf{A}\mathbf{y}^* = 0$, that is equilibria belong to the kernel of $\mathbf{A}$, $\mathbf{y}^* \in \ker \mathbf{A}$. If $\mathbf{A}$ is invertible, then we only have $\mathbf{y}^* = 0$.

i Example (FitzHugh-Nagumo model)

We consider the dynamical system defined by the ODE

$$\begin{cases} u' = f(u, r) = u(1 - u)(u - \alpha) - r, \\ r' = g(u, r) = \beta(u - \gamma r), \end{cases}$$

where $\alpha, \beta, \gamma \in \mathbb{R}$ are parameters of the model. This system is an important phenomenological model of excitability of cells and for modeling the action potential.

We define as *nullclines* associated to u' (resp. r') the implicit curve defined by the equation $f(u, r) = 0$ (resp. $g(u, r) = 0$). Equilibria are found as intersections between nullclines, because at those points all right hand sides simultaneously cancel. For the FitzHugh-Nagumo system, nullclines are:

$$\begin{cases} f(u, r) = 0, \Leftrightarrow r = u(1 - u)(u - \alpha), \\ g(u, r) = 0, \Leftrightarrow r = \gamma^{-1}u. \end{cases}$$

The first nullcline is a cubic function with zeros at 0, α , and 1, and such that $r \rightarrow -\infty$ for $u \rightarrow \infty$. The second nullcline is a line through the origin and with slope γ^{-1} . One equilibrium is certainly the point $(u_1, r_1) = (0, 0)$, for all choices of the parameters. The other equilibria depend on the choice of the parameters, specifically on whether the cubic function and the line intersect outside the origin. Those are such that

$$u(1 - u)(u - \alpha) = \gamma^{-1}u,$$

so, besides $u = 0$, the number of zeros depends on the sign of the discriminant:

$$\Delta = (\alpha + 1)^2 - 4\gamma^{-1}.$$

In conclusion, we have three equilibria for $\Delta > 0$, two equilibria for $\Delta = 0$ and only one for $\Delta < 0$.

A.5.3 Lyapunov stability and attractors

Consider a dynamical system $(\mathbf{f}, \Omega)$ with $\Omega \subseteq \mathbb{R}^n$, with an equilibrium $\mathbf{y}_0 \in \Omega$, that is $\mathbf{f}(\mathbf{y}_0) = 0$. How does the system behave in a neighborhood of the equilibrium? Say, if we start close to $\mathbf{y}_0$, do the stay close to it for long? Please note that the interest is purely qualitative: we are not interested in the specific form of the trajectory, but rather its behavior for $t \rightarrow \infty$.

Definition A.13 (Lyapunov stability). We say that $\mathbf{y}_0$ is a *locally stable equilibrium* if for each $\varepsilon > 0$ there exists $\delta = \delta(\varepsilon, \mathbf{y}_0)$ such that for all $\mathbf{y}_1 \in \Omega$ with $\|\mathbf{y}_1 - \mathbf{y}_0\| < \delta$ we have that $\|\Phi(t, \mathbf{y}_1) - \mathbf{y}_0\| < \varepsilon$ for all $t \geq 0$, where $\Phi(t, \mathbf{y})$ is the flow of the system.

Moreover, we say that $\mathbf{y}_0$ is *asymptotically stable* if in the above definition we have $\|\phi(t; \mathbf{y}_1) - \mathbf{y}_0\| \rightarrow 0$ for $t \rightarrow \infty$.

The difference between simple stability and asymptotic stability is that in the former case orbits stay close to the equilibrium without necessarily approaching it for $t \rightarrow \infty$. For instance, the vertical downward position of the frictionless pendulum is only stable, because the orbits of the systems (oscillations at given amplitude) stay close to it with distance equal to the amplitude but never approach it. On the other hand, in the presence of friction, the equilibrium becomes asymptotically stable. (Thermodynamically speaking, asymptotically stable equilibria are quite boring: it often means extinction, death.)

Finally, an *unstable equilibrium* is defined by (logically) negating the above definition. That is, $\mathbf{y}_0 \in \Omega$ is an *unstable equilibrium* if there exists $\varepsilon > 0$ such that for all $\delta > 0$ there exists $\mathbf{y}_1 \in \Omega$ with the property that $\|\mathbf{y}_1 - \mathbf{y}_0\| < \delta$ but $\|\phi(t_n, \mathbf{y}_1) - \mathbf{y}_0\| \geq \varepsilon$ for a sequence $\{t_n\}_n \rightarrow \infty$ and all $n \in \mathbb{N}$. In other words, there exists at least one initial condition that brings the associated trajectory arbitrarily far from the equilibrium at some time points t_n approaching infinity. (It would be too much to ask the same for all $t \geq M$, because we could have a diverging trajectory that periodically comes very close to the equilibrium, without really approaching it. Take for instance the function $\phi(t) = t \sin^2 t + \varepsilon/2$, where for $t = t_n = n\pi$ is equal to $\varepsilon/2$, but for $t_n = n\pi/2$ is diverging: it is clearly unstable.)

An interesting concept associated with equilibria is that of *basin of attraction*.

Definition A.14 (Basin of attraction). If an equilibrium is only *locally asymptotically stable*, then we have the basin of attraction defined as

$$\mathcal{B}(\mathbf{y}_0) := \left\{ \tilde{\mathbf{y}} \in \Omega : \lim_{t \rightarrow +\infty} \Phi(t, \mathbf{y}_0) = \mathbf{y}_0 \right\}.$$

An equilibrium is *globally asymptotically stable* when $\mathcal{B}(\mathbf{y}_0) = \Omega$.

Remark. Lyapunov stability is different from the concept of stability to perturbation, or zero stability (see Corollary A.1). In the latter, the aim is check whether we are able to recover the original solution (equilibrium or not) as the perturbation in the dynamical system goes

to zero. The original and perturbed solutions may diverge from each other for long time, but we can always reduce the gap between them by reducing the initial error (the perturbation): when that's not possible, the system is not (zero) stable. Lyapunov stability, on the other hand, concerns the *structural* stability properties of the system, that is we study the long-term behavior of solutions *without* controlling the initial perturbation. Actually, all initial conditions in the basin of attraction of an equilibrium yield solutions that converge to the same equilibrium, irrespective of the gap between them.

i Example (stability of the test problem)

The stability of equilibrium $y = 0$ of $y' = \lambda y$ depends clearly on λ . Suppose that $\lambda \in \mathbb{C}$. Then, the full solution is of the form

$$\phi(t) = e^{\lambda t} y_0.$$

We can expand the exponential to get more insights:

$$\phi(t) = e^{\lambda t} y_0 = e^{t \operatorname{Re} \lambda} e^{t \operatorname{Im} \lambda} y_0 = e^{t \operatorname{Re} \lambda} (\cos(t \operatorname{Im} \lambda) + i \sin(t \operatorname{Im} \lambda)) y_0.$$

Since the equilibrium is 0, we just need to check whether this trajectory, in modulus, stays close (or even approaches) zero over time. That is,

$$|\phi(t)|^2 = \phi^*(t) \phi(t) = e^{2t \operatorname{Re} \lambda} (\cos(t \operatorname{Im} \lambda)^2 + \sin(t \operatorname{Im} \lambda)^2) |y_0|^2,$$

so we have:

$$|\phi(t)| = e^{t \operatorname{Re} \lambda} |y_0|.$$

From this expression, we easily deduce that

1. If $\operatorname{Re} \lambda < 0$, then $|\phi(t)| \rightarrow 0$ for all $y_0 \in \mathbb{C}$. The equilibrium is therefore globally asymptotically stable.
2. If $\operatorname{Re} \lambda > 0$, then $|\phi(t)| \rightarrow \infty$ for at least one non trivial $y_0 \in \mathbb{C}$. The equilibrium is therefore unstable.
3. If $\operatorname{Re} \lambda = 0$, then $|\phi(t)| = |y_0|$. The equilibrium is stable.

i Example (stability of logistic equation)

The logistic equation $y' = y(1 - y)$ has two equilibria, $y_1 = 0$ and $y_2 = 1$. We could study their stability by taking advantage of the analytical solution, available in this case, but we will not. We will proceed more generally. The dynamics is determined by the sign of $f(y) = y(1 - y)$.

1. If $y_0 = y_1$ or $y_0 = y_2$, there is no dynamics over time.
2. If $y_0 \in (0, 1)$ (boundaries excluded), then the solution $\phi(t; y_0)$ will never leave the

interval $(0, 1)$, because 0 and 1 are barriers (equilibria are also orbits, and orbits cannot intersect). Moreover, in this region $f(y) > 0$. Therefore $y' > 0$, that the solution $\phi(t; y_0)$ increases over time. Since the equilibrium $y_2 = 1$ cannot be crossed, $\phi(t; y_0)$ indefinitely approaches y_2 (from the left) without crossing it.

3. If $y_0 > 1$, with the same reasoning as above we conclude that $\phi(t; y_0) > 1$ indefinitely. But here $f(y) < 0$, so $y' < 0$ and again $\phi(t; y_0)$ approaches y_2 (from the right).
4. If $y_0 < 0$, $y' = f(y) < 0$ and $\phi(t; y_0) \rightarrow -\infty$.

With the above analysis, we can easily conclude that

1. $y_1 = 0$ is *unstable*, because orbits diverge from it.
2. $y_2 = 0$ is *asymptotically stable*, but only *locally*, with $\mathcal{B}(y_2) = \{y > 0\}$.

When restricting the dynamical system to $\Omega = \mathbb{R}^+$, as in the case of population dynamics, the equilibrium y_2 is almost globally attractive, except when we start from $y_1 = 0$.

Please note that the above argument is very general, as it can be straightforwardly applied to *any* 1-D dynamical system: it is enough to study the sign and the zeros of $f(y)$.

A.5.4 Stability of linear ODEs

A.5.4.1 Computation of the matrix exponential

The computation of the matrix exponential is not a trivial task, in general. In some cases, however, it is practical. When $\mathbf{A}$ is diagonal we have seen that the matrix exponential is trivially the element-wise exponential.

When $\mathbf{A}$ is diagonalizable, that is there exists a matrix $\mathbf{S}$ such that

$$\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \Lambda,$$

with Λ diagonal matrix, the exponential matrix follows immediately from the following observation:

$$(\mathbf{S}\mathbf{A}\mathbf{S}^{-1})^k = (\mathbf{S}\mathbf{A}\mathbf{S}^{-1})(\mathbf{S}\mathbf{A}\mathbf{S}^{-1}) \dots (\mathbf{S}\mathbf{A}\mathbf{S}^{-1}) = \mathbf{S}\Lambda^k\mathbf{S}^{-1}.$$

In fact, we have

$$e^{\mathbf{A}} = \sum_{k=0}^{\infty} \frac{\mathbf{A}^k}{k!} = \sum_{k=0}^{\infty} \frac{(\mathbf{S}\mathbf{A}\mathbf{S}^{-1})^k}{k!} = \mathbf{S} \sum_{k=0}^{\infty} \frac{\Lambda^k}{k!} \mathbf{S}^{-1} = \mathbf{S}e^{\Lambda}\mathbf{S}^{-1}.$$

The matrix e^{Λ} is easy to compute, using the above formula (at least when it is easy to compute eigenvalues and eigenvectors.)

In the general case, when $\mathbf{A}$ is not diagonalizable, the computation is not straightforward. It is based on the so-called [Jordan canonical form](#). In practice, it is always possible to find a matrix $\mathbf{S}$ such that $\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \mathbf{J}$ is in the Jordan canonical form, that is $\mathbf{J}$ is a block diagonal matrix, each block with a specific structure. The matrix exponential of the Jordan canonical form is again a block diagonal matrix. The matrix exponential of each block can be computed explicitly.

The Jordan canonical form is as follows:

$$\mathbf{J} = \begin{pmatrix} \mathbf{J}_1 & & \\ & \ddots & \\ & & \mathbf{J}_r \end{pmatrix}$$

where $\mathbf{J}_i$ is a matrix of the form:

$$\mathbf{J}_i = \begin{pmatrix} \lambda_i & 1 & & \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{pmatrix}$$

The eigenvalues λ_i of the matrix $\mathbf{A}$ appears on the diagonal of the Jordan block $\mathbf{J}_i$. If the matrix is diagonalizable, then there are exactly n Jordan blocks each of dimension 1. In fact, the Jordan canonical form is diagonal. If some eigenvalues have geometric multiplicity strictly less than algebraic multiplicity, then the matrix is not diagonalizable. The Jordan blocks compensate for the difference in multiplicity. For instance, consider the matrix

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}.$$

The eigenvalues are $\lambda_1 = \lambda_2 = 1$. The algebraic multiplicity is 2, but the geometric multiplicity, that is the dimension of the eigenspace $E_\lambda = \ker(\mathbf{A} - \lambda\mathbf{I})$ associated to $\lambda = 1$, is only 1. The Jordan canonical form is

$$\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The matrix exponential of the Jordan block follows from the property (for a 2×2 block):

$$\begin{pmatrix} \lambda_i & 1 \\ 0 & \lambda_i \end{pmatrix}^k = \begin{pmatrix} \lambda_i^k & k\lambda_i^{k-1} \\ 0 & \lambda_i^k \end{pmatrix}.$$

So putting everything together we have

$$e^{t\mathbf{J}} = \begin{pmatrix} e^{\lambda_i t} & te^{\lambda_i t} \\ 0 & e^{\lambda_i t} \end{pmatrix}.$$

Please note that the appearance of the term $te^{\lambda_i t}$.

For the sake of completeness, we just recall that the algebraic multiplicity is associated with the characteristic polynomial $\mathcal{P}(\lambda) = \det(\lambda \mathbf{I} - \mathbf{A})$, and corresponds to the number of times a zero of $\mathcal{P}(\lambda)$ appears. More precisely, we can always write $\mathcal{P}(\lambda) = (\lambda - \lambda_1)^{\mu_1} \cdot (\lambda - \lambda_2)^{\mu_2} \cdots (\lambda - \lambda_r)^{\mu_r}$, $\{\lambda_i\}_{i=1}^r$, $r \leq n$, are the eigenvalues, and $1 \leq \mu_i \leq n$ the algebraic multiplicity of λ_i . The geometrical multiplicity μ_i is the dimension of the eigenspace $V_i = \ker(\lambda_i \mathbf{I} - \mathbf{A}) = \{\mathbf{v} \in \mathbb{R}^n : \lambda_i \mathbf{v} = \mathbf{A} \mathbf{v}\}$. Since $\nu_i \leq \mu_i$, when at least one eigenvalue has $\nu_i < \mu_i$, the direct sum of all V_i s does not fill the whole $\mathbb{R}^n$, and additional (generalized) eigenvectors are required. These are taken from $\mathbf{A}^k$, for some $k = 2, 3, \dots$, and leads to terms of the form $t^{k-1}e^{t\lambda_i}$ in the matrix exponential.

A.5.4.2 Stability

For the linear ODEs $\mathbf{y}' = \mathbf{A} \mathbf{y}$, with $\mathbf{A} \in \mathbb{R}^{n \times n}$ invertible, we only have the equilibrium $\mathbf{y}^* = \mathbf{0}$. In the more general case $\mathbf{y}' = \mathbf{A} \mathbf{y} + \mathbf{b}$, the equilibrium is $\mathbf{y}^* = \mathbf{A}^{-1} \mathbf{b}$. Note that in this case we can define $\mathbf{z}(t) = \mathbf{y}(t) - \mathbf{y}^*$ that satisfies the ODE $\mathbf{z}' = \mathbf{A} \mathbf{z}$ with equilibrium $\mathbf{z}^* = \mathbf{0}$. Thus, we can focus on the homogeneous case with no loss of generality.

We now try to characterize the stability of the equilibrium $\mathbf{y}^* = \mathbf{A}$ for $\mathbf{y}' = \mathbf{A} \mathbf{y}$.

Let's first consider the case of $\mathbf{A}$ diagonalizable. Here, there exists an invertible matrix $\mathbf{V} \in \mathbb{R}^{n \times n}$ such that $\mathbf{V}^{-1} \mathbf{A} \mathbf{V} = \Lambda$ is a diagonal matrix. The entries of this matrix are the eigenvalues of $\mathbf{A}$. The general solution of the ODE reads:

$$\phi(t) = e^{t\mathbf{A}} \mathbf{y}_0 = \mathbf{V} e^{t\Lambda} \mathbf{V}^{-1} \mathbf{y}_0.$$

We need to compute

$$\lim_{t \rightarrow \infty} \|\phi(t) - \mathbf{y}^*\|,$$

for an arbitrary initial condition $\mathbf{y}_0$.

Since the matrix exponential of a diagonal matrix is just the component-wise exponentiation, we have that the components of $\mathbf{V}^{-1} \phi$ are a linear combination of terms of the form $e^{\lambda_i t}$, being λ_i the i -th eigenvalue of $\mathbf{A}$. Thus, we have that

1. If $\operatorname{Re} \lambda_i < 0$ for all $i = 1, \dots, n$, then 0 is the only globally attractive equilibrium.
2. If there exists at least one eigenvalue such that $\operatorname{Re} \lambda_i > 0$, the equilibrium 0 is unstable.
3. If $\operatorname{Re} \lambda_i \leq 0$ for all $i = 1, \dots, n$, then 0 is stable.

For a generic $\mathbf{A} \in \mathbb{R}^{n \times n}$ (diagonalizable or not), given the set of eigenvalues λ_i , the solution is some linear combination of terms of the form:

$$e^{\lambda t}, te^{\lambda t}, \dots, t^m e^{\lambda t},$$

depending on the geometric multiplicity of λ . So we have that 1. and 2. above still applies, because the exponential is stronger than any polynomial. The non-trivial case is when $\operatorname{Re} \lambda_i = 0$ for some λ_i . If λ_i is such that we need extra terms of the form $t^j e^{\lambda_i t}$, $j \geq 1$, to complete the solution space, then the equilibrium is clearly unstable, because:

$$t^j e^{\lambda_i t} = t^j (\cos(t \operatorname{Im} \lambda_i) + i \sin(t \operatorname{Im} \lambda_i)) \rightarrow \infty$$

as $t \rightarrow \infty$. Otherwise, the equilibrium is stable (not asymptotically). The extra polynomial terms are required when the geometrical multiplicity of λ_i is strictly lower than its algebraic multiplicity. So, in the general case we replace 3. above with

3. If $\operatorname{Re} \lambda_i \leq 0$ for all $i = 1, \dots, n$, and for those with $\operatorname{Re} \lambda_i = 0$ the algebraic and geometrical multiplicity coincide, then the equilibrium is stable.

Concerning asymptotic stability, it is possible to find algebraic conditions on the coefficients of $\mathbf{A}$ such that $\operatorname{Re} \lambda_i < 0$ for all eigenvalues, *without* actually computing them.

A.5.4.3 The $n = 2$ case

The characteristic polynomial of the matrix

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is as follows

$$\mathcal{P}(\lambda) = \lambda^2 - (a + b)\lambda + (ad - bc) = \lambda^2 - \operatorname{tr}(\mathbf{A})\lambda + \det(\mathbf{A}).$$

where we introduced the *trace* $\operatorname{tr}(\mathbf{A}) = a + b$ and the *determinant* $\det(\mathbf{A}) = ad - bc$.

Proposition A.5. *The condition $\operatorname{Re} \lambda_i < 0$ for all $i = 1, \dots, n$ is equivalent to*

$$\operatorname{tr}(\mathbf{A}) < 0, \quad \text{and} \quad \det(\mathbf{A}) > 0.$$

The proof is simple. We also know that:

$$\begin{aligned} \operatorname{tr}(\mathbf{A}) &= \lambda_1 + \lambda_2, \\ \det(\mathbf{A}) &= \lambda_1 \lambda_2, \end{aligned}$$

because the above polynomial has always 2 roots on $\mathbb{C}$, so it admits a factorization $\mathcal{P}(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2)$ that expanded gives the equality.

Proposition A.5 is one of the most important results for the course, and we will use it very often.

The case $n = 2$ is also very common in applications, thus it is worth studying in depth the equilibria in the phase space. We suppose that the equilibrium is the origin, that is we study the equation $\mathbf{y}' = \mathbf{A}\mathbf{y}$.

i Example (real and distinct eigenvalues)

Suppose that the eigenvalues λ_1 and λ_2 of the matrix $\mathbf{A}$ are real, with eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$. Then the solution is:

$$\mathbf{y}(t) = c_1 e^{t\lambda_1} \mathbf{v}_1 + c_2 e^{t\lambda_2} \mathbf{v}_2,$$

where $\mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \mathbf{V}^{-1} \mathbf{y}_0$ and $\mathbf{V}$ is matrix with the eigenvectors. Thus, the solution is a linear combination of the eigenvectors.

When $\mathbf{y}_0 = \alpha \mathbf{v}_1$, then $c_1 = \alpha$ and $c_2 = 0$, so we stay along the line with direction $\mathbf{v}_1$. Along this line, the solution is

$$\mathbf{y}(t) = \alpha e^{t\lambda_1} \mathbf{v}_1,$$

thus $\mathbf{y}(t)$ stays along the line and, when $\lambda_1 < 0$, approaches the equilibrium for $t \rightarrow \infty$. In this case, we call the space generated by $\mathbf{v}_1$, the *stable manifold* of the equilibrium. Viceversa, for $\lambda_1 > 0$, the trajectory diverges from the equilibrium; in this case, $\mathbf{v}_1$, is the *unstable manifold*.

The same applies to $\mathbf{v}_2$. So, when $\lambda_1 < 0$ and $\lambda_2 < 0$, the equilibrium is globally stable, with stable manifold is simply $\mathbf{R}^2$. The equilibrium is a *stable node*. When the eigenvalues are both positive, we have the opposite behavior, and the equilibrium is an *unstable node*. Finally, when they have opposite sign, the equilibrium is a *saddle*.

A.5.4.4 The $n > 2$ case: Routh-Hurwitz criteria

In general, for $n > 2$, we can find a set of algebraic conditions ensuring the asymptotic stability, *without* computing the eigenvalues.

We have the following

Theorem A.7. *Given the polynomial*

$$\mathcal{P}(\lambda) = \lambda^n + a_1 \lambda^{n-1} + \dots + a_{n-1} \lambda + a_n,$$

where the coefficients a_i are real constants, $i = 1, \dots, n$, define the n Hurwitz matrices using the coefficients a_i of the characteristic polynomial:

$$\mathbf{H}_1 = (a_1), \quad \mathbf{H}_2 = \begin{pmatrix} a_1 & 1 \\ a_3 & a_2 \end{pmatrix}, \quad \mathbf{H}_3 = \begin{pmatrix} a_1 & 1 & 0 \\ a_3 & a_2 & a_1 \\ a_5 & a_4 & a_3 \end{pmatrix},$$

and in general

$$\mathbf{H}_n = \begin{pmatrix} a_1 & 1 & 0 & 0 & \cdots & 0 \\ a_3 & a_2 & a_1 & 1 & \cdots & 0 \\ a_5 & a_4 & a_3 & a_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & a_n \end{pmatrix},$$

where $a_j = 0$ if $j > n$. All the roots of the polynomial $\mathcal{P}(\lambda)$ are negative or have negative real part if and only if the determinants of all Hurwitz matrices are positive:

$$\det \mathbf{H}_j > 0, \quad j = 1, 2, \dots, n.$$

We can specialize the Theorem for low n . We have

- $n = 2$: $\det \mathbf{H}_1 = a_1 > 0$ and $\det \mathbf{H}_2 = a_1 a_2 > 0$. This is equivalent to $a_1 > 0$ and $a_2 > 0$. Note that $a_1 = -\operatorname{tr} \mathbf{A}$ and $a_2 = \det \mathbf{A}$.
- $n = 3$: as above, $a_1 > 0$, $a_1 a_2 - a_3 > 0$, and $\det \mathbf{H}_3 = a_3 \det \mathbf{H}_2 > 0$. So, this is equivalent to $a_i > 0$ and $a_1 a_2 - a_3 > 0$.
- $n = 4$: it is possible to show that we need $a_1 > 0$, $a_3 > 0$, $a_4 > 0$ and $a_1 a_2 a_3 > a_3^2 + a_1^2 a_4$.

We will make use of the case $n = 3$.

A.5.4.5 Other useful cases

Sometimes we are going to deal with 4×4 or larger matrices. Very often, though, they have a block structure. For instance:

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_1 & \mathbf{C} \\ \mathbf{0} & \mathbf{A}_2 \end{pmatrix},$$

where $\mathbf{A}_1$ and $\mathbf{A}_2$ can differ in dimension. Then, the eigenvalues of $\mathbf{A}$ are the union of the eigenvalues of $\mathbf{A}_1$ and $\mathbf{A}_2$. A specific case is the triangular matrix, where eigenvalues are on the diagonal.

A.5.5 Linearization method

The analysis of stability quickly becomes impractical for complex non-linearities. The study of *local* stability properties, however, can be carried out fairly easily. Suppose that $\mathbf{y}^* \in \mathbb{R}^n$ is an equilibrium of the dynamical system $(\mathbf{f}, \Omega)$, with $\mathbf{f} \in \mathcal{C}^1(\Omega)$. Then

$$\mathbf{f}(\mathbf{y}) = \mathbf{f}(\mathbf{y}^*) + D\mathbf{f}(\mathbf{y}^*)(\mathbf{y} - \mathbf{y}^*) + \dots,$$

where $D\mathbf{f}(\mathbf{y}^*)$ is the Jacobian of $\mathbf{f}$. Since $\mathbf{f}(\mathbf{y}^*) = 0$, we have the following linear approximation of the dynamical system:

$$\mathbf{y}' = D\mathbf{f}(\mathbf{y}^*)(\mathbf{y} - \mathbf{y}^*) + \dots$$

We now consider $\mathbf{z}(t) = \mathbf{y}(t) - \mathbf{y}^*$, and notice that $\mathbf{z}' = \mathbf{y}'$, to obtain:

$$\mathbf{z}' = \mathbf{A}\mathbf{z},$$

where we set $\mathbf{A} = D\mathbf{f}(\mathbf{y}^*)$. The dynamical system in $\mathbf{z}$ is linear, and we know how to analyze the stability of the equilibrium $\mathbf{z} = 0$. (This corresponds to the equilibrium $\mathbf{y}^*$ in the original variables.) In fact, let $\{\lambda_i\}$ be the eigenvalues of $D\mathbf{f}(\mathbf{y}^*)$. Then:

1. If $\operatorname{Re} \lambda_i < 0$ for all $i = 1, \dots, r$, then $\mathbf{y}^*$ is locally asymptotically stable.
2. If there exists $i \in \{1, \dots, r\}$ such that $\operatorname{Re} \lambda_i > 0$, the equilibrium is locally unstable.

We cannot conclude anything regarding the case $\operatorname{Re} \lambda_i = 0$, because in this case higher order terms in the Taylor expansion dictates the local dynamics of the system.

i Example (logistic equation, alternative analysis)

We consider again the ODE $y' = y(1 - y)$. Given $f(y) = y(1 - y)$, we have $f'(y) = 1 - 2y$. At the first equilibrium, $f'(0) = 1 > 0$, so it is locally unstable. For the second one, $f'(1) = -1 < 0$, so it is locally asymptotically stable.

A.6 Periodic orbits

So far, we went in more great detail in the study of equilibria of a dynamical system. Equilibria as “simple”, in the sense that we can find all of them as solution of the nonlinear system $\mathbf{f}(\mathbf{y}) = \mathbf{0}$. Their local stability follows from the linearization of the ODE around the equilibrium.

Periodic orbits are more difficult to characterize, but they are extremely important in applications.

A.6.1 Hamiltonian systems

Consider a system of 2 ODEs, written in general as

$$\begin{cases} y_1' = f_1(y_1, y_2), \\ y_2' = f_2(y_1, y_2). \end{cases}$$

The phase space is planar. Let us introduce the vector field

$$\mathbf{f}(y_1, y_2) = \begin{pmatrix} f_1(y_1, y_2) \\ f_2(y_1, y_2) \end{pmatrix},$$

and suppose that given $\Omega \subset \mathbb{R}^2$ we have a dynamical system $(\mathbf{f}, \Omega)$. We introduce another vector field:

$$\mathbf{G}(y_1, y_2) = \begin{pmatrix} -f_2(y_1, y_2) \\ f_1(y_1, y_2) \\ 0 \end{pmatrix},$$

with the following associated [differential form](#)

$$\omega(y_1, y_2) = -f_2(y_1, y_2)dy_1 + f_1(y_1, y_2)dy_2.$$

Recall the following interpretation of a differential form: given an infinitesimal displacement $d\mathbf{y}$ in $\mathbb{R}^3$, the quantity $\omega = \langle \mathbf{G}, d\mathbf{y} \rangle$ is the infinitesimal work done by $\mathbf{G}$ along $d\mathbf{y}$. A $\mathcal{C}^1(\Omega)$ differential form is *exact* when there exists a $\mathcal{C}^2(\Omega)$ function $H : \Omega \rightarrow \mathbb{R}$ such that $dH = \omega$ in Ω . In particular,

$$dH = \langle \nabla H, d\mathbf{y} \rangle = \langle \mathbf{G}, d\mathbf{y} \rangle = \omega,$$

so we conclude that $\nabla H = \mathbf{G}$. We call H potential or [Hamiltonian](#) function.

A simple necessary condition for exactness is that the curl of $\mathbf{G}$ cancels, since $\nabla \times \mathbf{G} = \nabla \times \nabla H = \mathbf{0}$, so

$$\nabla \times \mathbf{G}(y_1, y_2) = \begin{pmatrix} 0 \\ 0 \\ \frac{\partial f_1(y_1, y_2)}{\partial y_1} + \frac{\partial f_2(y_1, y_2)}{\partial y_2} \end{pmatrix} = \mathbf{0}.$$

The condition

$$\frac{\partial f_1(y_1, y_2)}{\partial y_1} + \frac{\partial f_2(y_1, y_2)}{\partial y_2} = 0,$$

is also sufficient for ω being exact when Ω is [simply connected](#). In conclusion, if $\nabla \times \mathbf{G} = \mathbf{0}$ and Ω is simply connected, then there exists a Hamiltonian function $H(y_1, y_2)$ such that:

$$\begin{cases} y_1' = \frac{\partial H}{\partial y_2}(y_1, y_2), \\ y_2' = -\frac{\partial H}{\partial y_1}(y_1, y_2). \end{cases}$$

ODEs of the above form are called *Hamiltonian systems*. They are very common in mechanics.

For a Hamiltonian system, orbits are level sets of the function H . Indeed, given a trajectory $\mathbf{y}(t)$ of the ODE, we have that the Hamiltonian function along the trajectory is constant:

$$\frac{d}{dt}H(y_1(t), y_2(t)) = \frac{\partial H}{\partial y_1}y_1'(t) + \frac{\partial H}{\partial y_2}y_2'(t) = \frac{\partial H}{\partial y_1}\frac{\partial H}{\partial y_2} - \frac{\partial H}{\partial y_2}\frac{\partial H}{\partial y_1} = 0.$$

Thus, $H(y_1(t), y_2(t)) = \text{constant}$ for $t \geq 0$. Given $H(y_1, y_2)$, we can easily draw the phase portrait.

i Example (conservative systems)

Consider the ODE

$$mx'' = F(x),$$

where m is the mass of a particle and $F(x)$ is a force. We can recast this into a system:

$$\begin{cases} y_1' = \frac{1}{m}y_2, \\ y_2' = F(y_1), \end{cases}$$

where $y_1 = x$ is the position and $y_2 = mx'$ is the momentum.

When the force is conservative, there exists a potential energy $U(x)$ such that $F = -U'$. This implies the existence of a Hamiltonian function

$$H(y_1, y_2) = \frac{1}{2m}y_2^2 + \frac{1}{m}U(y_1),$$

which is exactly the total energy, the sum of kinetic and potential energy. This quantity is conserved and the orbits are level sets of $H = H_0$, where $H_0 = H(y_1(0), y_2(0))$ is the initial energy of the particle.

Take for instance the [Lennard-Jones potential](#) below:

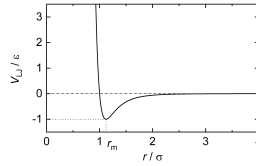


Figure A.4: Lennard-Jones potential

We observe that if the initial energy is between -1 and 0 , then we have closed orbits. These corresponds to periodic solutions. When the initial energy is positive, then orbits are open.

! Exercise (Double-well potential)

Find a Hamiltonian for the ODE

$$x'' = x - x^3,$$

and show how orbits vary as H_0 increases.

i Example (Lotka-Volterra system)

Consider the system:

$$\begin{cases} y_1' = y_1(a - by_2), \\ y_2' = y_2(-c + dy_1). \end{cases}$$

where $y_1(t)$ and $y_2(t)$ are respectively the prey and the predator density. We will study this system more in depth. We can find the Hamiltonian in the positive quadrant as follows. Note that

$$\frac{dy_1}{dy_2} = \frac{y_1(a - by_2)}{y_2(-c + dy_1)},$$

so we can separate the variables:

$$\frac{(-c + dy_1)dy_1}{y_1} = \frac{(a - by_2)dy_2}{y_2},$$

and integrate:

$$\int_{y_1(0)}^{y_1(t)} \left(-\frac{c}{y} + d\right) dy = \int_{y_2(0)}^{y_2(t)} \left(\frac{a}{y_2} - b\right) dy_2,$$

we obtain that $H(y_1(t), y_2(t)) = H(y_1(0), y_2(0))$ with

$$H(y_1, y_2) = -c \ln y_1 + dy_1 - a \ln y_2 + by_2.$$

This function has a minimum at $(y_1^*, y_2^*) = (\frac{a}{b}, \frac{c}{d})$, since $\nabla H(y_1^*, y_2^*) = 0$ and the Hessian is positive definite. For y_1 or $y_2 \rightarrow 0$ we have $H \rightarrow +\infty$, as well for $y_{1,2} \rightarrow +\infty$. So, the function has level set that are closed curves, and those are associated with periodic orbits.

A.6.2 Isolated periodic orbits

We have seen that periodic orbits are possible in planar systems. In all the above cases, however, periodic orbits are packed, that is if we have an orbit for an energy level H_0 , then we can have another periodic orbit infinitesimally close to the original one. Is it possible to have *isolated* periodic orbits? In other words, is it possible that there exists an open set $\Omega' \subseteq \Omega$ such that Ω' contains only *one* periodic orbit? If so, we call this orbit a *limit cycle*.

i Example

The ODE

$$\begin{cases} y_1' = y_2, \\ y_2' = y_2(1 - y_1^2 + y_2^2) - y_1. \end{cases}$$

has an isolated limit cycle for $y_1(t)^2 + y_2(t)^2 = 1$. To see this, first note that $(y_1(t), y_2(t)) = (\cos t, \sin t)$ is a (periodic) solution. Now we check the stability.

Take the function

$$E(y_1, y_2) = \frac{1}{2}(y_1^2 + y_2^2),$$

and note that

$$\frac{d}{dt}E(y_1(t), y_2(t)) = (1 - y_1(t)^2 - y_2(t)^2)y_2(t)^2.$$

Thus, if $y_1^2 + y_2^2 \leq 1$ we have $E'(t) \geq 0$, whereas if $y_1^2 + y_2^2 \geq 1$ we have $E'(t) \leq 0$. Hence, the limit cycle is attracting orbits.

A.7 Limit cycles

We have seen that periodic orbits are possible in planar systems. In all the above cases, however, periodic orbits are packed, that is if we have an orbit for an energy level H_0 , then we can have another periodic orbit infinitesimally close to the original one. Is it possible to have *isolated* periodic orbits? In other words, is it possible that there exists an open set $\Omega' \subseteq \Omega$ such that Ω' contains only *one* periodic orbit? If so, we call this orbit a *limit cycle*.

A.7.1 Dulac's criterion

We have the following result of non-existence of limit cycles:

Theorem A.8 (Dulac's criterion). *Let $\mathbf{y}' = \mathbf{f}(\mathbf{y})$ a planar dynamical system with $\mathbf{f} \in \mathcal{C}^1(\Omega)$, $\Omega \subset \mathbb{R}^2$ open. If there exists a function $h \in \mathcal{C}^1(\Omega)$ such that $\text{div}(h\mathbf{f})$ does not change sign in $\Omega' \subseteq \Omega$ simply connected, then there exists no limit cycle in Ω' .*

Note that the theorem applies also with $h \equiv 1$. The proof is simple. Take a closed orbit entirely lying in the region Ω' . We denote by $C \subset \Omega'$ the interior of the orbit whose boundary ∂C is the orbit itself. Then

$$\int_C \text{div}(h\mathbf{f}) d\mathbf{y} = \int_{\partial C} h \langle \mathbf{f}, \mathbf{n} \rangle ds.$$

Since ∂C is an orbit, $\langle \mathbf{f}, \mathbf{n} \rangle = 0$, so the second integral is zero, no matter of the choice of h . But the first integral cannot be zero, because $\text{div}(h\mathbf{f})$ has a constant sign in C . The contradiction yields to the non-existence of such a periodic orbit.

Unfortunately there is no algorithm for finding the function h . Usually is of the form $(y_1 y_2)^{-1}$, e^{y_1} , e^{y_2} , ...

i Example

Consider the ODE:

$$\begin{cases} y_1' = y_1(2 - y_1 - y_2), \\ y_2' = y_2(4y_1 - y_1^2 - 3). \end{cases}$$

We can show that the system has no closed orbits in the positive quadrant. We take $h(y_1, y_2) = 1/(y_1 y_2)$. Then:

$$\nabla \cdot (h\mathbf{f}) = \frac{\partial}{\partial y_1} \left(\frac{2 - y_1 - y_2}{y_2} \right) + \frac{\partial}{\partial y_2} \left(\frac{4y_1 - y_1^2 - 3}{y_1} \right) = -\frac{1}{y_2} < 0.$$

The region $\mathbb{R}_+^2$ is simply connected, and the functions are smooth, so there are no limit cycles in it.

A.7.2 Poincaré-Bendixson existence theorems

Theorem A.9 (Poincaré-Bendixson). *Suppose that $(\mathbf{f}, \Omega)$ is a planar dynamical system, $\Omega \subset \mathbb{R}^2$, with $\mathbf{f} \in \mathcal{C}^1(\Omega)$. If there exists a closed and bounded region $R \subseteq \Omega$ that does not contain any equilibrium and there exists a trajectory C that is “confined” in D , in the sense that it starts in D and stays in D for all future time, then either C is a closed orbit, or it spirals toward a closed orbit at $t \rightarrow \infty$. In either case, D contains a closed orbit.*

In order to apply the theorem, we need to find a “trapped” orbit in D . How to proceed? The idea is the following: we can construct a region D that is closed and connected, and such that its boundary is *inflow* for the dynamical system. That is, the vector field $\mathbf{f}$ only enters into D . So, orbits cannot leave.

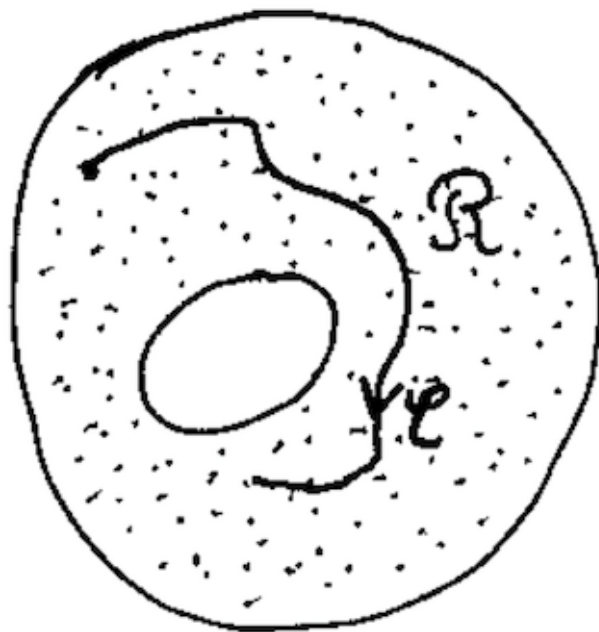


Figure A.5: Trapping region

Some comments:

- The requirement for no equilibria is important. Very often we do have an equilibrium inside (as we are going to see in a moment, a limit cycle always has an equilibrium inside of it), but we can remove it by “digging” a hole around it.
- Equilibria cannot be on the border of the region. Consider a homoclinic orbit as in the figure below. We see that the “punctured” region traps an orbit C , but there are no limit cycles inside. The orbit tends to the homoclinic.

We can be more precise. Let us define

Definition A.15 (α -limit and ω -limit sets). We define the α -limit and the ω -limit of an initial condition $\mathbf{y}_0$ as the set of Ω that the trajectory approaches as $t \rightarrow -\infty$ and $t \rightarrow \infty$, respectively:

$$\alpha(\mathbf{y}_0) = \lim_{t \rightarrow -\infty} \Phi(t, \mathbf{y}_0),$$

$$\omega(\mathbf{y}_0) = \lim_{t \rightarrow +\infty} \Phi(t, \mathbf{y}_0).$$

Then, we have the following

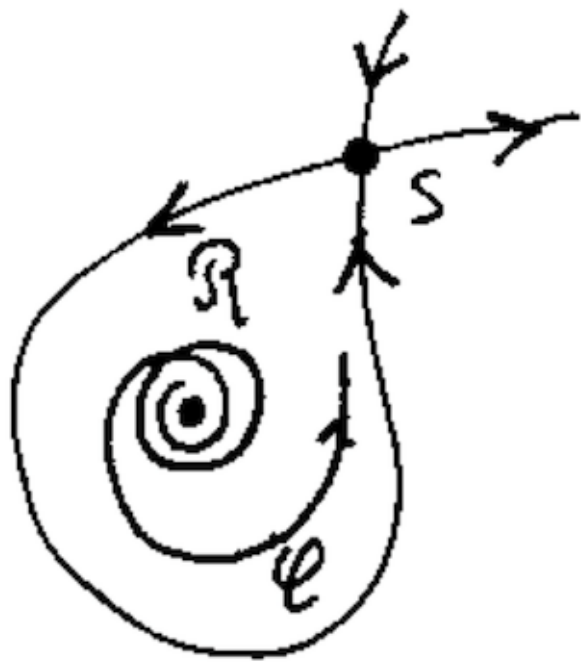


Figure A.6: Homoclinic orbit

Theorem A.10 (Poincaré-Bendixson Trichotomy). *Suppose that $(\mathbf{f}, \Omega)$ is a planar dynamical system, $\Omega \subset \mathbb{R}^2$, with $\mathbf{f} \in \mathcal{C}^1(\Omega)$. Let $C^+(\mathbf{y}_0) = \{\Phi(t, \mathbf{y}_0), t \geq 0\}$ a positive orbit that remains in a closed, bounded region $R \subseteq \Omega$ that contains only a finite number of equilibria. Then the ω -limit set takes one of the following three forms:*

1. $\omega(\mathbf{y}_0)$ is an equilibrium,
2. $\omega(\mathbf{y}_0)$ is a periodic orbits,
3. $\omega(\mathbf{y}_0)$ is a singular cycle, that is $\omega(\mathbf{y}_0)$ contains a finite number of equilibria and a set of orbits whose α - and ω -limit sets consist of one these equilibria for each orbit.

The point 3. is for instance a heteroclinic, which contains two saddle equilibria connected by the stable and unstable manifolds, or a homoclinic orbit, which is a single saddle equilibrium where the stable and unstable manifolds correspond (for $t \geq 0$).

The proofs of these theorems heavily rely on concepts from topology. Here, we only recall *index theory* for a planar dynamical system.

i Example (glycolysis)

We consider a model for glycolysis as proposed by Sel'kov (1968). In yeast cells glycolysis can proceed in a oscillatory fashion. Here is a model:

$$\begin{cases} x' = -x + ay + x^2y, \\ y' = b - ay - x^2y, \end{cases}$$

where $x(t)$ is the concentration of ADP (adenosine diphosphate) and $y(t)$ is the concentration of F6P (fructose-6-phosphate). The parameters $a > 0$ and $b > 0$ comes from the kinetic of the reaction.

The phase portrait is as follows:

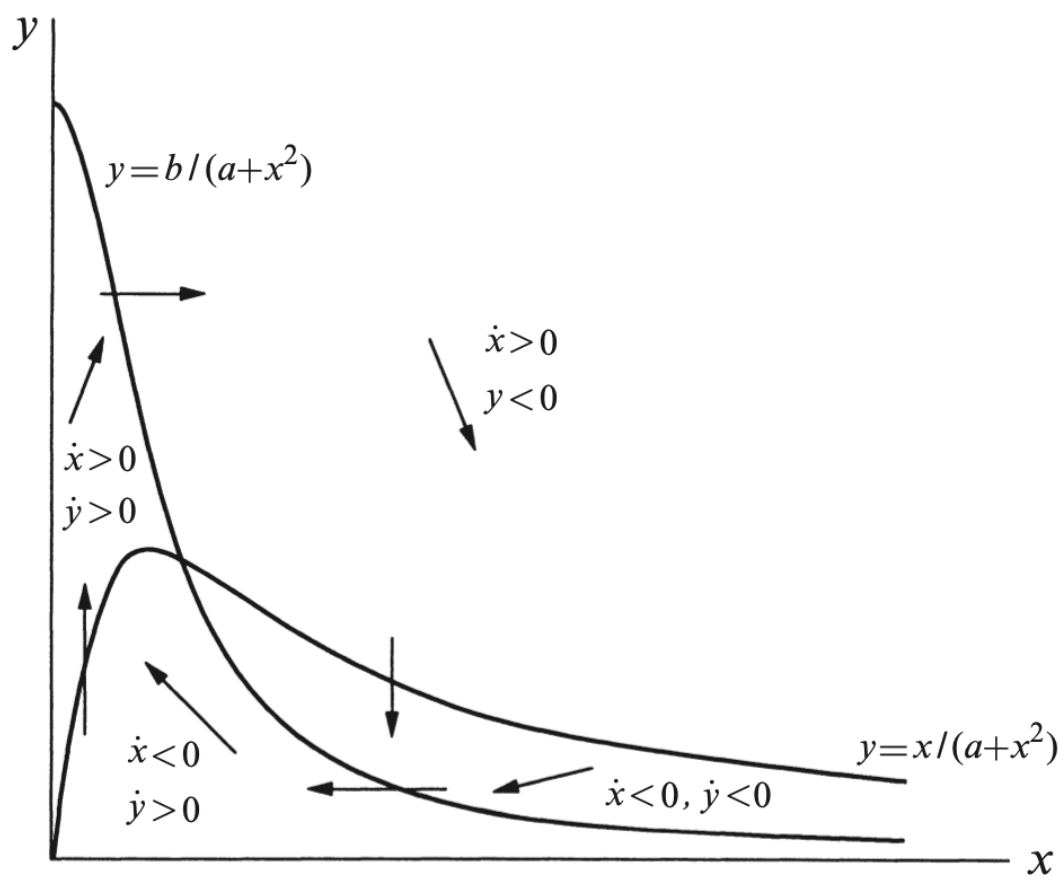


Figure A.7: Trapping region

By looking at the arrows, it looks like we could have a limit cycle or some stable equilibrium. We construct the trapping region as follows:

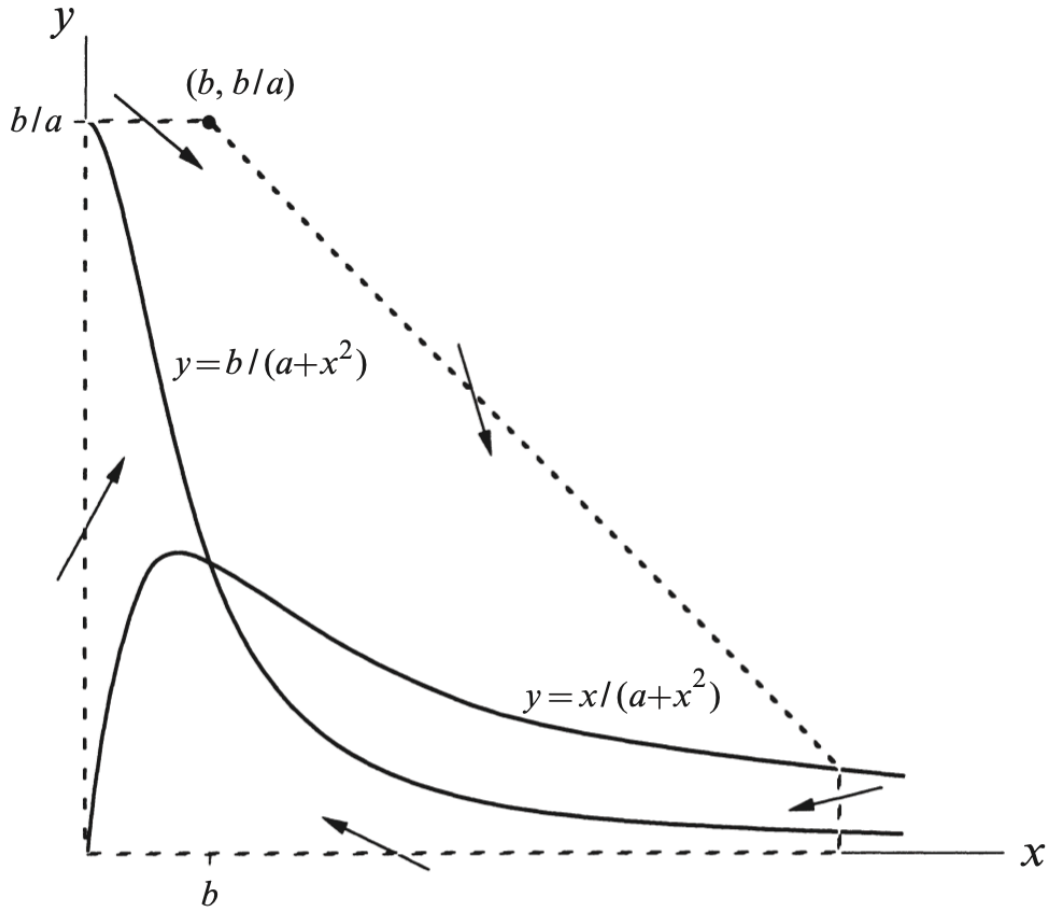


Figure A.8: Trapping region

Now we check. Graphically, we note that

- Above the nullcline $x' = 0$ we have $x' > 0$, below $x' < 0$.
- On the left of the nullcline $y' = 0$ we have $y' > 0$, on the right $y' < 0$.

Thus, the horizontal and vertical parts in of the region are easy to check, just by looking at the arrows. The diagonal part is more subtle. Intuitively, when x and y are large, the ODE is approximately $x' \approx x^2y$ and $y' \approx -x^2y$. So, $y'/x' \approx -1$, which is tangent to the trajectories. This suggests to take:

$$x' - (-y') = -x + ay + x^2y + (b - ay - x^2y) = b - x.$$

Hence, for $x > b$ we have $-y' > x'$, and trajectories are entering into the diagonal part, because their slope is more negative than -1 .

We conclude that R is a trapping region. However, we cannot apply the Theorem, because we still need to rule out equilibria. We have only one, at the intersection between nullclines. We can do so just by digging a hole around it, obtaining a “punctured” region.

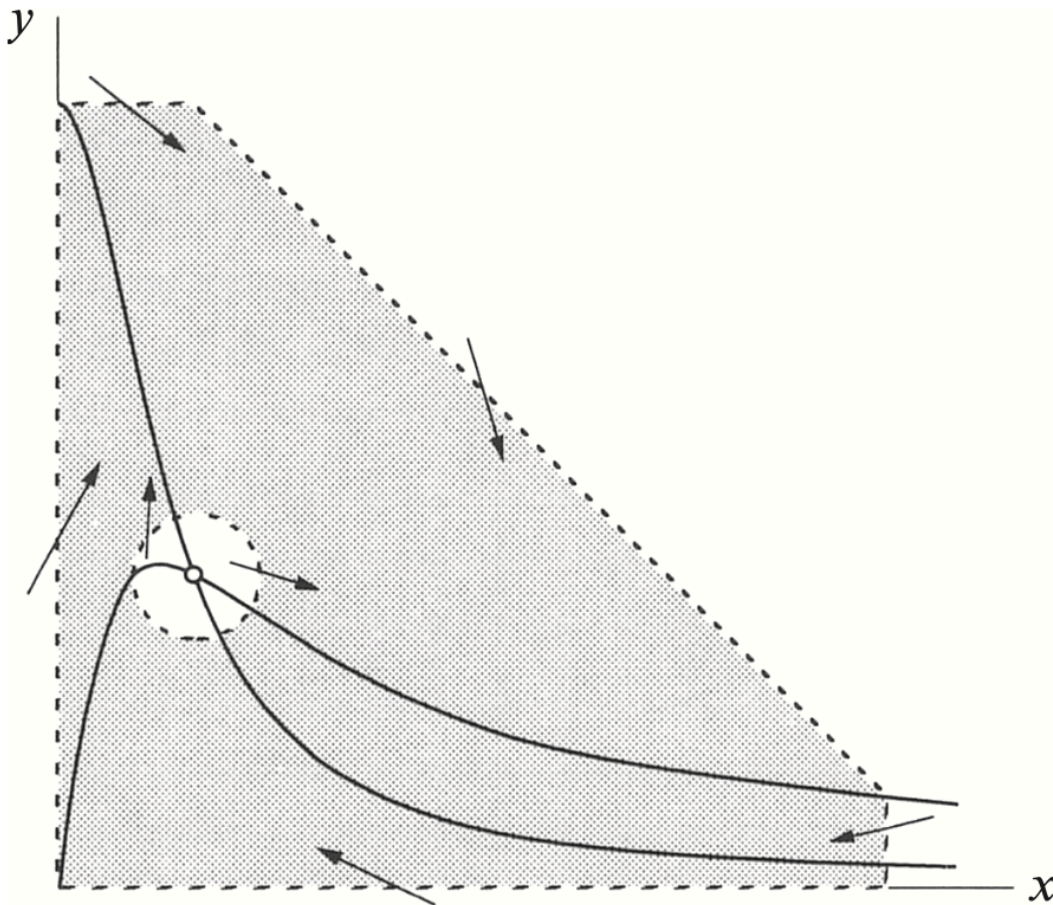


Figure A.9: Punctured region

For a sufficiently small hole, we can ensure that orbits are leaving or entering the region just by looking at the stability of the equilibrium. If it is unstable, then orbits are diverging, and thus entering into R . That is, R traps orbits and it does not contain equilibria. Therefore, there must exist a closed orbit in it!

The stability of the equilibrium is easy to check. First, the equilibrium is

$$(x^*, y^*) = \left(b, \frac{b}{a + b^2}\right).$$

The Jacobian is:

$$J_{\mathbf{f}}(x, y) = \begin{pmatrix} -1 + 2xy & a + x^2 \\ -2xy & -a - x^2 \end{pmatrix}.$$

The trace and the determinant at $(x, y) = (x^*, y^*)$:

$$\det J = a + b^2 > 0, \quad \text{tr } J = -\frac{b^4 + (2a - 1)b^2 + (a + a^2)}{a + b^2}.$$

Depending on a and b , the trace can be either positive or negative. Solving the numerator for b^2 we have that the trace is zero for

$$b^2 = \frac{1}{2}(1 - 2a \pm \sqrt{1 - 8a}).$$

For choice of the parameters such that the trace is positive, the equilibrium is unstable and the region contains a closed orbit.

A.7.3 Index theory

Definition A.16 (index of a closed curve). We define the *index* of a closed curve C , denoted by i_C , as the net number of counterclockwise revolutions made by the vector field $\mathbf{f}$ around C . More precisely, take the angle $\phi(t)$ between the tangent to the curve C and the vector field $\mathbf{f}$. Then, $\phi(t)$ has changed by an integer multiple of 2π . Such integer is the index.

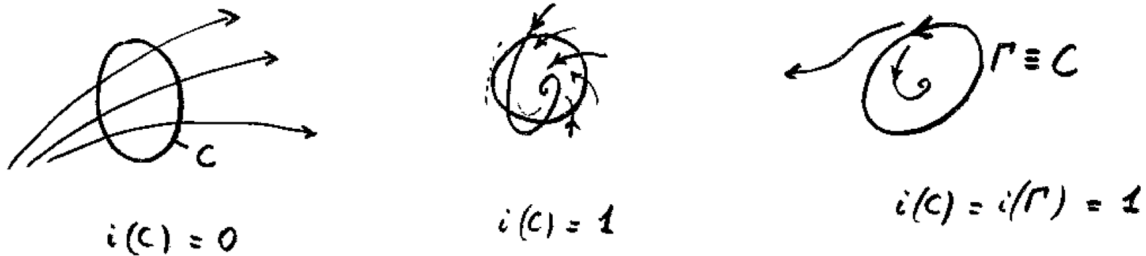


Figure A.10: Index of a curve

The index closely resembles residue theory in complex analysis. For instance, we have that

1. If C can be continuously deformed into C' without passing through an equilibrium, then $i_C = i_{C'}$. (Note that i_C is an integer-valued function of C . The only way i_C can be a continuous function of C is being constant!)

2. If C does not enclose any equilibrium, then $i_C = 0$. (Since C does not contain any zero of $\mathbf{f}$, we can shrink C to a tiny circle where $\mathbf{f}$ is almost constant, so the index is zero.)
3. If we reverse the arrows of the vector field, that is $t \mapsto -t$, the index is unchanged.
4. Suppose that C is a closed orbit. Then $i_C = +1$.

Mathematically, given the vector field $\mathbf{f}(x, y) = (f(x, t), g(x, y))$ in $\mathbb{R}^2$, the angle given by

$$\phi = \tan^{-1}(g/f),$$

so in terms of differentials we have:

$$d\phi = d(\tan^{-1}(g/f)) = \frac{f dg - g df}{f^2 + g^2}.$$

The index is, therefore,

$$i_C = \frac{1}{2\pi} \int_C d\phi = \frac{1}{2\pi} \int_C \frac{f dg - g df}{f^2 + g^2}.$$

Definition A.17 (Index of an equilibrium). The index of an equilibrium $\mathbf{y}^*$, denoted by $i_{\mathbf{y}^*}$ is the index of a curve containing the equilibrium.

Note that thanks to 1. the definition of $i_{\mathbf{y}^*}$ does not depend on the choice of C .

Theorem A.11. *All hyperbolic equilibria, that is the Jacobian of $\mathbf{f}$ is non-singular at the equilibrium, have $i_{\mathbf{y}^*} = \pm 1$. In particular, only the saddle has $i_{\mathbf{y}^*} = -1$.*

A non-hyperbolic equilibrium can also have index 0, for instance a saddle-node.

Theorem A.12. *If a closed curve C surrounds n isolated equilibria $\mathbf{y}_1^*, \mathbf{y}_2^*, \dots, \mathbf{y}_n^*$, then*

$$i_C = \sum_{k=1}^n i_{\mathbf{y}_k^*}.$$

The proof is familiar from multivariate calculus. We shrink the curve C so to surround the equilibria with tiny circles and corridors connecting them. The index does not change. Then we show that the contribution of corridors cancels out, because of symmetry reasons. So, we are only left with circles around the equilibria, which is the sum of the indices of each equilibrium.

Theorem A.13 (Poincaré). *Any closed periodic orbit in the phase plane must enclose equilibria whose indices sum to +1. If all equilibria are hyperbolic, then the limit cycle contains exactly $2s + 1$ equilibria, s saddles and $s + 1$ non saddles, with $s \geq 0$ integer.*

Example

Consider the system:

$$\begin{cases} y_1' = -y_1 - y_1^2 y_2^2, \\ y_2' = y_1 + y_2. \end{cases}$$

We show now that there exists no limit cycle around the origin.

First, notice that the divergence of $\mathbf{f}$ is:

$$\operatorname{div} \mathbf{f} = -2y_1 y_2^2,$$

which changes sign in $\mathbb{R}^2$, so we cannot exclude the existence of a limit cycle.

We have 2 equilibria, $\mathbf{y}_1^* = (0, 0)$ and $\mathbf{y}_2^* = (-1, 1)$. In particular, $\mathbf{y}_1^*$ is a saddle. A limit cycle around the origin must contain the saddle, but the index of the cycle is $+1$, whereas the index of the saddle is -1 . It cannot be. Even the case of a limit cycle surrounding both equilibria is not possible, because the sum of the indices is 0. We conclude that we cannot have a limit cycle around the origin.

A.7.4 Poincaré maps

We still need to discuss the *stability* of limit cycles. Since they are isolated by definition, they can attract or repulse nearby orbits.

Consider a periodic orbit $\Gamma \subset \mathbb{R}^n$, and take any point $\mathbf{y}_0 \in \Gamma$. We introduce a cross-section Σ to the cycle at this point as a smooth hypersurface of dimension $n - 1$ that crosses Γ at non-zero angle. A non-zero angle means that the normal of the hypersurface at $\mathbf{y}_0$ is not orthogonal to the tangent of Γ at $\mathbf{y}_0$.

Note that the limit cycle reduces to a point on the cross-section Σ . Other closeby orbits will also intersect the cross-section Σ at some angle (by continuity the angle remains non-zero in a neighborhood of $\mathbf{y}_0$). A point $\mathbf{z} \in \Sigma$, close to $\mathbf{y}_0$, will give some orbit through the flow of the ODE, and eventually return to Σ after some time at a point $\tilde{\mathbf{z}}$. This procedure is encoded in a map $\mathbf{z} \mapsto \tilde{\mathbf{z}} = \mathcal{P}(\mathbf{z})$.

Definition (Poincaré map). We call $\mathcal{P} : \Sigma \rightarrow \Sigma$ the Poincaré map of the periodic orbit Γ .

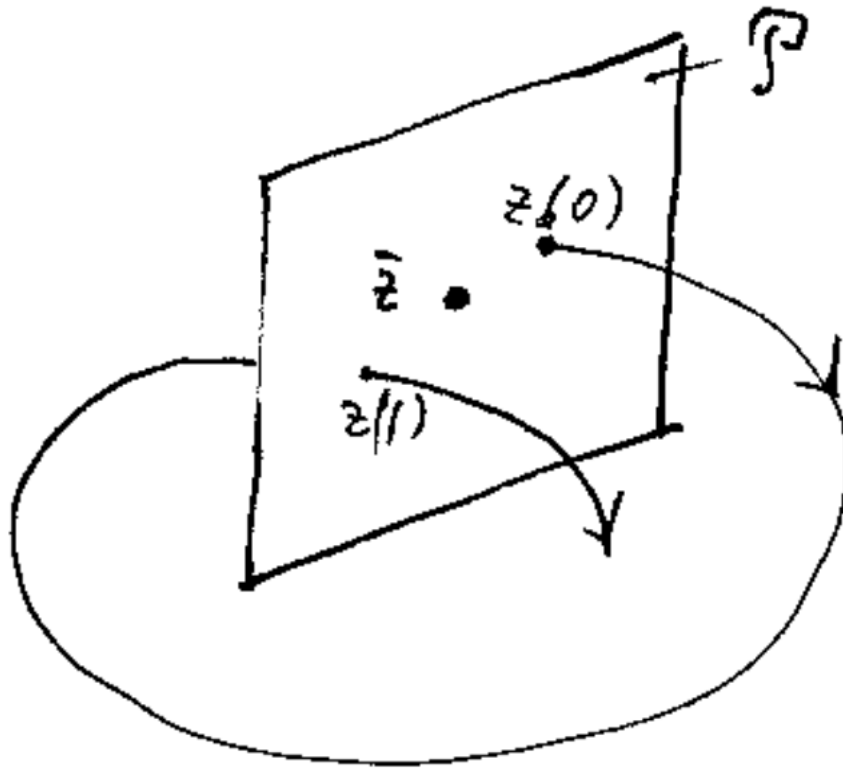


Figure A.11: Poincaré maps

A fixed point $\mathbf{z}^*$ of the map $\mathcal{P}$ corresponds to a limit cycle for the original system. Specifically, if $\mathbf{z}^*$ is a stable equilibrium for the *discrete map*

$$\mathbf{z}^{(k+1)} = \mathcal{P}(\mathbf{z}^{(k)}),$$

that is, $\mathbf{z}^* = \mathcal{P}(\mathbf{z}^*)$, then the limit cycle is stable. If $\mathbf{z}^*$ is asymptotically stable, then the limit cycle is asymptotically stable. Otherwise, it is unstable.

i Example

Consider the following ODE in polar coordinates:

$$\begin{cases} r' = r(1 - r^2), \\ \theta' = 1. \end{cases}$$

Since $\theta' = 1$, we make a turn after a period of 2π . Thus, if we start from r_0 , we return

at r_1 after $\Delta t = 2\pi$. Hence:

$$\int_{r_0}^{r_1} \frac{dr}{r(1-r^2)} = \int_0^{2\pi} dt = 2\pi.$$

Integrating we get:

$$\mathcal{P}(r) = \frac{1}{\sqrt{1 + e^{-4\pi}(r^{-2} - 1)}}.$$

For $r^* = 1$ we have a fixed point. We can show (graphically with the cobweb plot) that the equilibrium is attractive.

Given a parametrization of $\mathcal{P}$ in some local coordinate system $\xi = (\xi_1, \dots, \xi_{n-1})$ such that $\xi = \mathbf{0}$ corresponds to $\mathbf{z}^*$, the origin is a fixed point of the map: $\mathcal{P}(\mathbf{0}) = \mathbf{0}$. Thanks to the contraction Theorem, the fixed-point is asymptotically stable if $\mathcal{P}$ is a contraction, which means that

$$\mathbf{B} = \left. \frac{d\mathcal{P}}{d\xi} \right|_{\xi=\mathbf{0}}$$

has eigenvalues (multipliers) $\mu_1, \dots, \mu_{n-1}$ with modulus strictly less than 1, that is $|\mu_i| < 1$. It is possible to show that the multipliers are independent on the choice of the cross-section, the local parametrization, and the point $\mathbf{y}_0 \in \Gamma$. In other words, the multipliers are a characteristic of the limit cycle, from which we can deduce the stability.

We can now formalize the stability for a limit cycle. Consider a periodic solution $\phi(t)$ of the dynamical system

$$\mathbf{y}' = \mathbf{f}(\mathbf{y}),$$

that is $\phi(t)$ is a solution and $\phi(t + T_0) = \phi(t)$ for some $T_0 > 0$, called the *period* of the limit cycle. Now we perturb the periodic orbit:

$$\tilde{\phi}(t) = \phi(t) + \mathbf{u}(t),$$

for some perturbation $\mathbf{u}(t)$. Now notice that:

$$\mathbf{u}'(t) = \tilde{\phi}'(t) - \phi'(t) = \mathbf{f}(\phi(t) + \mathbf{u}(t)) - \mathbf{f}(\phi(t)) = \mathbf{A}(t)\mathbf{u}(t) + \mathcal{O}(\|\mathbf{u}(t)\|^2),$$

where $\mathbf{A} = J_{\mathbf{f}}(\phi(t))$, and $\mathbf{A}(t + T_0) = \mathbf{A}(t)$. The system $\mathbf{u}' = \mathbf{A}(t)\mathbf{u}$ is non-autonomous and linear. The set of solutions form the Wronskian matrix $\mathbf{W}(t)$ and are such that

$$\mathbf{W}' = \mathbf{A}(t)\mathbf{W},$$

with initial condition $\mathbf{W}(0) = \mathbf{I}$. Any solution of $\mathbf{u}' = \mathbf{A}(t)\mathbf{u}$ is of the form $\mathbf{u}(t) = \mathbf{W}(t)\mathbf{u}(0)$. In particular,

$$\mathbf{u}(T_0) = \mathbf{W}(T_0)\mathbf{u}(0).$$

We call $\mathbf{W}(T_0)$ *monodromy matrix* for the limit cycle Γ .

Theorem A.14. *The monodromy matrix has eigenvalues $1, \mu_1, \dots, \mu_{n-1}$, where μ_i are the multipliers of the Poincaré map associated to the limit cycle Γ .*

Thus, the size of the perturbation $\mathbf{u}(t)$, measured with $\|\mathbf{u}(t)\|$, will tend to zero if $|\mu_i| < 1$. The unitary eigenvalue corresponds to the tangent direction to the limit cycle.

B Introduction to bifurcations

B.1 Background

As we have seen, dynamical system often depend on parameters. For instance, the spruce budworm model

$$u' = \rho \left(1 - \frac{u}{q}\right) - \frac{u^2}{1 + u^2},$$

depends on two parameters, $\rho > 0$ and $q > 0$. The number of equilibria and their stability will depend on the parameters. However, most of the time a change of a parameter has little effect on the phase portrait: equilibria barely move, their stability is unchanged. In this case we say that the system is *topologically stable*. Otherwise, like for $\rho = \rho_1$ or $\rho = \rho_2$, we have a *bifurcation*.

Another example we have seen is the Gause-type prey-predator model:

$$\begin{cases} u' = \rho u(1 - u) - \frac{\alpha \delta uv}{1 + \delta u}, \\ v' = -v + \frac{\alpha \delta uv}{1 + \delta u}, \end{cases}$$

where we have 3 parameters. For $\delta = \frac{1}{\alpha-1}$ or $\delta = \frac{\alpha+1}{\alpha-1}$ we also have bifurcations.

In general, a parametric dynamical system is an equation of the form

$$\mathbf{y}' = \mathbf{f}(\mathbf{y}, \mathbf{p}),$$

where $\mathbf{p} \in \mathbb{R}^m$ is the set of parameters. For the spruce budworm equation, $\mathbf{p} = (\rho, q)$. Here, we are only focusing on the case $m = 1$, when $\mathbf{p}$ is a scalar. Bifurcations occurring in this case are of co-dimension 1.

B.2 Structural stability

Consider the one-parameter ODE

$$\mathbf{y}' = \mathbf{f}(\mathbf{y}, p),$$

and suppose that the problem is well-posed for $p \in \mathcal{P} \subseteq \mathbb{R}$ and $\mathbf{f}$.

Given $p = \bar{p}$, the system will have some phase portrait. Is it stable to small perturbations of the parameter p ? In other words, if we consider a new system:

$$\tilde{\mathbf{y}}' = \mathbf{f}(\tilde{\mathbf{y}}, p),$$

with $p \in (\bar{p} - \varepsilon, \bar{p} + \varepsilon)$, $\varepsilon \ll 1$, is the “error” between $\tilde{\mathbf{y}}(t)$ and $\mathbf{y}(t)$ going to zero as $\varepsilon \rightarrow 0$ for any choice of the initial condition? That is, is the phase portrait of the perturbed system “converging” to the phase portrait of the original ODE? This is typically the case.

Theorem B.1 (stability to perturbations). *Suppose that $\mathbf{f}(\mathbf{y}, p)$ is continuous and locally Lipschitz in $\mathbf{y}$, uniformly in p , and that the ODE is well-posed for any choice of p in the interval. Then the solution of the ODE $\phi(t, \mathbf{y}_0, p) \rightarrow \phi(t, \mathbf{y}_0, \bar{p})$ uniformly as $p \rightarrow \bar{p}$.*

This theorem gives a zero-stability result, in the limit $p \rightarrow \bar{p}$. How about the case of $p \in (\bar{p} - \varepsilon, \bar{p} + \varepsilon)$ but $\varepsilon > 0$? Is the perturbed ODE still “close” to the original one? We have the following definition

Definition B.1 (structural stability). The original ODE is *structurally stable* in some (closed) region of the phase space if and only if there exists $\varepsilon > 0$ such that the perturbed system is *topologically equivalent* to the original ODE for all $p \in (\bar{p} - \varepsilon, \bar{p} + \varepsilon)$.

Definition B.2 (topological equivalence). Two ODEs are *topologically equivalent* if there exists a diffeomorphism that smoothly maps the phase portrait of one system into the other, preserving the direction of time.

For instance, the two phase portraits below are topologically equivalent, because we can smoothly deform one into the other. (Put a fork at the center, then twist like you would do with spaghetti.)

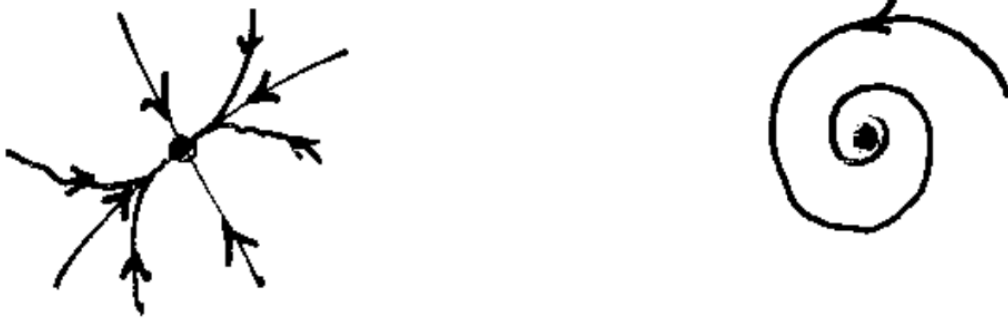


Figure B.1: Topological equivalence

B.3 Bifurcations

On the other hand, these below are *not* topologically equivalent, because if we shrink the limit cycle down to a point, we collide with another equilibrium at the center, making the map non-invertible.



Figure B.2: Phase portraits that are not topologically equivalent

When we are a situation like the one above, we have a *bifurcation*.

Definition B.3 (bifurcation). If for some value $p = \bar{p}$ a system is *not* structurally stable, then for $p = \bar{p}$ we have a *bifurcation*.

Bifurcations result from the *collision* of invariant sets of the dynamical system, for instance 2 equilibria or 2 limit cycles colliding one into the other, for some value of the parameter. We can visualize this in a *bifurcation plot*, like the one below:

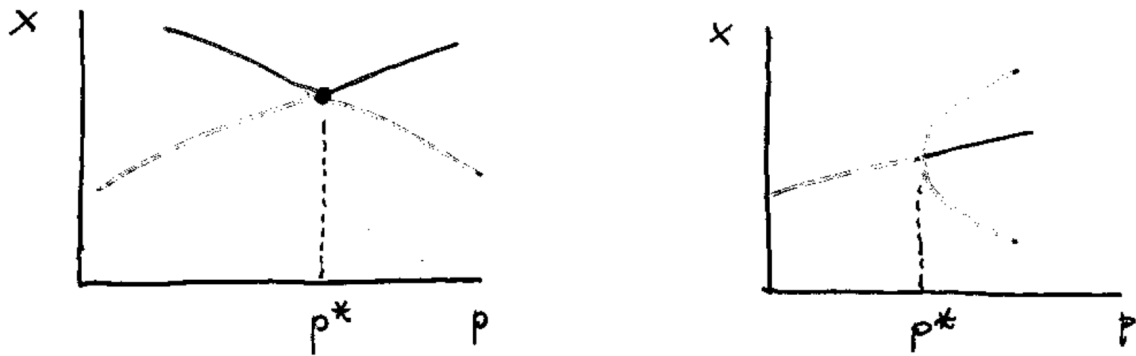


Figure B.3: bifurcation in 1D

where we have on the abscissa the parameter, and on the ordinate the phase space. The curves are equilibria, and for $\bar{p}$ we have a bifurcation.

The bifurcation plot is just a “stack” of phase portraits for various values of p . In some sense, a smooth transition between one layer to the other implies topological equivalence, otherwise we have bifurcations. For instance, for planar system we could have:

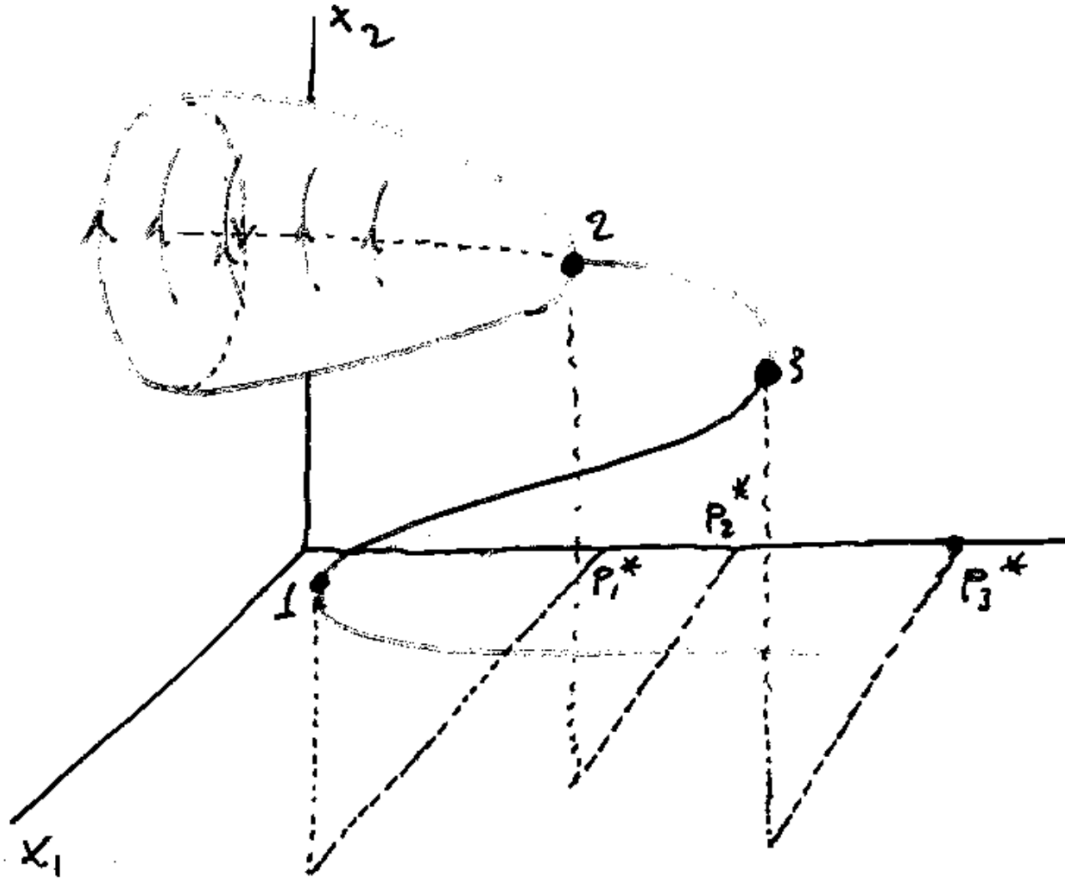


Figure B.4: bifurcation in 2D

The “cone” is formed by limit cycles. Here, we have 3 bifurcations.

B.4 Continuation of equilibria

Consider the dynamical system in 1D:

$$y' = f(y, p),$$

with $p \in \mathbb{R}$. An equilibrium $\bar{y}$ of the ODE for a given parameter $\bar{p}$ is a solution to the equation:

$$0 = f(\bar{y}, \bar{p}).$$

Suppose that f is a smooth function in (y, p) . Thanks to the implicit function theorem, if $\partial_p f(\bar{y}, \bar{p}) \neq 0$ we can locally define a represent the curve $f(\bar{y}, \bar{p}) = 0$ in the space $(y, p) \in \mathbb{R}^2$ with a function $y = \phi(p)$ such that $f(\phi(p), p) = 0$ for $p \in \mathcal{B}_\varepsilon(\bar{p})$, some neighborhood of $\bar{p}$.

The curve $(\phi(p), p)$ represents the *continuation* of the equilibrium $\bar{y}$ as we change the parameter $p = \bar{p}$. All points along $(\phi(p), p)$ are equilibria, by construction, for different values of the parameter p .

We can extend the same construction in several dimensions, say $\mathbf{y} \in \mathbb{R}^n$ and $p \in \mathbb{R}$. Let us by $\mathbf{f}_y(\mathbf{y}, p)$ the Jacobian of $\mathbf{f}$ with respect to $\mathbf{y}$. Suppose that $\bar{\mathbf{y}}$ is an equilibrium for the ODE

$$\mathbf{y}' = \mathbf{f}(\mathbf{y}, \bar{p}),$$

that is, $\mathbf{f}(\bar{\mathbf{y}}, \bar{p}) = \mathbf{0}$, and that

$$\det \mathbf{f}_y(\bar{\mathbf{y}}, \bar{p}) \neq 0,$$

then there exists a function $\mathbf{y} = \phi(p)$ such that $\mathbf{f}(\phi(p), p) = \mathbf{0}$ for $p \in \mathcal{B}_\varepsilon(\bar{p})$.

The *stability* of the equilibria along the curve is also preserved: the eigenvalues of $\mathbf{f}_y(\mathbf{y}, p)$ depends continuously on p so the branch of equilibria $\mathbf{y} = \phi(p)$ will inherit the same stability properties of $\bar{\mathbf{y}}$.

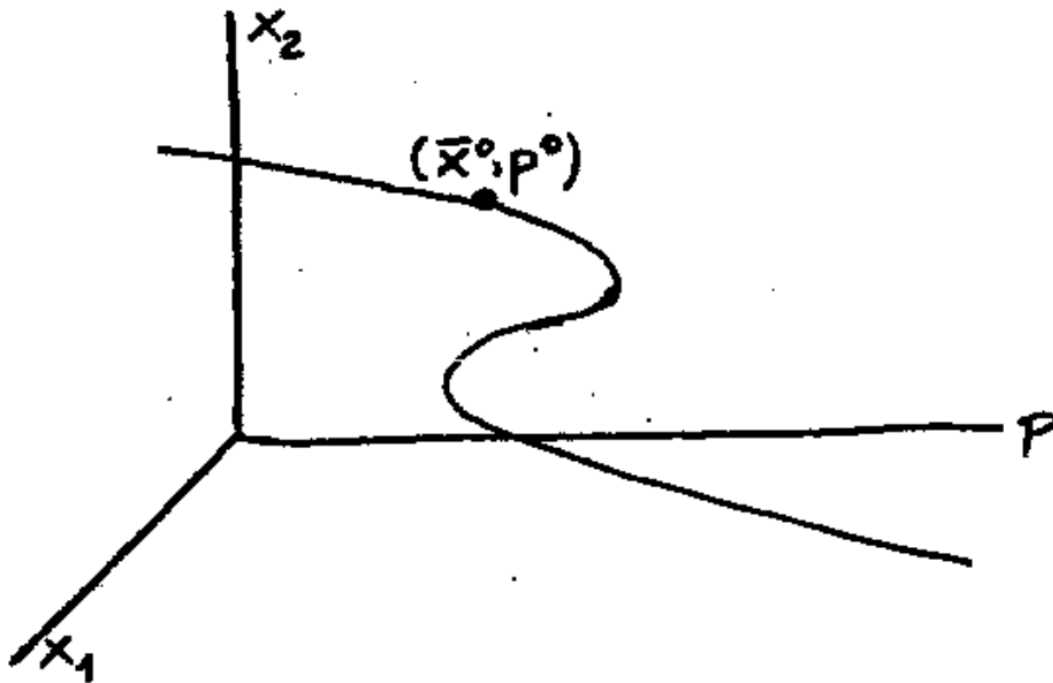


Figure B.5: Curve of equilibria

What if an eigenvalue of $\mathbf{f}_y(\phi(p), p)$ will end up with zero real part along the curve?

Then we have a bifurcation. We still denote this point with $\bar{p}$ and the corresponding equilibrium with $\bar{y}$. We have different options:

- **Tangent bifurcation.** $\mathbf{f}_y(\bar{y}, \bar{p})$ as eigenvalue 0 and $\mathbf{f}_p(\bar{y}, \bar{p}) \neq 0$.
- **Transcritical bifurcation.** $\mathbf{f}_y(\bar{y}, \bar{p})$ as eigenvalue 0 and $\mathbf{f}_p(\bar{y}, \bar{p}) = 0$.
- **Hopf bifurcation.** $\mathbf{f}_y(\bar{y}, \bar{p})$ as eigenvalue $\pm i\omega$ (only $n \geq 2$).

Note that if one real eigenvalue is zero, then we cannot apply the implicit function theorem, and the curve ϕ cannot be defined at $p = \bar{p}$ (this is the case of the first 2 bifurcations.)

B.5 Tangent bifurcation

The simplest bifurcation is perhaps the tangent bifurcation. Consider the following ODE:

$$y' = f(y, p) = p - y^2,$$

for $p \in \mathbb{R}$. We have the following situation:

- If $p > 0$ there are 2 equilibria, $\bar{y}^\pm = \pm\sqrt{p}$. Since $\partial_y f = -2y$, we have that $\bar{y}^-$ is unstable and $\bar{y}^+$ is asymptotically stable.
- If $p < 0$ there is no real equilibrium.
- If $p = \bar{p} = 0$ we have a single equilibrium $\bar{y} = 0$. The stability cannot be deduced from the linearization (why?), but from the sign of f we see that $\bar{y}$ is attractive for $y > 0$ and repulsive for $y < 0$. It is called *saddle-node*.

Since $\partial_y f(\bar{y}, \bar{p}) = 0$ but $\partial_p f(\bar{y}, \bar{p}) = 1 \neq 0$, for $p = \bar{p} = 0$ we have a *tangent bifurcation*. Tangent bifurcations are also called *limit points* (in MatCont denoted by LP). The reason for the name should be clear from the bifurcation plot.

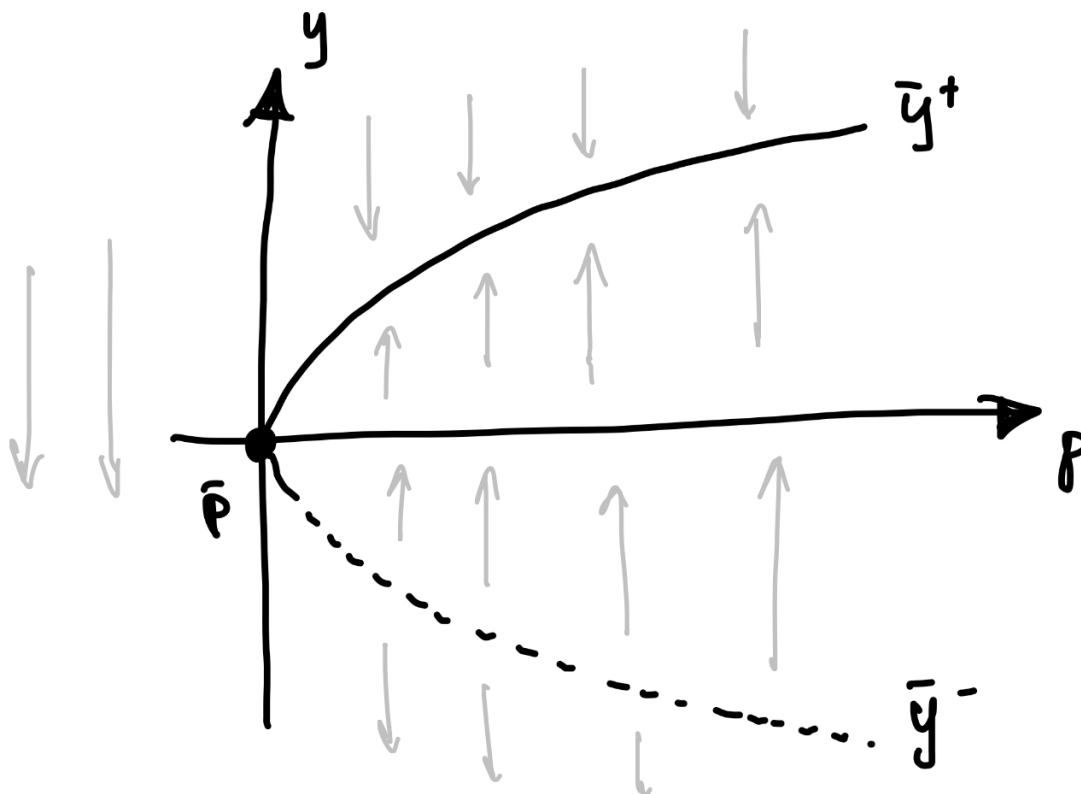


Figure B.6: tangent bifurcation

Tangent bifurcations are *catastrophic*, in the sense that a point at equilibrium right before the bifurcation will diverge from its state after the bifurcation (imagine to move a point along the stable branch from right to left.) Note that the curve is smooth, because $\partial_p f(\bar{y}, \bar{p}) \neq 0$. Check the next figure, where we visualize $f(y, p)$ around the bifurcation (in black the zero levelset, that is the curve of equilibria):

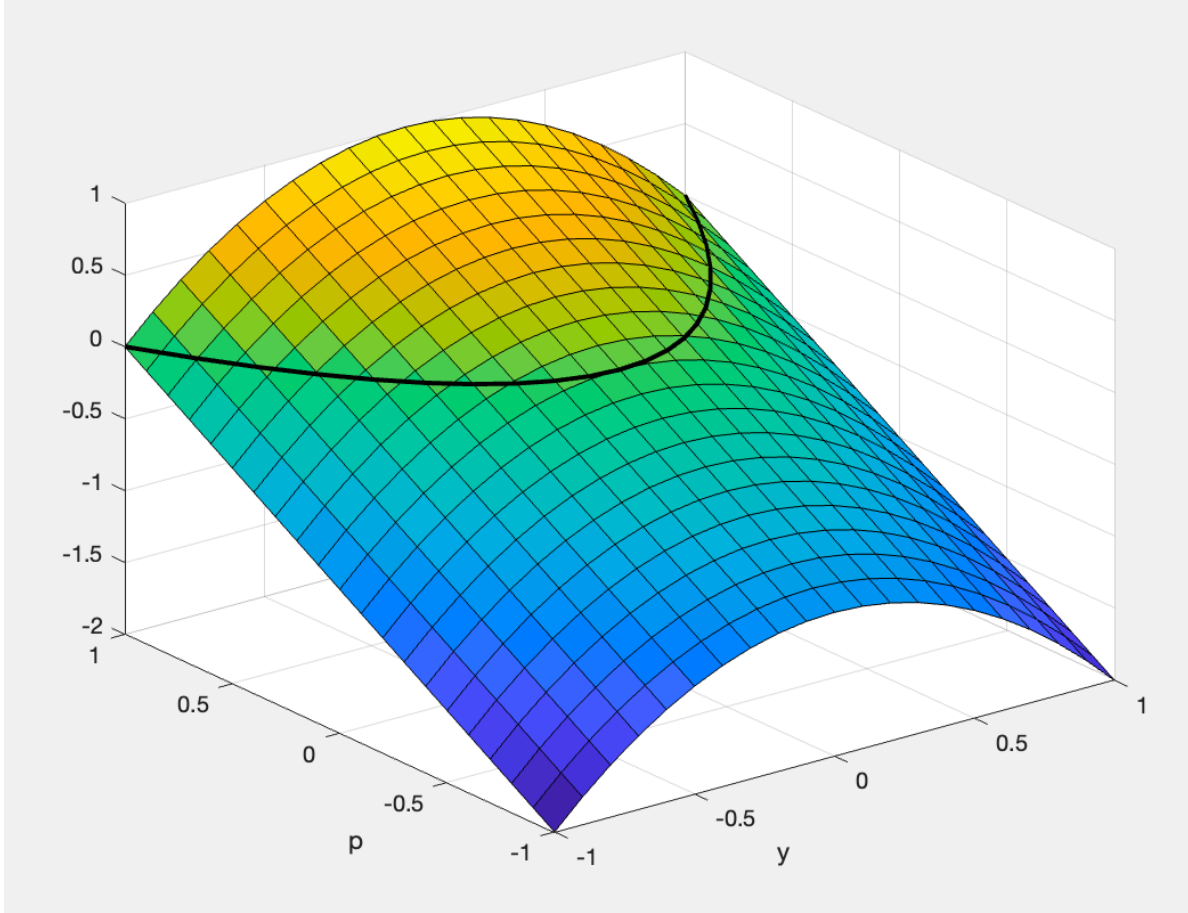


Figure B.7: image

B.6 Transcritical bifurcation

The transcritical bifurcation occurs when an equilibrium has 0 real part and also $\partial_p f(\bar{y}, \bar{p}) = 0$. The normal form is:

$$y' = py - y^2.$$

The derivative of f is $f_y(y, p) = p - 2y$. We have 2 equilibria,

- $y_0 = 0$, stable for $p > 0$ and unstable for $p < 0$.
- $\bar{y} = p$, unstable for $p > 0$ and stable for $p < 0$.

For $p = 0$ we only have one equilibrium $\bar{y} = 0$, which is a saddle-node because it is attractive for $y < 0$ and repulsive for $y > 0$.

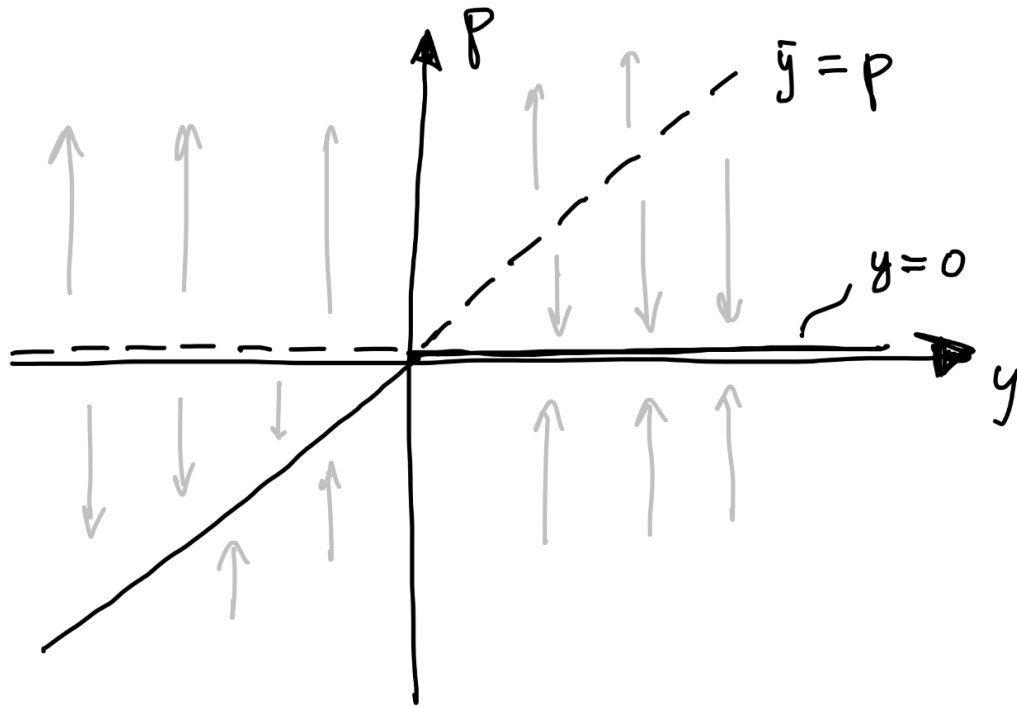


Figure B.8: image

Transcritical bifurcations are very common in biological modeling, because some equilibria (like the origin) do not depend on the parameters and are always present. Therefore, in the bifurcation plot their curve do not change, and can be intersected by other curves of equilibria. In fact, if $f(0, p) = 0$ for all p , then also $f_p(0, p) = 0$. Note that the above model is the logistic equation.

From the 3d visualization of f :

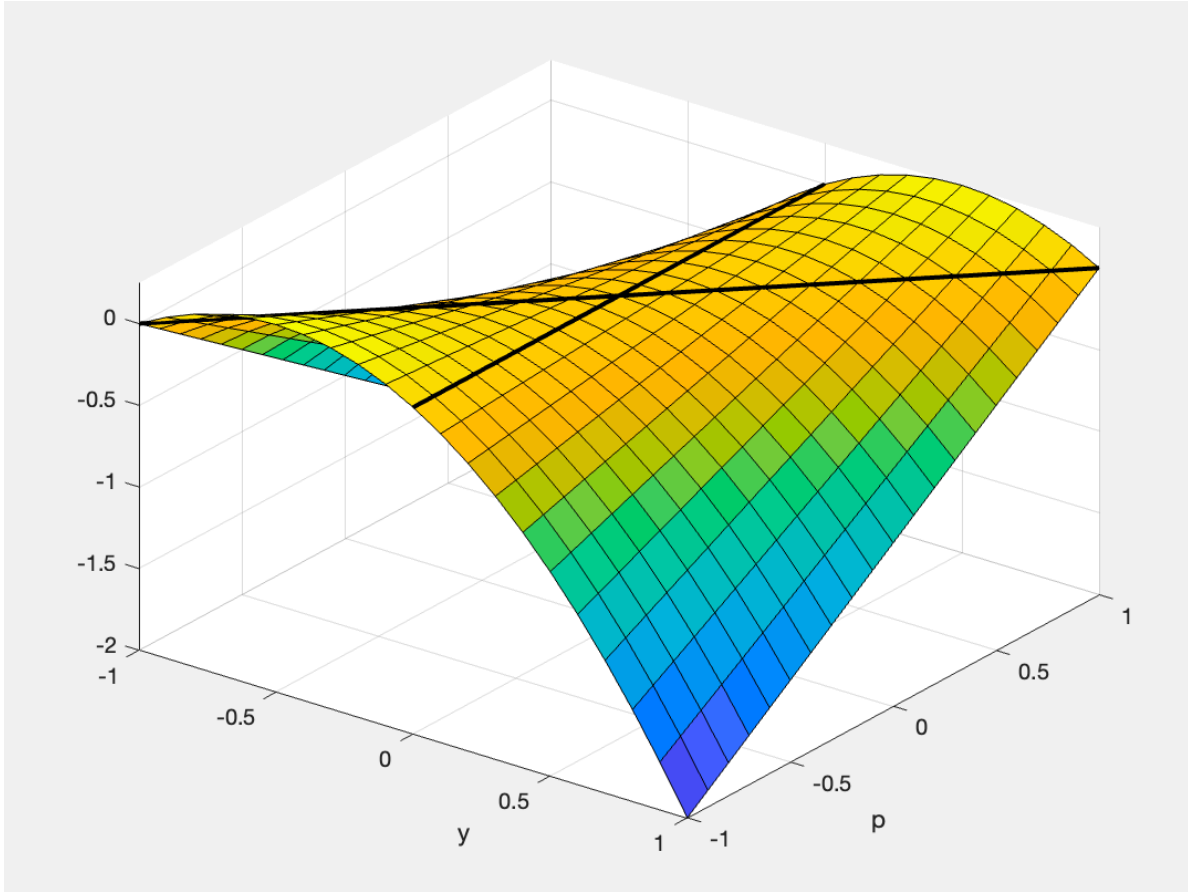


Figure B.9: image

we see a critical point at the bifurcation point. The eigenvectors give the tangents to the 2 curves of equilibria. In this way, it is possible to switch curve at the bifurcation point.

B.7 Hopf bifurcation

The Hopf bifurcation occurs when an equilibrium changes stability and a limit cycle appears. The normal form is:

$$\begin{cases} x' = \sigma x + \omega y + cx(x^2 + y^2), \\ y' = \sigma y - \omega x + cy(x^2 + y^2), \end{cases}$$

Suppose that σ , μ , and c depend on some parameter $p \in \mathbb{R}$.

We have that $(x, y) = (0, 0)$ is an equilibrium for all choices of p . The Jacobian at $(0, 0)$ has eigenvalues

$$\lambda^{\pm} = \sigma \pm i\omega,$$

When $\sigma(p) = 0$, the real part of the complex conjugate eigenvalues becomes zero: we have a Hopf bifurcation. To check the existence of a limit cycle, we consider the system in polar coordinates:

$$\begin{cases} x = r \cos \theta, \\ y = r \sin \theta, \end{cases} \Rightarrow \begin{cases} r = \sqrt{x^2 + y^2}, \\ \theta = \arctan(y/x). \end{cases}$$

To change the variables, simply note that:

$$\begin{aligned} r dr &= \frac{1}{2} d(r^2) = x dx + y dy \\ &= (\sigma(x^2 + y^2) + c(x^2 + y^2)(x^2 + y^2)) dt = \\ &= (\sigma r^2 + c r^4) dt, \end{aligned}$$

and

$$\begin{aligned} d\theta &= \frac{dy/x - ydx/x^2}{1 + (y/x)^2} = \frac{xdy - ydx}{x^2 + y^2} \\ &= \frac{x(\sigma y - \omega x + cy(x^2 + y^2)) - y(\sigma x + \omega y + cx(x^2 + y^2))}{x^2 + y^2} dt \\ &= -\frac{\omega(x^2 + y^2)}{x^2 + y^2} dt = -\omega dt. \end{aligned}$$

The system in polar coordinates is:

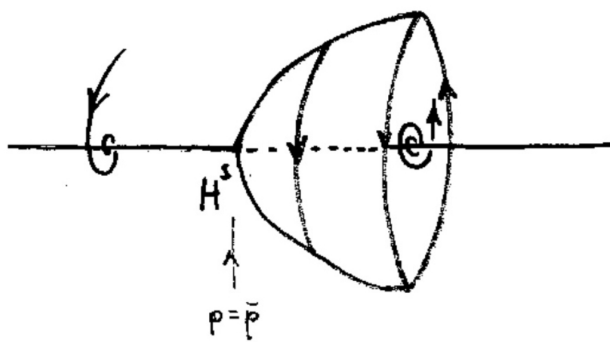
$$\begin{cases} r' = \sigma r + cr^3, \\ \theta' = -\omega. \end{cases}$$

Assuming $\omega(p) \neq 0$, we have two equilibria:

- $r = 0$, which corresponds to the origin and we know it is stable for $\sigma(p) < 0$ and unstable for $\sigma(p) > 0$.
- $r = \bar{r} = \sqrt{-\sigma/c}$, which exists only for $c(p) \neq 0$ and $\sigma(p)c(p) < 0$. When there exists, it is stable if $\sigma(p) > 0$ and unstable if $\sigma(p) < 0$. This corresponds to a limit cycle of radius $\bar{r}$.

In conclusion, we have 2 types of Hopf bifurcations.

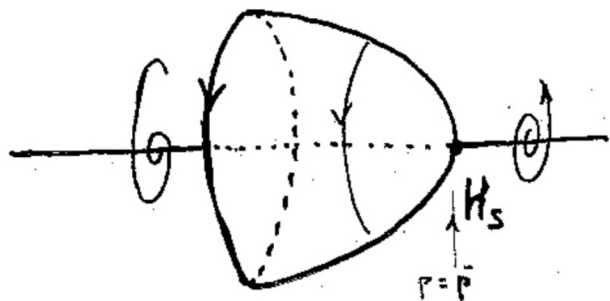
- *Supercritical Hopf bifurcation* H^S : $\sigma'(\bar{p}) > 0$ and $c(\bar{p}) < 0$. The stable equilibrium at the origin becomes unstable and a stable limit cycle is born. The bifurcation is not catastrophic.



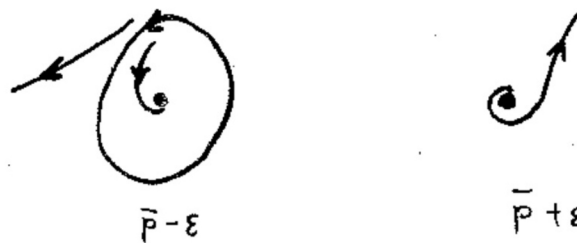
Supercritical Hopf
(non catastrophic)



- *Subcritical Hopf bifurcation* H_S : $\sigma'(\bar{p}) < 0$ and $c(\bar{p}) > 0$. The unstable equilibrium at the origin becomes stable and an unstable limit cycle is born. The bifurcation is catastrophic.



Subcritical Hopf
(catastrophic)



Note that for $c = 0$ at the bifurcation point we have an infinite number of limit cycles (non-isolated periodic orbit), like the classical oscillator. This is a *degenerate Hopf bifurcation* so the condition $c(\bar{p}) \neq 0$ is called *non-degeneracy condition*. The condition $\sigma'(\bar{p}) \neq 0$ is called *transversality condition*.

For a general system of ODEs:

$$\begin{cases} x' = f(x, y, p), \\ y' = g(x, y, p), \end{cases}$$

such that $(\bar{x}, \bar{y}) = (0, 0)$ is an equilibrium (with no loss of generality we can take the equilibrium at the origin), thanks to linearization we can always write the system around the equilibrium in the form:

$$\begin{cases} x' = \sigma(p)x + \omega(p)y + \tilde{f}(x, y, p), \\ y' = \sigma(p)y - \omega(p)x + \tilde{g}(x, y, p), \end{cases}$$

with $\tilde{f}(0, 0, p) = \tilde{g}(0, 0, p) = 0$. It is possible to further expand $\tilde{f}$ and $\tilde{g}$ so to find a term of the form cr^3 as above. The coefficient c is called *first Lyapunov exponent* and it reads:

$$16c = f_{xxx} + f_{xyy} + g_{xxy} + g_{yyy} + \frac{1}{\omega} \left(f_{xy}(f_{xx} + f_{yy}) - g_{xy}(g_{xx} + g_{yy}) - f_{xx}g_{xx} - f_{yy}g_{yy} \right).$$

Then we have the following:

Theorem B.2 (Hopf). *Suppose that $\sigma'(p) \neq 0$ (trasversality) and $c \neq 0$ (non-degeneracy). Then there exists a branch Γ_p of periodic solutions with period $T(p)$ for p such that $|p - \bar{p}|$ is small and $p > \bar{p}$ if $\sigma'(\bar{p})c < 0$ (resp. $p < \bar{p}$ if $\sigma'(\bar{p})c > 0$). Furthermore, $\Gamma_p \rightarrow \bar{y}$ for $p \rightarrow \bar{p}$ and $T(p) \rightarrow \frac{2\pi}{\omega(\bar{p})}$. If $c < 0$ Γ_p are attracting; if $c > 0$ Γ_p are repelling.*

We observed the Hopf bifurcation in Gause-type prey-predator models and in negative feedback networks.

B.8 Limit cycle tangent bifurcation

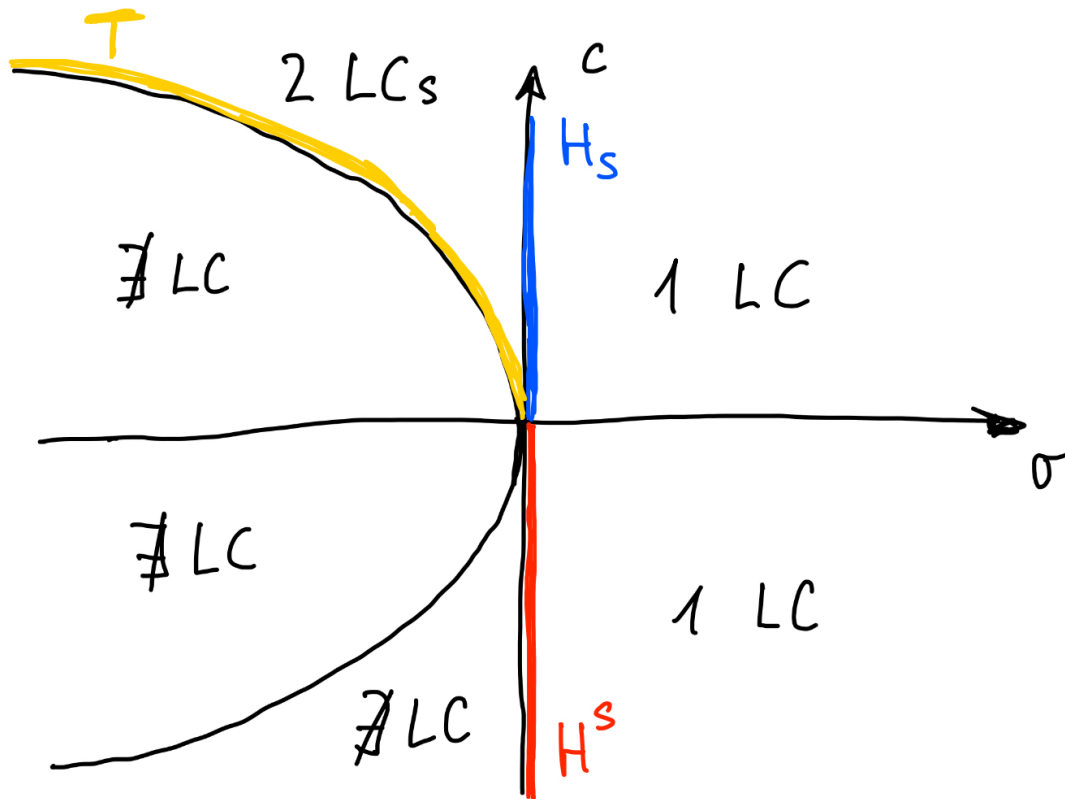
Consider the following model:

$$\begin{cases} r' = r\sigma(1 + cr^2 - r^4), \\ \theta' = -\omega. \end{cases}$$

We have 3 equilibria (note that necessarily $r \geq 0$): the origin $r = 0$ and:

$$\bar{r} = \sqrt{\frac{c \pm \sqrt{c^2 + 4\sigma}}{2}}.$$

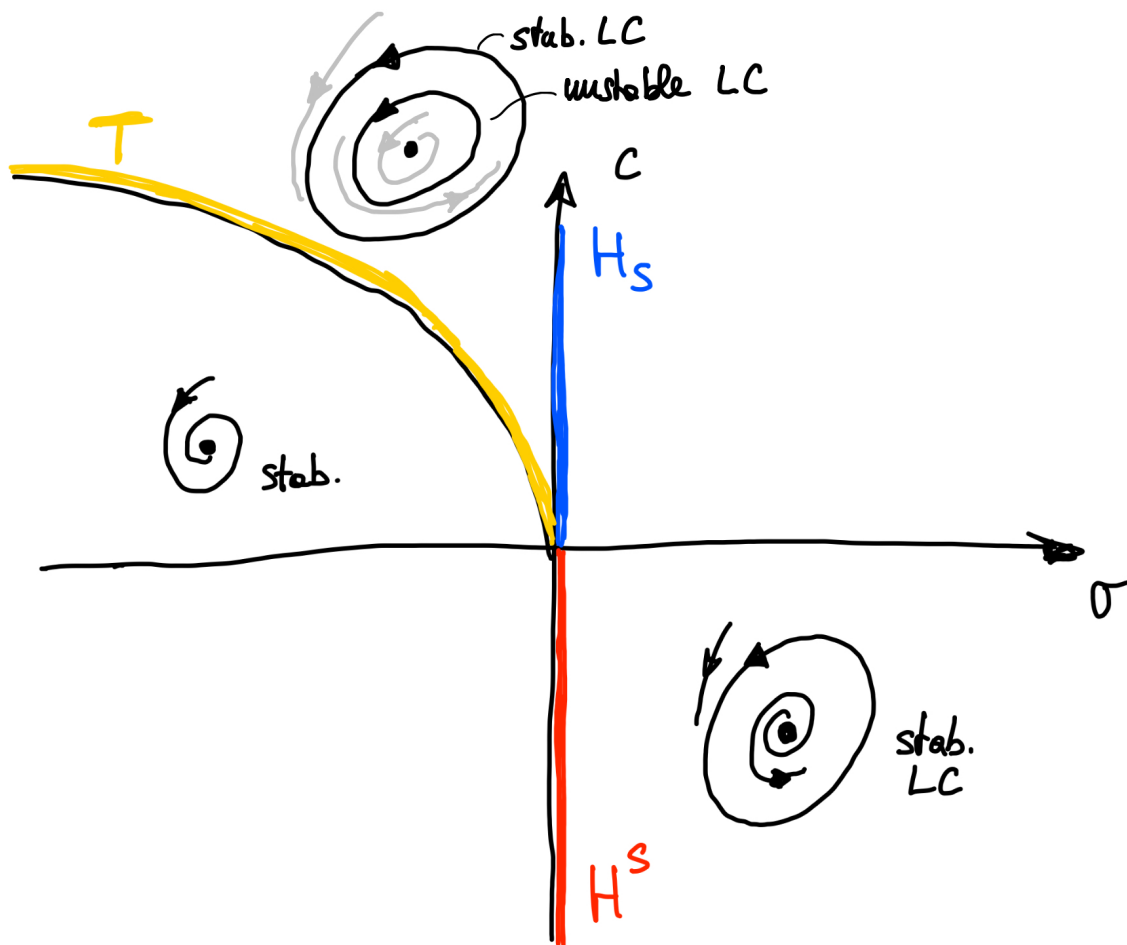
If we study the existence of limit cycles (LCs) in the (σ, c) plane we have:



where:

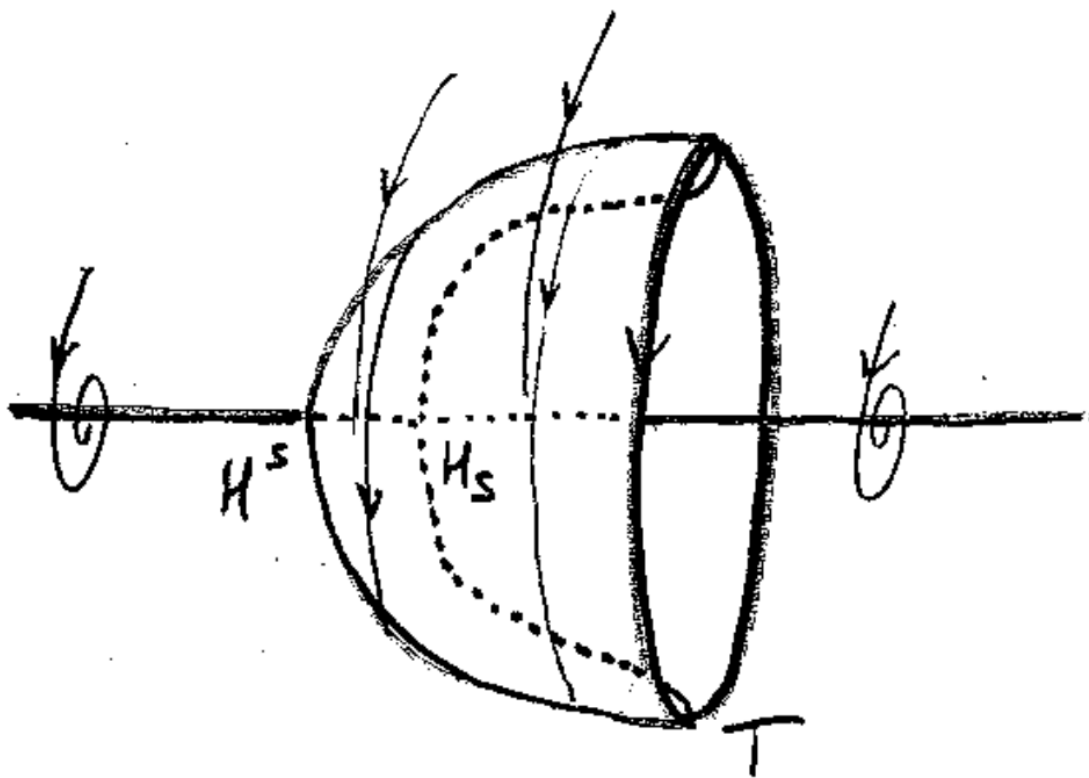
- H^S : supercritical Hopf bifurcation,
- H_S : subcritical Hopf bifurcation,
- T : tangent bifurcation between limit cycles.

In terms of stability we have:



As we cross T we have 2 limit cycles colliding and disappearing, exactly like the tangent bifurcation between equilibria. (In fact, this is what happens in polar coordinates.)

The tangent bifurcation can be visualized as follows:



We have seen this type of bifurcation in a negative feedback loop. It is catastrophic.

C Introduction to MATLAB

C.1 Overview

MATLAB is a software for scientific computing. It is a self-contained environment, with thousands of optimized functions for many tasks. MATLAB is a proprietary software, so you need a license to use it. Most universities offer an educational license (see this [website](#) for UniTrento.)

There are free alternatives too:

- [GNU Octave](#), an open-source software aiming at 1:1 compatibility with Matlab. You can use it, but beware that some MATLAB packages may not work (e.g., [MatCont](#)).
- [Python](#), probably the most popular programming language, has many libraries for scientific computing, such as [numpy](#), [scipy](#), [matplotlib](#), [jax](#), and many others. It would be the best alternative to Matlab, if it weren't for MatCont, that we will extensively use.
- [Julia](#), it is a python competitor, with a similar syntax but much better performance. Many libraries have a Julia wrapper, but again not MatCont.

You are free to use any Matlab alternative, and rely on it only for MatCont.

C.2 Installation

Matlab installation is straightforward on most OSes. Please follow the instruction on the website. We do not need special toolboxes. In case, you can install them later.

You can also use the [online version of MATLAB](#). It comes with an online storage called MATLAB Drive that you can use to sync your files.

C.3 Quick tour

Once open, Matlab will look like this:

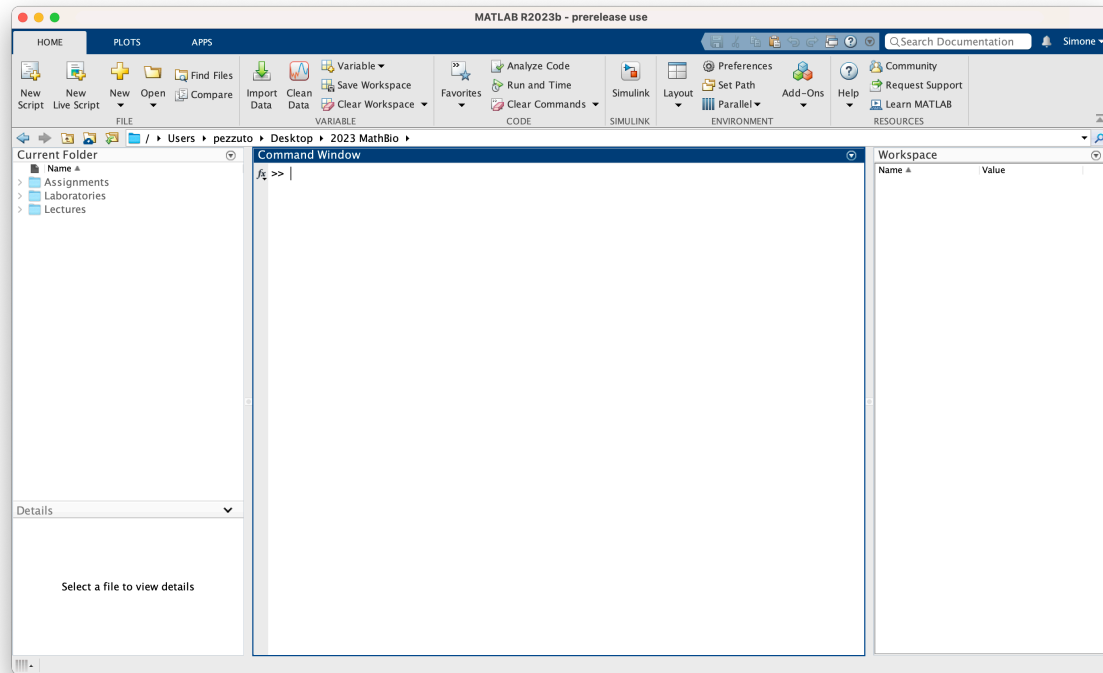


Figure C.1: The MATLAB interface

You can notice different panels:

- **Command Window** is where we type commands.
- **Current Folder** is the current working directory.
- **Workspace** is a summary of all variables.

The system is waiting for an input `>>`.

C.3.1 Basics

MATLAB stands for **MA**trix **LAB**oratory, so its philosophy is to store everything numeric as a matrix. (This is not entirely true anymore, recent versions of Matlab introduced many new types.) A scalar is a 1×1 matrix. Matlab is a *weakly typed* (and interpreted) programming language. A variable is generic, and no type is needed like in C:

```
>> a = 2.45
a =
    2.4500
```

As we input `a = 2.45`, the system responds with its representation. Note that this is not the internal precision! All numbers with a period are stored as `double`. To suppress the output, just put a semicolon:

```
>> a = 2.45;
```

We can execute more commands per line (note the comma):

```
>> a = 2.45 , b = 3.1; A = 1.2;
```

Warning

Matlab is [case-sensitive](#): the variable `a` and `A` are different. This is the standard behavior also for file names in POSIX systems like UNIX, but not for Windows. Also on macOS you need to be careful: by default the file system is case-insensitive like Windows.

If a number is assigned to nothing, it will go to the variable `ans`:

```
>> 2.45;
>> whos
  Name      Size      Bytes  Class  Attributes
  ans       1x1         8    double
>> ans
ans =
    2.4500
```

Some variables are predefined, like `pi`, π , and `1i` or `1j`, the imaginary unit i . Note that also `i` and `j` are valid for Matlab, but dangerous: `i` is often an iteration variable in a for-loop:

```
>> a = 5 + 2*i
a =
    5.0000 + 2.0000i

>> i = 2;
>> b = 5 + 2*i
b =
     9

>> c = 5 + 2*1i
c =
    5.0000 + 2.0000i
```

You can clear a variable by using `clear myvar`, or all variables with `clear`. The command `clc` clears the screen.

If you are not sure of a command, use tab-completion: type the beginning of a command, then type “TAB”. You can also use `help` and `doc`.

C.3.2 Output format

By default, numbers are visualized in short format. The command `format` can change the behavior:

```
>> format long
>> 1/7
ans =
    0.142857142857143
```

Below in the table different options:

Format	ans
format rat	1/7
format short	0.1429
format short e	1.4286e-01
format short g	0.14286
format long	0.142857142857143
format long e	1.428571428571428e-01
format long g	0.142857142857143

C.3.3 Vector and matrices

We have many ways to generate a vector in Matlab. The most direct is:

```
>> b = [1 5 6 7];
>> c = [1, 5, 6, 7];
```

The two vectors are identical: the comma is optional. The dimension is 1×4 , that is 1 row and 4 columns. For column vector, we need to use the semi-colon (now mandatory):


```
>> d = [1; 5; 6; 7];
>> whos
```

Name	Size	Bytes	Class	Attributes
b	1x4	32	double	
c	1x4	32	double	
d	4x1	32	double	

Note that the memory requirement is 32 bytes, that is $4 \cdot 8$ bytes, where 8 bytes is a 64-bit floating-point number (double).

We can generate a vector also as follows:

`<start> : <increment> : <end>`

If we omit the increment, it is assumed one:

```
>> 1:8
ans =
     1     2     3     4     5     6     7     8

>> 1:2:8
ans =
     1     3     5     7

>> 0.5:-0.1:0
ans =
    0.5000    0.4000    0.3000    0.2000    0.1000         0
```

The command above is quite powerful: it also compensates for rounding off errors. Alternatively, we can use `linspace`:

```
>> linspace(0,1,5)
ans =
     0    0.2500    0.5000    0.7500    1.0000
```

Matrices easily follows:

```
>> A = [ 1 2 3 4; 5 6 7 8 ; 9 10 11 12; 13 14 15 16 ]
A =
     1     2     3     4
     5     6     7     8
     9    10    11    12
    13    14    15    16
```

which is returning a 4×4 matrix. More complex matrices can be created with *ad-hoc* functions like `eye`, `ones`, `zeros`, `reshape`, and so on.

```
>> reshape(A, 2, 8)
ans =

     1     9     2    10     3    11     4    12
     5    13     6    14     7    15     8    16

>> B = [ 1:3 0; 4:-1:1 ]
B =

     1     2     3     0
     4     3     2     1

>> eye(2,3)
ans =

     1     0     0
     0     1     0

>> ones(2,2)
ans =

     1     1
     1     1
```

Exercise

Try to create the following matrix:

$$\begin{array}{c}
 \left. \begin{array}{c} n \\ n \end{array} \right\} \left[\begin{array}{cc}
 \overbrace{\begin{array}{cccc} 1 & 1 & \dots & 1 \end{array}}^n & \overbrace{\begin{array}{cccc} 0 & 0 & \dots & 0 \end{array}}^n \\
 \begin{array}{cccc} 1 & 1 & \dots & 1 \\ \vdots & \vdots & & \vdots \\ 1 & 1 & \dots & 1 \end{array} & \begin{array}{cccc} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 0 \end{array} \\
 \begin{array}{cccc} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{array} & \begin{array}{cccc} 1 & 2 & \dots & n \\ n & n & \dots & n \\ \vdots & \vdots & & \vdots \\ n & n & \dots & n \end{array}
 \end{array} \right]
 \end{array}$$

Figure C.2: image

💡 Solution

```
>> n = 3;
>> A = [ones(n), zeros(n); ...
        eye(n), [1:n; n*ones(n-1,n) ];
```

C.3.4 Matrix manipulation

We can access and modify matrices quite naturally:

```
>> v = linspace(0, 1, 5);
>> v(2)
ans =
    0.2500
```

The operator () is much more powerful: we can use slices:

```
>> v([1 3 5])
ans =
    0    0.5000    1.0000

>> v(1:3)
ans =
    0    0.2500    0.5000

>> v(2:end)
ans =
    0.2500    0.5000    0.7500    1.0000

>> v([1 2; 3 2])
ans =
    0    0.2500
    0.5000    0.2500
```

We also can use vectors or matrices of indices. For instance, if indices are [1 3 5], the output is [v(1) v(3) v(5)]. With matrices is similar:

```
>> A = [ 1 2 3 ; 4 5 6 ; 7 8 9]
A =
    1    2    3
    4    5    6
    7    8    9

>> A(1,1)
ans =
    1

>> A(1:3,1:2)
ans =
    1    2
    4    5
    7    8
```

```
>> A(1:end-1,2:end)
ans =
     2     3
     5     6

>> A([1 2],[2 3])
ans =
     2     3
     5     6
```

In the last case we extract the first and the second row, and then crossed with the second and third columns.

```
>> A(1:end)
ans =
     1     4     7     2     5     8     3     6     9

>> A([2 3; 9 6])
ans =
     4     7
     9     8
```

The operation $A(1:end)$ is equivalent to $A(:)$, except for returning a column vector instead of a row vector.

C.3.5 Operations on Matrices

Let's now discuss algebraic operations on matrices. The basic rules are as follows, for two matrices $\mathbf{A} \in \mathbb{C}^{n \times n}$ and $\mathbf{B} \in \mathbb{C}^{p \times q}$:

- Addition is defined if and only if $n = p$ and $m = q$, and $\mathbf{A} + \mathbf{B} \in \mathbb{C}^{n \times m}$.
- Multiplication is defined if and only if $m = p$, and $\mathbf{AB} \in \mathbb{C}^{n \times q}$.
- Transpose $\mathbf{A}^T \in \mathbb{C}^{m \times n}$, with $(\mathbf{A}^T)_{ij} = (\mathbf{A})_{ji}$.
- Conjugate transpose $\mathbf{A}^H \in \mathbb{C}^{m \times n}$, with $(\mathbf{A}^H)_{ij} = \overline{(\mathbf{A})_{ji}}$.

The last two operations can be performed as follows:

```
>> A = [ 1+1i, 3; 1, 1-2i ];
>> A.'
ans =
 1.0000 + 1.0000i 1.0000
```

```

      3.0000      1.0000 - 2.0000i
>> A'
ans =
      1.0000 - 1.0000i      1.0000
      3.0000      1.0000 + 2.0000i

```

If the matrix is real, these two operations are the same. In fact, it is common to use ' as the transpose of a matrix:

```

>> a = [ 1 2 5 ];
>> b = [ 4 6 4 ];
>> H = [ 1 2 3 ; 2 4 7 ; 1 4 3 ];
>> G = [ 4 6 2 ; 8 4 1 ; 3 2 9 ];
>> a + b
ans =
      5      8      9
>> a - b
ans =
     -3     -4      1
>> H + G
ans =
      5      8      5
     10      8      8
      4      6     12
>> H * G
ans =
     29     20     31
     61     42     71
     45     28     33
>> G * a
Error using *
Inner matrix dimensions must agree.

```

This is the most common error. In this case, the vector **a** is 1×3 , while **H** is 3×3 , which is not valid.

```

>> G * a'
ans =
     26
     21
     52

```

```

>> a * G
ans =
    35    24    49
>> 2 * a
ans =
     2     4    10
>> a' * b
ans =
     4     6     4
     8    12     8
    20    30    20
>> a * b'
ans =
    36
>> A^2
ans =
     8    22    26
    17    48    55
    12    30    40
>> a^2
Error using ^
Inputs must be a scalar and a square matrix.
To compute elementwise POWER, use POWER (.^) instead.

```

The last message tells us that exponentiation is intended as matrix multiplication and is only defined for square matrices. It suggests using the dot version instead. In fact, scalar binary operations like multiplication, division, and exponentiation become element-wise operations for vectors and matrices when you put a dot in front of the operation symbol:

```

>> a .* b
ans =
     4    12    20
>> a ./ b
ans =
    0.2500    0.3333    1.2500
>> a .^ 2
ans =
     1     4    25
>> a .^ b
ans =
     1    64   625

```

The only thing to keep in mind is that the two operands should have the same dimensions, or at least one of them should be a scalar (in which case it is repeated).

Exercise

Try to create the following vector, for arbitrary n :

$$\left[\underbrace{1, 1, \dots, 1}_n, \underbrace{2, 2, \dots, 2}_n, \dots, \underbrace{n, n, \dots, n}_n \right].$$

Solution

```
>> n = 5;  
>> v = reshape(ones(n,1)*(1:n),1,[])
```

C.3.6 Elementary Mathematical Functions

One of the strengths of MATLAB is the vast number of mathematical functions (both elementary and advanced) it offers. Unless specified otherwise, almost all of these functions operate element-wise. For example, given a matrix **A**, **exp(A)** is not the matrix exponential (which is calculated with **expm(A)**), but rather a matrix whose elements are the exponentials of the original elements. Here are some elementary mathematical functions:

- **abs(A)**: Absolute value of the elements of **A**
- **sqrt(A)**: Square root of the elements of **A**
- **exp(A)**: Exponential function applied to the elements of **A**
- **log(A)**: Natural logarithm of the elements of **A**
- **log10(A)**: Base 10 logarithm of the elements of **A**
- **log2(A)**: Base 2 logarithm of the elements of **A**
- **sin(A)**: Sine of the elements of **A**
- **cos(A)**: Cosine of the elements of **A**
- **tan(A)**: Tangent of the elements of **A**
- **asin(A)**: Arcsine of the elements of **A** (in radians)
- **acos(A)**: Arccosine of the elements of **A** (in radians)
- **atan(A)**: Arctangent of the elements of **A** (in radians)
- **sinh(A)**: Hyperbolic sine of the elements of **A**
- **cosh(A)**: Hyperbolic cosine of the elements of **A**
- **tanh(A)**: Hyperbolic tangent of the elements of **A**

Exercise

1. Compute $\frac{e^5 + \sin(\pi)}{\sqrt{\log_2 30} - 10}$ and $e^{\log_{10} 50} + e^{\log 30} + e^{\log_2 40}$.
2. Define $\mathbf{x} = [1, 3, 4]$ and $\mathbf{y} = [1, 1, 2]$. Compute $2x_i \log_2(|y_i| + 1) - y_i \log_{10}(x_i + 2)$ and $\arctan\left(\frac{x_i}{y_i}\right) - \sin^2\left(x_i \sqrt[3]{|y_i|^2}\right)$.

Solution

```
>> % Point 1
>> (exp(5) + sin(pi))/(sqrt(log2(30))-10)
>> exp(log10(50)) + exp(log(30)) + exp(log2(40))
>> % Point 2
>> x = [ 1 3 4 ];
>> y = [ 1 1 2 ];
>> 2*x.*log2(abs(y)+1) - y.*log10(x+2)
>> atan(x./y) - sin(x.*abs(y).^(2/3)).^2
```

C.3.7 Elementary Mathematical Functions (Continued)

Let $\mathbf{A}$, $\mathbf{B}$ be two matrices, and $\mathbf{b}$ be a vector.

- `size(A)`: Returns a two-element vector, where the first element is the number of rows in $\mathbf{A}$, and the second element is the number of columns.
- `size(A, 1)`: Returns the first element of `size(A)`, which is the number of rows.
- `size(A, 2)`: Returns the second element of `size(A)`, which is the number of columns.
- `length(b)`: Returns the number of elements in the vector $\mathbf{b}$.
- `max(b)`: Returns the largest element in the vector $\mathbf{b}$.
- `min(b)`: Returns the smallest element in the vector $\mathbf{b}$.
- `max(A)`: Returns a row vector containing the maximum element of each column of $\mathbf{A}$.
- `min(A)`: Returns a row vector containing the minimum element of each column of $\mathbf{A}$.
- `max(A, B)`: Returns a matrix of the same dimensions as $\mathbf{A}$ and $\mathbf{B}$, containing the element-wise maximum.
- `min(A, B)`: Returns a matrix of the same dimensions as $\mathbf{A}$ and $\mathbf{B}$, containing the element-wise minimum.
- `max(A, [], 2)`: Returns a column vector containing the maximum element of each row of $\mathbf{A}$. If you replace 2 with 1, you get `max(A)`.
- `min(A, [], 2)`: Returns a column vector containing the minimum element of each row of $\mathbf{A}$. If you replace 2 with 1, you get `min(A)`.

- `sum(b)`: Returns a scalar equal to the sum of the elements in the vector `b`.
- `sum(A)`: Returns a row vector whose elements are the column-wise sum of the elements in matrix `A`.
- `sum(A, 2)`: Returns a column vector whose elements are the row-wise sum of the elements in matrix `A`.
- `diag(A)`: Returns the vector of the diagonal elements of matrix `A`.
- `diag(A, k)`: Returns the vector of the k -th super-diagonal of matrix `A`.
- `diag(A, -k)`: Returns the vector of the k -th sub-diagonal of matrix `A`.
- `diag(b)`: Returns a matrix with the elements of vector `b` on its diagonal.
- `diag(b, k)`: Returns a matrix with the elements of vector `b` on its k -th super-diagonal.
- `diag(b, -k)`: Returns a matrix with the elements of vector `b` on its k -th sub-diagonal.
- `tril(A)`: Returns the lower triangular part of matrix `A`, making all elements strictly above the diagonal zero (even in a rectangular matrix).
- `triu(A)`: Returns the upper triangular part of matrix `A`, making all elements strictly below the diagonal zero (even in a rectangular matrix).

There are many other important functions such as `det` (determinant of a square matrix), `trace` (trace of a matrix), `norm` (for calculating norms), and so on. We will introduce these functions as needed.

Exercise

Try to use the function `magic(n)`. Then sum by row, column, diagonal, etc. What do you observe?

Solution

```
>> n = 4;
>> A = magic(n);
>> sum(A, 2)
>> sum(A, 1)
>> sum(diag(A))
```

We notice that the sum is always 34, at least for $n = 4$. In general, the magic square contains numbers from 1 to n^2 , and the magic number is $\frac{1}{2}n(n^2 + 1)$. This is true even if we sum over the antidiagonal

```
>> sum(diag(fliplr(A)))
```

The function `fliplr` (*flip left-right*) inverts the right with the left in a given matrix.

C.3.8 Functions and Scripts

Many of the commands we have introduced are defined as functions, which are procedures that take input data and provide output. In MATLAB, there are several ways to define a function.

Suppose we want to create a function `fun(x)` that returns the value of $f(x) = x \sin(x) + \cos^2(x)$ for a given x . The “legacy” method involves defining a string corresponding to the function and then evaluating it using the `eval` command:

```
>> fun = 'x*sin(x) + (cos(x))^2';  
>> x = 1.0;  
>> eval(fun)  
ans =  
    1.1334
```

However, there is an issue when x is a vector:

```
>> x = [ 1 2 3 ];  
>> eval(fun)  
Error using *  
Inner matrix dimensions must agree.
```

To handle vector input correctly, we need to “vectorize” the expression by using element-wise operations:

```
>> fun = 'x.*sin(x) + cos(x).^2';  
>> x = [ 1 2 3 ];  
>> eval(fun)  
ans =  
    1.1334    1.9918    1.4034
```

Alternatively, you can use the `vectorize` command to transform the function into its vectorized version.

However, a more convenient way is to define functions using “anonymous functions”:

```
>> fun = @(x) x.*sin(x) + cos(x).^2;  
>> x = [ 1 2 3 ];  
>> fun(x)  
ans =  
    1.1334    1.9918    1.4034
```

The `@` symbol is called a “function handle.” Note that the `vectorize` command does not work with anonymous functions.

There are at least two other methods for defining functions, such as using `inline` and the Symbolic Math Toolbox. We won’t delve into these methods here.

Instead, let’s explore using scripts to define functions. An “script” is a text file with a `.m` extension containing a list of commands that MATLAB will execute when the file is called. You can edit the script using any text editor or with MATLAB’s built-in editor by running the following command:

```
>> edit
```

Suppose, for example, that the file `myscript.m` contains the following:

```
clear; clc;
% This is a comment:
% sum of squares of the diagonal
% of the magic matrix
n = 4;
A = magic(n);
summa = sum(diag(A).^2)
```

If the file is in the working directory, you can run it as follows:

```
>> myscript
summa =
    414
```

The script is executed as if you had entered the commands directly in the terminal.

You can also have scripts that contain only a function with the following syntax:

```
function y = myfun(x)
    y = x.*sin(x) + (cos(x)).^2;
end
```

In this case, you must save the file with the same name as the function (e.g., `myfun.m`).

Now you can use the function:

```
>> x = [ 1 2 3 ];
>> myfun(x)
ans =
    1.1334    1.9918    1.4034
```

Function scripts can have multiple arguments and return multiple outputs:

```
function [A, b] = vecmat(n, m)
    % Generate a matrix A of size m x n and
    % a vector b of size n x 1 with random values.
    A = rand(n, m);
    b = rand(n, 1);
end
```

You can call this function as follows:

```
>> [K, f] = vecmat(2, 3)
K =
    0.4387    0.7655    0.1869
    0.3816    0.7952    0.4898
f =
    0.4456
    0.6463
```

Script functions can contain at most one function. It is also a good practice to use lowercase names without spaces. You can use functions with multiple arguments and return values as needed.

There are many other advanced features and techniques for creating and using functions in MATLAB, but these are the basics to get you started.

C.3.9 Loops and Control Structures

MATLAB is a full-fledged interpreted programming language, and it provides many control flow operations.

```
>> for i = 1:3, magic(i), end;
ans =
     1
ans =
     1     3
     4     2
ans =
     8     1     6
     3     5     7
     4     9     2
```

In the example above, a `for` loop iterates from 1 to 3, and for each iteration, the `magic` function is called.

```
>> n = 1; while n < 4, magic(n), n = n + 1; end;
ans =
     1
ans =
     1     3
     4     2
ans =
     8     1     6
     3     5     7
     4     9     2
```

In this example, a `while` loop is used. It continues to execute as long as the condition `n < 4` is true. Inside the loop, the `magic` function is called, and `n` is incremented.

```
>> if rand(1) > 0.5, disp('Greater than 0.5'); else, disp('Less than 0.5'); end;
Greater than 0.5
```

The `if` statement is used to perform conditional execution. In this example, it checks if a random number is greater than 0.5 and displays a message accordingly.

As we need more details, we will add them gradually. Typically, writing these control structures directly in the command line can be cumbersome and less readable. It's preferable to write everything in a script with proper indentation and execute the script as needed.

i Exercise

- Write a function named `cubediag` that takes a matrix `A` as input and returns the sum of the cubes of the diagonal elements.
- Write another function named `cubeantidiag` that returns the sum of the cubes of the anti-diagonal elements (the diagonal that is symmetric to the main diagonal).

You can define these functions in separate script files or directly in the MATLAB environment.

Solution

```
function s = cubediag(A)
    s = sum(diag(A).^2);
end

function s = cubeantidiag(A)
    % we could also use fliplr
    Aflip = A(:,end:-1:1);
    s = cubediag(Aflip);
end

function s = cubeantidiag(A)
    % case of square matrices
    n = size(A,1);
    s = sum(A(end-n+1:1-n:n));
end
```

C.3.10 Plots

To create a plot of a real-valued function of a real variable in MATLAB, you can use the `plot` command. In its basic form, you provide two vectors, `x` and `y`, of the same size that represent the data points.

```
>> x = linspace(-1, 1, 100);
>> parab = @(x) x.^2;
>> plot(x, parab(x));
```

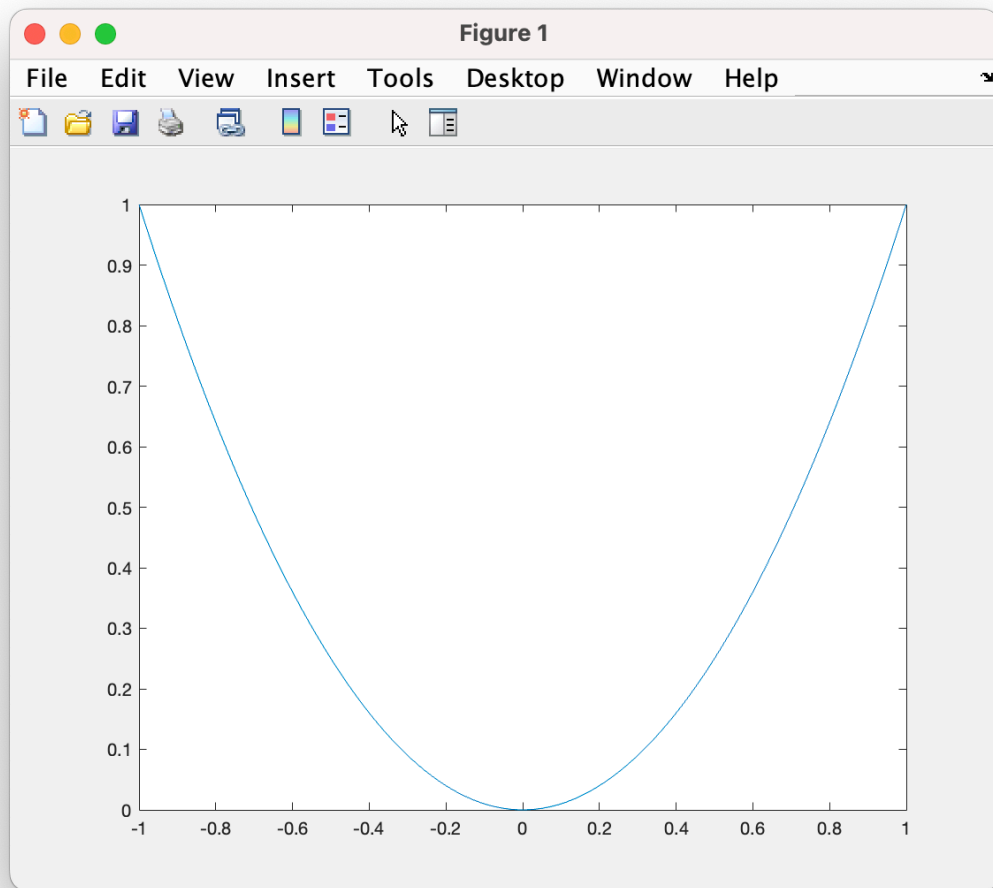


Figure C.3: Parabola Plot

You can customize the style, size, and color of the line using various options typically provided after the data points:

```
>> x = linspace(-1, 1, 20);  
>> plot(x, parab(x), 'r*--', 'LineWidth', 2.0, 'MarkerSize', 15.0);  
>> grid on;  
>> xlabel('X-Axis');  
>> ylabel('Y-Axis');  
>> title('A Parabola', 'FontWeight', 'bold');
```

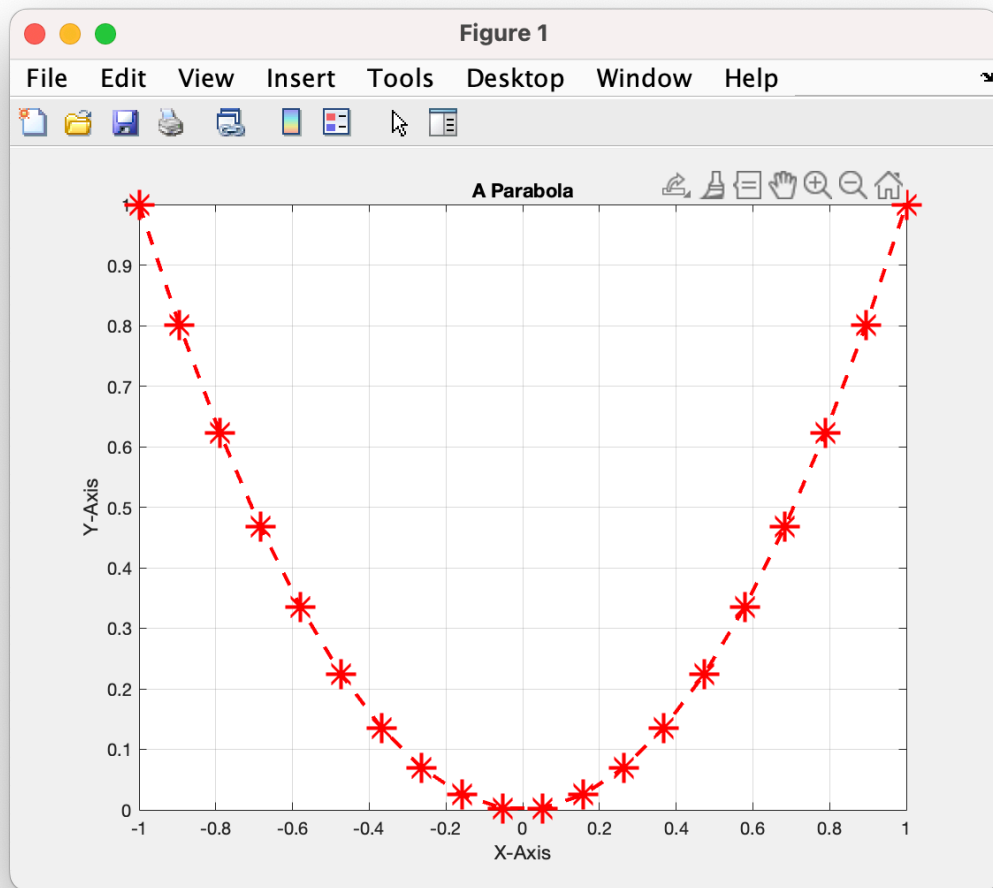



Figure C.4: Styled Parabola Plot

To see the possible combinations of line styles and markers, you can type `help plot` or `doc plot`. However, the most commonly used is the third argument, which is a string like `r*--`, which translates to:

- `r`: Red line
- `*`: Asterisk-shaped markers
- `--`: Dashed line

The order of styles is fixed, but each of them is optional. For example, by simply typing `y`, you will get a yellow line.

To overlay multiple plots, you can follow several approaches:

1. Direct Overlay:

```
>> x = linspace(-1, 1, 20);  
>> parab = @(x) x.^2;  
>> cubic = @(x) x.^3;  
>> plot(x, parab(x), x, cubic(x));
```

2. Creating a Data Matrix:

```
>> plot(x, [parab(x); cubic(x)]);
```

3. Using the hold Command:

```
>> plot(x, parab(x));  
>> hold all;  
>> plot(x, cubic(x));  
>> hold off;
```

You can also add a legend to distinguish the plots:

```
>> legend('Parabola', 'Cubic');
```

You can add multiple plots to the same graph by using the hold command:

```
>> plot(x, parab(x));  
>> hold on;  
>> plot(x, cubic(x));  
>> hold off;
```

You can add a legend to differentiate between the plots:

```
>> legend('Parabola', 'Cubic');
```

Another important type of plot, especially when evaluating the behavior of errors concerning parameters, is the use of logarithmic or semilogarithmic scales. For a given function $f(x)$ with the graph $y = f(x)$:

- **semilogx** represents the points after changing the variable $x \mapsto \log x$.
- **semilogy** represents the points after changing the variable $y \mapsto \log y$.
- **loglog** performs both of the above changes.

For example, the graph $y = e^{\alpha x}$, in a y -logarithmic scale, becomes $\tilde{y} := \log y = \alpha x$, which is a straight line.

```
>> x = linspace(1, 10, 100);  
>> semilogy(x, exp(x), x, exp(2*x), 'LineWidth', 2.0);  
>> grid on;  
>> legend('Slope 1', 'Slope 2', 'Location', 'NorthWest');
```

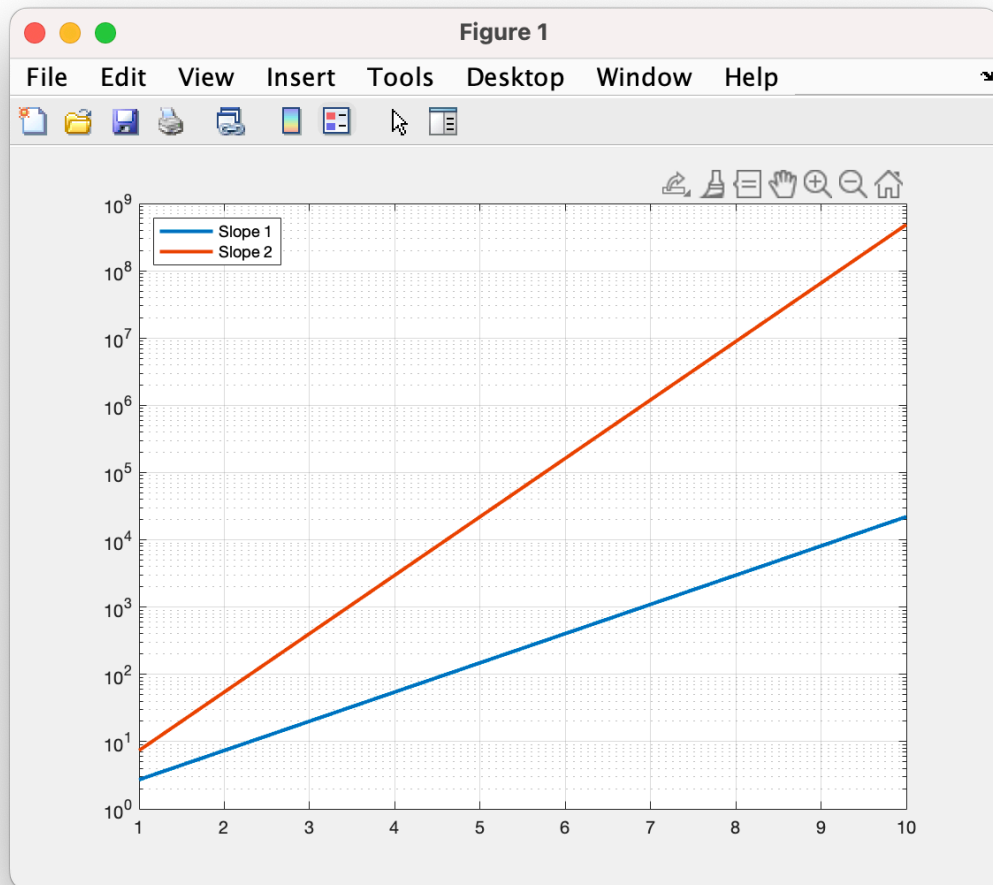


Figure C.5: Logarithmic Plot

i Exercise

1. Plot the graph of the function $f(x) = 2 + (x-3) \sin(5(x-3))$ for $0 \leq x \leq 6$. Overlay dashed lines that bound this function.
2. Consider the function $f(x) = (\log x)^2$ for $0.1 \leq x \leq 10$. What do you expect from the graph in logarithmic scaling? Plot and verify.

💡 Solution

```
>> x = linspace(0, 6, 100);
>> f = 2 + (x-3).*sin(5*(x-3));
>> r = [ 2 + (x-3); 2 - (x-3) ];
>> plot(x, f, 'k-', x, r, 'k--', 'LineWidth', 2.0);
```

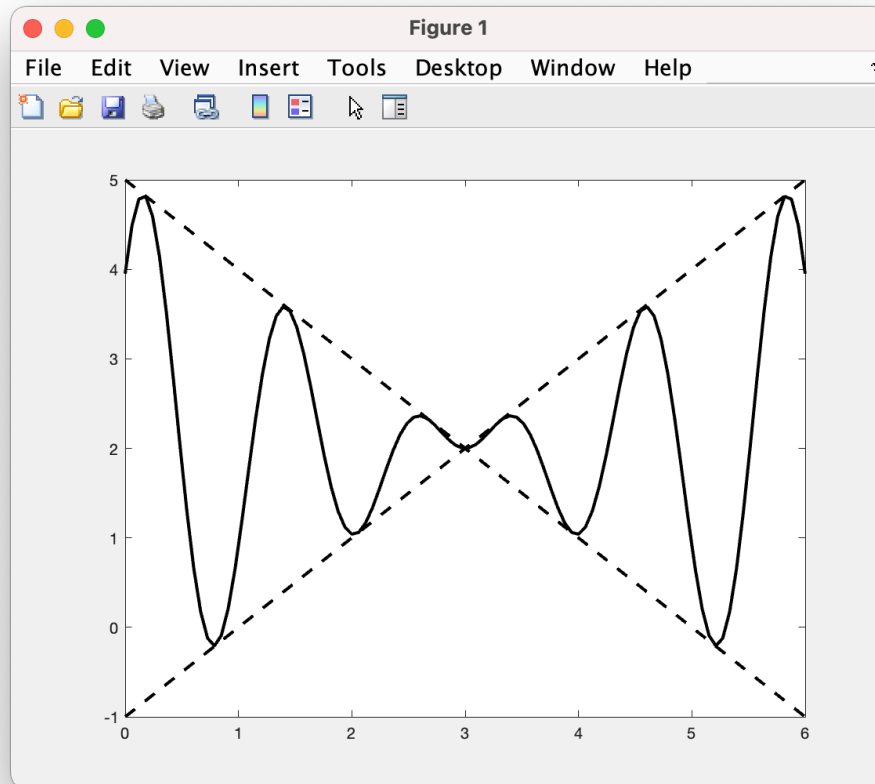
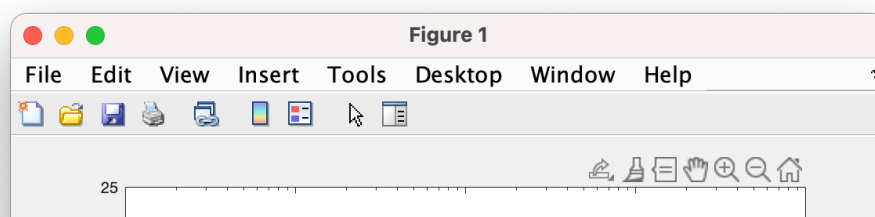


Figure C.6: image

Given $\hat{x} = \log x$ and $\hat{y} = \log y$, the plot in x -log scale is $y = (\log 2)^2 = \hat{x}^2$. We expect a parabola.

```
>> x = 10.^linspace(-2,2,100);
>> y = log(x).^2;
>> semilogx(x, y, 'LineWidth', 2.0);
```



C.4 Approximation error

When working with approximations, it's essential to measure the accuracy of an approximation $\hat{x}$ to a number $x \in \mathbb{R}$. Two common metrics are:

- Absolute Error: $E_{\text{abs}}(\hat{x}) = |x - \hat{x}|$
- Relative Error: $E_{\text{rel}}(\hat{x}) = \frac{|x - \hat{x}|}{|x|}$ for $x \neq 0$

Significant figures are a way to represent the accuracy of an approximation. Given an approximation $\hat{x}$ to x , it has p significant figures if:

$$E_{\text{abs}}(\hat{x}) \leq \frac{1}{2} \times 10^{s-p+1}$$

Where s is the largest integer such that $|x| \geq 10^s$.

When x is not known (common in many practical applications), significant figures can be determined by considering successive approximations. For a sequence $\{x_i\}_{i=0}^{\infty}$ that converges to x , you can calculate the absolute error at each step $|x_i - x_{i-1}|$.

Exercise

Complete the following table (use `format short` for relative error):

x	$\hat{x}$	Relative Error	Absolute Error	Significant Figures
1.6925	1.69285	—	—	—
23.130	23.129	—	—	—
23.130	23.1299	—	—	—
23.130	23.129999	—	—	—
0.00345	0.00343	—	—	—
0.01008	0.01012	—	—	—
0.01008	0.01002	—	—	—
0.01008	0.0102	—	—	—

💡 Solution

x	$\hat{x}$	Relative Error	Absolute Error	Significant Fig- ures
1.6925	1.69285	$0.21 \cdot 10^{-3}$	$0.4 \cdot 10^{-3}$	4
23.130	23.129	$0.43 \cdot 10^{-4}$	$0.1 \cdot 10^{-2}$	4
23.130	23.1299	$0.43 \cdot 10^{-5}$	$0.1 \cdot 10^{-3}$	5
23.130	23.129999	$0.43 \cdot 10^{-7}$	$0.1 \cdot 10^{-5}$	7
0.00345	0.00343	$0.58 \cdot 10^{-2}$	$0.1 \cdot 10^{-4}$	2
0.01008	0.01012	$0.40 \cdot 10^{-2}$	$0.4 \cdot 10^{-4}$	3
0.01008	0.01002	$0.60 \cdot 10^{-2}$	$0.6 \cdot 10^{-4}$	2
0.01008	0.0102	$0.12 \cdot 10^{-1}$	$0.1 \cdot 10^{-3}$	1

C.4.1 Floating-Point Arithmetic

Natural numbers ($\mathbb{N}$), although infinite, can be represented exactly on a computer within a predefined range of values. For example, a 32-bit integer can represent numbers from 0 to $2^{32} - 1$. Similarly, it can represent signed integers ($\mathbb{Z}$) from -2^{31} to $2^{31} - 1$.

Real numbers ($\mathbb{R}$), on the other hand, are too numerous to be represented exactly within any chosen range. There are at least two possibilities: the first involves fixing the number of digits after the decimal point (fixed-point). The second considers the proper subset $\mathbb{F} \subset \mathbb{R}$ of floating-point numbers:

$$y = \pm m \times \beta^{e-t},$$

where β is the base, t is the precision, and $e \in [e_{\min}, e_{\max}]$ is the exponent. For example, with $\beta = 2$, $t = 3$, $e_{\min} = -1$, and $e_{\max} = 3$, you can represent numbers like:

0, 0.25, 0.3125, 0.3750, 0.4375, 0.5, 0.625, 0.750, 0.875,
1.0, 1.25, 1.50, 1.75, 2.0, 2.5, 3.0, 4.0, 5.0, 6.0, 7.0.

The following graph illustrates the spacing of these numbers:

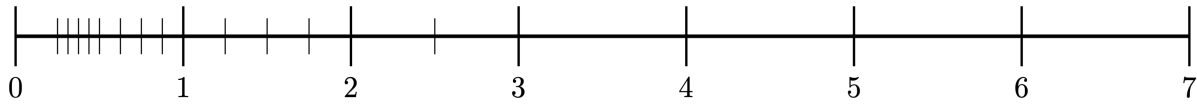


Figure C.8: Graphical representation of floats

The IEEE-754 standard sets values for these parameters in two significant cases: single precision (`float` in C), where $\beta = 2$, $t = 24$, $e_{\min} = -125$, and $e_{\max} = 128$, and double precision (`double` in C), where $\beta = 2$, $t = 53$, $e_{\min} = -1021$, and $e_{\max} = 1024$.

Machine epsilon (ε_M) is the smallest positive number such that $1 + \varepsilon_M \neq 1$ in the floating-point system. It represents the distance between 1 and the next representable number in the system. For single precision, ε_M is approximately 1.19×10^{-7} , while for double precision, it's approximately 2.22×10^{-16} .

In MATLAB, you can check these values using the `eps` function:

```
>> eps
ans =
    2.2204e-16
```

The limits of representable numbers can also be checked with `realmin` and `realmax`:

```
>> realmax
ans =
    1.7977e+308
>> realmin
ans =
    2.2251e-308
```

As observed from the previous graph, the spacing between numbers in $\mathbb{F}$ is not constant. MATLAB's `eps` command allows you to determine these spacings:

```
>> eps(1)
ans =
    2.2204e-16
>> eps(10)
ans =
    1.7764e-15
>> eps(100)
ans =
    1.4211e-14
>> eps(1000)
```



```
ans =  
1.1369e-13
```

The set $\mathbb{F}$ also includes some exceptional cases:

```
>> 1e400  
ans =  
    Inf  
>> 1e-400  
ans =  
    0  
>> 1/0  
ans =  
    Inf  
>> 0/0  
ans =  
    NaN
```

- The number `Inf` represents infinity, meaning it's beyond `realmax`. It's not considered an error, and arithmetic operations involving infinity are still valid, although they can often lead to undesirable results.
- When dealing with numbers smaller than `realmin`, they are approximated to 0.
- The situation `NaN` (Not-a-Number) indicates that the result is undefined or has no meaningful value.

Floating-point arithmetic has its limitations, including a lack of associativity for some operations, which can lead to significant errors in some cases. For example, when subtracting two nearly equal numbers, you can experience loss of precision due to the finite precision of the representation:

```
>> x = 1.0e-15;  
>> (1+x)-1  
ans =  
1.1102e-15  
>> (1-1)+x  
ans =  
1.0000e-15
```

This issue is known as **cancellation error** and should be avoided whenever possible in numerical computations. The problem is indeed quite serious, and it's a common issue in numerical computations, especially when dealing with small numbers. For example, you have two functions:

$$f(x) = \frac{1 - \cos x}{x^2}, \quad g(x) = \frac{1}{2} \left(\frac{\sin(x/2)}{x/2} \right)^2.$$

Both of these functions are identical. However, when you evaluate them for $x = 1.2 \times 10^{-8}$ in MATLAB, you get different results:

```
>> x = 1.2e-8;
>> (1 - cos(x)) / x^2;
ans =
    0.7710

>> 0.5 * (sin(x / 2) / (x / 2))^2;
ans =
    0.5000
```

The first result is clearly incorrect, and it violates the bounds of the function, which should be between 0 and 0.5. This is a classic example of the problem of numerical precision and the limitations of finite-precision arithmetic. Small values of x lead to significant errors due to rounding and truncation.

To address such issues, numerical analysts often use techniques like Taylor series expansions, higher precision arithmetic, or specialized algorithms to improve the accuracy of computations involving small numbers.

It is good practice to avoid cases like these as much as possible, as they are quite common. For example:

$$\begin{aligned} f(x) &= \frac{e^x - 1}{x}, & \text{for } x \approx 1, \\ x_{1,2} &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, & \text{for } b^2 \approx 4ac \text{ and } b \approx \sqrt{b^2 - 4ac}, \\ s_n^2 &= \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2 \right) \end{aligned}$$

In the last case, when calculating variance, it is even possible to obtain negative results, which have no mathematical sense.

i Exercise (Archimedes' Method for π Approximation)

The strategy used by Archimedes to approximate π involves considering regular polygons inscribed and circumscribed within the unit circle. In fact, if n is the number of sides, the perimeter P_n approaches 2π as $n \rightarrow \infty$.

Starting with a hexagon and successively doubling the number of sides, it can be found that as $i \rightarrow \infty$, $6 \cdot 2^i \cdot t_i$ approaches π , where t_i satisfies the following relation:

$$t_0 = \frac{1}{\sqrt{3}}, \quad t_{i+1} = \frac{\sqrt{t_i^2 + 1} - 1}{t_i}.$$

Here are the tasks related to this approximation:

1. Implement this algorithm in MATLAB and compare the approximation with π provided by the `pi` constant. What do you observe?
2. Replace the recursive formula with its equivalent:

$$t_0 = \frac{1}{\sqrt{3}}, \quad t_{i+1} = \frac{t_i}{\sqrt{t_i^2 + 1} + 1}.$$

Comment on any differences observed.

Note: Use `format long` to display the differences.

Solution

The value given by Matlab for π is:

```
>> format long
>> pi
ans =
    3.141592653589793
```

We compute now π with method 1. pia and method 2. pib:

```
>> format long;
>> n = 30;
>> ta = 1/sqrt(3);
>> tb = 1/sqrt(3);
>> for i = 1:n
    % first method
    ta = (sqrt(ta^2+1)-1)/ta;
    pia = 6*2^i*ta;
    % second method
    tb = tb/(sqrt(tb^2+1)+1);
    pib = 6*2^i*tb;
>> end
>> fprintf('Method 1, pi = %f, err = %e\n', pia, abs(pia-pi));
>> fprintf('Method 2, pi = %f, err = %e\n', pib, abs(pib-pi));
```

We see that the first method suffers of cancellation errors.

Exercise (Polynomial Expansion)

Consider the polynomial $p(x) = (x - 1)^7$ for $x \in [0.998, 1.012]$. Expand the polynomial using the binomial formula and compare it to the original unexpanded polynomial. Comment on any differences.

💡 Solution

Expanding we have

$$p(x) = x^7 - 7x^6 + 21x^5 - 35x^4 + 35x^3 - 21x^2 + 7x - 1.$$

Now we compare

```
>> x = 0.998:0.0001:1.012;  
>> plot(x, x.^7-7*x.^6+21*x.^5-35*x.^4+35*x.^3-21*x.^2+7*x-1, ...  
        x, (x-1).^7, 'LineWidth', 2.0);  
>> grid on;  
>> legend('Expanded', 'Original');
```

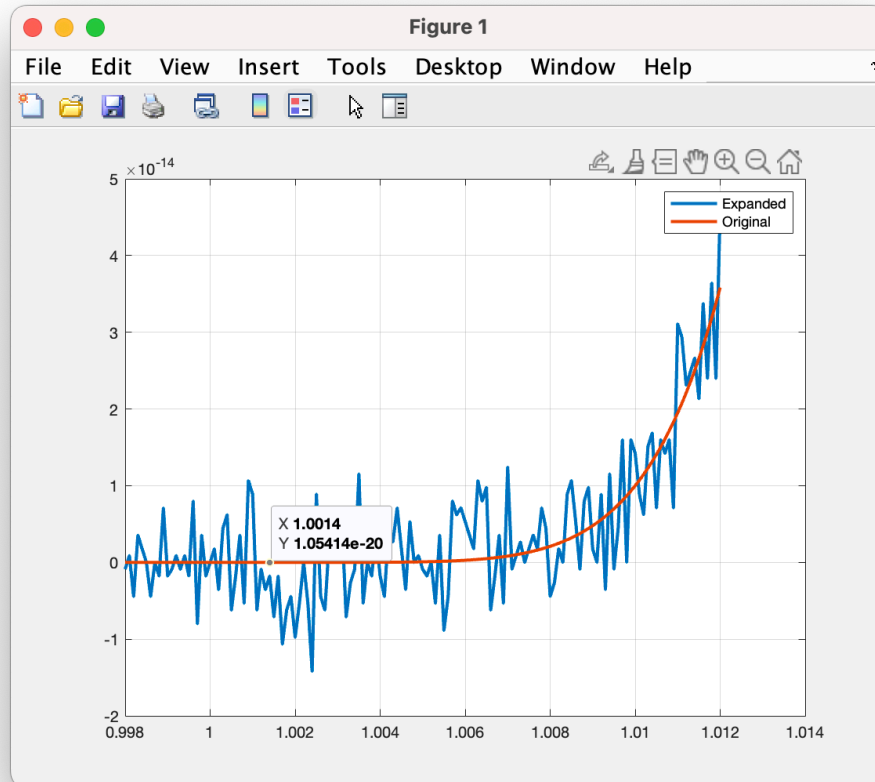


Figure C.9: image